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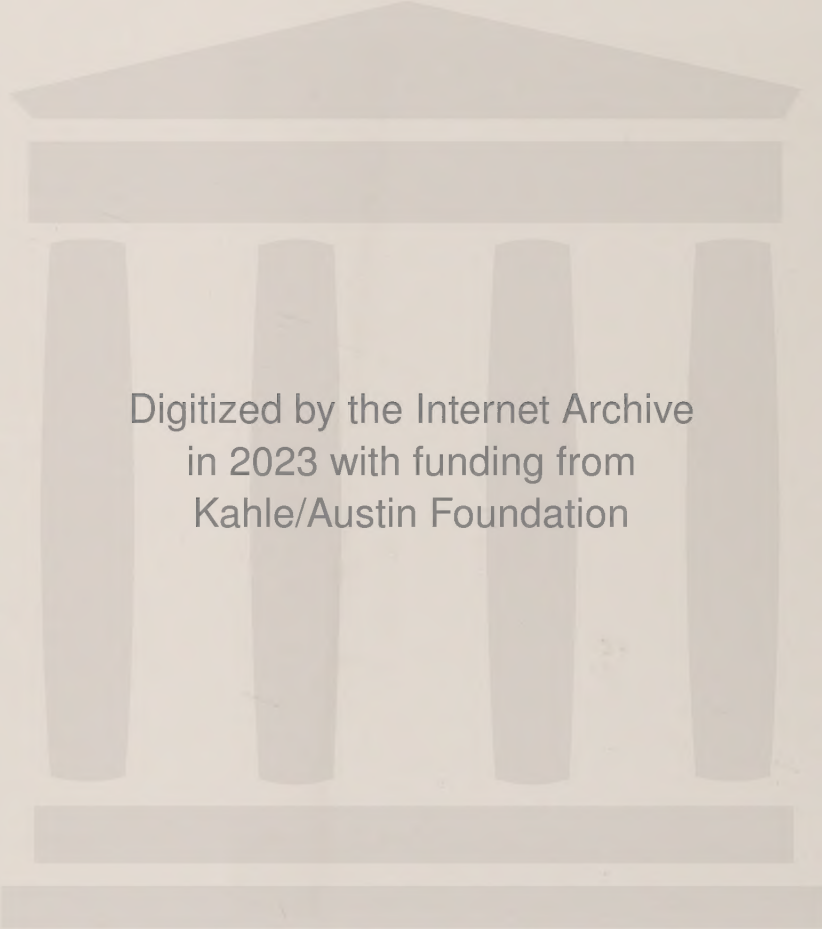
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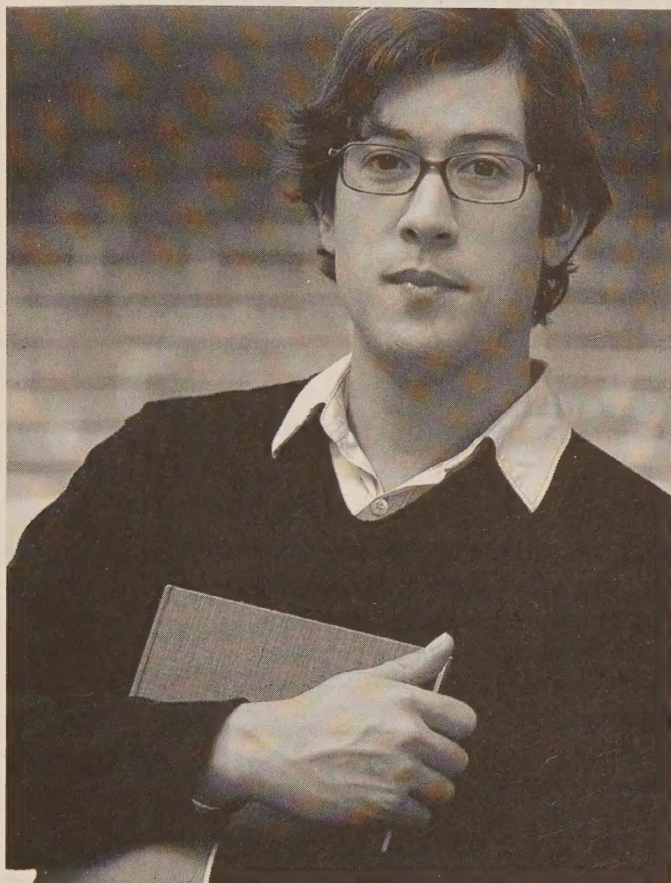
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GRE[®] MATHEMATICS TEST



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GRE® Mathematics Test

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ABOUT RESEARCH & EDUCATION ASSOCIATION

Founded in 1959, Research & Education Association is dedicated to publishing the finest and most effective educational materials—including software, study guides, and test preps—for students in middle school, high school, college, graduate school, and beyond.

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The Graduate Record Examination in MATHEMATICS

Introduction

ABOUT THIS BOOK

This book provides an intensive, complete review of all the math essentials you need to know to master the GRE Mathematics Subject Test. It is our intention to provide practice that is somewhat more demanding than the test itself, so that you can be confident come Test Day. Our goal is to help the student reach a high level of math competence. Inside this book you'll find six full-length REA practice tests based on the format of the most recently administered GRE test, but even more challenging. Like the GRE test itself, each of our model tests lasts two hours and 50 minutes. Each one includes every type of question you can expect to encounter on the actual exam. We also give you an answer key complete with detailed explanations that spell out how the correct answers were arrived at. By completing all six practice exams and studying the explanations that follow, you can discover your strengths and weaknesses and thereby become well prepared for the actual exam.

To help you brush up on the subject matter, the book also features 281 pages of topical reviews, summarizing the essentials of the various topics covered on the GRE Mathematics test. Again, our approach is to cover all the ground necessary to fully frame the test. Our reviews are designed to work as an at-a-glance overview of the many topics the test covers. Should you require more detailed coverage, you'll want to consult a textbook whose contents parallel the topical coverage of the GRE.

ABOUT THE TEST

The GRE Mathematics Test is offered three times a year by Educational Testing Service, under the direction of the Graduate Record Examinations Board. Applicants for graduate school submit GRE test results together with other undergraduate records as part of a highly competitive admission process.

The questions on the test are composed by a committee of specialists who are selected from various undergraduate and graduate faculties across the U.S. This test consists of 66 multiple-choice questions. Some questions are grouped together under a particular diagram and/or graph. Emphasis is placed on these areas:

I. Calculus (50%)

Material learned in the usual sequence of elementary calculus courses—differential and integral calculus of one and of several variables—includes calculus-based applications and connections with coordinate geometry, trigonometry, differential equations, and other branches of mathematics.

II. Algebra (25%)

A. Elementary algebra: basic algebraic techniques and manipulations acquired in high school and used throughout mathematics

B. Linear algebra: matrix algebra, systems of linear equations, vector spaces, linear transformations, characteristic polynomials, eigenvalues and eigenvectors

C. Abstract algebra and number theory: elementary topics from group theory; the theory of rings and modules, field theory, and number theory

III. Additional Topics (25%)

A. Introductory real analysis: sequences and series of numbers and functions, continuity, differentiability and integrability, elementary topology of $\mathbb{R}$ and $\mathbb{R}^n$

B. Discrete mathematics: logic, set theory, combinatorics, graph theory, and algorithms

C. Other topics: general topology, geometry, complex variables, probability and statistics, and numerical analysis

This list is by no means all-embracing. (You may, for example, face questions asking you to find the correct mathematical model to represent a “real-life” situation.) The test requires more than simple recall: It gauges both your understanding of fundamental concepts and your ability to apply these concepts in specific situations. Because of the wide gamut of coursework in undergraduate mathematics, not all the material that you may have studied will be presented on the exam. Rather, the questions are based on the courses of study most commonly offered in an undergraduate mathematics curriculum.

SCORING YOUR PRACTICE TESTS

Each correct answer receives one “raw score” point, each incorrect answer deducts one-quarter of a point, and any omissions are not counted.

Use this formula to calculate your raw score:

$$\frac{\text{Number right}}{\text{Number right}} - \left(\frac{\text{Number wrong}^*}{\text{Number right}} \times 1/4 \right) = \frac{\text{Raw score}^\dagger}{\text{Raw score}^\dagger}$$

By using the score conversion table on page xii (which is based on the new score formulas used for the version of the test known as Mathematics (Rescaled), which replaced the old scoring standards in 2001), your raw score can be converted to a scaled score, and a percentile ranking can be determined. Your scaled score is what is sent on to the colleges, and your estimated percentile rank shows how you stack up against other students who have taken the exam in a recent three-year period. Let's say your raw score turns out to be 46. Then, according to the conversion table, your scaled score would be 730 and your percentile rank would be 76. This means that you would have scored better than 76% of the students taking the GRE Mathematics Test in that three-year period.

While taking REA's practice tests, it is critical to observe the same conditions that you will find during the actual test administration.

- Work in a quiet place, free from distractions and interruptions.
- Do not use any reference materials, calculators, or slide rules since these items cannot be used during the test.
- Work straight through each test. There are no breaks during the actual exam.
- Time yourself accurately, if possible with a wristwatch, stopwatch or clock alarm. Looking up to check the time will act as a distraction (watches with alarms or flashing lights are not permitted at the testing center).
- Since incorrect answers are each penalized one-quarter of a point, and unanswered questions are not counted, it is not *generally* a good idea to guess. However, if you are able to eliminate one or more of the answers, guessing may be to your advantage statistically (that is, your odds of choosing the correct answer will improve).

* Do not include unanswered questions.

† Round to nearest whole number.

GRE MATHEMATICS (RESCALED) PRACTICE-TEST SCORE CONVERSION TABLE[‡]

RAW SCORE	SCALED SCORE	PERCENTILE RANK	RAW SCORE	SCALED SCORE	PERCENTILE RANK
66	890	99	29	590	43
65	880	98	28	580	41
64	870	97	26-27	570	37
62-63	860	97	25	560	35
61	850	95	24	550	33
60	840	94	23	540	31
59	830	93	21-22	530	29
57-58	820	92	20	520	27
56	810	90	19	510	24
55	800	88	18	500	22
54	790	87	17	490	20
52-53	780	85	15-16	480	17
51	770	83	14	470	15
50	760	82	13	460	13
49	750	80	12	450	11
47-48	740	78	10-11	440	09
46	730	76	09	430	08
45	720	74	08	420	06
44	710	72	07	410	05
43	700	70	05-06	400	03
41-42	690	68	04	390	02
40	680	66	03	380	01
39	670	63	02	370	01
38	660	61	00-01	360	01
36-37	650	58			
35	640	56			
34	630	53			
33	620	51			
31-32	610	48			
30	600	46			

[‡]This table is provided to enable you to figure your six REA practice-test scores. While it will yield results that should strongly approximate your scoring range on the actual test, statistical formulas used by Educational Testing Service foreclose the possibility that the score you achieve on any REA practice test could precisely replicate or predict your score on the actual GRE Mathematics Test.

GRE MATHEMATICS PRO STUDY PLAN

Making the Most of REA's 12-Week Pro Study Plan

As a busy student or professional, you may be wondering how you will find time to work, have a social life, take care of family obligations, and prepare for the GRE Mathematics Subject Test. But don't worry, our GRE Pro Study Plan gives you a complete road map from now until test day. This flexible schedule allows you to work at your own pace and shows you how to prepare for the GRE Mathematics as efficiently as possible. You'll get organized, stay on track, and make the most of your valuable study time.

You will need to make time *every day* to study for the GRE Mathematics Test. How and when you study is up to you, but consistency is the key to completing your GRE preparation. Make studying a priority and consider it a "job" until you take the GRE.

Keep in mind that a high score on the GRE doesn't just help you get into a better grad school; it helps move your career forward. Therefore, a solid commitment to daily study is worth every minute of your time. The results will pay off in the long run!

This study schedule will help you become thoroughly prepared for the GRE Mathematics. Although the schedule is designed as a 12-week study program, it can be compressed into a 6-week plan by combining two weeks of study into one. If you choose the 12-week schedule, you should plan to study for *at least* one hour a day. GRE examinees who are following a 6-week program need to set aside *at least* two hours for studying every day.

Find a study routine that works for you and stick to it! Some people like to get up early and study for an hour or two before going to work. Others might choose to study while commuting, on their lunch hour, at the library, or at home after work. Whatever schedule you choose, make a commitment to study every day – even on weekends.

Remember, daily focused concentration on the GRE subject matter will help you retain more information, fully grasp the material tested, and improve your overall GRE score.

This GRE test prep with offers you plenty of practice and reinforcement. We've included six practice tests with answer explanations for every test question, so you can "practice for real" and know what to expect on the GRE Mathematics *before* exam day.

Week	Activity
1	Read and study our introduction to the GRE Mathematics Subject Test. Be sure you understand the format of the GRE Mathematics and know exactly what is tested on the exam. Now would be a good time to plan a study schedule. When and where will you study? How will you balance your responsibilities with studying? Decide when and where you study best, and get ready to tackle the GRE Mathematics! Start your study routine by taking the GRE Mathematics Practice Test 1 to determine your strengths and weaknesses. Make sure you give yourself several hours after work, on a weekend, or at another convenient time to take the exam. You will need to focus and concentrate, so take the practice test at a time and place where you will not be disturbed. When you take the test, try to do your best, even on sections where you may be confused. After you have finished the test, record your scores. This will help you track your progress as you study. Later in the week, study the answer explanations for the questions you answered incorrectly. In the cases where you erred, find out why. Take notes and pay attention to sections where you missed a lot of questions. You will need to spend more time reviewing the related material.
2	Make a firm commitment to study for at least an hour a day, every day for the next few weeks. It may seem hard to find time in your busy schedule, but remember: the more you study, the better prepared you will be for the GRE Mathematics. This week, focus your study on the Algebra Review (chapters 1-4) Although these are short chapters, pay close attention to the formulas, theorems, and equations; they will show you how to solve questions you may encounter on the actual GRE Mathematics exam. Be sure you are comfortable with the material before continuing to next week's study agenda.
3	Study the Calculus Review. These nine chapters cover everything you will need to know about Calculus when you take the GRE Mathematics. There are many equations and figures included with our review, so go through each section slowly and make sure you understand everything. Try to solve the sample problems on your own, and then check to see how well you did. If you find yourself in need of extra review or clarification on a topic, you may want to consult your math textbook or ask a classmate or professor for additional help.
4	Keep working your way through the review. This week, spend time studying the topics included in our "Miscellaneous" section. Probability and Statistics, Real and Complex Variables, Set Theory, and Topology are just some of the must-know GRE subjects you'll encounter on the test. As you go through each chapter, focus on the examples and practice problems and make sure you understand how each problem was solved. This review section contains a lot of formulas, figures, theorems, and equations, so take your time and make sure you understand everything before continuing to next week's study plan.

Week	Activity
5	It's time to test what you've learned! This week, set aside a few hours where you won't be disturbed and take Practice Test 2 in the book. Give yourself plenty of time to take the exam and try to do your best, even on sections where you may feel weak. After you've finished, grade your exam and compare your results to your score from test 1. Are there certain topics or sections where you missed a lot of questions? If so, take note of those sections so you can increase your study in those areas.
6	Congratulations! You've reached the halfway point! Success is just around the corner, but keep going, there's more work to be done. Take Practice Test 3 in the book. Record your score and see how well you did. After you've evaluated your test results, go back through each review chapter and brush up on the topics you need to study. Later this week, re-read the answer explanations for any questions you answered incorrectly. Make sure you understand why you answered the question wrong, so you can improve your test-taking skills.
7	Your hard work is paying off! By the time you've finished studying for the GRE Mathematics, you'll have an excellent understanding of all the topics tested on the exam – and that will help you get a high score. Take time this week to take Practice Test 4. You should be comfortable with the format and content of the GRE Mathematics now. Note any questions you missed and review the correct answers to see where you erred. If you're having difficulty with any mathematics topic, ask a classmate or professor to help you with your studies.
8	Spend Week 8 reviewing your notes on the GRE Mathematics topics. Now is a good time to go back and review any topics where you feel you need improvement. Quiz yourself on the commonly used formulas and theorems, so you're ready for any question on the GRE. Remember, your primary goal right now is to get a high score on the GRE, so keep working!
9	This week, review any questions you answered incorrectly on the previous practice tests. Spend time studying the answer explanations and re-read the relevant chapters for extra review. Work out any problems that you answered wrong on the practice tests, so you understand how to obtain the right answer. If you feel you need extra GRE Mathematics practice, why not review your notes during your lunch hour or on your commute?
10	Are you ready for even more practice? It's time to take Practice Test 5. Read each question carefully and take your time when solving the math problems. Try to remember everything you've learned in the reviews and in previous practice tests, so you can do your best. When you've completed the test, tally your score and see how well you've improved since the last practice test. In a few days, review any questions you answered incorrectly, and focus on the answer explanations to these questions.

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Week	Activity
11	You have just about reached your goal! After a day or two of additional study to reinforce any areas of weakness, take Practice Test 6. Remember what you've learned and answer every question to the best of your ability. How much has your score improved since you took the first practice exam? After the test, thoroughly review all the explanations for the question you answered incorrectly. Remember, your primary goal right now is to get a high score on the GRE, so keep working!
12	Use this time to restudy any sections in which you need improvement. Why not re-read the review chapters or retake the practice tests to help you boost your skills? Just for fun, take Practice Test 1 again. After you've completed the exam, compare your two scores and see how much you've improved in just a matter of weeks!

**Congratulations! You've worked hard and you're ready for
the GRE Mathematics Subject Test!**

ALGEBRA REVIEW

- **Polynomials and Equations**
- **Inequalities**
- **Linear Algebra**
- **Abstract (Modern) Algebra**

Chapter 1

POLYNOMIALS AND EQUATIONS

POLYNOMIALS

A polynomial in x , denoted $P(x)$, consists of one or more terms such that the terms are either an integral constant or the product of an integral constant and a positive integral power of x .

Example: $5x^3 + 2x^2 + 3$ is a polynomial in x .

$2x^2 + x^{\frac{1}{2}} - 1$ is not a polynomial in x .

$9x^3 + 3x^{-2} + 4$ is not a polynomial in x .

The degree of a monomial is the sum of the exponents of the variables. The degree of a monomial with no variables is 0.

Example: $5x^2$ has degree 2.

$3x^3y^2z$ has degree 6.

9 has degree 0.

The degree of a polynomial in x is equal to the exponent of that term with the highest power of x whose coefficient is not 0.

Example: $5x^4 + 7x + 12$ has degree 4.

Algebraic Operations with Polynomials

The following operations are usually performed with polynomials:

- 1) Addition: Combine the like terms which differ only in the numerical coefficient.

$$\begin{aligned}
 \text{Example: } & (x^2 - 3x + 5) + (4x^2 + 6x - 3) \\
 &= (x^2 + 4x^2) + (6x - 3x) + (5 - 3) \\
 &= 5x^2 + 3x + 2
 \end{aligned}$$

- 2) Subtraction: First, change the signs of all the terms in the expression which is being subtracted; then, add this result to the other expression.

$$\begin{aligned}
 \text{Example: } & (5x^2 + 4y^2 + 3z^2) - (4xy + 7y^2 - 3z^2 + 1) \\
 &= 5x^2 + 4y^2 + 3z^2 - 4xy - 7y^2 + 3z^2 - 1 \\
 &= (5x^2) + (4y^2 - 7y^2) + (3z^2 + 3z^2) - 4xy - 1 \\
 &= (5x^2) + (-3y^2) + (6z^2) - 4xy - 1 \\
 &= 5x^2 - 3y^2 + 6z^2 - 4xy - 1
 \end{aligned}$$

- 3) Multiplication: Multiplication of a polynomial by a polynomial is achieved by multiplying each of the terms of one polynomial by each of the terms of the other polynomial and combining the result.

$$\begin{aligned}
 \text{Example: } & (5y + z + 1)(y^2 + 2y) \\
 &= [(5y)(y^2) + (5y)(2y)] + [(z)(y^2) + (z)(2y)] \\
 &\quad + [(1)(y^2) + (1)(2y)] \\
 &= (5y^3 + 10y^2) + (y^2z + 2yz) + (y^2 + 2y) \\
 &= (5y^3) + (10y^2 + y^2) + (y^2z) + (2yz) + (2y) \\
 &= 5y^3 + 11y^2 + y^2z + 2yz + 2y .
 \end{aligned}$$

POLYNOMIAL FACTORIZATION

The factors of an algebraic expression consist of two or more algebraic expressions which, when multiplied together, produce the given algebraic expression.

A prime factor is a polynomial with no factors other than itself and 1. The least common multiple for a set of numbers is the smallest quantity divisible by every number of the set. For algebraic expressions, the least common

multiple is the polynomial of lowest degree and smallest numerical coefficients for which each of the given expressions will be a factor.

The greatest common factor for a set of numbers is the largest factor that is common to all members of the set. For algebraic expressions, the greatest common factor is the polynomial of highest degree and largest numerical coefficients which is a factor of all the given expressions.

Some important formulae, useful for the factoring of polynomials, are listed below.

$$a(c + d) = ac + ad$$

$$(a + b)(a - b) = a^2 - b^2$$

$$(a + b)(a + b) = (a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)(a - b) = (a - b)^2 = a^2 - 2ab + b^2$$

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

$$(ax + b)(cx + d) = acx^2 + (ad + bc)x + bd$$

$$(a + b)(c + d) = ac + bc + ad + bd$$

$$(a + b)(a + b)(a + b) = (a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a - b)(a - b)(a - b) = (a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$(a - b)(a^2 + ab + b^2) = a^3 - b^3$$

$$(a + b)(a^2 - ab + b^2) = a^3 + b^3$$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$$

$$(a - b)(a^3 + a^2b + ab^2 + b^3) = a^4 - b^4$$

$$(a - b)(a^4 + a^3b + a^2b^2 + ab^3 + b^4) = a^5 - b^5$$

$$(a-b)(a^5 + a^4b + a^3b^2 + a^2b^3 + ab^4 + b^5) = a^6 - b^6$$

$$(a-b)(a^{n-1} + a^{n-2}b + a^{n-3}b^2 + \dots + ab^{n-2} + b^{n-1}) = a^n - b^n$$

where n is any positive integer greater than 1 (2, 3, 4, ...).

$$(a+b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \dots - ab^{n-2} + b^{n-1}) = a^n + b^n$$

where n is any positive odd integer greater than 1 (3, 5, 7, ...).

The procedure for factoring a polynomial completely is as follows.

Step 1: First find the greatest common factor, if there is any. Then examine each factor remaining for other factors.

Step 2: Continue factoring the factors obtained in Step 1 until all factors other than monomial factors are prime.

Example: Factoring $4 - 16x^2$,

$$\begin{aligned} 4 - 16x^2 &= 4(1 - 4x^2) \\ &= 4(1 + 2x)(1 - 2x) . \end{aligned}$$

EQUATIONS

An equation is defined as a statement of equality of two separate expressions known as members.

A statement of equality between two expressions containing rational coefficients and whose exponents are integers is called a rational integral equation. The degree of the equation is given by the term with the highest power, as shown below:

$$a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x + a_0 = 0 ,$$

where $a_n \neq 0$, the a_i , $i = 1 \dots n$ are rational constant coefficients, and n is a positive integer. The degree of the equation is n .

In order to simplify the equations, the following basic laws hold:

- 1) The addition or subtraction of the same expression on both

sides of an equation results in an equivalent equation to the original one:

Example: Given the equation

$$y + 6 = 10 ,$$

we can add (-6) to both sides

$$y + 6 + (-6) = 10 + (-6)$$

to get $y + 0 = 10 - 6 \Rightarrow y = 4$.

So, $y + 6 = 10$ is equivalent to $y = 4$.

- 2) The multiplication or division on both sides of an equation by the same expression results in an equivalent equation to the original. (The divisor may not equal zero.)

Example: $3x = 6 \Rightarrow \frac{3x}{3} = \frac{6}{3} \Rightarrow x = 2$.

So, $3x = 6$ is equivalent to $x = 2$.

- 3) If both members of an equation are raised to the same power, then the resultant equation is equivalent to the original equation.

Example: $x^2 = 3y^4$.

If we raise both members to the -2 power, we get

$$(x^2)^{-2} = (3y^4)^{-2}$$

$$\frac{1}{(x^2)^2} = \frac{1}{(3y^4)^2}$$

$$\frac{1}{x^4} = \frac{1}{9y^8} .$$

If we raise both members to the $\frac{1}{2}$ power, which is the same

as taking the square root, we get:

$$(x^2)^{\frac{1}{2}} = (3y^4)^{\frac{1}{2}}$$

$$x = \pm\sqrt{3}y^2 .$$

- 4) The reciprocal of both members of an equation is equivalent to the original equation.

$$\frac{2x+y}{z} = \frac{5}{2} \qquad \frac{z}{2x+y} = \frac{2}{5} .$$

Note: The reciprocal of zero is undefined.

- 5) When evaluating equations containing absolute values, proceed as follows:

Example: $|5 - 3x| = 7$ is valid if either

$$\begin{array}{ll} 5 - 3x = 7 & \text{or} \quad 5 - 3x = -7 \\ -3x = 2 & -3x = -12 \\ x = \frac{-2}{3} & x = 4 . \end{array}$$

Therefore, the solution set is $x = \left(\frac{-2}{3}, 4\right)$.

Chapter 2

INEQUALITIES

INEQUALITIES

An inequality is a statement that the value of one quantity or expression is greater than or less than that of another.

Example: $5 > 4$ means that the value of 5 is greater than the value of 4.

A conditional inequality is an inequality whose validity depends on the values of the variables in the sentence. That is, certain values of the variables will make the sentence true, and others will make it false. For example, $3 - y > 3 + y$ is a conditional inequality for the set of real numbers, since it is true for any replacement less than 0 and false for all others.

Another example, $x + 5 > x + 2$, is an absolute inequality for the set of real numbers, meaning that for any real value x , the expression on the left is greater than the expression on the right.

A third example, $5y < 2y + y$, is inconsistent for the set of non-negative real numbers. For any x greater than 0, the sentence is always false. A sentence is inconsistent if it is always false when its variables assume allowable values.

The solution of a given inequality in one variable x consists of all values of x for which the inequality is true.

The following facts are useful in finding the solution set of the inequalities.

1) If a , b , and c are real numbers, then

a) if $a < b$ and $b < c$ then $a < c$;

b) if $a > b$ and $b > c$ then $a > c$.

2) If a , b , and c are real numbers and $a > b$, then

$a + c > b + c$ and $a - c > b - c$.

- 3) If a , b , and c are real numbers and $a > b$, and if c is positive, then

$$ac > bc . \text{ If } c \text{ is negative, then } ac < bc .$$

- 4) If a and b are real numbers and $a > b$, and if n is a positive integer, then

$$a^n > b^n \text{ and } a^{-n} < b^{-n} .$$

- 5) If $x > y$ and $q > p$, then $x + q > y + p$.

$$\text{If } x > y > 0 \text{ and } q > p > 0, \text{ then } xq > yp .$$

- 6) The solution set of $|x| < a$, $a > 0$, is $\{x | -a < x < a\}$.

$$\text{The solution set of } |x| > a, a > 0, \text{ is } \{x | x > a \text{ or } x < -a\} .$$

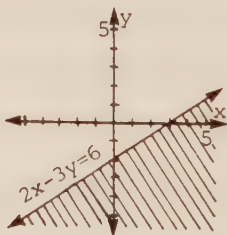
INEQUALITIES IN TWO VARIABLES

An inequality in the form $ax + by < c$ is a linear inequality in two variables. The equation for the boundary of the solution set is given by $ax + by = c$.

To graph a linear inequality, first graph the boundary.

Next, choose any point off the boundary and substitute its coordinates into the original inequality. If the resulting statement is true, the graph lies on the same side of the boundary as the test point. A false statement indicates that the solution set lies on the other side of the boundary.

Problem: Solve $2x - 3y \geq 6$.



Solution: The statement $2x - 3y \geq 6$ means $2x - 3y$ is greater than or equal to 6. Symbolically, we have $2x - 3y \geq 6$ or $2x - 3y = 6$. Consider the corresponding equality and graph $2x - 3y = 6$. To find the x -intercept, set $y = 0$:

$$2x - 3y = 6$$

$$2x - 3(0) = 6$$

$$2x = 6$$

$$x = 3 .$$

$(3, 0)$ is the x -intercept.

To find the y -intercept, set $x = 0$:

$$2x - 3y = 6$$

$$2(0) - 3y = 6$$

$$-3y = 6$$

$$y = -2 .$$

$(0, -2)$ is the y -intercept.

A line is determined by two points. Therefore, draw a straight line through the two intercepts, $(3, 0)$ and $(0, -2)$. Since the inequality is mixed, a solid line is drawn through the intercepts. This line represents the part of the statement $2x - 3y = 6$.

We must now determine the region for which the inequality $2x - 3y > 6$ holds true.

Choose two points (to be safe) to decide on which side of the line the region $2x - 3y > 6$ lies. We shall try the points $(0, 0)$ and $(5, 1)$.

For $(0, 0)$

$$2x - 3y > 6$$

$$2(0) - 3(0) > 6$$

$$0 - 0 > 6$$

$$0 > 6$$

False

For $(5, 1)$

$$2x - 3y > 6$$

$$2(5) - 3(1) > 6$$

$$10 - 3 > 6$$

$$7 > 6$$

True

The inequality, $2x - 3y > 6$, holds true for the point $(5, 1)$. We shade this region of the xy -plane, i.e., the area lying below the line $2x - 3y = 6$ and containing $(5, 1)$.

Therefore, the solution contains the solid line, $2x - 3y = 6$, and the part of the plane below this line.

Chapter 3

LINEAR ALGEBRA

SYSTEM OF LINEAR EQUATIONS AND MATRICES

A system of linear equations can be written in an equivalent matrix notation. For example, let us have the following three equations in three unknowns, x_1 , x_2 and x_3 :

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

This can be written as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix},$$

$$\text{or } \mathbf{Ax} = \mathbf{b} .$$

In the above equation, $\mathbf{A}$ is a matrix, while $\mathbf{x}$ and $\mathbf{b}$ are column vectors. Matrix $\mathbf{A}$ is a 3×3 matrix, i.e., it has 3 rows and 3 columns.

MATRIX ARITHMETIC

- 1) Addition and Subtraction: When adding two matrices, $\mathbf{A}$ and $\mathbf{B}$, add the corresponding elements of $\mathbf{A}$ and $\mathbf{B}$. Therefore, the i, j entry of $\mathbf{A} + \mathbf{B}$ is the i, j entry of $\mathbf{A}$ plus the i, j entry of $\mathbf{B}$. For subtracting $\mathbf{B}$ from $\mathbf{A}$, subtract the i, j elements of $\mathbf{B}$ from the i, j elements of $\mathbf{A}$.

- 2) **Multiplication of a Matrix by a Scalar:** Multiply the scalar value by all the entries of the matrix.
- 3) **Product of Two Matrices:** When multiplying two matrices, $\mathbf{A}$ and $\mathbf{B}$, the matrices must be of the sizes $M \times N$ and $N \times P$ (the number of columns of $\mathbf{A}$ must be equal to the number of rows of $\mathbf{B}$); to obtain the i, j entry of $\mathbf{AB}$, multiply the entries in row i of $\mathbf{A}$ by the corresponding entries in column j of $\mathbf{B}$. Add up the resulting products; this sum is the i, j entry of $\mathbf{AB}$. If $\mathbf{AB} = \mathbf{C}$, then

$$C_{ij} = \sum_{k=1}^N a_{ik} b_{kj} .$$

Example: If $\mathbf{A} = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 3 & 3 \\ 7 & 2 \end{bmatrix}$, then

$$\mathbf{AB} = \begin{bmatrix} 27 & 12 \\ 47 & 22 \end{bmatrix} .$$

- 4) **Inverse of Matrix:** If $\mathbf{A}$ is a square matrix and a matrix exists such that $\mathbf{AB} = \mathbf{BA} = \mathbf{I}$, where $\mathbf{I}$ is an identity matrix in which all the entries except main diagonal are zero, and the entries on main diagonal are all 1, then $\mathbf{A}$ is invertible and $\mathbf{B}$ is the inverse of $\mathbf{A}$, $\mathbf{A}^{-1}$. The method of finding the inverse of a matrix will be described later.

Following are a few facts about matrices.

- A) If $\mathbf{A}$ and $\mathbf{B}$ are invertible matrices of the same size, then
 - i) $\mathbf{AB}$ is invertible;
 - ii) $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.
- B) If $\mathbf{A}$ is an invertible matrix, then
 - i) $\mathbf{A}^{-1}$ is invertible, $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$;
 - ii) $\mathbf{A}^n$ is invertible, $(\mathbf{A}^n)^{-1} = (\mathbf{A}^{-1})^n$, n is an integer.
- C) If $\mathbf{A}$ is a square matrix and x and y are integers, then

i) $A^x A^y = A^{x+y}$;

ii) $(A^x)^y = A^{xy}$.

D) Trace: The trace of a matrix is the sum of the main diagonal entries.

Example: If $A = \begin{bmatrix} 4 & 5 & -6 \\ 11 & -18 & 19 \\ -2 & 5 & -20 \end{bmatrix}$,

then

$$\text{trace}(A) = 4 + (-18) + (-20) = -34 .$$

DETERMINANTS

The determinant of a matrix is a scalar number. The determinant of a 2×2 matrix, $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$, is $a_{11}a_{22} - a_{21}a_{12}$.

We will describe how to find the determinant of any matrix by using the cofactor expansion method.

If A is a square matrix, then the minor of entry a_{ij} , denoted (M_{ij}) , is defined to be the determinant of the submatrix remaining after the i th row and j th column of A are removed.

Example:

$$A = \begin{bmatrix} 7 & 1 & 3 \\ 1 & 3 & 5 \\ 17 & 4 & 20 \end{bmatrix}, \quad M_{11} = \det \begin{bmatrix} 3 & 5 \\ 4 & 20 \end{bmatrix} = 40 .$$

If A is a square matrix, then the cofactor of entry a_{ij} , denoted c_{ij} , is defined to be the scalar $(-1)^{i+j}M_{ij}$.

Example:

$$A = \begin{bmatrix} 7 & 1 & 3 \\ 1 & 3 & 5 \\ 17 & 4 & 20 \end{bmatrix}, c_{11} = (-1)^{1+1}M_{11} = (-1)^2 40 = 40$$

$$\begin{bmatrix} + & - & + & - & \dots \\ - & + & - & + & \dots \\ + & - & + & - & \dots \\ - & + & - & + & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

Example: $C_{11} = +M_{11}$, $C_{12} = -M_{12}$, $C_{43} = -M_{43}$, etc.

If A is a square matrix, $\det(A)$ can be found by cofactor expansion along the i th row or j th column of A . This is done by multiplying the entries in the i th row or the j th column of A by their cofactors and summing the resulting products. Thus,

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots$$

or

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \dots$$

Example: expansion along first row:

$$\det \begin{bmatrix} 2 & 4 & 3 \\ 7 & 2 & 12 \\ 1 & 3 & 29 \end{bmatrix} = (2)(22) - 4(191) + (3)(19) = -663.$$

If A is a square matrix and c_{ij} is the cofactor of a_{ij} , then the matrix of cofactors from A is:

$$\begin{bmatrix} c_{11} & c_{12} & \dots \\ c_{21} & c_{22} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}.$$

The transpose of this matrix is called the adjoint of A ($\text{adj}(A)$).

THEOREM 1:

If A is an invertible matrix, then

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) .$$

Determinant Properties

If A is an $M \times N$ matrix, then the transpose of A , denoted by (A^t) , is defined as the $N \times M$ matrix, where the rows and columns of A are switched.

Example:

$$\text{If } A = \begin{bmatrix} 1 & 2 \\ 2 & 7 \\ 5 & 23 \end{bmatrix}, \text{ then } A^t = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 7 & 23 \end{bmatrix} .$$

Properties of the transpose operation:

- a) $(A^t)^t = A$
- b) $(A + B)^t = A^t + B^t$
- c) $(kA)^t = kA^t$ (where k is scalar)
- d) $(AB)^t = B^t A^t$

THEOREM 2:

If A is a square matrix, then $\det(A) = \det(A^t)$. Because of this theorem, all determinant theorems concerning the rows of a matrix also apply to the columns of a matrix.

If A and B are square matrices of the same size, and k is scalar, then:

- a) $\det(kA) = k^N \det(A)$ (N is the number of rows of A .)
- b) $\det(AB) = \det(A) \det(B)$

THEOREM 3:

A square matrix A is invertible if and only if $\det(A) \neq 0$.

If A is invertible, then $\det(A^{-1}) = 1/\det(A)$.

CHARACTERISTIC POLYNOMIAL

The differential equation

$$y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1y' + a_0y = 0$$

can be expressed as the characteristic polynomial

$$D^n + a_{n-1}D^{n-1} + \dots + a_1D + a_0$$

where the linear differential operator is factored in the form $(D - r)$. For instance, the differential equation

$$(D - r_1)(D - r_2) \dots (D - r_n)y = 0$$

has the general solution

$$y = c_1e^{r_1x} + c_2e^{r_2x} + \dots + c_ne^{r_nx}.$$

If values of r_x are equal, then values of e^{r_nx} are replaced by xe^{r_nx} , $x^2e^{r_nx}$, and so on.

Example: Solve

$$y''' - 4y'' + 4y' = 0.$$

The corresponding characteristic polynomial is

$$D^3 - 4D^2 + 4D,$$

which can be written as

$$D(D^2 - 4D + 4) = D(D - 2)^2.$$

Here, the roots are 0 and 2, which is repeated. Thus, the solution is

$$y = c_1 + c_2e^{2x} + c_3xe^{2x}.$$

Note that c_1 was $c_1e^{0x} = c_1$. Similarly, linear functions (L) of the form $L(y)$

$= f$ can be solved by determining the homogenous and particular solutions and through superposition.

Finally, the determinant

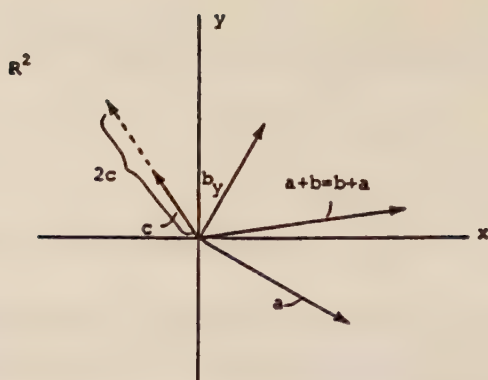
$$D(\lambda) = \begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix}$$

produces a polynomial of the n th degree in λ and is called the characteristic polynomial. This aids in the determination of eigenvalues and eigenvectors, which will be discussed later.

VECTOR SPACES

Let V be a set of objects on which two operations are defined, addition, and multiplication by scalars (real numbers). Let it also be given that a , b and c are members of V , and x and y are scalars. If V conforms to the following rules, then V is called a vector space, and a , b and c are called vectors:

- a) $a + b \in V$
- b) $a + b = b + a$
- c) $a + (b + c) = (a + b) + c$
- d) The zero vector 0 is an element of V such that $0 + a = a$
- e) The negative of a , $-a$, is an element of V such that $a + (-a) = 0$
- f) $xa \in V$
- g) $x(a + b) = xa + xb$
- h) $(x + y)a = xa + ya$
- i) $x(ya) = (xy)a$
- j) $1a = a$



R^2 is a vector space.

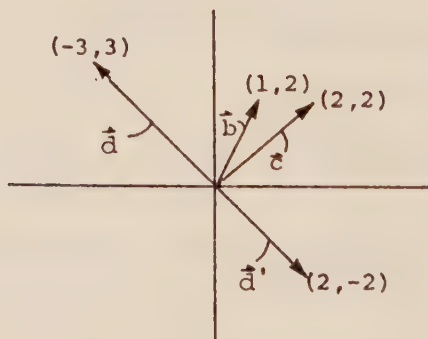
A vector space can be in the form of N -space, with its vectors in the form of N -tuples. The operations of addition and subtraction of vectors, and multiplication of a vector with a scalar are the same as for matrices.

Linear Independence

Let V be a vector space, and let $v_1, \dots, v_n$ be elements of V . Vectors $v_1, \dots, v_n$ are linearly dependent if there exist numbers $a_1, \dots, a_n$ not all equal to zero such that

$$a_1 v_1 + \dots + a_n v_n = 0.$$

If such numbers do exist, then $v_1, \dots, v_n$ are linearly dependent.



d and d' are linearly dependent since $d + \frac{3}{2}d' = 0$

d, b, c (or d', b, c) are linearly dependent sets of vectors since $b = \frac{3}{4}c + \frac{1}{6}d$.

If $Z = \{Z_1, Z_2, \dots, Z_R\}$ is a set of vectors in R^N , and $R > N$, then z is linearly dependent.

Rank of a Matrix: The number of linearly independent rows or columns of a matrix is the rank of the matrix. For example, let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

Then row number 1 is $a_1 = [1 \ 2]$, and row number 2 is $a_2 = [2 \ 4]$. But we can check that $2a_1 + (-1)a_2 = [0 \ 0]$. Therefore, only one row is linearly independent and $\text{rank}(A) = 1$.

If the determinant of a matrix is non-zero, then that matrix has full rank.

A square matrix is invertible if it has full rank.

Basis and Dimension

If V is any vector space and $x = \{v_1, v_2, v_3, \dots, v_R\}$ is a finite set of vectors in V , then x is called a basis for V if:

- a) x is linearly independent;
- b) x spans V , i.e., any vector in V can be obtained as a linear combination of vectors in x .

The number of vectors in a basis of a finite dimensional vector space is called the dimension of that vector space.

An inner product on a vector space V produces a real number $\langle v_1, v_2 \rangle$ from each pair of vectors $v_1, v_2 \in V$, which satisfies the following axioms for all vectors $v_1, v_2, v_3 \in V$ and for all scalars L :

- a) $\langle v_1, v_2 \rangle = \langle v_2, v_1 \rangle$

$$b) \langle \mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_3 \rangle = \langle \mathbf{v}_1, \mathbf{v}_3 \rangle + \langle \mathbf{v}_2, \mathbf{v}_3 \rangle$$

$$c) \langle \mathbf{v}_1, \mathbf{v}_1 \rangle \geq 0, \text{ and } \langle \mathbf{v}_1, \mathbf{v}_1 \rangle = 0 \text{ if and only if } \mathbf{v} = 0.$$

If $\mathbf{X} = (x_1, x_2, \dots, x_N)$ and $\mathbf{Y} = (y_1, y_2, \dots, y_N)$ are vectors in R^N , then the Euclidean inner product is defined as

$$\langle \mathbf{X}, \mathbf{Y} \rangle = x_1 y_1 + x_2 y_2 + \dots + x_N y_N$$

Example: If $\mathbf{X} = (2, 4, 6)$ and $\mathbf{Y} = (7, 1, 1)$, then

$$\langle \mathbf{X}, \mathbf{Y} \rangle = 14 + 4 + 6 = 24.$$

A vector space with an inner product is called an inner product space.

Null Space: Null space W of vector space V is the space in which the inner product of any vector in W with any vector in V is equal to zero. That is, if $\mathbf{w}_i \in W$ and $\mathbf{v}_j \in V$, then $\langle \mathbf{w}_i, \mathbf{v}_j \rangle = 0$.

EIGENVALUES AND EIGENVECTORS

If $\mathbf{A}$ is an $N \times N$ matrix, then the non-zero vector $\mathbf{X} \in R^N$ is called an eigenvector of $\mathbf{A}$ if $\mathbf{AX} = \lambda \mathbf{X}$, where λ is a scalar called the eigenvalue of $\mathbf{A}$.

If $\mathbf{A}$ is a square matrix, then the characteristic equation of $\mathbf{A}$ is defined to be $\det(\lambda \mathbf{I} - \mathbf{A}) = 0$. When expanded, the $\det(\lambda \mathbf{I} - \mathbf{A})$ is called the characteristic polynomial of $\mathbf{A}$.

If $\mathbf{A}$ is a square matrix and there exists a matrix $\mathbf{B}$ such that $\mathbf{B}^{-1} \mathbf{A} \mathbf{B}$ is diagonal, then $\mathbf{A}$ is diagonalizable and the similarity transformation matrix $\mathbf{B}$ diagonalizes $\mathbf{A}$.

The following basic laws hold:

- 1) If $\mathbf{A}$ is an $N \times N$ matrix and $\mathbf{A}$ is diagonalizable, then $\mathbf{A}$ has N linearly independent eigenvectors.
- 2) If an $N \times N$ matrix has N distinct eigenvalues, then it is diagonalizable.
- 3) A symmetric matrix (a matrix $\mathbf{A}$ such that $\mathbf{A} = \mathbf{A}^t$) has all real eigenvalues.

Chapter 4

ABSTRACT (MODERN) ALGEBRA

Modern algebra provides methods to study both scientific and artistic representations of nature. Thus, while modern algebra is abstract, it has important applications in a wide variety of disciplines.

GROUPS

A class G of objects $a, b, c, \dots$ is a group of objects if and only if it is possible to define a binary operation which associates a result $a \circ b$ with every pair of elements a, b , of G , such that the following conditions hold:

- 1) $(a \circ b) \in G$
- 2) $(a \circ b) \circ c = a \circ (b \circ c)$ for $a, b, c \in G$.
- 3) There exists an element e of G such that $a \circ e = e \circ a$ for every $a \in G$.
- 4) There exists an element x of G such that $a \circ x = x \circ a = e$ for every $a \in G$.

An Abelian group, or commutative group, is the result of $a \circ b = b \circ a$ for all $a, b \in G$ if $(G, \circ)$ is a group.

Example: The set whose only member is the integer 1 with the usual operation of multiplication is a group. This is an Abelian group.

THEOREM 1

Let $(G, \circ)$ be a group. Group $(G, \circ)$ has the following properties:

- a) The identity is unique.

- b) If $a, b, c \in G$ and $a \circ b = a \circ c$, then $b = c$.
- c) If $a, b, c \in G$ and $b \circ a = c \circ a$, then $b = c$.
- d) Each element a of G has a unique inverse, denoted by a^{-1} .
- e) If $a, b \in G$, there exists a unique element x of G such that $a \circ x = b$, and a unique element y of G such that $y \circ a = b$.
- f) If $a, b \in G$, then $(a \circ b)^{-1} = b^{-1} \circ a^{-1}$.

The order of a group G denoted $\circ(G)$ is defined as the number of elements G . G is said to be a finite group when $\circ(G)$ is a finite number.

THEOREM 2

Let $(G, \circ)$ and $(H, *)$ be groups. If π is the operation on $G \times H$ defined by

$$(a_1, b_1) \pi (a_2, b_2) = (a_1 \circ a_2, b_1 * b_2)$$

for all $a_1, a_2 \in G$, and $b_1, b_2 \in H$, then $(G \times H, \pi)$ is a group called the direct product.

A permutation group is any group whose elements are permutations of n symbols. It is called the symmetric group and denoted S_n .

A homomorphism h is a mapping from group G with the operation $*$ into a group H with the operation $\circ$ such that for all elements $a, b \in G$, $h(a * b) = h(a) \circ h(b)$. It is an epimorphism if h is onto; it is a monomorphism if h is one-to-one; and it is an isomorphism if h is one-to-one and onto.

Example: The function $h: Y \rightarrow Y$ defined by $h(n) = 0$ for each $n \in Y$ is a homomorphism.

RINGS

A class R of object $a, b, c, \dots$ is a ring if and only if it is possible to define two binary operations, usually denoted as (abstract) addition and multiplication, such that the following hold true:

- 1) R is a commutative group with respect to addition; R is closed under addition.

$$a + b = b + a$$

$$a + (b + c) = (a + b) + c$$

$$a + 0 = a$$

$$a + (-a) = a - a = 0$$

2) ab is an element of R .

3) $a(bc) = (ab)c$.

4) $a(b + c) = ab + ac$; $(b + c)a = ba + ca$.

A ring with identity is a ring containing a multiplicative identity ε (left) such that $\varepsilon a = a$ for all a . A similar right identity is necessary. An element a of a ring with identity may have a multiplicative inverse a^{-1} ; however, if one exists, it is necessarily unique for both left and right inverses.

FIELDS

A field is a ring with identity containing the following:

- 1) at least one element different from zero;
- 2) a multiplicative inverse a^{-1} for each element $a \neq 0$.

Fields and rings are commutative if and only if $ab = ba$ for all a, b .

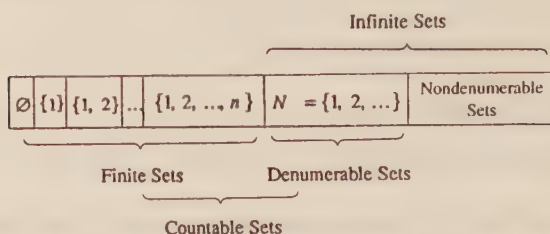
NUMBER THEORY

A set A is called equivalent (or equipotent or of the same power) to a set B , denoted by $A \sim B$, if there exists a function $f: A \rightarrow B$ which is one-to-one and onto. We classify the sets in the following manner:

A set is finite if it is empty or equivalent to $\{1, 2, \dots, n\}$ for some $n \in \mathbb{N}$; otherwise, it is said to be infinite.

$\emptyset$	$\{1\}$	$\{1,2\}$	$\dots$	$\{1,2,\dots,n\}$	?
Finite Sets					Infinite Sets

A set is called denumerable if it is equivalent to the set of positive integers. A set is called countable if it is finite or denumerable.



The denumerable sets are the “smallest” in size among the infinite sets.

CARDINAL NUMBERS

The relation $\sim$, defined in a previous section, is an equivalence relation among sets. This, it enables us to divide the collection of sets into disjoint classes.

The sets can be classified with respect to their “power.” To each set X we assign a cardinal number (or its power), denoted by $\text{card } X$ (or $\bar{x}$), in such a way that the same cardinal number is assigned to two distinct sets A and B if and only if $A \sim B$.

The rules concerning cardinal numbers are as follows:

- 1) Each set A is associated with a cardinal number, denoted by $\text{card } A$, and for each cardinal number there exists a set A such that $a = \text{card } A$.
- 2) For any two sets A and B , $\text{card } A = \text{card } B$ if and only if $A \sim B$.
- 3) $\text{Card } A = 0$ if and only if $A = \emptyset$.
- 4) For any nonempty finite set A ,
i.e., $A \sim \{1, 2, \dots, k\}$,
 $\text{card } A = k$.

Thus, the cardinal number of a finite set is the number of its elements.

Ordering of Cardinal Numbers

The cardinal number of a finite set is called a finite cardinal number and the cardinal number of an infinite set is called a transfinite cardinal number. The order of the finite cardinal numbers is

$$0 < 1 < 2 \dots < n < n + 1 < \dots,$$

which is the order of the nonnegative integers.

Let A and B be sets. Then $\text{card } A$ is said to be less than $\text{card } B$, i.e.,

$$\text{card } A < \text{card } B,$$

provided that A is equipotent to a subset of B but B is not equipotent to any subset of A .

Note that this definition also applies to the finite cardinal numbers and leads to the usual ordering of numbers.

Example: $\text{card } N < \text{card } R$, for N the set of non-negative integers and R the set of real numbers.

The set N is a subset of R . Thus, N is equipotent to a subset of R ,

$$N \sim (N \subset R).$$

The set R of all real numbers is nondenumerable, and N is denumerable. Hence, R is not equipotent to any subset of N .

We have $\text{card } N < \text{card } R$.

Let A and B be sets. We write $\text{card } A \leq \text{card } B$ if the set A has the same power as some subset of the set B . Hence,

$$(A \subset B) \Rightarrow (\text{card } A \leq \text{card } B).$$

If $\text{card } A \leq \text{card } B$ and $\text{card } A \neq \text{card } B$, then we write $\text{card } A < \text{card } B$.

THEOREM 3 (Schröder-Bernstein Theorem)

If $\text{card } A \leq \text{card } B$ and $\text{card } B \leq \text{card } A$, then $\text{card } A = \text{card } B$.

THEOREM 4

If p , r , and s are cardinal numbers, then

1) if $p \leq r$ and $r \leq s$, then $p \leq s$.

2) if $p \leq r$, then $p + s \leq r + s$.

From Theorem 3 we conclude that if $A \subset B \subset C$ and $\text{card } A = \text{card } C$, then
 $\text{card } A = \text{card } B = \text{card } C$.

THEOREM 5

If X is the domain of f , then

$$\text{card } f(X) \leq \text{card } X.$$

Suppose $y \in f(X)$. Let $F(y)$ be an arbitrary element of the set $f^{-1}(y)$.

$$F(y) \in f^{-1}(y)$$

Here we make use of the axiom of choice, which will be discussed later. For various y 's, the sets $f^{-1}(y)$ are disjoint. Thus, F determines a one-to-one mapping of the set $f(X)$ onto a subset of the set X . Thus,

$$\text{card } f(X) \leq \text{card } X.$$

The Schröder-Bernstein Theorem is sometimes formulated as follows: If A and B are sets such that A is equipotent to a subset of B , and B is equipotent to a subset of A , then $A \sim B$.

Addition of Cardinal Numbers, Continuum

The arithmetic for finite cardinal numbers is simple. For instance, if k and l are two finite cardinal numbers, their sum $k + l$ and their product kl have the usual meaning. The problem is how to define the sum and product in a general manner that would apply to the transfinite cardinal numbers and that would preserve the usual meanings and properties of the arithmetic of finite cardinal numbers.

Cardinal Sum

The sum $a + b$ of two cardinal numbers a and b is defined as the power of the union of two disjoint sets which have powers a and b , respectively.

We have $\text{card } A + \text{card } B = \text{card } (A \cup B)$ if $A \cap B = \emptyset$.

The sum of two cardinal numbers is well-defined because:

- 1) if a and b are distinct cardinal numbers, then there exist disjoint sets A and B such that $\text{card } A = a$ and $\text{card } B = b$.
- 2) if $A, B, A',$ and B' are sets such that $\text{card } A = \text{card } A', \text{card } B = \text{card } B',$ and $A \cap B = A' \cap B' = \emptyset$, then $\text{card } (A \cup B) = \text{card } (A' \cup B')$.

THEOREM 6

Let $x, y,$ and z be arbitrary cardinal numbers. Then

$$\begin{aligned}x + y &= y + x \\x + (y + z) &= (x + y) + z.\end{aligned}$$

The power of the set of natural numbers is denoted by $\aleph_0$ (read aleph-null)

$$\text{card } N = \aleph_0$$

Example: Find the cardinal sum $2 + 5$ of the two finite cardinal numbers 2 and 5. Let

$$\begin{aligned}N_2 &= \{1, 2\} \quad \text{and} \\N_5 &= \{3, 4, 5, 6, 7\}.\end{aligned}$$

Then $\text{card } N_2 = 2$ and $\text{card } N_5 = 5$.

The sets N_2 and N_5 are disjoint.

Hence,

$$2 + 5 = \text{card } (N_2 \cup N_5) = \text{card } (\{1, 2, 3, 4, 5, 6, 7\}) = 7,$$

which agrees with the usual sum of two integers.

Example: Compute $\aleph_0 + \aleph_0$.

Let N_e denote the set of even natural numbers and N_o the set of odd natural numbers. Then

$$\text{card } N_e = \aleph_0$$

$$\text{card } N_0 = \aleph_0$$

$$\text{and } N_e \cap N_0 = \emptyset, N_e \cup N_0 = N.$$

We have

$$\aleph_0 + \aleph_0 = \text{card } N_e + \text{card } N_0 = \text{card}(N_e \cup N_0) = \text{card } N = \aleph_0.$$

Example: Find $n + \aleph_0$.

Consider sets $\{1, 2, \dots, n\}$ and $\{n+1, n+2, \dots\}$.

Sets $\{n+1, n+2, \dots\}$ and N are equivalent, since $f(k) = k + n$ is a one-to-one mapping of N onto $\{n+1, n+2, \dots\}$. Hence,

$$\text{card } \{1, 2, \dots, n\} = n$$

$$\text{and } \text{card } \{n+1, n+2, \dots\} = \aleph_0.$$

$$\text{Also, } \{1, \dots, n\} \cap \{n+1, \dots\} = \emptyset.$$

We have

$$n + \aleph_0 = \text{card } [\{1, \dots, n\} \cup \{n+1, \dots\}] = \text{card } N = \aleph_0.$$

Power of Continuum

A set X is said to have the power of the continuum or to have cardinality c if and only if it is equivalent (equipotent) to the unit interval $[0, 1]$.

Example: Interval $(0, 1)$ is equivalent to $[0, 1]$.

Define the set

$$A = [0, 1] - \left\{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\right\}.$$

Then

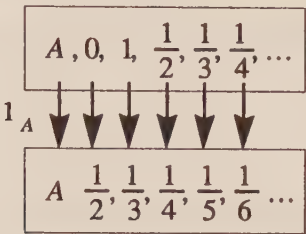
$$[0, 1] = A \cup \left\{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\right\}$$

and

$$(0, 1) = A \cup \left\{ \frac{1}{2}, \frac{1}{3}, \dots \right\}.$$

The following diagram defines the functions

$$f: [0, 1] \rightarrow (0, 1).$$



Obviously, f is one-to-one and onto. Thus, $[0, 1] \sim (0, 1)$.

Example: The set of real numbers $\mathfrak{R}$ is equivalent to $(-1, 1)$. The function

$$f(x) = \frac{x}{1 - |x|}, \quad -1 < x < 1$$

is one-to-one and onto.

THEOREM 7

The unit interval $[0, 1]$ is nondenumerable. Assuming the contrary, then

$$[0, 1] = \{x_1, x_2, x_3, \dots\}.$$

That is, the elements of $[0, 1]$ can be written in a sequence. Each of the elements of this sequence can be written in the form of an indefinite decimal:

$$x_1 = 0.a_{11}a_{12}a_{13}\dots a_{1k}\dots$$

$$x_2 = 0.a_{21}a_{22}a_{23}\dots a_{2k}\dots$$

$\vdots$

$$x_k = 0.a_{k1}a_{k2}a_{k3}\dots a_{kk}\dots$$

$\vdots$

Each $a_{ij} \in \{0, 1, 2, \dots, 9\}$.

Each decimal expansion contains an infinite number of non-zero elements. In cases like

$$\frac{1}{2} = 0.5000\dots = 0.4999\dots$$

we choose $0.4999\dots$

Construct the real number

$$y = 0.b_1b_2b_3\dots$$

in the following way:

choose b_1 so $b_1 \neq a_{11}$ and $b_1 \neq 0$

choose b_2 so $b_2 \neq a_{22}$ and $b_2 \neq 0$

$\vdots$

choose b_k so $b_k \neq a_{kk}$ and $b_k \neq 0$

$\vdots$

For each x_k , $k \in N$,

$$y \neq x_k$$

because, since $b_1 \neq a_{11}$, $y \neq x$, etc.

Hence, $y \notin [0, 1]$, which is a contradiction.

We conclude that $[0, 1]$ is nondenumerable.

Cardinal numbers $\aleph_0$ and c are frequently used transfinite cardinal numbers.

Example: Compute $\aleph_0 + c$.

$$(-1, 1) \sim \mathfrak{R}$$

By the relationship

$$g(x) = \frac{1}{2}(x + 1), -1 < x < 1, (-1, 1) \sim (0, 1).$$

So, $(0, 1) \sim \mathfrak{R}$; hence, $\text{card } (0, 1) = \text{card } \mathfrak{R} = c$.

Consider the set $N \cup (0, 1)$. Since N and $(0, 1)$ are disjoint, $\text{card } (N \cup (0, 1)) = \text{card } N + \text{card } (0, 1) = \aleph_0 + c$.

By the Schröder-Bernstein theorem,
 since $\aleph \sim (0, 1) \subset (N \cup (0, 1))$
 and $(N \cup (0, 1)) \subset \aleph$,
 we have $(N \cup (0, 1)) \sim \aleph$. Hence,
 $\text{card } [N \cup (0, 1)] = \text{card } \aleph = c$.

Thus,

$$\aleph_0 + c = c.$$

Multiplication of Cardinal Numbers

The product $a \times b$ of two cardinal numbers a and b is the power of the Cartesian product of two sets having powers a and b , respectively; i.e., $\text{card } A \times \text{card } B = \text{card } A \times B$.

It is easy to see that this definition is independent of the choice of representatives A and B .

THEOREM 8

Let x , y , and z be arbitrary cardinal numbers. Then

- 1) $xy = yx$.
- 2) $(xy)z = x(yz)$.

THEOREM 9

Let x , y , and z be cardinal numbers. Then

$$x \times (y + z) = (x \times y) + (x \times z).$$

Let $x = \text{card } X$, $y = \text{card } Y$, and $z = \text{card } Z$, where

$$Y \cap Z = \emptyset.$$

We have

$$X \times (Y \cup Z) = (X \times Y) \cup (X \times Z)$$

and

$$(X \times Y) \cap (X \times Z) = X \times (Y \cap Z) = \emptyset .$$

Hence,

$$\begin{aligned} x \times (y + z) &= \text{card } [X \times (Y \cup Z)] \\ &= \text{card } (X \times Y) + \text{card } (X \times Z) \\ &= (x \times y) + (x \times z) . \end{aligned}$$

From Theorem 9 we obtain by induction the following:

$$n \times \text{card } X = \underbrace{\text{card } X + \text{card } X + \dots + \text{card } X}_{n \text{ times}} .$$

This result can be generalized to the sum of an infinite number of terms as follows.

THEOREM 10

Let

$$\text{card } T = x$$

and let

$$F: t \rightarrow 2^x$$

be a set-valued mapping such that

$$\text{card } F(t) = y$$

$$F(t) \cap F(t') = \emptyset \text{ for } t \neq t' .$$

Then

$$\text{card } \left[\bigcup_t^N F(t) \right] = x \times y .$$

Example: Compute $\aleph_0 \times \aleph_0$.

The function

$$f: N \times N \rightarrow N$$

is defined by

$$f(m, n) = 2^m 3^n$$

for all $(m, n) \in N \times N$.

Since f is injective,

$$N \times N \sim f(N \times N) \subset N.$$

Since $N \times N$ is infinite, so is $f(N \times N)$. Every infinite subset of a denumerable set is denumerable.

Thus, $f(N \times N)$ is denumerable and so is $N \times N$. We have

$$N \times N \sim N$$

and $\aleph_0 \times \aleph_0 = \aleph_0$.

Example: Show that $c \times c = c$.

First we prove that $cc \leq c$.

Since the set $\mathfrak{R}$ of real numbers and $(0, 1)$ are equivalent, to prove that $cc \leq c$, it is enough to show that there is an injection from $(0, 1) \times (0, 1)$ to $(0, 1)$.

Each $x \in (0, 1)$ is expressed by its infinite decimal expansion, i.e.,

$$\frac{1}{2} = 0.49999 \dots,$$

but not 0.5. For each number $x \in (0, 1)$, there is a unique decimal expansion. The function

$$f: (0, 1) \times (0, 1) \rightarrow (0, 1)$$

defined by

$$f(0.x_1x_2x_3\dots, 0.y_1y_2y_3\dots) = 0.x_1y_1x_2y_2x_3y_3\dots$$

is injective.

Hence, $cc \leq c$.

It is easy to show that $cc \geq c$.

Hence, $cc = c$.

ORDINAL NUMBERS

Order-Isomorphism

Two well-ordered sets $(A, \leq)$ and $(B, \leq')$ are said to be order-isomorphic if there exists a bijection

$$f: A \rightarrow B$$

such that if $a_1, a_2 \in A$ and $a_1 \leq a_2$, then

$$f(a_1) \leq' f(a_2).$$

Function $f: A \rightarrow B$ is called an order-isomorphism if $(A, \leq)$ and $(B, \leq')$ are order-isomorphic; we write

$$(A, \leq) \approx (B, \leq') \text{ or } A \approx B.$$

THEOREM 11

Relation $\approx$ is an equivalence relation.

We shall introduce finite and infinite ordinal numbers as follows:

- 1) To each well-ordered set $(A, \leq)$, an ordinal number is assigned, denoted by $\text{ord}(A, \leq)$. Conversely, if α is an ordinal number, then there is a well-ordered set $(A, \leq)$ such that

$$\alpha = \text{ord}(A, \leq).$$

- 2) Let $(A, \leq)$ and $(B, \leq')$ be well-ordered sets. Then

$$\text{ord}(A, \leq) = \text{ord}(B, \leq'),$$

if and only if

$$(A, \leq) \approx (B, \leq').$$

THEOREM 12

Any two finite, well-ordered sets which have the same number of elements are order-isomorphic.

We shall apply the notation:

- 1) $\text{ord}(A, \leq) = 0$ if and only if $A = \emptyset$.
- 2) if $(A, \leq)$ is a well-ordered set such that A contains n elements, $n \in N$, then

$$\text{ord}(A, \leq) = n .$$

The ordinal number of the set N of natural numbers, with the usual less than or equal to relation $\leq$, is usually denoted by ω , i.e.,

$$\omega = \text{ord}\{1, 2, 3, \dots\}.$$

Note that a given well-ordered set has only one cardinal number. However, it is possible that, under different well-orderings, the set may have distinct ordinal numbers.

Ordering of Ordinal Numbers

“Less than or Equal to”

Let α and β be ordinal numbers and let $(A, \leq)$ and $(B, \leq')$ be well-ordered sets such that

$$\alpha = \text{ord}(A, \leq) \text{ and } \beta = \text{ord}(B, \leq') .$$

We say that α is less than or equal to β , i.e., $\alpha \leq \beta$, if and only if $(A, \leq)$ is order-isomorphic to a segment of $(B, \leq')$. If $\alpha \leq \beta$ and $\alpha \neq \beta$, then we write $\alpha < \beta$.

THEOREM 13

The only order-isomorphism of a well-ordered set $(A, \leq)$ onto a segment of $(A, \leq)$ is the identity function of A onto A .

THEOREM 14

If α and β are ordinal numbers such that $\alpha \leq \beta$ and $\beta \leq \alpha$, then $\alpha = \beta$.

THEOREM 15

For any ordinal numbers α and β , either $\alpha \leq \beta$ or $\beta \leq \alpha$.

THEOREM 16

Let α and β be any two ordinal numbers. Then exactly one of the following is true:

- 1) $\alpha < \beta$
- 2) $\alpha = \beta$
- 3) $\beta < \alpha$

Addition and Multiplication of Ordinal Numbers

Let $(A, \leq)$ and $(B, \leq')$ be any two disjoint, well-ordered sets. The total order relation $\leq^*$ on $A \cup B$ is defined as follows:

- 1) if a and b are both in A (or in B), then $a \leq^* b$ if and only if $a \leq b$ (or $a \leq' b$).
- 2) if $a \in A$ and $b \in B$, then we write $a \leq^* b$.

The set $(A \cup B, \leq^*)$ is a well-ordered set.

Ordinal Sum

Let α and β be ordinal numbers. The ordinal sum of α and β , denoted by $\alpha + \beta$, is the ordinal number

$$\text{ord}(A \cup B, \leq^*),$$

where $(A, \leq)$ and $(B, \leq')$ are disjoint, well-ordered sets such that

$$\text{ord}(A, \leq) = \alpha \text{ and } \text{ord}(B, \leq') = \beta.$$

THEOREM 17

Let α , β , and γ be any ordinal numbers. Then

- 1) $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$.
- 2) $\alpha < \beta$ implies $\gamma + \alpha < \gamma + \beta$.
- 3) $\alpha + \beta = \alpha + \gamma$ implies $\beta = \gamma$.

Lexicographic Ordering

Let $(A, \leq)$ and $(B, \leq')$ be well-ordered sets. The lexicographic ordering $\leq^*$ on $A \times B$ is defined by the following:

- 1) If $a \leq x$, then $(a, b) \leq^*(x, y)$ for any $b, y \in B$.
- 2) If $a = x$ and $b \leq' y$, then $(a, b) \leq^*(x, y)$.

THEOREM 18

Let $(A, \leq)$ and $(B, \leq')$ be well-ordered sets. Then the lexicographic ordering $\leq^*$ on $A \times B$ is a well-ordered relation on $A \times B$.

Ordinal Product

The ordinal product $\beta\alpha$ of two ordinal numbers α and β is defined by

$$\beta\alpha = \text{ord}(A \times B, \leq^*),$$

where $(A, \leq)$ and $(B, \leq')$ are well-ordered sets such that $\text{ord}(A, \leq) = \alpha$, $\text{ord}(B, \leq') = \beta$, and $\leq^*$ is the lexicographic ordering of $A \times B$.

THEOREM 19

Let α , β , and γ be any ordinal numbers. Then

- 1) $(\alpha\beta)\gamma = \alpha(\beta\gamma)$.
- 2) $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$.

THEOREM 20

Any set of ordinal numbers is well-ordered.

PRIME NUMBERS

Two non-zero integers a and b are relatively prime provided the greatest common divisor $\gcd(a, b) = 1$.

THEOREM 21

If p is a prime and a is an integer such that p does not divide a , the a and p are relatively prime.

THEOREM 22

There are infinitely many primes.

THEOREM 23 — The Fundamental Theorem of Arithmetic

Let a be a natural number larger than 1. Then

- 1) a is a prime or a is the product of finitely many primes.
- 2) If $a = p_1 p_2 \dots p_r = q_1 q_2 \dots q_s$, where $p_1 p_2 \dots p_r, q_1 q_2 \dots q_s$ are primes, then $r = s$ and each $p_i (1 \leq i \leq r)$ can be paired with a $q_j (1 \leq j \leq s)$ so that the paired primes are equal.

An equation whose solutions are required to be integers is called a diophantine equation.

If a, b , and c are integers, the diophantine equation $ax + by = c$ is called a linear diophantine equation in two variables.

THEOREM 24

Let a, b , and $c \in Y$ and let $d = \gcd(a, b)$

let $Y =$ the set of all integers

- 1) If d does not divide c , then $ax + by = c$ has no integral solution.

- 2) If d divides c , then $ax + by = c$ has an infinite number of integral solutions.

CALCULUS REVIEW

- **Functions**
- **Coordinate Geometry**
- **Trigonometric Functions**
- **Limits**
- **Derivatives**
- **Application of Derivatives**
- **Integrals**
- **Sequences and Series**
- **Differential Equations**

Chapter 1

FUNCTIONS

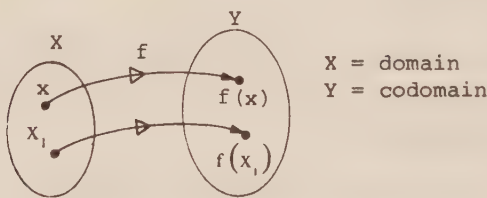
FUNCTIONS

A function is a correspondence between two sets, the domain and the range, such that for each value in the domain there corresponds exactly one value in the range.

A function has three distinct features:

- 1) the set X , which is the domain.
- 2) the set Y , which is the co-domain or range.
- 3) a functional rule, f , that assigns only one element $y \in Y$ to each $x \in X$. We write $y = f(x)$ to denote the functional value y at x .

Consider the figure below. The “machine” f transforms the domain X , element by element, into the co-domain Y .



COMBINATION OF FUNCTIONS

Let f and g represent functions; then

- 1) the sum $(f + g)(x) = f(x) + g(x)$.
- 2) the difference $(f - g)(x) = f(x) - g(x)$.

3) the product $(fg)(x) = f(x)g(x)$.

4) the quotient $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$, $g(x) \neq 0$.

5) the composite function $(g \circ f)(x) = g(f(x))$, where $f(x)$ must be in the domain of g .

PROPERTIES OF FUNCTIONS

Among the many properties of functions, we list here those that are most useful in the GRE Test.

A. Periodicity of a Function

A function f with domain X is periodic if there exists a positive real number p such that $f(x + p) = f(x)$ for all $x \in X$.

The smallest number p with this property is called the period of f .

Over any interval of length p , the behavior of a periodic function can be completely described.

B. Inverse of a Function

Assuming that f is a one-to-one function with domain X and range Y , then a function g having domain Y and range X is called the inverse function of f if:

$$f(g(y)) = y \text{ for every } y \in Y; \text{ and}$$

$$g(f(x)) = x \text{ for every } x \in X.$$

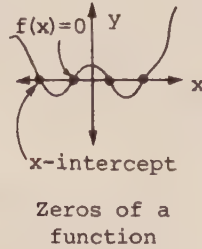
The inverse of the function f is denoted f^{-1} .

To find the inverse function f^{-1} , solve the equation $y = f(x)$ for x in terms of y .

Note: This solution must be a function.

C. Zeros of a Function

The “zeros” of an arbitrary function $f(x)$ are particular values of x for which $f(x) = 0$.



D. Fixed Points of a Function

Fixed points of a function $f(x)$ are those values of x which satisfy the equation $f(x) = x$.

Problem: Find the fixed points of the inverse of the function

$$f(x) = \frac{3x-2}{x-1}.$$

Solution: In order to find the inverse of $f(x)$, we have to solve the following for x :

$$y = f(x) = \frac{3x-2}{x-1}$$

$$y(x-1) = 3x-2$$

$$xy - y = 3x - 2$$

$$xy - 3x = y - 2$$

$$x(y-3) = y-2$$

$$x = \frac{y-2}{y-3} = g(y)$$

Thus, $g(y) = \frac{y-2}{y-3}$ is the inverse function of $f(x)$.

It can be easily verified that

$$\begin{aligned}
 f(g(y)) &= \frac{3g(y)-2}{g(y)-1} \\
 &= \frac{3\frac{y-2}{y-3}-2}{\frac{y-2}{y-3}-1} \\
 &= \frac{3(y-2)-2(y-3)}{y-2-(y-3)} \\
 &= \frac{3y-6-2y+6}{y-2-y+3} \\
 &= \frac{y+0}{0+1} = y \\
 &= \frac{y}{1} = y
 \end{aligned}$$

The fixed points of $g(y)$ can be obtained by setting

$$y = \frac{y-2}{y-3},$$

which implies that

$$y^2 - 3y = y - 2$$

$$y^2 - 4y + 2 = 0,$$

or
$$y = \frac{4 \pm \sqrt{16-8}}{2} = \frac{4 \pm \sqrt{8}}{2} = 2 \pm \sqrt{2}.$$

FUNCTION OF SEVERAL VARIABLES

Similar to functions of one variable, there are many instances which involve functions of two or more variables. For example, the volume of a right circular cone is

$$V(r, h) = \frac{1}{3} \pi r^2 h.$$

The function of two variables can be defined as a mapping from the domain of points (x, y) to a unique number w . We say that w is the image, under the mapping of f , of (x, y) by writing

$$w = f(x, y) .$$

Similarly, the notation $w = f(x, y, z)$ means that when values are assigned to the three variables x , y , and z , the value w is uniquely determined. An example of a function of three variables is

$$f(x, y, z) = x^2 + y^2 + z^2 .$$

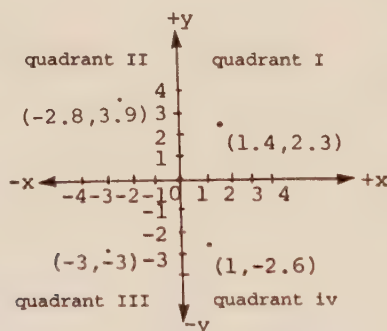
Chapter 2

COORDINATE GEOMETRY

INTRODUCTION, DISTANCE AND MIDDLE POINT FORMULAS

Coordinate geometry refers to the study of geometric figures using algebraic principles.

The Cartesian coordinate plane consists of a pair of perpendicular lines called the coordinate axes. The vertical axis is the y -axis and the horizontal axis is the x -axis. The intersection of the x -axis and y -axis is the origin. Positive is right of the origin on the x -axis and above the origin on the y -axis. Negative is in the respective opposite direction.



The abscissa is the coordinate of the projection of a given point onto the x -axis, and the ordinate is the coordinate of the projection of a given point on the y -axis.

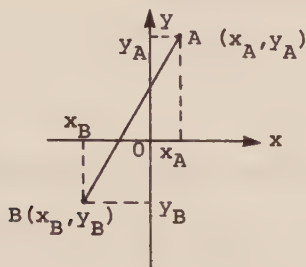
The first quadrant contains two positive coordinates; the second, third, and fourth quadrants are counterclockwise in order.

THEOREM 1

For any two point A and B with coordinates (x_A, y_A) and (x_B, y_B) , respectively, the distance between A and B is represented by

$$AB = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2}.$$

This is commonly known as the Distance Formula.

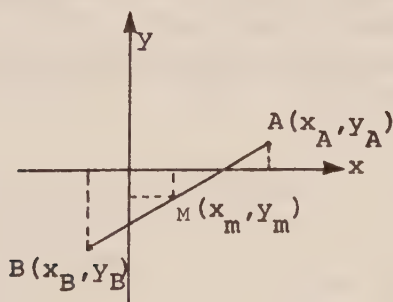


THEOREM 2

Given a line segment with endpoints $A(x_A, y_A)$ and $B(x_B, y_B)$, the coordinates of the midpoint of this line segment are $M(x_m, y_m)$:

$$x_m = \frac{x_A + x_B}{2}, \quad y_m = \frac{y_A + y_B}{2}.$$

This is commonly known as the Midpoint Formula.



The distance between a given point $A(x_0, y_0)$ and a line $l, ax + by + c = 0$, is given by the formula:

$$d = \left| \frac{ax_0 + by_0 + c}{\sqrt{a^2 + b^2}} \right|.$$

SLOPE AND LINE EQUATIONS

Slope

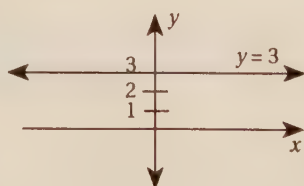
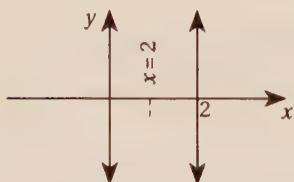
The slope m of a non-vertical line segment determined by points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by $m = \frac{y_2 - y_1}{x_2 - x_1}$, and is equal to the slope of any segment on that non-vertical line.

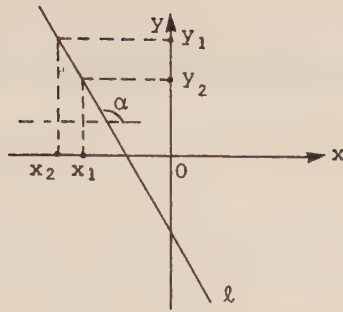
Inclination

The slope m of a line l equals the tangent of its inclination, where the inclination α is the angle above the x -axis and is included between the line and the positive direction of the x -axis.

Intercept

If the graph of a line crosses a coordinate axis, the point at which this occurs is called an intercept. If a line crosses the x -axis, this point is called the x -intercept and always has the y value, $y = 0$, e.g., $x = 2$. Similarly, if a line crosses the y -axis, this point is called the y -intercept and always has the x value, $x = 0$, e.g., $y = 3$.





The slope of l :

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan \alpha$$

THEOREM 3

Two non-vertical lines are perpendicular if and only if their slopes are negative reciprocals.

THEOREM 4

Two non-vertical lines are parallel if and only if they have the same slope. All vertical lines are parallel.

THEOREM 5

A linear equation is any equation which can be written in the form of $Ax + By + C = 0$. The graph of every linear equation in x and y is a straight line.

THEOREM 6

The equation of the line passing through the origin and having slope m is

$$y = mx.$$

THEOREM 7

The equation of the line whose y -intercept is b , and whose x -intercept is a , is

$$\frac{x}{a} + \frac{y}{b} = 1.$$

THEOREM 8

The equation of the line with slope m and y -intercept b is

$$y = mx + b.$$

THEOREM 9

The equation of the line passing through two points, $A(x_1, y_1)$ and $B(x_2, y_2)$, is

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}.$$

THEOREM 10

The equation of the line with slope m which passes through a point $A(x_0, y_0)$ is

$$y - y_0 = m(x - x_0).$$

Problem: A line which passes through the origin has an inclination angle of 45° . One point on the line is given by $(x, 1)$. Find the value of x at that point.

Solution: $\tan \alpha = m = \frac{y_2 - y_1}{x_2 - x_1}$ where $y_1 = 0$

$$x_1 = 0$$

$$\tan 45 = 1 = m = \frac{y_2 - 0}{x_2 - 0} \quad \text{if} \quad y_2 = 1$$

then $x_2 = 1$.

LOCUS

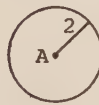
A locus is the set of points, and only those points, that satisfy a given condition or a set of given conditions.

To prove that a locus is correct, we must prove both of the following statements:

- 1) If a point is on the locus, then the point satisfies the given conditions.
- 2) If a point satisfies the given conditions, then it is on the locus.

THEOREM 11

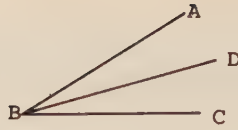
The locus of points at a given distance from a fixed point is a circle, with the fixed point as its center and with the radius equal to the given distance. The equation of a circle of radius r with its center at the origin is $x^2 + y^2 = r^2$.



This circle represents all points that are 2 cm. from point A.

THEOREM 12

The locus of all points equidistant from the sides of an angle is the angle bisector.



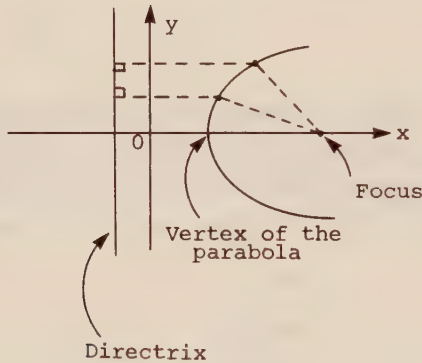
Line BD represents all points equidistant from sides AB and BC of angle ABC .

THEOREM 13

The locus of the vertex of the right angle of a right triangle with a fixed hypotenuse is a circle with the hypotenuse as diameter.

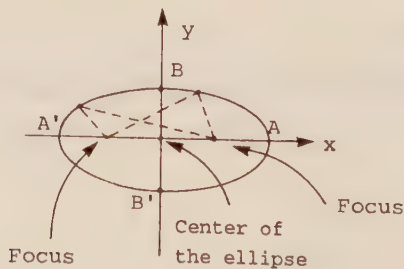
Parabola

A parabola is the locus of the points equally distant from a fixed line (called the directrix) and a fixed point (called the focus).



Ellipse

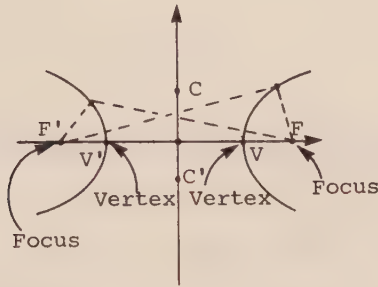
An ellipse is the locus of all points for which the sum of their distances from two fixed points is a constant.



The line segment BB' , which represents the perpendicular bisector of the line connecting A and A' , is called the minor axis, while the line segment AA' is called the major axis. The major axis is longer than the minor axis.

Hyperbola

A hyperbola is the locus of all points for which the difference of their distances from two fixed points (called the foci) is a constant.



The line segment VV' is the hyperbola's transverse axis, while segment CC' is the conjugate axis.

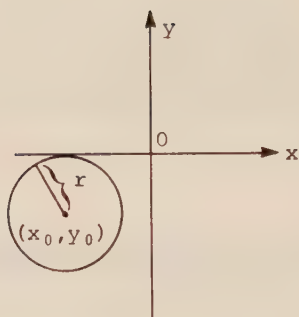
CONIC SECTIONS

THEOREM 14

The equation of a circle is

$$(x - x_0)^2 + (y - y_0)^2 = r^2,$$

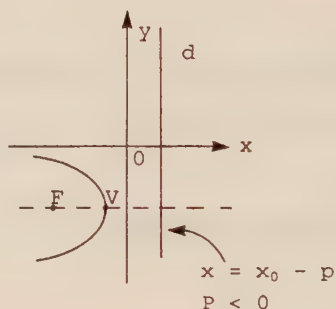
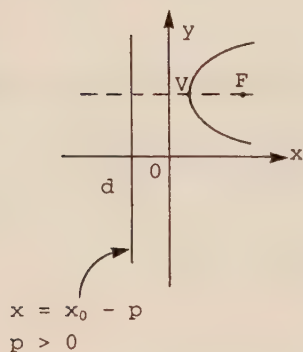
where (x_0, y_0) is the coordinate of the center of the circle and r is the length of the radius.



THEOREM 15

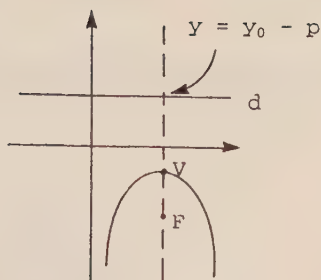
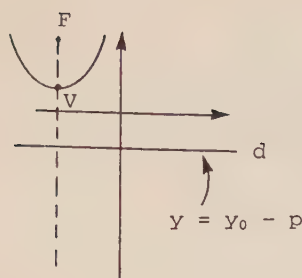
- 1) The equation of a parabola with vertex $V(x_0, y_0)$ and directrix d at $x = x_0 - p$ is

$$(y - y_0)^2 = 4p(x - x_0) .$$



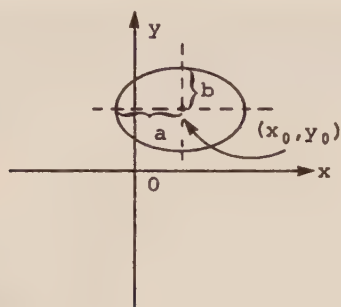
- 2) The equation of a parabola with vertex $V(x_0, y_0)$ and directrix d at $y = y_0 - p$ is

$$(x - x_0)^2 = 4p(y - y_0) .$$

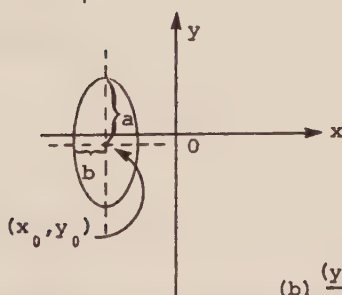


THEOREM 16

The equation of an ellipse with center $C(x_0, y_0)$ and the major axis parallel to the x -axis or y -axis, respectively, is:



$$(a) \frac{(x-x_0)^2}{a^2} + \frac{(y-y_0)^2}{b^2} = 1$$



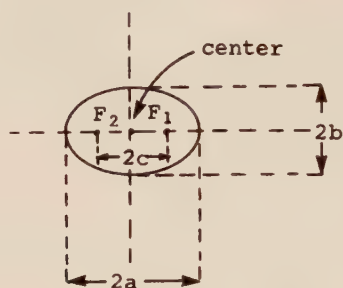
$$(b) \frac{(y-y_0)^2}{a^2} + \frac{(x-x_0)^2}{b^2} = 1$$

In both cases, the length of the major axis is $2a$ and the length of the minor axis is $2b$.

$C = \sqrt{a^2 - b^2}$ is the distance between the center of the ellipse and either focus.

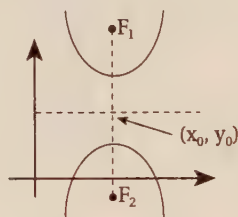
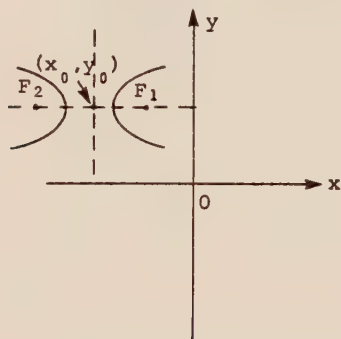
Eccentricity

$e = \frac{c}{a}$ is called the eccentricity.



THEOREM 17

The equation of a hyperbola with center $C(x_0, y_0)$ and transverse axis parallel to the x -axis or y -axis, respectively, is:



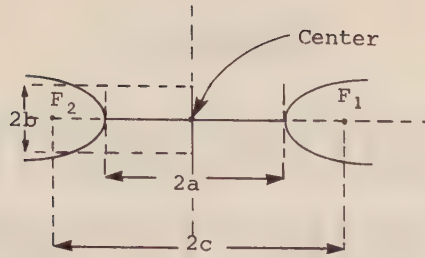
$$(a) \quad \frac{(x-x_0)^2}{a^2} - \frac{(y-y_0)^2}{b^2} = 1$$

$$(b) \quad \frac{(y-y_0)^2}{a^2} - \frac{(x-x_0)^2}{b^2} = 1$$

In both cases, the length of the transverse axis is $2a$ and the length of the conjugate axis is $2b$.

$C = \sqrt{a^2 + b^2}$ is the distance between the center and either focus.

$e = \frac{c}{a}$ is the eccentricity.



Problem: Find the coordinates of the focus and the equation of the directrix of the parabola $y^2 = 12x$.

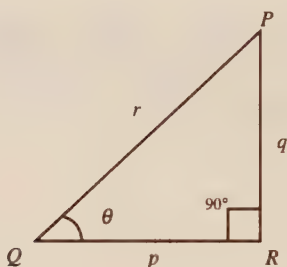
Solution: $(y - y_0)^2 = 4p(x - x_0)$. It appears that y_0 and x_0 are (0) respectively, thus leaving $y^2 = 4p(x)$. Therefore, $p = 3$ and the focus is given by $(p, 0) = (3, 0)$. The directrix is $x = -p = -3$.

Chapter 3

TRIGONOMETRIC FUNCTIONS

TRIGONOMETRIC FUNCTIONS

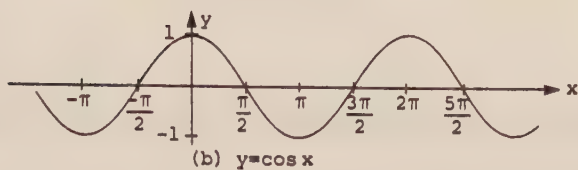
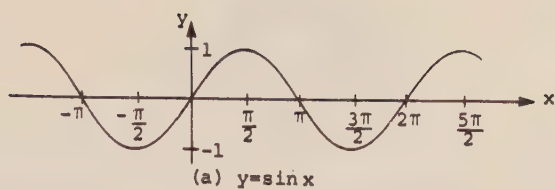
For the right triangle PQR , as shown in the diagram, the trigonometric functions are defined as follows:



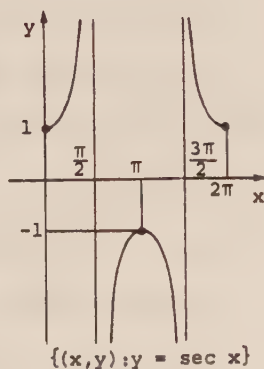
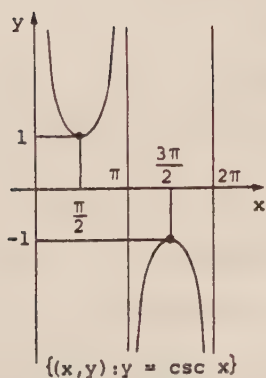
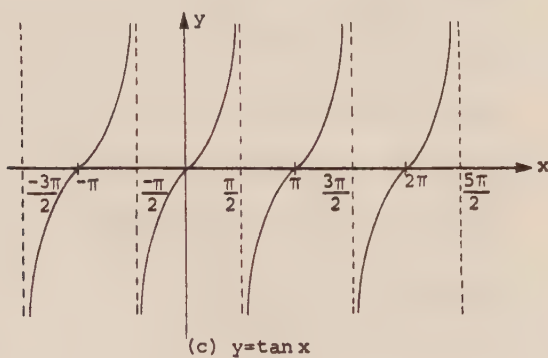
$$\sin \theta = \frac{q}{r}, \quad \cos \theta = \frac{p}{r}, \quad \tan \theta = \frac{q}{p}$$

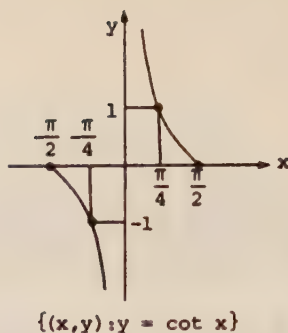
Provided the denominators are not zero, the following relationships exist:

$$\csc \theta = \frac{1}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \cot \theta = \frac{1}{\tan \theta}$$



The figures above and below show the graphs of each of the trigonometric functions. Notice that the x -axis is measured in radians.





SOME USEFUL TRIGONOMETRIC IDENTITIES

Some useful trigonometric formulas are listed below. They can be used to simplify different trigonometric functions.

$$\sin \theta \csc \theta = \cos \theta \sec \theta = \tan \theta \cot \theta = 1$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sec^2 \theta = 1 + \tan^2 \theta, \quad \csc^2 \theta = 1 + \cot^2 \theta$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin(\theta \pm \phi) = \sin \theta \cos \phi \pm \cos \theta \sin \phi$$

$$\cos(\theta \pm \phi) = \cos \theta \cos \phi \mp \sin \theta \sin \phi$$

$$\tan(\theta \pm \phi) = \frac{\tan \theta \pm \tan \phi}{1 \mp \tan \theta \tan \phi}$$

$$\cot(\theta \pm \phi) = \frac{\cot \theta \cot \phi \mp 1}{\cot \phi \pm \cot \theta}$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$

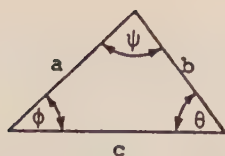
$$\sin 4\theta = 8 \cos^3 \theta \sin \theta - 4 \cos \theta \sin \theta$$

$$\cos 2\theta = 2 \cos^2 \theta - 1 = \cos^2 \theta - \sin^2 \theta$$

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

$$\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$$

$$\sin \theta \pm \sin \phi = 2 \sin \frac{1}{2}(\theta \pm \phi) \cos \frac{1}{2}(\theta \mp \phi)$$



$$\cos \theta + \cos \phi = 2 \cos \frac{1}{2}(\theta + \phi) \cos \frac{1}{2}(\theta - \phi)$$

$$\cos \theta - \cos \phi = -2 \sin \frac{1}{2}(\theta + \phi) \sin \frac{1}{2}(\theta - \phi)$$

$$\tan \theta \pm \tan \phi = \frac{\sin(\theta \pm \phi)}{\cos \theta \cos \phi}, \quad \cot \phi \pm \cot \theta = \frac{\sin(\theta \pm \phi)}{\sin \theta \sin \phi}$$

$$\sin \frac{\theta}{2} = \pm \left(\frac{1 - \cos \theta}{2} \right)^{\frac{1}{2}}, \quad \cos \frac{\theta}{2} = \pm \left(\frac{1 + \cos \theta}{2} \right)^{\frac{1}{2}}$$

$$\tan \frac{\theta}{2} = \frac{1 - \cos \theta}{\sin \theta} = \frac{\sin \theta}{1 + \cos \theta} = \pm \left(\frac{1 - \cos \theta}{1 + \cos \theta} \right)^{\frac{1}{2}}$$

$$\frac{\sin \theta \pm \sin \phi}{\cos \theta + \cos \phi} = \tan \frac{1}{2}(\theta \pm \phi)$$

Sine law:
$$\frac{a}{\sin \theta} = \frac{b}{\sin \phi} = \frac{c}{\sin \psi}$$

Cosine law:
$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc \cos \theta \\ b^2 &= c^2 + a^2 - 2ca \cos \phi \\ c^2 &= a^2 + b^2 - 2ab \cos \psi \end{aligned}$$

Problem: Express the following in terms of the trigonometric functions of x only.

$$\frac{1}{16} [2 \sin x + \sin 3x - \sin 5x]$$

Solution: The first term $\sin x$ is already a function of x only. The second term involves $3x$, which is given by

$$\sin 3x = 3\sin x - 4\sin^3 x .$$

The third term, $\sin 5x$, can be expressed as

$$\begin{aligned}\sin 5x &= \sin(2x + 3x) = \sin 2x \cos 3x + \cos 2x \sin 3x \\ &= (2\sin x \cos x)(4\cos^3 x - 3\cos x) + (\cos^2 x - \sin^2 x)(3\sin x - 4\sin^3 x) \\ &= 8\sin x \cos^4 x - 6\sin x \cos^2 x + 3\cos^2 x \sin x - 4\cos^2 x \sin^3 x \\ &\quad - 3\sin^3 x + 4\sin^5 x .\end{aligned}$$

Converting all terms in terms of $\sin x$ by using the identity $\cos^2 x = 1 - \sin^2 x$, we obtain

$$\begin{aligned}\sin 5x &= 8\sin x (1 - \sin^2 x)^2 - 6\sin x(1 - \sin^2 x) + 3(1 - \sin^2 x) \sin x \\ &\quad - 4(1 - \sin^2 x) \sin^3 x - 3\sin^3 x + 4\sin^5 x \\ &= 16\sin^5 x - 20\sin^3 x + 5\sin x .\end{aligned}$$

Thus,

$$\begin{aligned}&\frac{1}{16} [2\sin x + \sin 3x - \sin 5x] \\ &= \frac{1}{16} [2\sin x + 3\sin x - 4\sin^3 x - 16\sin^5 x + 20\sin^3 x - 5\sin x] \\ &= \frac{1}{16} [2\sin x + 3\sin x - 4\sin^3 x - 16\sin^5 x + 20\sin^3 x - 5\sin x] \\ &= \frac{1}{16} [16\sin^3 x - 16\sin^5 x] \\ &= \sin^3 x - \sin^5 x.\end{aligned}$$

Chapter 4

LIMITS

DEFINITIONS AND THEOREMS OF LIMITS

Let f be a function that is defined on an open interval containing a , but possibly not defined at a itself. Let L be a real number. The statement

$$\lim_{x \rightarrow a} f(x) = L$$

defines the limit of the function $f(x)$ at the point a . Very simply, L is the value that the function approaches as the point a is approached.

THEOREMS OF LIMITS

The following are important properties of limits:

Consider $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = K$. Then

a) uniqueness — if $\lim_{x \rightarrow a} f(x)$ exists, then it is unique.

$$\text{b) } \lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x) = L \pm K.$$

$$\text{c) } \lim_{x \rightarrow a} [f(x) \times g(x)] = \lim_{x \rightarrow a} f(x) \times \lim_{x \rightarrow a} g(x) = L \times K.$$

$$\text{d) } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L}{K}, \text{ provided that } K \neq 0.$$

$$\text{e) } \lim_{x \rightarrow a} \frac{1}{g(x)} = \frac{1}{K}, K \neq 0.$$

$$f) \quad \lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n \text{ for } n > 0.$$

$$g) \quad \lim_{x \rightarrow a} [cf(x)] = c \left[\lim_{x \rightarrow a} f(x) \right], \quad c \in R.$$

$$h) \quad \lim_{x \rightarrow a} cx^n = c \lim_{x \rightarrow a} x^n = ca^n, \quad c \in R.$$

i) If f is a polynomial, then

$$\lim_{x \rightarrow a} f(x)^n = f(a)^n \text{ for all } a \in R.$$

$$j) \quad \lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a} \text{ when } a \geq 0 \text{ and } n \text{ is a positive integer or when } a \leq 0 \text{ and } n \text{ is an odd integer.}$$

k) If $f(x) \leq h(x) \leq g(x)$ for all x in an open interval containing a , except possibly at a , and if

$$\lim_{x \rightarrow a} f(x) = L = \lim_{x \rightarrow a} g(x), \text{ then } \lim_{x \rightarrow a} h(x) = L.$$

If the function $f(x)$ has jumps at some values of x , then one-sided limits are defined. The statement $\lim_{x \rightarrow a^+} f(x) = R$ tells us that as x approaches a from the right or from positive infinity, the function f has the limit R . Similarly, the statement $\lim_{x \rightarrow a^-} f(x) = L$ says that as x approaches a from the left-hand side or from negative infinity, the function f has the limit L .

SOME SPECIAL RULES

$$a) \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

$$b) \quad \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e, \quad \lim_{n \rightarrow 0} (1 + n)^{\frac{1}{n}} = e$$

c) For $a > 1$,

$$\lim_{x \rightarrow \infty} a^x = +\infty, \quad \lim_{x \rightarrow -\infty} a^x = 0$$

$$\lim_{x \rightarrow \infty} \log_a x = +\infty, \quad \lim_{x \rightarrow 0} \log_a x = -\infty$$

d) For $0 < a < 1$,

$$\lim_{x \rightarrow +\infty} a^x = 0, \quad \lim_{x \rightarrow -\infty} a^x = +\infty$$

$$\lim_{x \rightarrow +\infty} \log_a x = -\infty, \quad \lim_{x \rightarrow 0} \log_a x = +\infty$$

e) $\lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty$

but f) $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist, because as x approaches zero from the left, the function approaches $-\infty$ and from the right, it approaches $+\infty$.

Problem: Evaluate

1) $\lim_{x \rightarrow \infty} \sqrt{x^2 + 2x} - x$;

2) $\lim_{x \rightarrow \infty} \frac{(e^x + e^{-x})}{2 - 3e^x}$.

Solution:

1) Note that

$$\sqrt{x^2 + 2x} - x \times \frac{\sqrt{x^2 + 2x} + x}{\sqrt{x^2 + 2x} + x} = \frac{2}{\sqrt{1 + \frac{2}{x}} + 1},$$

and the last quantity tends to 1 as $x \rightarrow \infty$.

2) Dividing the numerator and denominator by e^{-x} , we obtain

$$\frac{(e^x + e^{-x})e^{-x}}{(2 - 3e^x)e^{-x}} = \frac{1 + e^{-2x}}{2e^{-x} - 3}$$

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{1 + e^{-2x}}{2e^{-x} - 3} &= \frac{\lim_{x \rightarrow +\infty} (1 + e^{-2x})}{\lim_{x \rightarrow +\infty} (2e^{-x} - 3)} \\ &= \frac{1 + 2 \lim_{x \rightarrow +\infty} e^{-2x}}{2 \lim_{x \rightarrow +\infty} e^{-x} - 3} \\ &= \frac{1 + 2 \times 0}{2 \times 0 - 3} = -\frac{1}{3}. \end{aligned}$$

Chapter 5

DERIVATIVES

DEFINITION OF DERIVATIVE

The derivative of a function expresses its rate of change with respect to an independent variable. The derivative is also the slope of the tangent line to the curve. Assume $y = f(x)$ is a function of x . If the limit

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

exists and is finite, this limit is called the derivative of f at x . One common notation for $f'(x)$ is $\frac{dy}{dx}$.

RULES FOR FINDING DERIVATIVES

General Rules

1) If f is a constant function, $f(x) = c$, then $f'(x) = 0$.

2) If $f(x) = x$, then $f'(x) = 1$.

3) If f is differentiable, then

$$cf(x)' = cf'(x).$$

4) Power Rule: If $f(x) = x^n$, $n \in \mathbb{Z}$, then

$$f'(x) = nx^{n-1}; \text{ if } n < 0, \text{ then } x^n \text{ is not defined at } x = 0.$$

5) If f and g are differentiable on the interval (a, b) , then:

$$\text{a) } (f + g)'(x) = f'(x) + g'(x).$$

b) Product Rule: $(fg)'(x) = f(x)g'(x) + g(x)f'(x)$.

Example:

Find $f'(x)$ if $f(x) = (x^3 + 1)(2x^2 + 8x - 5)$.

$$\begin{aligned} f'(x) &= (x^3 + 1)(4x + 8) + (2x^2 + 8x - 5)(3x^2 + 0) \\ &= 4x^4 + 8x^3 + 4x + 8 + 6x^4 + 24x^3 - 15x^2 \\ &= 10x^4 + 32x^3 - 15x^2 + 4x + 8 \end{aligned}$$

c) Quotient Rule: $\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$.

Example:

Find $f'(x)$ if $f(x) = \frac{3x^2 - x + 2}{4x^2 + 5}$.

$$\begin{aligned} f'(x) &= \frac{-(3x^2 - x + 2)(8x) + (4x^2 + 5)(6x - 1)}{(4x^2 + 5)^2} \\ &= \frac{-(24x^3 - 8x^2 + 16x) + (24x^3 - 4x^2 + 30x - 5)}{(4x^2 + 5)^2} \\ &= \frac{4x^2 + 14x - 5}{(4x^2 + 5)^2} \end{aligned}$$

6) If $f(x) = x^{\frac{m}{n}}$, then $f'(x) = \frac{m}{n} x^{\frac{m}{n}-1}$, where $m, n \in \mathbb{Z}$ and $n \neq 0$.

7) Polynomials: If $f(x) = (a_0 + a_1x + a_2x^2 + \dots + a_nx^n)$, then

$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1} .$$

This employs the power rule and rules concerning constants.

8) Chain Rule: Let $f(u)$ be a composite function, where $u = g(x)$.

Then $f'(u) = f'(a)g'(x)$ or, if $y = f(u)$ and $u = g(x)$, then

$$1 \ D_x y = (D_u y)(D_x u) = f'(u)g'(x) .$$

The short form of this is

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}$$

The three most basic trigonometric derivatives are:

$$\frac{d}{dx}(\sin x) = \cos x,$$

$$\frac{d}{dx}(\cos x) = -\sin x, \text{ and}$$

$$\frac{d}{dx}(\tan x) = \sec^2 x .$$

The exponential function e^x has the simplest meaning of all derivatives. Its derivative is itself.

$$\frac{d}{dx}e^x = e^x, \text{ and}$$

$$\frac{d}{dx}e^u = e^u \frac{du}{dx} .$$

Since the natural logarithmic function is the inverse of $y = e^x$ and $\ln e = 1$, it follows that

$$\frac{d}{dx} \ln y = \frac{1}{y} \frac{dy}{dx} , \text{ and}$$

$$\frac{d}{dx} \ln u = \frac{1}{u} \frac{du}{dx} .$$

If x is any real number and a is any positive real number, then

$$a^x = e^{x \ln a} .$$

From this definition, we can obtain the following:

$$1) \quad \frac{d}{dx} a^x = a^x \ln a \quad \text{and} \quad \frac{d}{dx} a^u = a^u \ln a \frac{du}{dx} .$$

$$2) \quad \frac{d}{dx} (\log_a x) = \frac{1}{x \ln a} \quad \text{and} \quad \frac{d}{dx} \log_a |u| = \frac{1}{u \ln a} \frac{du}{dx} ,$$

where $u \neq 0$.

Sometimes it is useful to take the logs of a function and then differentiate, since the computation becomes easier (as in the case of a product).

Problem: Find the derivative of the following function:

$$y = (x + 5)(x^4 + 1) .$$

Solution:

$$\ln y = \ln \left[(x + 5)(x^4 + 1) \right] = \ln(x + 5) + \ln(x^4 + 1)$$

$$\frac{d}{dx} \ln y = \frac{d}{dx} \ln(x + 5) + \frac{d}{dx} \ln(x^4 + 1)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x + 5} + \frac{4x^3}{x^4 + 1}$$

$$\frac{dy}{dx} = (x + 5)(x^4 + 1) \left[\frac{1}{x + 5} + \frac{4x^3}{x^4 + 1} \right]$$

$$= (x^4 + 1) + 4x^3(x + 5)$$

$$= 5x^4 + 20x^3 + 1$$

This is the same result as that obtained by using the product rule.

Problem: Find $\frac{dy}{dt}$ for $y = x^3 - 3x^2 + 5x - 4 = f(x)$,
 where $x = t^2 + t = g(t)$.

Solution:

$$\begin{aligned} f'(x) &= \frac{dy}{dx} = 3x^2 - 6x + 5 \\ &= 3(t^2 + t)^2 - 6(t^2 + t) + 5 \\ g'(t) &= \frac{dx}{dt} = 2t + 1 \end{aligned}$$

Applying the chain rule,

$$\begin{aligned} \frac{dy}{dt} &= \frac{dy}{dx} \cdot \frac{dx}{dt} \\ &= [3(t^2 + t)^2 - 6(t^2 + t) + 5](2t + 1) . \end{aligned}$$

PARTIAL DIFFERENTIATION

Partial Differentiation

Let f represent a function of two variables. The first partial derivative of f with respect to x and y are the functions f_x and f_y , respectively. These functions are defined as

$$f_x(x, y) = \lim_{h \rightarrow 0} f \frac{(x + h, y) - f(x, y)}{h}$$

and

$$f_y(x, y) = \lim_{h \rightarrow 0} f \frac{(x, y + h) - f(x, y)}{h} .$$

This holds true provided that limits exist.

The first two partial derivatives are found in the manner prescribed below:

- 1) The derivative $f_x(x, y)$ is found by differentiating $f(x, y)$ with respect to x in the usual manner, while y is considered to be constant.
- 2) The derivative $f_y(x, y)$ is found by differentiating $f(x, y)$ with respect to y , keeping x constant.

CHAIN RULE

Partial derivatives can also be calculated by using the chain rule.

Let $z = f(x, y)$ represent a continuous function and let $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ also be continuous.

If $x = x(a, b)$ and $y = y(a, b)$ are functions of a and b so that $\frac{\partial x}{\partial a}$, $\frac{\partial x}{\partial b}$,

$\frac{\partial y}{\partial a}$, $\frac{\partial y}{\partial b}$ all exist, then z is a function of a and b and the following are true:

$$\frac{\partial z}{\partial a} = \left(\frac{\partial f}{\partial x} \right) \left(\frac{\partial x}{\partial a} \right) + \left(\frac{\partial f}{\partial y} \right) \left(\frac{\partial y}{\partial a} \right), \text{ and}$$

$$\frac{\partial z}{\partial b} = \left(\frac{\partial f}{\partial y} \right) \left(\frac{\partial y}{\partial b} \right) + \left(\frac{\partial f}{\partial x} \right) \left(\frac{\partial x}{\partial b} \right).$$

THEOREM 1

Suppose that $z = f(x, y)$ and x and y are functions of the same variable. Then

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy.$$

This is similar to the results obtained for the function $w = f(x, y, z)$.

$$dw = \frac{\partial w}{\partial x} dx + \frac{\partial w}{\partial y} dy + \frac{\partial w}{\partial z} dz$$

This equation holds true when x , y , and z are either independent or intermediate variables.

Problem: Find the derivative $\frac{dy}{dx}$ of the following equation by methods of partial differentiation:

$$e^{xy} + \sin xy + 1 = 0$$

Solution:

$$f_x = ye^{xy} + y \cos xy$$

$$f_y = xe^{xy} + x \cos xy$$

$$\frac{dy}{dx} = \frac{ye^{xy} + y \cos xy}{xe^{xy} + x \cos xy} = -\frac{y(e^{xy} + \cos xy)}{x(e^{xy} + \cos xy)}$$

$$\frac{dy}{dx} = \frac{-y}{x}$$

$\frac{dy}{dx}$ could also be found by implicit differentiation as follows:

$$\frac{d}{dx}(e^{xy} + \sin xy + 1) = 0,$$

i.e.,
$$e^{xy}\left(y + x \frac{dy}{dx}\right) + \cos xy\left(y + x \frac{dy}{dx}\right) = 0.$$

Collecting terms with $\frac{dy}{dx}$ factor to one side, we get

$$(xe^{xy} + x \cos xy) \frac{dy}{dx} = -(ye^{xy} + y \cos xy)$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{-(ye^{xy} + y \cos xy)}{xe^{xy} + x \cos xy} = \frac{-y}{x} \times \frac{e^{xy} + \cos xy}{e^{xy} + \cos xy} \\ &= \frac{-y}{x}. \end{aligned}$$

Chapter 6

APPLICATION OF DERIVATIVES

MAXIMA AND MINIMA

A function f is said to have a relative, or local, maximum at $x = a$ if

$$f(a) \geq f(x)$$

for all x in some interval about a . For a local minimum at $x = l$,

$$f(l) \leq f(x)$$

for all values of x in some open interval about c .

THEOREM 1

Let the function f be defined for $a \leq x \leq l$ and have a relative maximum or minimum at $x = c$, where $a < c < l$. If the derivative $f'(x)$ exists as a finite number $x = c$, then

$$f'(c) = 0 .$$

Solving Maxima and Minima Problems

- Step 1: Determine which variable is to be maximized or minimized (i.e., the dependent variable y).
- Step 2: Find the independent variable x .
- Step 3: Write an equation involving x and y . All other variables can be eliminated by substitution.

Step 4: Differentiate with respect to the independent variable.

Step 5: Set the derivative equal to zero to obtain critical values.

Step 6: Determine maxima and minima.

In order to determine whether that point is a minimum or a maximum, evaluate the second derivative at that point.

If the second derivative is positive, then it is a minimum; if the second derivative is negative, then it is a maximum.

Problem: Suppose a manufacturer can sell x items per week at a price $P = 200 - 0.01x$ cents, and that it costs $y = 50x + 20,000$ cents to produce x items. Find the production level for maximum profits.

Solution: Total revenue per week on x items is

$$xP = 200x - 0.01x^2.$$

The manufacturer's profit T is revenue minus cost.

$$\begin{aligned} T = xP - y &= (200x - 0.01x^2) - (50x + 20,000) \\ &= 150x - 0.01x^2 - 20,000 \end{aligned}$$

$$\frac{dT}{dx} = 150 - 0.02x, \text{ which is zero when } x = 7,500.$$

Since the second derivative $\frac{d^2T}{dx^2} = -0.02$ is negative, this production level gives the maximum profit.

Points of Inflection

Points which satisfy $f''(x) = 0$ may be positions where concavity changes. These points are called the points of inflection. The point of inflection is the point at which the curve crosses its tangent line.

ROLLE'S THEOREM

Let the function f be defined and continuous on the closed interval $[a, b]$ and differentiable in the open interval (a, b) . Furthermore, let

$$f(a) = f(b) = 0.$$

Then there is at least one number c between a and b where $f'(x)$ is zero; that is,

$$f'(c) = 0.$$

Problem: Check the existence of minima or maxima of the following function:

$$y = x^3 - 4x = f(x).$$

Solution: The function $f(x)$ has zeros at 0, +2, and -2. Therefore, at least one maxima or minima occurs in the interval $[0, 2]$ and at least one in $[-2, 0]$. Setting $f'(x) = 0$ gives

$$3x^2 - 4 = 0, \text{ or}$$

$$x = \pm \frac{2}{\sqrt{3}}.$$

Therefore, maxima or minima occur at $x = -\frac{2}{\sqrt{3}}$ and $x = \frac{2}{\sqrt{3}}$. Taking the second derivative,

$$f''(x) = 6x.$$

Since $f''\left(-\frac{2}{\sqrt{3}}\right)$ is negative, $x = -\frac{2}{\sqrt{3}}$ is a maxima, and since $f''\left(\frac{2}{\sqrt{3}}\right)$ is positive, $x = \frac{2}{\sqrt{3}}$ is a minima.

MEAN VALUE THEOREM

If $y = f(x)$ is continuous at every point on the closed interval $[a, b]$ and has a derivative at every point in the interior interval (a, b) , then there is at least one number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Problem: Let $f(x) = x^3$, $a = -2$, and $b = +2$. Find the points where the slope of $f(x)$ is equal to the slope of the secant.

Solution: The secant has the slope $\frac{f(b) - f(a)}{b - a}$.

$$f'(x) = 3x^2, \text{ or } f'(c) = 3c^2.$$

Therefore, according to the mean value theorem,

$$3c^2 = \frac{f(b) - f(a)}{b - a} = \frac{8 - (-8)}{2 - (-2)} = \frac{16}{4} = 4, \text{ or}$$

$$c^2 = \frac{4}{3}, \text{ or } c = \pm \frac{2}{\sqrt{3}}.$$

Therefore, $\pm \frac{2}{\sqrt{3}}$ are the two points where the tangent to the curve $y = x^3$ is parallel to the chord through the points $(-2, -8)$ and $(+2, +8)$.

L'HOPITAL RULE

An application of the mean value theorem is in the evaluation of

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}, \text{ where } f(a) = 0 \text{ and } g(a) = 0.$$

The L'Hopital rule states that if the $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is an indeterminate form (i.e., $\frac{0}{0}$ or $\frac{\infty}{\infty}$), then we can differentiate the numerator and the denominator separately and arrive at an expression that has the same limit as the original problem.

Thus,
$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} .$$

Problem: Evaluate the following limits:

$$1) \lim_{t \rightarrow 0} \frac{\sqrt{1+t} - \left(1 + \frac{t}{2}\right)}{t^2}$$

$$2) \lim_{t \rightarrow \infty} \frac{t^2 + t}{2t^2 + 1}$$

Solution:

1) Directly substituting the limit yields $\frac{0}{0}$. Applying the L'Hopital rule,

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{\sqrt{1+t} - \left(1 + \frac{t}{2}\right)}{t^2} &= \lim_{t \rightarrow 0} \frac{\frac{1}{2}(1+t)^{-\frac{1}{2}} - \frac{1}{2}}{2t} \\ &= \lim_{t \rightarrow 0} \frac{-\frac{1}{4}(1+t)^{-\frac{3}{2}}}{2} = -\frac{1}{8} . \end{aligned}$$

2) This assumes the form $\frac{\infty}{\infty}$ as $t \rightarrow 0$. Applying the L'Hôpital rule,

$$\lim_{t \rightarrow \infty} \frac{t^2 + t}{2t^2 + 1} = \lim_{t \rightarrow \infty} \frac{2t + 1}{4t} = \lim_{t \rightarrow \infty} \frac{2}{4} = \frac{1}{2}.$$

Note that we differentiated the numerator and denominator twice.

TAYLOR'S THEOREM WITH REMAINDER

The expansion of the function $f(x)$ with $n + 1$ derivatives on an interval containing the value x_0 is

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \dots + \frac{f^{(n)}(x_0)(x - x_0)^n}{n!} + R_n.$$

R_n is the remainder represented by the formula

$$R_n = \frac{f^{(n+1)}(\epsilon_0)(x - x_0)^{n+1}}{(n+1)!},$$

where ϵ_0 is some number between x and x_0 .

Let f be a continuous function of two variables, and let all of its partial derivatives be continuous about the point $P(a, b)$.

The expansion for f is

$$f(x, y) = f(a, b) + \sum_{1 \leq r+s \leq p} \frac{\partial^{r+s} f(a, b)}{\partial x^r \partial y^s} \frac{(x-a)^r}{r!} \frac{(y-b)^s}{s!} + R_p.$$

The remainder R_p is represented by the formula

$$R_p = \sum_{r+s \geq p+1} \frac{\partial^{r+s} f(\epsilon, \eta)}{\partial x^r \partial y^s} \frac{(x-a)^r}{r!} \frac{(y-b)^s}{s!}.$$

The value of (ϵ, η) is located on the line segment between points (a, b) and (x, y) .

The remainder term demonstrates that if $0 < r < d$, then

$$\varepsilon = a + \frac{(x-a)}{d} \tau, \quad \eta = b + \frac{(y-b)}{d} \tau.$$

Another mode of expression of Taylor's theorem is

$$f(x, y) = f(a, b) + \sum_{g=1}^p \frac{1}{g!} \left[\sum_{r=0}^q \frac{g!}{(g-r)!r!} \frac{\partial^g f(a, b)}{\partial x^{g-r} \partial y^r} (x-a)^{g-r} (y-b)^r \right] + R_p.$$

If P tends to infinity for some function f , the remainder approaches 0 ($R_p \rightarrow 0$), and we obtain a representation of f as an infinite series in x and y . This series is called a double series and f is said to have expanded about the point (a, b) .

MISCELLANEOUS DEFINITIONS

Jacobian: Let $x = f(u, v)$, $y = g(u, v)$ be a mapping of the xy -plane into the uv -plane. Then the "Jacobian" of this mapping is the determinant

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}.$$

This can be generalized to any number of variables.

Wronskian: Let $f_1(x)$ and $f_2(x)$ be two differentiable functions. Then the "Wronskian" $W(x)$ is defined as

$$W(x) = \begin{vmatrix} f_1(x) & f_2(x) \\ f_1'(x) & f_2'(x) \end{vmatrix}.$$

This can also be generalized to arbitrary number functions.

Chapter 7

INTEGRALS

INDEFINITE INTEGRALS

The indefinite integral of $f(x)$, denoted by $\int f(x)dx$, is the most general integral of $f(x)$; that is,

$$\int f(x)dx = F(x) + C.$$

$F(x)$ is any function such that $F'(x) = f(x)$. C is an arbitrary constant.

Integration Formulas

$$1) \int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

$$2) \int \frac{dx}{x} = \ln|x| + C$$

$$3) \int \frac{dx}{x-a} = \ln|x-a| + C$$

$$4) \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$5) \int \frac{xdx}{x^2 + a^2} = \frac{1}{2} \ln|x^2 + a^2| + C$$

$$6) \int \frac{dx}{(a^2 - x^2)^{\frac{1}{2}}} = \sin^{-1} \frac{x}{a} + C$$

$$7) \int \sin ax dx = -\frac{1}{a} \cos ax + C$$

$$8) \int \cos ax dx = \frac{1}{a} \sin ax + C$$

$$9) \int \sec^2 x dx = \tan x + C$$

$$10) \int e^{ax} dx = \frac{e^{ax}}{a} + C$$

$$11) \int \sinh ax dx = \frac{1}{a} \cosh ax + C$$

$$12) \int \cosh ax dx = \frac{1}{a} \sinh ax + C$$

Algebraic Simplification

Certain seemingly complicated integrals can be made simple by simple algebraic manipulations.

Example: Find $\int \frac{x}{x+1} dx$.

Write

$$\frac{x}{x+1} = \frac{x+1-1}{x+1} = 1 - \frac{1}{x+1}$$

$$\int \frac{x}{x+1} dx = \int dx - \int \frac{dx}{x+1} = x - \ln|x+1| + C$$

Substitution of Variables

Suppose $F(x)$ is expressed as a composite function, $F(x) = f(u(x))$; then the differential $F'(x)dx = f'(u)du$.

Therefore, $\int F'(x)dx = \int f'(u)du = f(u) + c = f(u(x)) + c = F(x) + c$.

Example: Evaluate $\int \frac{x}{x^2 + a^2} dx$.

Let $u = x^2 + a^2$

$$du = 2x \, dx$$

$$\frac{1}{2} du = x dx$$

$$\int \frac{x}{x^2 + a^2} dx = \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln|u| + c = \frac{1}{2} \ln|x^2 + a^2| + c.$$

Change of Variables

Example: Evaluate $\int_0^1 x(1+x)^{\frac{1}{2}} dx$.

Let $u = 1 + x$, $du = dx$, $x = u - 1$

$$\int_0^1 x(1+x)^{\frac{1}{2}} = \int_1^2 (u-1)u^{\frac{1}{2}} du.$$

* Notice the change in the limits for $x = 0$, $u = 1$ and for $x = 1$, $u = 2$.

$$\int_1^2 (u-1)u^{\frac{1}{2}} du = \int_1^2 u^{\frac{3}{2}} - u^{\frac{1}{2}} du$$

$$= \frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} \Big|_1^2$$

$$= \left[\frac{2}{5} \sqrt{32} - \frac{2}{3} \sqrt{8} \right] - \left(\frac{2}{5} - \frac{2}{3} \right)$$

$$= \frac{4\sqrt{2}}{15} - \frac{4}{15} = \frac{4}{15}(\sqrt{2} - 1).$$

Integration by Parts

This method is based on the formula

$$d(uv) = u \, dv + v \, du .$$

The corresponding integration formula,

$$uv = \int u \, dv + \int v \, du , \text{ is applied in the form}$$

$$\int u \, dv = uv - \int v \, du .$$

This procedure involves the identification of u and dv and their manipulation into the form of the latter equation. v must be easily determined. If a definite integral is involved, then

$$\int_a^b u \frac{dv}{dx} dx = uv \Big|_a^b - \int_a^b v \frac{du}{dx} dx .$$

Example: Evaluate $\int x \cos x \, dx$.

$$u = x \qquad dv = \cos x \, dx$$

$$du = dx \qquad v = \sin x$$

$$\begin{aligned} \int x \cos x \, dx &= x \sin x - \int \sin x \, dx \\ &= x \sin x - (-\cos x) + c \\ &= x \sin x + \cos x + c \end{aligned}$$

Trigonometric Integrals

Integrals of the form $\int \sin^n x \, dx$ or $\int \cos x \, dx$ can be evaluated without resorting to integration by parts. This is done in the following manner:

$$\text{write } \int \sin^n x \, dx = \int \sin^{n-1} x \sin x \, dx , \text{ if } n \text{ is odd.}$$

Since the integer $n - 1$ is even, we may then use the fact that $\sin^2 x$

$= 1 - \cos^2 x$ to obtain a form that is easier to integrate.

Example:

$$\begin{aligned}\int \sin^5 x \, dx &= \int \sin^4 x \sin x \, dx \\ &= \int (\sin^2 x)^2 \sin x \, dx ,\end{aligned}$$

but $\sin^2 x = 1 - \cos^2 x$.

Hence,

$$\begin{aligned}\int \sin^5 x \, dx &= \int (1 - \cos^2 x)^2 \sin x \, dx \\ &= \int (1 - 2\cos^2 x + \cos^4 x) \sin x \, dx .\end{aligned}$$

Substitute $u = \cos x$, $du = -\sin x \, dx$

$$\begin{aligned}&= \int (1 - 2u^2 + u^4) (-du) \\ &= -u + \frac{2}{3}u^3 - \frac{u^5}{5} \\ &= -\cos x + \frac{2}{3}\cos^3 x - \frac{1}{5}\cos^5 x + c .\end{aligned}$$

A similar technique can be employed for odd powers of $\cos x$.

If the integrand is $\sin^n x$ or $\cos^n x$ and n is even, then the half angle formulas,

$$\begin{aligned}\sin^2 x &= \frac{1 - \cos 2x}{2} \quad \text{or} \\ \cos^2 x &= \frac{1 + \cos 2x}{2} ,\end{aligned}$$

may be used to simplify the integrand.

Example:

$$\int \cos^2 x \, dx = \frac{1}{2} \int (1 + \cos 2x) \, dx$$

$$= \frac{1}{2}x + \frac{1}{4}\sin 2x + c .$$

DEFINITE INTEGRALS

The definite integral of $f(x)$, denoted by $\int_a^b f(x)dx$, is given by

$$\int_a^b f(x)dx = F(b) - F(a),$$

where $F(x)$ is any function such that $F'(x) = f(x)$.

Chapter 8

SEQUENCES AND SERIES

INFINITE SEQUENCES

An infinite sequence is a function whose domain is the set of positive integers.

A sequence $\{a_n\}$ has the limit ℓ (denoted by $\lim_{n \rightarrow \infty} a_n = \ell$) if for every $c > 0$, there exists a positive number N such that if $n > N$, then $|a_n - \ell| < c$.

If the limit $\lim_{n \rightarrow \infty} a_n$ does not exist, then the sequence $\{a_n\}$ has no limit.

The following facts are useful for sequences:

Suppose that $\lim_{n \rightarrow \infty} a_n = \ell$ and $\lim_{n \rightarrow \infty} b_n = m$, where m and ℓ are real numbers; then it can be proven that

$$1) \lim_{n \rightarrow \infty} (a_n + b_n) = \ell + m .$$

$$2) \lim_{n \rightarrow \infty} (a_n - b_n) = \ell - m .$$

$$3) \lim_{n \rightarrow \infty} a_n b_n = \ell \times m .$$

$$4) \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\ell}{m} .$$

5) If $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ are infinite sequences such that $a_n \leq b_n \leq c_n$ for all n , and if

$$\lim_{n \rightarrow \infty} a_n = \ell = \lim_{n \rightarrow \infty} c_n ,$$

then $\lim_{n \rightarrow \infty} b_n = \ell .$

6) If $\{a_n\}$ represents a sequence and $\lim_{n \rightarrow \infty} |a_n| = 0$, then

$$\lim_{n \rightarrow \infty} a_n = 0 .$$

A monotonic sequence is a sequence with successive terms that are non-decreasing, i.e.,

$$(a_1 \leq a_2 \leq \dots \leq a_n),$$

or non-increasing, i.e.,

$$(a_1 \geq a_2 \geq \dots \geq a_n) .$$

A sequence is said to be bounded if there is a real positive number α such that $|a_m| \leq \alpha$ for all m .

CONVERGENCE

An infinite series $\sum_{k=0}^{\infty} a_k$, sometimes written $a_0 + a_1 + a_2 + \dots$, is the sequence $\{s_0, s_1, s_2, \dots\}$ of partial sums.

An infinite series converges if $\lim_{n \rightarrow \infty} s_n = s$ for some real number s . The series diverges if the sequence of the partial sums diverges (i.e., the limit does not exist).

Geometric series have the form $a + ar + r^2 + \dots + ar^{n-1}$, where a and r are real numbers and $a \neq 0$.

Tests of Convergence

- 1) For every $c > 0$, if there exists an integer N such that $|s_k - s_\ell| < c$ whenever $k, \ell > N$, then the infinite series $\sum a_n$ is convergent.

- 2) If $\sum a_n$ is a positive term series (i.e., all the terms are positive) and if there exists a number n such that $S_n < m$ for every n , then the series converges and has the sum $s \leq m$. If no such m exists, then the series diverges.
- 3) The p-series expressed as $\sum_{n=1}^{\infty} \frac{1}{p^n}$ converges if $p > 1$ and diverges if $p \leq 1$.
- 4) **Comparison Test:** Let $\sum a_n$ and $\sum b_n$ represent positive term series; then
- If $a_n \leq b_n$ and the series $\sum b_n$ converges, then $\sum a_n$ converges.
 - If $a_n \geq b_n$ and the series $\sum b_n$ diverges, then $\sum a_n$ diverges.
- 5) Let $\{a_k\}$ represent a decreasing series of positive terms.

If $a_k \geq 0$, then $\sum_{k=0}^{\infty} (-1)^k a_k$ converges.

The series $\sum a_k$ is absolutely convergent if the series $\sum |a_k| = |a_1| + |a_2| + \dots + |a_n|$, obtained by taking the absolute value of each term, is convergent.

The series $\sum a_k$ is conditionally convergent if $\sum |a_k|$ diverges while $\sum a_k$ converges.

- 6) **Ratio Test:** Suppose that in the series $\sum a_k$, every $a_k \neq 0$,

$$\lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \rho \quad \text{or} \quad \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = +\infty.$$

Then:

- a) If $\rho < 1$, the series $\sum a_k$ converges absolutely.
- b) If $\rho > 1$, or if $\lim_{k \rightarrow \infty} \frac{|a_{k+1}|}{|a_k|} = +\infty$, the series diverges.
- c) If $\rho = 1$, the test gives no information.

7) **Root Test:** Let $\sum a_k$ represent an infinite series.

- a) If $\lim_{k \rightarrow \infty} k\sqrt[k]{|a_k|} = \ell < 1$, the series is absolutely convergent.
- b) If $\lim_{k \rightarrow \infty} k\sqrt[k]{|a_k|} = \ell > 1$ or $\lim_{k \rightarrow \infty} k\sqrt[k]{|a_k|} = \infty$, the series is divergent.
- c) If $\lim_{k \rightarrow \infty} k\sqrt[k]{|a_k|} = 1$, the test is not conclusive.

POWER SERIES

A power series is a series of the form

$$c_0 + c_1(x-a) + c_2(x-a)^2 + \dots + c_n(x-a)^n,$$

in which a and c_i , $i = 1, 2, 3$, etc. are constants.

The notations

$$\sum_{n=0}^{\infty} c_n(x-a)^n \quad \text{and} \quad \sum_{n=0}^{\infty} c_n x^n$$

are used to describe power series.

A power series $\sum c_n x^n$ is said to converge:

1) at x_1 if and only if $\sum c_n x_1^n$ converges;

2) on the set S if and only if $\sum c_n x^n$ converges for each $x \in S$.

If $\sum c_n x^n$ converges at $x_1 \neq 0$, then it converges absolutely whenever $|x| < |x_1|$. If $\sum c_n x^n$ diverges at x_1 , then it diverges for $|x| > |x_1|$.

SOME USEFUL SERIES

Taylor Series

Taylor series is the expansion of a function f into a power series about the point a and is given by the formula

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n .$$

$f^{(n)}(a)$ represents the n th derivative of f at point a .

Maclaurin Series

Maclaurin series is the special case of Taylor series when $a = 0$.

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

Maclaurin series expansions of $\sin x$ and $\cos x$ are given by

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} , \text{ and}$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} .$$

Binomial Series

For each real number m ,

$$(a+b)^m = \sum_{k=0}^m \frac{m!}{(m-k)!k!} a^k b^{m-k}.$$

Geometric Series

Geometric series is $a + ar + ar^2 + \dots + ar^{n-1} = \sum_{k=0}^{n-1} ar^k$, and it converges

to the sum $\frac{a}{1-r}$ if $|r| < 1$ and diverges if $|r| \geq 1$. The partial sum has the following formula:

$$\sum_{k=0}^{n-1} ar^k = \frac{a - ar^n}{1-r}.$$

Fourier Series

Fourier series is the expansion of any function in terms of sine and cosine.

$$f(x) = \sum_{k=-\infty}^{\infty} (a_k \cos kx + \ell_k \sin kx),$$

where

$$a_0 = \frac{1}{2\lambda} \int_0^{2\lambda} f(x) dx;$$

$$a_k = \frac{1}{\lambda} \int_0^{2\lambda} f(x) \cos kx dx, \quad k = 1, 2, \dots, \infty; \text{ and}$$

$$b_k = \frac{1}{\lambda} \int_0^{2\lambda} f(x) \sin kx dx, \quad k = 1, 2, \dots, \infty.$$

Chapter 9

DIFFERENTIAL EQUATIONS

The use of differential equations to solve significant and important problems in engineering, the physical sciences, and the social sciences allows formulation of functions satisfying equations containing derivatives of the unknown function. Differential equations are an important part of the calculus, the fundamentals of which are presented here.

BASIC DEFINITIONS AND CLASSIFICATIONS

- 1) A differential equation is an equation which involves derivatives of an unknown function.
- 2) An ordinary differential equation is an equation which involves only ordinary derivatives.
- 3) A partial differential equation is an equation which involves partial derivatives.
- 4) The order of a differential equation is the order of the highest derivative that appears in the equation.
- 5) The degree of a differential equation is the greatest exponent of the highest ordered derivative in the equation.
- 6) A linear ordinary differential equation has the general form

$$a_0(x) \frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1}(x) \frac{dy}{dx} + a_n(x)y = b(x),$$

where a_0 is not identically zero.

The conditions for a linear equation are as follows:

- a) The dependent variable and its derivatives occur only in the first degree.
- b) No product of the dependent variable and/or any of its derivatives are present.
- c) No transcendental function of the dependent variable and/or its derivatives occur.

FIRST-ORDER EQUATIONS

Linear Equations

Given a first-order, ordinary linear equation of the general form

$$\frac{dy}{dx} + P(x)y = Q(x) \quad (9.1)$$

multiply both sides by an integrating factor (see 9.2.4)

$$u(x) = e^{\int P(x) dx} \quad (9.2)$$

to get

$$u(x) \frac{dy}{dx} + u(x)P(x)y = u(x)Q(x)$$

since $\frac{du}{dx} = e^{\int P(x) dx} P(x) = u(x)P(x)$, we get

$$u(x) \frac{dy}{dx} + \frac{du}{dx} y = u(x)Q(x)$$

Then $\frac{d}{dx}[u(x)y] = u(x)Q(x)$

Integrate both sides to get $u(x)y = \int u(x)Q(x)dx$

so $y = \frac{1}{u(x)} \int u(x)Q(x)dx = e^{-\int P(x) dx} [\int e^{\int P(x) dx} Q(x)dx]$

When $P(x) = a$, where $a = \text{constant}$, $u(x) = e^{ax}$

For this situation, $y = e^{-\int a dx} [\int Q(x)e^{\int a dx} dx]$

Which becomes $y = e^{-ax} [\int Q(x)e^{ax} dx]$

The general solution for (9.1), then, is

$$y = \frac{1}{u(x)} \int u(x)Q(x)dx + c. \quad (9.3)$$

Simple solutions:

1) For $\frac{dy}{dx} + ay = 0$, where a is a constant,

$$y = ce^{-ax}.$$

2) For $\frac{dy}{dx} + ay = g(x)$,

$$y = e^{-ax} \int e^{ax} g(x)dx + ce^{-ax}$$

Several methods exist to aid in the solution of more complex differential equations, as discussed below.

Separable Equations

If an equation of the form

$$M(x, y) + N(x, y)\frac{dy}{dx} = 0$$

can be written

$$M(x)M(y)dx + N(x)N(y)dy = 0, \quad (9.4)$$

it is said to be separable.

Separating the variables gives

$$\frac{M(x)}{N(x)}dx + \frac{N(y)}{M(y)}dy = 0.$$

The general solution for (9.4) is

$$\int^x \frac{M(x)}{N(x)}dx + \int^y \frac{N(y)}{M(y)}dy = c. \quad (9.5)$$

Problem: Find the general solution of

$$xy^2dx - (x + 5)dy = 0.$$

Solution: The variables are separable, since the equation can be written as

$$\frac{xdx}{x+5} - \frac{dy}{y^2} = 0.$$

Assuming that $x \neq -5$ and $y \neq 0$, the general solution is

$$\int \frac{xdx}{x+5} - \int y^{-2} dy = c,$$

where c is an arbitrary constant. Thus,

$$x - 5 \ln|x+5| + \frac{1}{y} = c$$
$$y = \frac{1}{c - x + \ln|x+5|}.$$

Exact Equations

An exact equation has the form

$$M(x, y) dx + N(x, y) dy = 0, \quad (9.6)$$

where

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

When an equation is exact, there exists a function $\mu(x, y)$ such that

$$\frac{\partial \mu(x, y)}{\partial x} = M(x, y) \quad \text{and} \quad \frac{\partial \mu(x, y)}{\partial y} = N(x, y)$$

for all $(x, y) \in D$.

Hence, the equation may be written

$$\frac{\partial \mu(x, y)}{\partial x} dx + \frac{\partial \mu(x, y)}{\partial y} dy = 0,$$

and $\mu(x, y)$ may be determined.

Solution:

- 1) Find $\mu(x, y)$:

$$\mu(x, y) = \int M(x, y) dx + \phi(y), \quad (9.7)$$

holding y as a constant.

- 2) Set $\frac{\partial \mu}{\partial y} = N(x, y)$, holding x as a constant.

- 3) Solve for $\phi'(y)$:

$$\phi'(y) = N(x, y) - \int \frac{\partial M(x, y)}{\partial y} dx. \quad (9.8)$$

- 4) Substituting back into (9.7) gives the general solution

$$\mu(x, y) = \int M(x, y) dx + \int \phi'(y) dy, \quad (9.9)$$

where $\phi'(y)$ is given by (9.8).

Integrating Factors

If the equation

$$M(x, y)dx + N(x, y)dy = 0 \quad (9.10)$$

is not exact, then an integrating factor is sought to make (9.10) exact. Finding the integrating factor for a given equation can be very difficult. Some of the important rules and procedures follow.

- 1) If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$ is a function of x alone, then

$$\mu(x) = \exp \left[\int \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} dx \right] \quad (9.11)$$

is the integrating factor of (9.10).

- 2) If $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{-M}$ is a function of y alone, then

$$\mu(y) = \exp \left[\int \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{-M} dy \right] \quad (9.12)$$

is the integrating factor of (9.10).

- 3) If $\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \neq 0$, then

$$\mu(x, y) = \frac{1}{\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}} \quad (9.13)$$

is the integrating factor of (9.10).

- 4) If (9.10) can be written in the form

$$yf(x, y)dx + xg(x, y)dy = 0, \quad (9.14)$$

where

$$f(x, y) \neq g(x, y),$$

then

$$\mu(x, y) = \frac{1}{\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y}} \quad (9.15)$$

is the integrating factor of (9.14).

Sometimes an integrating factor may be found by inspection, after grouping the terms in the equation, and recognizing a certain group as being a part of an exact differential. For example:

Group of Terms	Integrating Factor	Exact Differential
$x \, dy - y \, dx$	$\frac{1}{x^2}$	$\frac{x \, dy - y \, dx}{x^2} = d\left(\frac{y}{x}\right)$
$x \, dy - y \, dx$	$\frac{1}{y^2}$	$-\frac{y \, dx - x \, dy}{y^2} = d\left(-\frac{x}{y}\right)$
$x \, dy - y \, dx$	$\frac{1}{xy}$	$\frac{dy}{y} - \frac{dx}{x} = d\left(\ln \frac{y}{x}\right)$
$x \, dy - y \, dx$	$\frac{1}{x^2 + y^2}$	$\frac{xdy - ydx}{x^2 + y^2} = \frac{\frac{y \, dy - y \, dx}{x^2}}{1 + \left(\frac{y}{x}\right)^2}$ $= d\left(\arctan \frac{y}{x}\right)$
$x \, dy + y \, dx$	$\frac{1}{(xy)^n}$	$\frac{x \, dy + y \, dx}{(xy)^n} = d\left(\frac{-1}{(n-1)(xy)^{n-1}}\right),$ if $n \neq 1$
		$\frac{x \, dy + y \, dx}{xy} = d(\ln(xy)),$ if $n = 1$

Group of Terms	Integrating Factor	Exact Differential
$x dx + y dy$	$\frac{1}{(x^2 + y^2)^n}$	$\frac{x dx + y dy}{(x^2 + y^2)^n} = d\left(\frac{-1}{2(n-1)(x^2 + y^2)^{n-1}}\right),$ if $n \neq 1$
$x dx + y dy$	$\frac{1}{(x^2 + y^2)^n}$	$\frac{x dx + y dy}{x^2 + y^2} = d\left(\frac{1}{2}\ln(x^2 + y^2)\right),$ if $n = 1$

Homogeneous Equations

If the equation

$$\frac{dy}{dx} = f(x, y) \quad (9.16)$$

depends only on the ratio $\frac{y}{x}$ or $\frac{x}{y}$, then it is said to be homogeneous.

Let $y = vx$ and $\frac{dy}{dx} = x \frac{dv}{dx} + v$. Then (9.16) becomes

$$x \frac{dv}{dx} + v = F(v), \quad (9.17)$$

a separable equation, which can be solved by direct integration. After solving, replace v by $\frac{y}{x}$ to get the solution to (9.16).

A function $f(x, y)$ is called homogeneous of degree n if

$$f(\lambda x, \lambda y) = \lambda^n f(x, y). \quad (9.18)$$

Non-Homogeneous Equations

Consider the equation

$$(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0, \quad (9.19)$$

where $a_1, b_1, c_1, a_2, b_2,$ and c_2 are all constants.

- 1) If $a_1b_2 - a_2b_1 = 0$, use the transformation

$$\begin{aligned} t &= a_1x + b_1y, \\ dy &= \frac{dt - a_1dx}{b_1} \end{aligned} \quad (9.20)$$

to reduce (9.19) to a separable form:

$$P(x, t)dx + Q(x, t)dt = 0. \quad (9.21)$$

- 2) If $a_1b_2 - a_2b_1 \neq 0$, use the transformation

$$\begin{aligned} x &= X + h, \\ y &= Y + k, \end{aligned} \quad (9.22)$$

where h and k are solutions of the system

$$\begin{aligned} a_1h + b_1k + c_1 &= 0, \\ a_2h + b_2k + c_2 &= 0 \end{aligned} \quad (9.23)$$

to reduce (9.19) to the homogeneous equation

$$(a_1X + b_1Y)dX + (a_2X + b_2Y)dY = 0. \quad (9.24)$$

For an equation of the form

$$yf(xy)dx + xg(xy)dy = 0, \quad (9.25)$$

use the transformation

$$\begin{aligned} z &= xy, \\ y &= \frac{z}{x}, \end{aligned}$$

$$dy = \frac{x dz - z dz}{x^2} \quad (9.26)$$

to reduce (9.25) to a separable form:

$$P(x, z)dx + Q(x, z)dz = 0. \quad (9.27)$$

Problem: Find the general solution of
 $(xy - y^2)dx - x^2dy = 0$.

Solution: Using $y = vx$ and substituting, one obtains

$$(x \cdot vx - v^2x^2)dx - x^2(vdx + xdv) = 0.$$

Divide by x^2 , producing

$$(v - v^2)dx - (vdx + xdv) = 0,$$

or

$$\frac{dv}{v^2} + \frac{dx}{x} = 0.$$

The variables x and v are separable with solutions

$$-\frac{1}{v} + \ln x = c \quad \text{or} \quad \ln x - \frac{x}{y} = c,$$

$$y = \frac{x}{\ln cx} \quad \text{or} \quad x = c_1 e^{\frac{x}{y}}.$$

SECOND-ORDER EQUATIONS

Basic Concepts

The problem of finding a solution for

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = G(x) \quad (9.28)$$

on the interval $x_0 < x < x_1$, while satisfying the conditions

$$y(x_0) = A \quad \text{and}$$

$$y(x_1) = B,$$

is known as a boundary value problem.

The homogeneous, or reduced, or complementary equation is

$$y'' + P(x)y' + Q(x)y = 0, \quad (9.29)$$

and is obtained from (9.28) by setting $G(x) = 0$.

Fundamental Solutions to Homogeneous Equations

The derivative operator is defined by

$$L[\phi] = \phi'' + p\phi' + q\phi$$

or

$$L = D^2 + pD + q,$$

in which D is the derivative operator.

- 1) If $y = y_1(x)$ and $y = y_2(x)$ are solutions to

$$L[y] = y'' + P(x)y' + Q(x)y = 0, \quad (9.30)$$

then

$$y = c_1y_1(x) + c_2y_2(x)$$

is also a solution to (9.30); c_1 and c_2 are arbitrary constants.

- 2) $y_1(x)$ and $y_2(x)$ are a set of fundamental solutions.
 3) The Wronskian is

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1y_2' - y_1'y_2. \quad (9.31)$$

In general,

$$W(f_1, f_2, \dots, f_n) = \begin{vmatrix} f_1 & f_2 \dots f_n \\ f_1' & f_2' \dots f_n' \\ \vdots & \vdots \\ f_1^{n-1} & f_2^{n-1} \dots f_n^{n-1} \end{vmatrix}.$$

4) $W(y_1, y_2)$ is either identically zero, or else is never zero.

Reduction of Order

Consider the second order equation

$$y'' + p(x)y' + q(x)y = 0. \quad (9.32)$$

If one solution,

$$y = y_1(x),$$

is known, a second solution can be determined:

$$y = y_2(x) = v(x)y_1(x). \quad (9.33)$$

Differentiating (9.33) and substituting into (9.32) gives

$$v'' + \left(p + 2 \frac{y_1'}{y_2'} \right) v' = 0, \quad (9.34)$$

which is a first-order linear equation in v' .

Solving (9.34) gives

$$v' = c \exp \left[- \int p(x) + \frac{y_1'(x)}{y_1(x)} dx \right] = cu(x).$$

Integrating again gives

$$v = c \int u(x) dx.$$

The general solution of (9.32) is

$$y = c_1 y_1(x) + c_2 y_1(x) \int u(x) dx. \quad (9.35)$$

Homogeneous Equations with Constant Coefficients

Consider the equation

$$ay'' + by' + cy = 0. \quad (9.36)$$

The characteristic or auxiliary equation of this is

$$ar^2 + br + c = 0. \quad (9.37)$$

The roots of (9.37) are

$$r_1 = \frac{-b + (b^2 - 4ac)^{\frac{1}{2}}}{2a}$$

and

$$r_2 = \frac{-b - (b^2 - 4ac)^{\frac{1}{2}}}{2a}. \quad (9.38)$$

- 1) If the roots are real and unequal ($b^2 - 4ac > 0$), then

$$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}. \quad (9.39)$$

- 2) If the roots are real and equal ($b^2 - 4ac = 0$), then

$$y = c_1 e^{r_1 x} + c_2 x e^{r_1 x}. \quad (9.40)$$

- 3) If the roots are complex ($b^2 - 4ac < 0$),

$$y = c_1 e^{\lambda x} \cos \mu x + c_2 e^{\lambda x} \sin \mu x \quad (9.41)$$

where

$$\lambda = \frac{b}{2a}; \quad \mu = \frac{(b^2 - 4ac)^{\frac{1}{2}}}{2a}.$$

Non-Homogeneous Equations

The solution to a homogeneous equation, (9.29),

$$y'' + P(x)y' + Q(x)y = 0, \quad (9.42)$$

is called a complementary solution and is denoted by y_c .

The solution of a non-homogeneous equation is

$$y = y_c + y_p,$$

where y_p is the particular solution of the non-homogeneous equation.

Two methods of finding a particular solution are:

- 1) The method of undetermined coefficients.
- 2) The method of variation of parameters.

Method of Undetermined Coefficients

Consider

$$y'' + ay' + by = f(x). \quad (9.43)$$

- 1) The characteristic solution, y_c , is determined by setting $f(x) = 0$.
- 2) Now use $f(x)$ to determine the particular solution, y_p (compare y_1 with y_c to make sure that they are not similar).
- 3) To find the undetermined coefficients, substitute y_p , y'_p , and y''_p into the differential equation for y_1 , $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$, respectively.
- 4) $y = y_c + y_p$.

$f(x)$	$y_p(x)$
$a_0x^n + a_1x^{n-1} + \dots + a_n = P_n(x)$	$x^s(A_0x^n + A_1x^{n-1} + \dots + A_n)$
$P_n(x)e^{\alpha x}$	$x^s(A_0x^n + A_1x^{n-1} + \dots + A_n)e^{\alpha x}$
$P_n(x)e^{\alpha x} \begin{cases} \sin \beta x \\ \cos \beta x \end{cases}$	$x^s(A_0x^n + A_1x^{n-1} + \dots + A_n)e^{\alpha x} \begin{cases} \cos \beta x + (B_0x^n + B_1x^{n-1} + \dots + B_n)e^{\alpha x} \sin \beta x \end{cases}$

s is the smallest nonnegative integer ($s = 0, 1$, or 2) to ensure that no term in $y_p(x)$ is also the solution of the corresponding homogeneous equation.

Method of Variation of Parameters

Also referred to as Lagrange's method, this method applies in all cases (including variable coefficients) in which the complementary function is known.

Consider the equation

$$y'' + a_1y' + a_2y = f(x), \quad (9.43)$$

where a_1 and a_2 are constants.

First solve the homogeneous equation $f(x) = 0$.

The particular solution is

$$y_p(x) = u_1y_1(x) + u_2y_2(x), \quad (9.44)$$

where u_1 and u_2 must simultaneously satisfy

$$u_1'(x)y_1(x) + u_2'(x)y_2(x) = 0$$

and

$$u_1'(x)y_1(x) + u_2'(x)y_2'(x) = f(x). \quad (9.45)$$

Using Cramer's rule,

$$u_1' = \frac{\begin{vmatrix} 0 & y_2(x) \\ f(x) & y_2'(x) \end{vmatrix}}{W(y_1, y_2)}$$

and

$$u_2' = \frac{\begin{vmatrix} y_1(x) & 0 \\ y_2'(x) & f(x) \end{vmatrix}}{W(y_1, y_2)}.$$

Therefore, the general solution is

$$y_p(x) = -y_1(x) \int \frac{y_2(x)f(x)}{W(y_1, y_2)} dx + y_2(x) \int \frac{y_1(x)f(x)}{W(y_1, y_2)} dx. \quad (9.46)$$

Problem: $\frac{d^2y}{dx^2} + y = \tan x$. Solve using variation of parameters.

Solution: The complementary function is

$$y_c = c_1 \sin x + c_2 \cos x,$$

and assume that

$$y_p = v_1 \sin x + v_2 \cos x,$$

where v_1 and v_2 will be determined, such that this is a particular integral of the original equation.

$$y_p' = v_1 \cos x - v_2 \sin x + v_1' \sin x + v_2' \cos x.$$

Imposing the condition $v_1' \sin x + v_2' \cos x = 0$,

$$y_p' = v_1 \cos x - v_2 \sin x$$

$$y_p'' = -v_1 \sin x - v_2 \cos x + v_1' \cos x + v_2' \sin x.$$

Substituting y_0 and y_p'' into the original equation produces

$$v_1' \cos x - v_2' \sin x = \tan x .$$

Now to solve v_1' and v_2' , we have

$$v_1' \sin x + v_2' \cos x = 0$$

$$v_1' \cos x - v_2' \sin x = \tan x$$

$$v_1' = \frac{\begin{vmatrix} 0 & \cos x \\ \tan x & -\sin x \\ \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix}}{\begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix}} = \frac{\cos x \tan x}{-1} = \sin x$$

$$v_2' = \frac{\begin{vmatrix} \sin x & 0 \\ \cos x & \tan x \\ \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix}}{\begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix}} = \frac{\sin x \tan x}{-1} = \frac{-\sin x}{\cos x}$$

$$= \frac{\cos^2 x - 1}{\cos x} = \cos x - \sec x .$$

Integration provides

$$v_1 = -\cos x + c_3$$

$$v_2 = \sin x - \ln|\sec x + \tan x| + c_4$$

$$\begin{aligned} y_p &= [-\cos x + c_3] \sin x + [\sin x - \ln|\sec x + \tan x| + c_4] \cos x \\ &= c_3 \sin x + c_4 \cos x - \cos x \ln|\sec x + \tan x| . \end{aligned}$$

Letting $A = c_3$ and $B = c_4$ and realizing $y = y_c + y_p$,

$$y = c_1 \sin x + c_2 \cos x + A \sin x + B \cos x - \cos x \ln|\sec x + \tan x| .$$

MISCELLANEOUS TOPICS

- **Probability and Statistics**
- **Combinational Analysis**
- **Numerical Analysis**
- **Real Variables**
- **Complex Variables**
- **Set Theory**
- **Topology**
- **Logic**
- **Algorithmic Processes**

Chapter 1

PROBABILITY AND STATISTICS

SAMPLE SPACE

Probability is defined in some sample space. A sample space of a random experiment is the collection of all possible outcomes of the experiment. For example, if two coins are tossed, then the sample space is given by

$$S = \{(H, H), (H, T), (T, H), (T, T)\}.$$

Each of (H, H) , (H, T) , (T, H) and (T, T) is a possible outcome of the experiment.

An event is a subject of a sample space. For example, in the above experiment, the event that the first coin turns out to be a head is $\{(H, H), (H, T)\}$.

AXIOMS OF PROBABILITY

To every event E in the sample space s , we assign a probability function P , which satisfies the following axioms:

- 1) For every event E of s , $0 \leq P(E) \leq 1$.
- 2) $P(S) = 1$, and $P(\phi) = 0$.
- 3) The probability of the union of a finite or countable number of pairwise exclusive events is the sum of the probabilities of the events; that is:

$$P(E_1 \cup E_2 \cup E_3 \cup \dots) = P(E_1) + P(E_2) + P(E_3) + \dots,$$

where

$$E_i \cap E_j = \phi \text{ for all } i \neq j.$$

COUNTING METHODS

If a sample $S = \{e_1, e_2, \dots, e_n\}$ contains n simple events, $E_i = \{e_i\}$, $i = 1, 2, \dots, n$, then, using a uniform probability model, we assign probability $\frac{1}{n}$ for each point in S ; that is, $P(E_i) = \frac{1}{n}$. To determine the probability of an

event A , we need:

- 1) The number of possible outcomes in S .
- 2) The number of outcomes in the event A .

Then,

$$P(A) = \frac{\text{number of outcomes in event } A}{\text{number of possible outcomes in } S}.$$

Two formulas are important in regard to counting the possible outcomes. Suppose we take r distinct objects from a set of n objects. If the ordering does not matter, then such arrangement is called permutation, and the total number of such ordering, denoted by nPr , is defined as

$$nPr = \frac{n!}{(n-r)!}.$$

For example, in tossing the two coins, the ordering does not matter if we treat (H, T) and (T, H) as the same outcome. If ordering matters, then such arrangement is called combination, and the total number of combinations,

denoted by nCr or $\binom{n}{r}$, is defined by

$$nCr = \binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

Problem: A committee of 5 is to be selected at random from a group of 10 men and 8 women. What is the probability that there will be exactly 3 men on the committee?

Solution: The 10 men and 8 women constitute a set of 18 elements with 10 elements of one kind and 8 elements of another kind. So, the sample space S consists of $\binom{18}{5}$ outcomes; that is, $N(S) = \binom{18}{5}$. The event A is the subset of S with 3 men and 2 women in the selection. Now this selection can be broken into two parts, with the selection of the men as one task and the selection of the women as a second task. The 3 men represent a combination of 10 objects taken 3 at a time; that is, $\binom{10}{3}$. The 2 women represent a combination of 8 objects taken 2 at a time; that is, $\binom{8}{2}$. Thus, there are $\binom{10}{3} \times \binom{8}{2}$ elements in event A and

$$P(A) = \frac{\binom{10}{3} \binom{8}{2}}{\binom{18}{5}} = \frac{20}{51} .$$

PROBABILITY DISTRIBUTION AND DENSITY FUNCTION

For discrete random values, we can define probability point function. The probability point function is the probability that the discrete random variable X takes some particular value x , or

$$Q(x_i) = P(X = x_i) .$$

If S_x is a finite sample space with n elements and Q is a probability point function, then

- 1) $0 \leq Q(x_i) \leq 1$ for all $x_i \in S_x$.
- 2) $Q(x_i) = 1$.

The function Q is sometimes called the frequency distribution of X , or the probability function of the discrete random variable X .

Sometimes probability point function is also called probability mass function, or *pmf*.

Probability Distribution Function

The function F , whose value for each real number x is given by

$$F_x(x) = P_x(X \leq x),$$

is called the probability distribution function for the random variable X .

The probability distribution function can be specified by the probability point function as follows:

$$F(x) = \sum_{x_i \leq x} Q(x_i),$$

where the sum is taken over all points $x_i \leq x$ such that $P(x_i) > 0$.

For continuous random variables, $F_x(x)$ is a continuous function of x and no probability point functions are defined. In this case, probability density function is defined as the probability that the continuous random variables fall in an infinitesimal interval, x and Sx . In terms of probability distribution function, the probability density function $f_x(x)$ is defined as

$$F_x(x) = \frac{dF_x(x)}{dx}.$$

The probability density function f_x has the following properties:

- 1) $f_x(x) \geq 0$ for all x .
- 2) f_x has at most a finite number of discontinuities in every finite interval of the real line.
- 3) $\int_{-\infty}^{\infty} f_x(u) du = 1$.
- 4) For every interval $[a, b]$,

$$P_x(a \leq X \leq b) = \int_a^b f_x(x) dx.$$

STATISTICS

Statistics is the science of assembling, organizing, and analyzing data, as well as drawing conclusions about what the data mean. Often these conclusions come in the form of predictions.

The first task of a statistician is to define the population and to choose the sample.

In most cases, it is impractical or impossible to examine the entire group, which is called the population or universe. This being the case, one must instead examine a part of the population, called a sample. The chosen sample has to reflect as closely as possible the characteristics of the population. For example, statistical methods were used in testing the Salk vaccine, which protects children against polio. The sample consisted of 400,000 children. Half the children, chosen at random, received the Salk vaccine; the other half received an inactive solution. After the study was completed, it turned out that 50 cases of polio appeared in the group that received the vaccine and 150 cases appeared in the group that did not receive the vaccine.

Based on that study of a sample of 400,000 children, it was decided that the Salk vaccine was effective for the entire population.

NOTATION AND DEFINITION OF MEANS

By

$$x_1, x_2, \dots, x_n,$$

we denote the measurements observed in a sample of size n . The letter i in x_i is called a subscript or index. It stands for any of the numbers 1, 2, ..., n .

Throughout the book we will be using the summation notation. The symbol

$$\sum_{i=1}^n x_i$$

denotes the sum of all x_i 's; that is,

$$\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_{n-1} + x_n .$$

Example: $\sum_{i=1}^4 x_i y_i = x_1 y_1 + x_2 y_2 + x_3 y_3 + x_4 y_4 .$

Example: Let a be a constant. Then

$$\begin{aligned} \sum_{i=1}^n a x_k &= a x_1 + a x_2 + \dots + a x_n \\ &= a(x_1 + x_2 + \dots + x_n) \\ &= a \sum_{k=1}^n x_k . \end{aligned}$$

In general,

$$\sum a x_k = a \sum x_k$$

and

$$\sum (a x_k + b y_k) = a \sum x_k + b \sum y_k .$$

Often, when no confusion can arise, we write $\sum_k x_k$ instead of $\sum_{k=1}^n x_k$.

Arithmetic Mean

The arithmetic mean, or mean, of a set of measurements is the sum of the measurements divided by the total number of measurements.

The arithmetic mean of a set of numbers, $x_1, x_2, \dots, x_n$ is denoted by $\bar{x}$ (read “ x bar”).

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Example: The arithmetic mean of the numbers 3, 7, 1, 24, 11, 32 is

$$\bar{x} = \frac{3 + 7 + 1 + 24 + 11 + 32}{6} = 13.$$

Example: Let $f_1, f_2, \dots, f_n$ be the frequencies of the numbers $x_1, x_2, \dots, x_n$ (i.e., the number x_i occurs f_i times). The arithmetic mean is

$$\bar{x} = \frac{f_1x_1 + f_2x_2 + \dots + f_nx_n}{f_1 + f_2 + \dots + f_n} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}.$$

Note that the total frequency, that is, the total number of cases, is $\sum_{i=1}^n f_i$.

Example: If the measurements, 3, 7, 2, 8, 0, 4 occur with frequencies 3, 2, 1, 5, 10, and 6, respectively, then the arithmetic mean is

$$\bar{x} = \frac{3 \times 3 + 7 \times 2 + 2 \times 1 + 8 \times 5 + 0 \times 10 + 4 \times 6}{3 + 2 + 1 + 5 + 10 + 6} \approx 3.3.$$

Keep in mind that the arithmetic mean is strongly affected by extreme values.

Example: Consider four workers whose monthly salaries are \$2,500, \$3,200, \$3,700, and \$48,000. The arithmetic mean of their salaries is

$$\frac{\$57,400}{4} = \$14,350.$$

The figure \$14,350 can hardly represent the typical monthly salary of the four workers.

Example: The deviation d_i of x_i from its mean $\bar{x}$ is defined to be

$$d_i = x_i - \bar{x}.$$

The sum of the deviations of $x_1, x_2, \dots, x_n$ from their mean $\bar{x}$ is equal to zero. Indeed,

$$\sum_{i=1}^n d_i = \sum_{i=1}^n (x_i - \bar{x}) = 0 .$$

Thus,

$$\begin{aligned} \sum_{i=1}^n (x_i - \bar{x}) &= \sum_{i=1}^n x_i - n\bar{x} = \sum x_i - n \frac{\sum x_i}{n} \\ &= \sum x_i - \sum x_i = 0 . \end{aligned}$$

Example: If $z_1 = x_1 + y_1, \dots, z_n = x_n + y_n$, then $\bar{z} = \bar{x} + \bar{y}$.

Indeed,

$$\bar{x} = \frac{\sum x}{n}, \quad \bar{y} = \frac{\sum y}{n}, \quad \bar{z} = \frac{\sum z}{n} .$$

We have

$$\bar{z} = \frac{\sum z}{n} = \frac{\sum (x + y)}{n} = \frac{\sum x}{n} + \frac{\sum y}{n} = \bar{x} + \bar{y} .$$

Example: Consider the numbers $x_1, x_2, \dots, x_n$, whose deviations from any number a are $d_1, d_2, \dots, d_n$, respectively. Then

$$\bar{x} = a + \bar{d} .$$

The deviation d_i of x_i from a is defined to be

$$d_i - x_i = a .$$

Hence,

$$\bar{x} = \frac{\sum x_i}{n} = \frac{\sum (d_i + a)}{n} = \frac{na}{n} + \frac{\sum d_i}{n} = a + \bar{d} .$$

Similarly, if $x_1, x_2, \dots, x_n$, have frequencies $f_1, f_2, \dots, f_n$, respectively, then

$$\bar{x} = a + \frac{\sum f_i d_i}{N}, \text{ where } N = \sum_{i=1}^n f_i.$$

Indeed,

$$\begin{aligned}\bar{x} &= \frac{\sum f_i x_i}{\sum f_i} = \frac{\sum f_i x_i}{N} = \frac{\sum f_i (a + d_i)}{N} \\ &= \frac{a \sum f_i}{N} + \frac{\sum f_i d_i}{N} = a + \frac{\sum f_i d_i}{N}.\end{aligned}$$

The arithmetic mean plays an important role in statistical inference.

We will be using different symbols for the sample mean and the population mean. The population mean is denoted by μ , and the sample mean is denoted by $\bar{x}$. The sample mean $\bar{x}$ will be used to make inferences about the corresponding population mean μ .

Example: Suppose a bank has 500 savings accounts. We pick a sample of 12 accounts. The balance on each account (in dollars) is

\$657	\$284	\$51
251	73	327
65	412	218
539	225	195

The sample mean $\bar{x}$ is

$$\bar{x} = \frac{\sum_{i=1}^{12} x_i}{12} = \$274.75.$$

The average amount of money in each of 12 sampled accounts is \$274.75. Using this information, we estimate the total amount of money in the bank to be

$$\$274.75 \times 500 = \$137,375.$$

Geometric Mean

The geometric mean g (or G) of a set of n numbers

$$x_1, x_2, \dots, x_n$$

is the n th root of the product of the numbers:

$$g = \sqrt[n]{x_1 \times x_2 \times \dots \times x_n}.$$

Example: The geometric mean of the numbers 3, 9, 27 is

$$g = \sqrt[3]{3 \times 9 \times 27} = 9.$$

Example: Find the geometric mean of the numbers 3, 5, 7, 8, 10, 13, 16.
We begin with

$$g = \sqrt[7]{3 \times 5 \times 7 \times 8 \times 10 \times 13 \times 16} = \sqrt[7]{1,747,200}.$$

Using common logarithm,

$$\log g = \frac{1}{7} \log 1,747,200 = \frac{1}{7} \times 6.2423 = 0.8918.$$

Hence,

$$g = 7.794.$$

Example: The frequencies of measurements $x_1, x_2, \dots, x_n$ are $f_1, f_2, \dots, f_n$, respectively. The geometric mean of the measurements is

$$\begin{aligned} g &= \sqrt[N]{\underbrace{x_1 \times x_1 \times \dots \times x_1}_{f_1 \text{ times}} \times \underbrace{x_2 \times x_2 \times \dots \times x_2}_{f_2 \text{ times}} \times \dots \times \underbrace{x_n \times x_n \times \dots \times x_n}_{f_n \text{ times}}} \\ &= \sqrt[N]{x_1^{f_1} \times x_2^{f_2} \times \dots \times x_n^{f_n}}, N = \sum_1^n f_i. \end{aligned}$$

If all the numbers are positive, then

$$\log g = \frac{1}{N} \log(x_1^{f_1} \times x_2^{f_2} \times \dots \times x_n^{f_n})$$

$$\begin{aligned}
 &= \frac{1}{N}(f_1 \log x_1 + \dots + f_n \log x_n) \\
 &= \frac{\sum f_i \log x_i}{\sum f_i}.
 \end{aligned}$$

Problem: The money deposited in an interest-bearing account increased from \$1,000 to \$5,000 in three years. Find the average percentage increase per year.

Solution: The increase is 500%. But the average percentage increase is not $\frac{500\%}{3}$.

Denote the average percentage increase as q . Then

amount deposited	1,000
amount after first year	$1,000 + q1,000 = 1,000(1 + q)$
amount after second year	$1,000(1 + q) + 1,000(1 + q)q$ $= 1,000(1 + q)^2$
amount after third year	$1,000(1 + q)^2 + 1,000(1 + q^2)q$ $= 1,000(1 + q)^3$ $= 5,000$

Hence,

$$(1 + q)^3 = 5,000$$

and

$$q = \sqrt[3]{5} - 1.$$

In general, if the initial amount is A and the yearly interest is q , then after n years, the amount is M .

$$M = A(1 + q)^n$$

This equation is called the compound interest formula.

Weighted Arithmetic Mean

With the numbers $x_1, x_2, \dots, x_n$ we associate weighting factors or weights $w_1, w_2, \dots, w_n$, depending on how significant each number is. The weighted arithmetic mean is defined by

$$\bar{x} = \frac{x_1 w_1 + x_2 w_2 + \dots + x_n w_n}{w_1 + w_2 + \dots + w_n} = \frac{\sum x_n w_n}{\sum w_n}.$$

Example: The pilot has to pass three tests. The second is weighted three times as much as the first and the third test is weighted four times as much as the first. The pilot reached a score of 40 on the first test, 45 on the second test, and 60 on the third test.

The weighted mean is

$$\frac{40 \times 1 + 45 \times 3 + 60 \times 4}{8} = 51.875.$$

Example: If f_1 numbers have mean m_1 , f_2 numbers have mean $m_2, \dots, f_n$ numbers have mean m_n , then the mean of all numbers is

$$\bar{x} = \frac{f_1 m_1 + f_2 m_2 + \dots + f_n m_n}{f_1 + f_2 + \dots + f_n}.$$

Harmonic Mean

The harmonic mean h of numbers $x_1, x_2, \dots, x_n$ is the reciprocal of the arithmetic mean of the reciprocals of the numbers:

$$h = \frac{1}{\frac{1}{n} \sum_{i=1}^n \frac{1}{x_i}} = \frac{n}{\sum \frac{1}{x_i}}.$$

Example: The harmonic mean of the numbers 2, 4, 9 is

$$h = \frac{3}{\frac{1}{2} + \frac{1}{4} + \frac{1}{9}} = 3.48 .$$

Problem: During four successive years, the prices of gasoline were 70, 75, 78, and 95 cents per gallon. Find the average cost of gasoline over the four-year period.

Solution: There are two methods of computing this average.

Method 1.

Suppose the car owner used 100 gallons each year. Then

$$\begin{aligned} \text{Average Cost} &= \frac{\text{total cost}}{\text{total number of gallons}} \\ &= \frac{0.7 \cdot 100 + 0.75 \cdot 100 + 0.78 \cdot 100 + 0.95 \cdot 100}{400} \\ &= 0.795 \text{ \$/gal} . \end{aligned}$$

Method 2.

Suppose the car owner spends \$100 on gasoline each year. Then

$$\begin{aligned} \text{Average Cost} &= \frac{\text{total cost}}{\text{total number of gallons}} \\ &= \frac{400}{509.66} \\ &= 0.785 \text{ \$/gal} . \end{aligned}$$

Both averages are correct. Depending on assumed conditions we obtain different answers.

Let g be the geometric mean of a set of positive numbers $x_1, x_2, \dots, x_n$, and let $\bar{x}$ be the arithmetic mean and h the harmonic mean. Then

$$h \leq g \leq \bar{x} .$$

The equality holds only if all the numbers $x_1, x_2, \dots, x_n$ are identical.

Root Mean Square

The root mean square or quadratic mean of the numbers $x_1, x_2, \dots, x_n$ is denoted by $\sqrt{x^2}$ and defined as

$$\sqrt{x^2} = \sqrt{\frac{\sum_{i=1}^n x_i^2}{n}} .$$

Example: The quadratic mean of the numbers 2, 3, 5, 7, is

$$\sqrt{x^2} = \sqrt{\frac{2^2 + 3^2 + 5^2 + 7^2}{4}} = 4.66 .$$

Example: The quadratic mean of two positive numbers, a and b , is not smaller than their geometric mean.

$$\sqrt{ab} \leq \sqrt{\frac{a^2 + b^2}{2}}$$

Indeed, since

$$0 \leq (a - b)^2 = a^2 - 2ab + b^2 ,$$

we obtain

$$ab \leq \frac{a^2 + b^2}{2}$$

and

$$\sqrt{ab} \leq \sqrt{\frac{a^2 + b^2}{2}}.$$

All means can also be computed for the grouped data, that is, when data are presented in a frequency distribution. Then all values within a given class interval are considered to be equal to the class mark or midpoint of the interval.

MEASURES OF VARIABILITY

The degree to which numerical data tend to spread about an average value is called the variation or dispersion of the data. We shall define various measures of dispersion.

The simplest measure of data variation is the range.

Range

The range of a set of numbers is defined to be the difference between the largest number and the smallest number of the set.

Example: The range of numbers 3, 6, 21, 24, 38 is $38 - 3 = 35$.

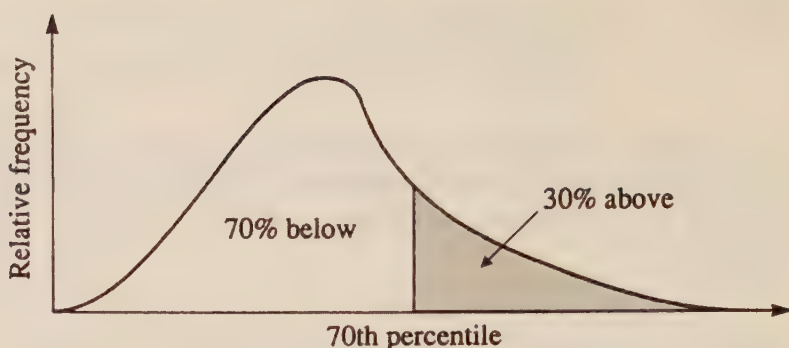
For grouped data, the range is the difference between the upper limit of the last interval and the lower limit of the first interval.

Next we shall define percentiles.

Percentiles

The n th percentile of a set of numbers arranged in order of magnitude is that value which has $n\%$ of the numbers below it and $(100 - n)\%$ above it.

Example:

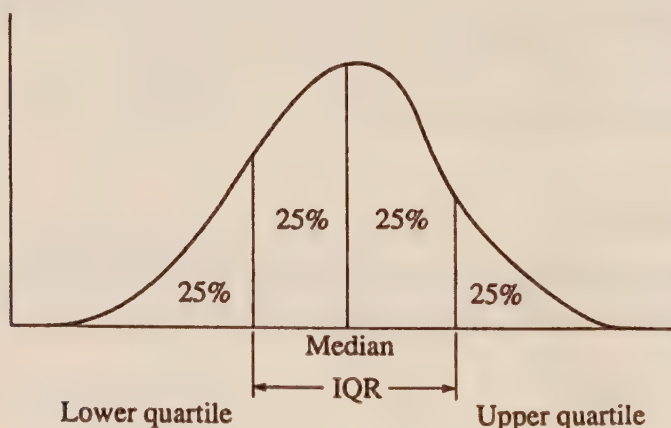


The 70th percentile of a set of numbers.

Percentiles are often used to describe the results of achievement tests. For example, someone graduates in the top 10% of the class. Frequently used percentiles are the 25th, 50th, and 75th percentiles, which are called the lower quartile, the middle quartile (median), and the upper quartile, respectively.

Interquartile Range

The interquartile range, denoted IQR, of a set of numbers is the difference between the upper and lower quartiles.



Now we shall introduce an important concept of deviation.

The deviation of a number x from its mean $\bar{x}$ is defined to be

$$x - \bar{x} .$$

Using deviations, we can construct many different measures of variability.

Observe that the mean deviation for any set of measurements is always zero. Indeed, let $x_1, x_2, \dots, x_n$ be measurements. Their mean is given by

$$\bar{x} = \frac{\sum x_i}{n} .$$

The deviations are $x_1 - \bar{x}, x_2 - \bar{x}, \dots, x_n - \bar{x}$, and their mean is equal to

$$\frac{\sum_{i=1}^n x_i - \bar{x}}{n} = \frac{\sum x_i}{n} - \bar{x} = 0 .$$

MEASURES OF DISPERSION

Average Deviation

The average deviation of a set of n numbers $x_1, x_2, \dots, x_n$ is defined by

$$\text{Average Deviation} = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n} = \overline{|x_i - \bar{x}|} ,$$

where $\bar{x}$ is the arithmetic mean of the numbers $x_1, x_2, \dots, x_n$.

Example: We find the average deviation of the numbers 3, 5, 6, 8, 13, 21 by

$$\bar{x} = \frac{3+5+6+8+13+21}{6} = 9.33$$

$$\begin{aligned}\text{Average Deviation} &= \frac{6.33 + 4.33 + 3.33 + 1.33 + 4.33 + 12.33}{6} \\ &= 5.33 .\end{aligned}$$

If the frequencies of the numbers $x_1, x_2, \dots, x_n$ are $f_1, f_2, \dots, f_n$, respectively, then the average deviation becomes

$$\text{Average Deviation} = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i} .$$

We will be using this formula for grouped data where x_i 's represent class marks and f_i represents class frequencies.

The sum $\sum_{i=1}^n |x_i - a|$ is the maximum when a is the median.

Definition of Standard Deviation

The standard deviation of a set $x_1, x_2, \dots, x_n$ of n numbers is defined by

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}} = \sqrt{(x_i - \bar{x})^2} .$$

The sample standard deviation is denoted by s , while the corresponding population standard deviation is denoted by σ .

A simple manual task was given to six children and the time each child took to complete the task was measured. Results are shown in the table.

	x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
	12	2.5	6.25
	9	-0.5	0.25
	11	1.5	2.25
	6	-3.5	12.25
	10	0.5	0.25
	9	-0.5	0.25
Total	57	0	21.5

For this sample, we shall find the standard deviation and variance.

The average $\bar{x}$ is 9.5.

$$\bar{x} = 9.5 .$$

The standard deviation is

$$s = \sqrt{\frac{21.5}{6}} = 1.893 ,$$

and the variance is

$$s^2 = 3.583.$$

SIMPLIFIED METHODS FOR COMPUTING STANDARD DEVIATION AND VARIANCE

By definition,

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n} .$$

Hence,

$$\begin{aligned}
 s^2 &= \frac{\sum (x_i - \bar{x})^2}{n} = \frac{\sum x^2 - 2x_i\bar{x} + \bar{x}^2}{n} \\
 &= \frac{\sum x^2}{n} - 2\bar{x} \frac{\sum x}{n} + \frac{n\bar{x}^2}{n} = \frac{\sum x^2}{n} - 2\bar{x}^2 + \bar{x}^2 \\
 &= \bar{x}^2 - \bar{x}^2 = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2,
 \end{aligned}$$

and

$$s^2 = \sqrt{\bar{x}^2 - \bar{x}^2} = \sqrt{\frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n} \right)^2}.$$

Similarly, for

$$s^2 = \frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{\sum f_i},$$

we find that

$$\begin{aligned}
 s^2 &= \frac{\sum f(x - \bar{x})^2}{\sum f} = \frac{\sum fx^2 - 2\bar{x} \sum fx + \bar{x}^2 \sum f}{\sum f} \\
 &= \frac{\sum fx^2}{\sum f} - \bar{x}^2 = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2.
 \end{aligned}$$

Example: Consider the example in Section 1.8. We shall use the formula

$$s^2 = \overline{x^2} - \bar{x}^2$$

to compute the value of s^2 .

$$s^2 = \frac{563}{6} - (9.5)^2 = 3.583$$

and

$$s = 1.893.$$

Remember that $\overline{x^2}$ denotes the mean of the squares of all values of x , and $\bar{x}^2$ denotes the square of the mean of all x 's.

Let D be an arbitrary constant and

$$d_i = x_i - D$$

be the deviations of x_i from D . Then

$$x_i = d_i + D \text{ and}$$

$$\bar{x} = \bar{d} + D.$$

Hence,

$$\begin{aligned} s^2 &= (x - \bar{x})^2 = (d - \bar{d})^2 = d^2 - 2d\bar{d} + \bar{d}^2 \\ &= \overline{d^2} - \bar{d}^2 = \frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f} \right)^2. \end{aligned}$$

Example: We find the standard deviation for the data shown below. The height of 100 students was measured and recorded.

Height (inches)	Class Mark x	x^2	Frequency f	fx^2	fx
60 - 62	61	3,721	7	26,047	427
63 - 65	64	4,096	21	86,016	1,344
66 - 68	67	4,489	37	166,093	2,479
69 - 71	70	4,900	26	127,400	1,820
72 - 74	73	5,329	9	47,961	657
			$\Sigma f =$	$\Sigma fx^2 =$	$\Sigma fx =$
			100	453,517	6,727

We apply the formula

$$s = \sqrt{\frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f}\right)^2}.$$

Hence,

$$s = \sqrt{4,535 - 4,525} = 3.16.$$

Using the empirical rule, we can interpret the standard deviation of a set of measurements.

Empirical Rule

If a set of measurements has a bell-shaped histogram, then:

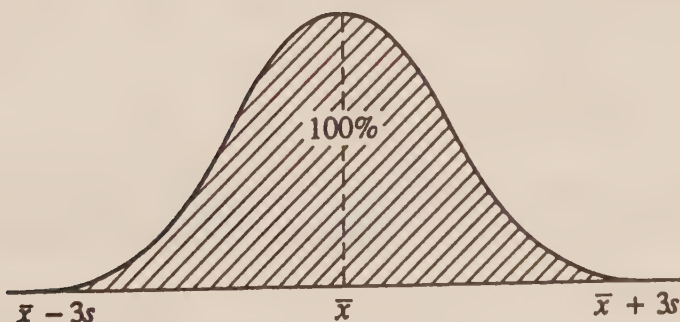
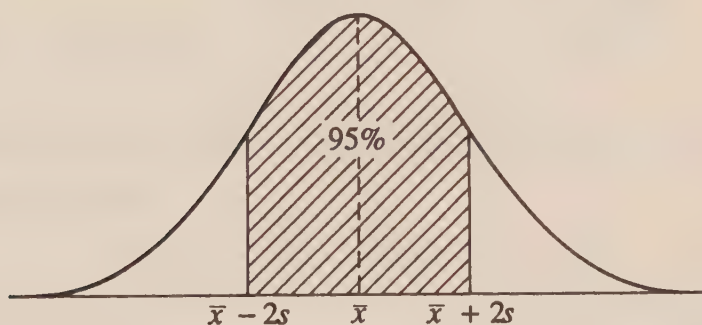
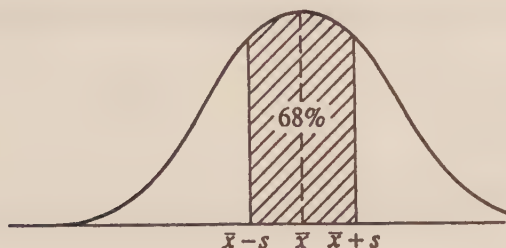
- 1) the interval $\bar{x} \pm s$ contains approximately 68% of the measurements.
- 2) $\bar{x} \pm 2s$ contains approximately 95% of the measurements.
- 3) $\bar{x} \pm 3s$ contains approximately all the measurements.

Note that from the empirical rule we can find the approximate value of the sample standard deviation s . Approximately 95% of all the measure-

ments are located in the interval $\bar{x} \pm 2s$. The length of this interval is $4s$. Hence, the range of the measurements is equal to approximately $4s$.

$$\text{Approximate Value of } s = \frac{\text{range}}{4} .$$

For normal (bell-shaped) distributions, we have the following:



We shall discuss some properties of the standard deviation.

In the definition of the standard deviation,

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}} .$$

We can replace $\bar{x}$ by any other average y which is not the arithmetic mean:

$$s = \sqrt{\frac{\sum (x_i - y)^2}{n}} .$$

Note that of all such standard deviations, the one with $y = \bar{x}$ is the minimum. Indeed,

$$\begin{aligned} \frac{\sum (x - y)^2}{n} &= \frac{\sum (x^2 - 2xy + y^2)}{n} \\ &= \frac{\sum x^2 - 2y \sum x + ny^2}{n} \\ &= y^2 - 2y \frac{\sum x}{n} + \frac{\sum x^2}{n} . \end{aligned}$$

The equation $y^2 + ay + b$, where a and b are constants, has a minimum value if and only if $y = -\frac{1}{2}a$. Hence,

$$y = \frac{\sum x}{n} = \bar{x} .$$

Consider two sets consisting of N_1 and N_2 measurements, or two frequency distributions with total frequencies N_1 and N_2 . The variances are s_1^2 and s_2^2 , respectively.

If both sets have the same mean $\bar{x}$, then the combined (or pooled) variance of both sets or both frequency distributions is given by

$$s^2 = \frac{N_1 s_1^2 + N_2 s_2^2}{N_1 + N_2} .$$

STUDENT'S t DISTRIBUTION

For small sample sizes, the test statistic

$$z = \frac{\bar{x} - \mu_0}{\sigma} \sqrt{n} \quad (1)$$

replacing σ by s falsely rejects the null hypothesis $\mu = \mu_0$ at a higher percentage than that specified by α .

W.S. Gosset found the distribution and percentage points of the test statistic

$$\frac{\bar{x} - \mu_0}{\sigma} \sqrt{n} , \quad n < 30 .$$

He published the results under the pen name Student.

Define the statistic

$$t = \frac{\bar{x} - \mu}{s} \sqrt{n-1} . \quad (2)$$

Now consider samples of size n drawn from a normal population with mean μ . For each sample, compute t using the sample mean $\bar{x}$ and sample standard deviation s . Thus, we obtain the sampling distribution for t , which is given by

$$Y = \frac{Y_0}{\left(1 + \frac{t^2}{n-1}\right)^{\frac{n}{2}}} + \frac{Y_0}{\left(1 + \frac{t^2}{v}\right)^{\frac{v+1}{2}}} , \quad (3)$$

where Y_0 is a constant depending on the sample size n and such that the area under the curve (3) is equal to one.

The constant

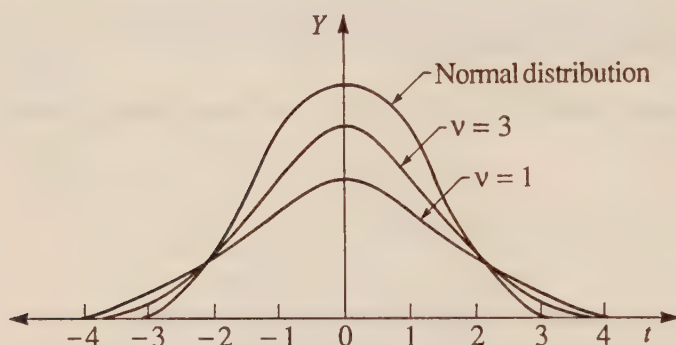
$$v = n - 1 \quad (4)$$

(sometimes denoted df) is called the number of degrees of freedom.

The distribution (3) is called Students' t distribution.

Note that for large values of n , curve (3) approaches the standardized normal curve

$$Y = \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}. \quad (5)$$



There are many different t distributions. We choose a particular one by specifying the parameter v (the number of degrees of freedom).

As n , and hence v , gets larger, the t distribution approaches the z distribution.

Confidence Intervals

Using the tables of the t distribution, we can define 90%, 95%, and 99%, and other confidence intervals as was done for the normal distributions. The population mean μ can be estimated within specified limits of confidence.

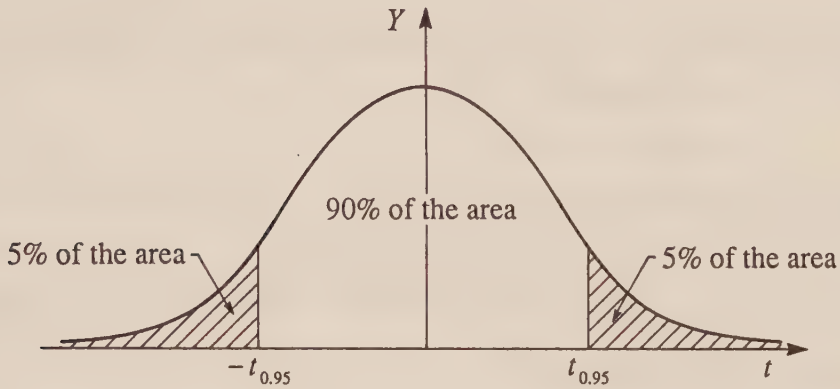
Let us denote by $-t_{0.95}$ and $t_{0.95}$ the values of t for which 5% of the area lies in each tail of the t distribution. Then a 90% confidence interval for t is

$$-t_{0.95} < \frac{\bar{x} - \mu}{s} \sqrt{n-1} < t_{0.95} . \quad (6)$$

From (6) we find that

$$\bar{x} - t_{0.95} \frac{s}{\sqrt{n-1}} < \mu < \bar{x} + t_{0.95} \frac{s}{\sqrt{n-1}} . \quad (7)$$

Hence, μ lies in the interval $\bar{x} \pm t_{0.95} \frac{s}{\sqrt{n-1}}$ with 90% confidence.



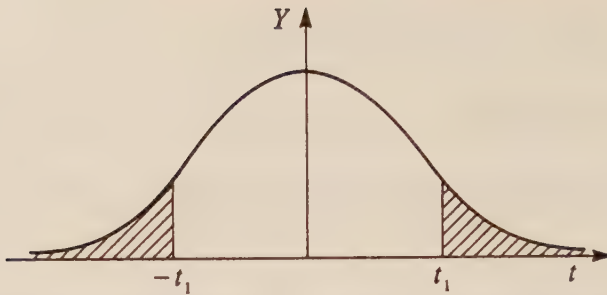
In general, the confidence limits for population means are defined by

$$\bar{x} \pm t_c \frac{s}{\sqrt{n-1}} , \quad (8)$$

where the values $\pm t_c$ are called critical values or confidence coefficients.

We choose t_c depending on the confidence desired and the sample size.

Example: The graph of Student's t distribution with 7 degrees of freedom ($\nu = 7$) is shown.



We want to find the value of t_1 for which the total shaded area is 0.05. Student's t distribution is symmetric. Thus, the shaded area to the right of t_1 is 0.025. The area to the left is

$$1 - 0.025 = 0.975 .$$

t_1 represents the 97.5th percentile $t_{0.975}$.

From the tables for $\nu = 7$, we find that

$$t_{0.975} = 2.36 .$$

The required value of t is 2.36 .

Example: A sample of 17 measurements of the diameter of a drill gave a mean $\bar{x} = 3.47$ inches and a standard deviation $s = 0.05$ inches.

We shall find 95% confidence limits for the actual diameter of the drill. The 95% confidence limits are given by

$$\bar{x} \pm t_{0.975} \frac{s}{\sqrt{n-1}} .$$

Since $n = 17$, the number of degrees of freedom ν is

$$\nu = n - 1 = 16 .$$

From the tables for $\nu = 16$ and $t_{0.975}$, we find that

$$t_{0.975} = 2.12 .$$

Substituting all values, we find that

$$\bar{x} \pm t_{0.975} \frac{s}{\sqrt{n-1}} = 3.47 \pm 2.12 \frac{0.05}{4}$$

$$= 3.47 \pm 0.0265 .$$

Hence, we are 95% confident that the actual mean lies between 3.4965 inches and 3.4435 inches.

THE CHI-SQUARE DISTRIBUTION

We define the statistic

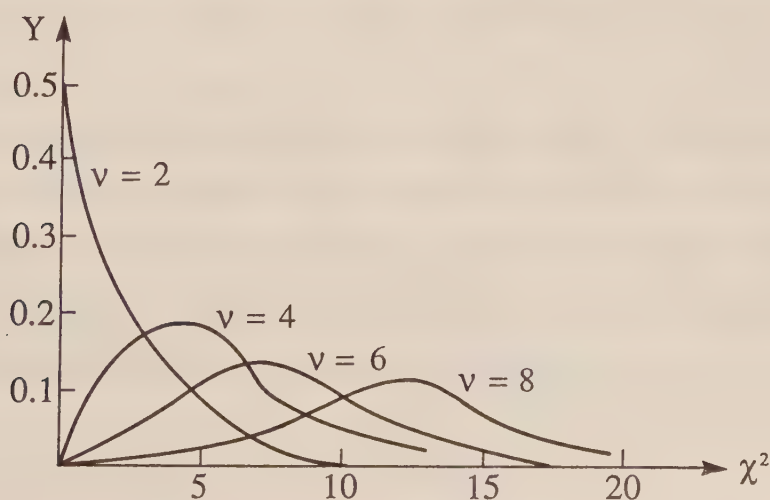
$$\chi^2 = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2}{\sigma^2}$$

χ is the Greek letter chi.

Samples of size n are drawn from a normal population with standard deviation σ . For each sample, the value χ^2 is computed. We obtain the chi-square distribution

$$Y = Y_0 \chi^{v-2} e^{-\frac{1}{2}\chi^2}$$

where $v = n - 1$ is the number of degrees of freedom, and Y_0 is a constant depending on v and such that the area under the curve is 1.



Chapter 2

COMBINATIONAL ANALYSIS

If an event can happen in any one of n ways and if, when this has occurred, another event can happen in any one of m ways, then the number of ways in which both events can happen in the specified order is

$$n \times m .$$

Example: A bank offers three types of checking accounts and four types of savings accounts. Any customer who wants to have both accounts has $3 \times 4 = 12$ possibilities.

FACTORIAL

Factorial n , denoted by $n!$, is defined as

$$n! = 1 \times 2 \times \dots \times (n-1) \times n \qquad 0! = 1.$$

Example: $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

$$2! = 2 \times 1 = 2$$

Problem: In how many ways can six children be arranged in a row?

Solution: Each of six children can fill the first position. Then each of five remaining children can fill the second place. Thus, there are 6×5 ways of filling the first two places. There are four ways of filling the third place, three ways of filling the fourth place, two ways of filling the fifth place, and one way of filling the sixth place. Hence,

$$\text{Number of arrangements} = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 6! = 720.$$

We have a general rule: the number of arrangements of n different objects in a row is $n!$

PERMUTATIONS

The number of ways k objects can be picked from among n objects with regard to order is denoted by ${}_nP_k$ or $P(n, k)$ or $P_{n, k}$ and is equal to

$${}_nP_k = n(n-1)(n-2) \dots (n-k+1) = \frac{n!}{(n-k)!}.$$

Example: Three letters are given: x, y, z . There are ${}_3P_2$ ways of choosing two letters at a time:

$${}_3P_2 = \frac{3!}{(3-2)!} = 2 \times 3 = 6.$$

Indeed, there are six possibilities:

$$xy, xz, yx, yz, zx, zy.$$

Example: From the set of numbers 1, 2, 3, 4, 5, 6, 7, 8, 9, one chooses three at a time. There are

$${}_9P_3 = \frac{9!}{6!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{6 \times 5 \times 4 \times 3 \times 2 \times 1} = 504$$

three-digit numbers.

Observe that the number of permutations of n objects, taken n at a time, is

$${}_nP_n = \frac{n!}{(n-n)!} = n!.$$

${}_nP_k$ is called the number of permutations of n different objects, taken k at a time.

Problem: In how many ways can 10 people be arranged in a circle?

Solution: Person number one can be placed anywhere. The remaining nine people can be arranged in

$$9! = 362,880$$

ways.

The number of permutations of n objects consisting of groups in which n_1 are alike, ... is

$$\frac{n!}{n_1!n_2!\dots},$$

where $n_1 + n_2 + \dots = n$.

Example: The number of permutations of letters in the word criss-cross is

$$\frac{10!}{4! \ 2! \ 2! \ 1! \ 1!} = 37,800.$$

COMBINATIONS

The number of ways one can choose k objects out of n objects, disregarding the order, is denoted by

$${}_nC_k, \ C(n, k) \text{ or } \binom{n}{k},$$

and is equal to

$${}_nC_k = \frac{n(n+1)\dots(n-k+1)}{k!} = \frac{n!}{k!(n-k)!} = \frac{{}_nP_k}{k!}.$$

${}_nC_k$ is the number of combinations of n objects taken k at a time.

Example: The number of combinations of the letters x , y , and z , taken two at a time, is

$${}_3C_2 = \frac{3!}{2! (3-2)!} = 3.$$

The combinations are xy , xz , and yz . Combinations xy and yx are the same, but permutations xy and yx are not the same.

Problem: A committee consists of seven people. There are 13 candidates. In how many ways can the committee be chosen?

Solution:

$${}_{13}C_7 = \frac{13!}{7! 6!} = \frac{8 \times 9 \times 10 \times 11 \times 12 \times 13}{2 \times 3 \times 4 \times 5 \times 6} = 1,716.$$

Problem: How many possible outcomes has a lotto game where a player chooses seven numbers out of 49 numbers?

Solution:

$$\begin{aligned} {}_{49}C_7 &= \frac{49!}{7! 42!} = \frac{43 \times 44 \times 45 \times 46 \times 47 \times 48 \times 49}{2 \times 3 \times 4 \times 5 \times 6 \times 7} \\ &= 85,900,584 \\ {}_nC_k &= {}_nC_{n-k}. \end{aligned}$$

The number of combinations of n objects taken 1, 2, ..., n at a time is

$${}_nC_1 + {}_nC_2 + \dots + {}_nC_n = 2^n - 1.$$

It is difficult to evaluate $n!$ for large numbers. In such cases, an approximate formula (called Stirling's Formula) is used:

$$n! \approx \sqrt{2\pi n} n^n e^{-n},$$

where e is the natural base of logarithms, $e = 2.718281828\dots$

Problem: Determine the probability of four 4's in six tosses of a die.

Solution: The result of each toss is the event 4 or non 4 ($\bar{4}$). Thus,

$$4, 4, 4, \bar{4}, 4, \bar{4}$$

or

$$\bar{4}, 4, 4, \bar{4}, 4, 4$$

are successes.

The probability of an event 4, 4, 4, $\bar{4}$, 4, $\bar{4}$ is

$$\begin{aligned} P(4, 4, 4, \bar{4}, 4, \bar{4}) &= \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{1}{6} \times \frac{5}{6} \\ &= \left(\frac{1}{6}\right)^4 \times \left(\frac{5}{6}\right)^2. \end{aligned}$$

All events in which four 4's and two non 4's occur have the same probability. The number of such events is

$${}_6C_4 = \frac{6!}{4! 2!} = 15,$$

and all these events are mutually exclusive. Hence,

$$P(\text{four 4's in six tosses}) = 15 \times \left(\frac{1}{6}\right)^4 \times \left(\frac{5}{6}\right)^2 = 0.008.$$

Problem: Thirty percent of the cars produced by a factory have some defect. A sample of 100 cars is selected at random. What is the probability that

- 1) exactly 10 cars will be defective?
- 2) 95 or more will be defective?

Solution: In general, if $p = p(E)$ and $q = p(\bar{E})$, then the probability of getting exactly m E 's in n trials is

$${}_nC_m p^m q^{n-m}.$$

$$1) \quad p(10 \text{ defective cars}) = {}_{100}C_{10} \left(\frac{3}{10}\right)^{10} \left(\frac{7}{10}\right)^{90}$$

$$2) \quad p(95 \text{ or more defective cars}) = p(95 \text{ defective}) + \\ p(96 \text{ defective}) + p(97 \text{ defective}) + p(98 \text{ defective}) + \\ p(99 \text{ defective}) + p(100 \text{ defective}) =$$

$${}_{100}C_{95} \left(\frac{3}{10}\right)^{95} \left(\frac{7}{10}\right)^5 + {}_{100}C_{96} \left(\frac{3}{10}\right)^{96} \left(\frac{7}{10}\right)^4 \\ + \dots + {}_{100}C_{100} \left(\frac{3}{10}\right)^{100} \left(\frac{7}{10}\right)^0$$

Chapter 3

NUMERICAL ANALYSIS

BASICS OF NUMERICAL ANALYSIS

Numerical analysis is often thought of as being synonymous with computer programming or data processing. However, this is not exactly so; these devices only make the execution of algorithms less cumbersome.

The necessary ingredients in any numerical analysis problem can be summarized as follows:

Input $\rightarrow$ Algorithm $\rightarrow$ Output

The form of the algorithm is based on the mathematical derivation of a method for solving a posed problem; once a method that will give a solution has been derived, then the algorithm shows the sequence of mathematical operations to follow in order to extract a numerical answer.

The data given is Input, the method is the Algorithm, and the results are Output.

Where computers are applied, the algorithms are usually laid out in flow-charts.

ORDER OF CONVERGENCE

The order of convergence is most useful in numerical analysis, in that it allows us to find the approximation error.

If $\{a_n\}_{n=1}^{\infty}$ is a sequence converging to a , then $\{a_n\}_{n=1}^{\infty}$ converges to a

with $O(\beta_n)$ rate of convergence, where $\{\beta_n\}_{n=1}^{\infty}$ is another sequence with $\beta_n \neq 0$ for each n ; if

$$\frac{|a - a_n|}{|\beta_n|} \leq K$$

for sufficiently large n , then K is independent of n .

Generally, if $\lim_{x \rightarrow 0} F(x) = L$, the convergence is called $O(G(x))$ if $\exists K > 0$, independent of $x \ni$

$$\frac{|F(x) - L|}{|G(x)|} \leq K$$

for sufficiently small $|x| > 0$ and

$$F(x) = L + O(G(x))$$

or $F(x) \rightarrow L$ with rate of convergence $O(G(x))$. If

$$a_n = a + O(\beta_n) \quad a'_n = a' + O(\beta'_n) \quad \beta'_n = \beta_n.$$

then

$$1) \quad a_n \pm a'_n = a \pm a' + O(\beta_n);$$

$$2) \quad a_n a'_n = aa' + O(\beta_n); \text{ and}$$

$$3) \quad \frac{a_n}{a'_n} = \frac{a}{a'} + O(\beta_n), \quad a' \neq 0.$$

The following sections pertain to numerical methods for equations of one variable. These algorithms are classics that are most often adapted for computer solution. Understanding the method facilitates algorithm and program development.

THE BISECTION METHOD

If f is continuous on $[a, b]$, with $f(a)$ and $f(b)$ having opposite signs, the

intermediate value theorem states that there is $p \in (a, b) \ni f(p) = 0$. By repeatedly halving the subintervals, keeping the half containing p , we solve $f(x) = 0$.

Step 1: Find endpoints a, b .

Step 2: Set $p = a + \frac{b-a}{2}$.

Step 3: If $f(p) = 0$, then we finish (the accuracy is a matter of choice).

Step 4: If $f(p) \neq 0$, then if $f(a)f(p) > 0$, let $a = p$. Otherwise, set $b = p$.

Step 5: Repeat procedure from Step 2.

The sequence generated while approximating p is one such that

$$|p_n - p| \leq \frac{b-a}{2^n}, \quad n \geq 1.$$

If we choose an error bound of ϵ , the minimum number of iterations N needed to ensure accuracy within ϵ is

$$N \geq \frac{\log_{10}(b-a) - \log_{10} \epsilon}{\log_{10} 2}.$$

FIXED POINT ITERATIVE METHOD

A fixed point of a continuous function h is a number x such that $h(x) = x$.

Basis for Fixed Point: Let $h(x) = x - f(x)$.

THEOREM 1

If f is defined on $[a, b]$, and $f(p) = 0$ for some $p \in [a, b]$, then h has a fixed point $p \in [a, b]$.

THEOREM 2

If $h \in C[a, b]$, and $h(x) \in [a, b] \forall x \in [a, b]$, then h has a fixed point in

$[a, b]$. If $h'(x)$ exists on $[a, b]$, with $|h'(x)| < 1 \forall x \in [a, b]$, then h has a unique fixed point in $[a, b]$.

We rewrite $f(x) = 0$ in the form $h(x) = x$ after we have tested h for a fixed point. This test is important because it ensures convergence.

Step 1: Find initial approximation p_0 .

Step 2: Set $p = h(p_0)$.

Step 3: Check $|p - p_0|$ to decide termination. Otherwise proceed to Step 4.

Step 4: Set $p_0 = p$.

Step 5: Repeat procedure from Step 2.

THEOREM 3

If the function $h(x) - x = 0$ satisfies a Lipschitz condition on $[a, b]$ with $h' < 1 \forall x \in [a, b]$, then the sequence generated by $p_n = h(p_{n-1})$, $n \geq 1$, converges to the unique fixed point in $[a, b]$.

NEWTON-RAPHSON METHOD

If a function $h \in C^2[a, b]$ with $h(p) = 0$, and $\hat{x}$ an approximation to p such that $h'(\hat{x}) \neq 0$, $|\hat{x} - p|$ small, the first degree Taylor polynomial for $h(x)$ about $\hat{x}$ is

$$h(x) = h(\hat{x}) + (x - \hat{x})h'(\hat{x}) + \frac{(x - \hat{x})^2 h''(\zeta(x))}{2}$$

$\zeta(x)$ between x and $\hat{x}$. With $x = p$,

$$0 = h(\hat{x}) + (p - \hat{x})h'(\hat{x}) + \frac{(p - \hat{x})^2}{2} h''(\zeta(p)).$$

Dropping the last term (since $|\hat{x} - p|$ is small) and solving for p ,

$$p \approx \hat{x} - \frac{h(\hat{x})}{h'(\hat{x})}.$$

We can generate the sequence $\{p_n\}$ until we are as close as we wish, where

$$p_n = p_{n-1} - \frac{h(p_{n-1})}{h'(p_{n-1})}, \quad n \geq 1.$$

To find the solution to $f(x) = 0$, follow these steps:

Step 1: Set $p = p_0 - \frac{f(p_0)}{f'(p_0)}$.

Step 2: Check $|p - p_0|$ to decide termination. Otherwise proceed to Step 3.

Step 3: Set $p_0 = p$.

Step 4: Repeat procedure from Step 1.

It is important to choose an initial approximation that is close to the root. Otherwise, convergence may be slow.

The Newton-Raphson method has quadratic convergence. If e_n is the n th absolute error, then, if p is a simple root,

$$|e_{n+1}| \approx \frac{1}{2} \frac{e_n^2 |f''(p)|}{|f'(p)|}$$

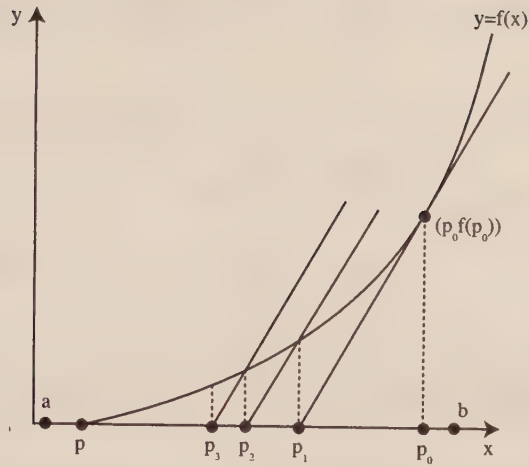
or as

$$\hat{x}_n \rightarrow p, \quad \frac{e_{n+1}}{e_n} \rightarrow \frac{f''(p)}{2f'(p)}, \quad f'(p) \neq 0.$$

The number of correct decimal places should double with each iteration.

FIXED TANGENT METHOD

This method is based upon the Newton-Raphson method. The slope of the first tangent at $(p_0, f(p_0))$ is used for all iterations.



The equation of the tangent at $f(p_0)$ is

$$y = f(p_0) + f'(p_0) (x - p_0).$$

This line intersects the x -axis at $x=p_1$, where $y=0$.

Solving the equation for p_1 ,

$$p_1 = p_0 - \frac{f(p_0)}{f'(p_0)}$$

In general, using successive lines parallel to the fixed tangent,

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_0)}, n \geq 1.$$

Iterate until p_n is satisfactory.

SECANT METHOD

This method is a variation of the Newton-Raphson method, applied to functions whose derivatives are cumbersome to evaluate.

We know that

$$f'(p_{n-1}) = \lim_{x \rightarrow p_{n-1}} \frac{f(x) - f(p_{n-1})}{x - p_{n-1}}$$

at $x = p_{n-2}$

$$\begin{aligned} f'(p_{n-1}) &\approx \frac{f(p_{n-2}) - f(p_{n-1})}{p_{n-2} - p_{n-1}} \\ &= \frac{f(p_{n-1}) - f(p_{n-2})}{p_{n-1} - p_{n-2}}. \end{aligned}$$

If we use this approximation of $f'(p_{n-1})$ in Newton's formula

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})},$$

we have

$$p_n = p_{n-1} - \frac{f(p_{n-1})(p_{n-1} - p_{n-2})}{f(p_{n-1}) - f(p_{n-2})}, \quad n \geq 2.$$

This method requires two initial approximations.

Step 1: Set $r_0 = f(p_0)$, $r_1 = f(p_1)$, (p_0, p_1 initial approximations).

Step 2: Find $p = P_1 - \frac{r_1(P_1 - P_2)}{r_1 - r_0}$.

Step 3: If $|p - p_1|$ is satisfactory, then terminate. Otherwise proceed to Step 4.

Step 4: Set $p_0 = p_1$

$$r_0 = r_1$$

$$p_1 = p$$

$$r_1 = f(p).$$

Step 5: Continue from Step 2.

The secant method converges to the root p with

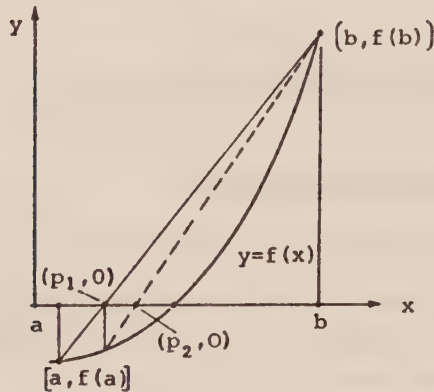
$$\left| \frac{e_{n+1}}{e_n e_{n-1}} \right| \rightarrow \frac{1}{2} \left| \frac{f''(p)}{f'(p)} \right|,$$

provided that f is twice continuously differentiable, with $f'(p) \neq 0$. The order of convergence has a value approximately equal to 1.62.

REGULA FALSI

Using the Regula Falsi method to find $x \ni f(x) = 0, x \in [a, b]$, the intervals $[a_i, b_i]$ are chosen in the same way as the bisection method; new approximate intervals are generated in a fashion similar to the secant method.

Let f be continuous on $[a, b]$ with $f(a)f(b) < 0$.



The secant through $(a, f(a))$ and $(b, f(b))$ has slope

$$\frac{f(b) - f(a)}{b - a}.$$

The equation of the secant is

$$y - f(a) = \frac{f(b) - f(a)}{b - a} (x - a).$$

Using point $(p_1, 0)$ gives

$$p_1 = a - \frac{f(a)(b-a)}{f(b)-f(a)} \rightarrow P_1 = a - \frac{f(a)(b-a)}{f(b)-f(a)}.$$

For the iterative procedure, we get

$$p_n = a_n - \frac{f(a_n)(b_n - a_n)}{f(b_n) - f(a_n)}, \quad n \geq 1.$$

For each new p , check whether $f(p_n)f(a) < 0$, as in the bisection method. This allows us to determine the interval containing the root.

Another important aspect of numerical analysis is the approximation of polynomials. Polynomial approximation is essential in experimental studies, especially in engineering, where the physics of the problem is complex.

BASIC THEOREMS OF ALGEBRA

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

a_i 's constants, $a_n \neq 0$ is a polynomial of degree n .

Fundamental Theorem of Algebra

If P is a polynomial of degree $n \geq 1$, then $P(x) = 0$ has at least one root.

If $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ is a polynomial of degree $n \geq 1$, then there exist unique constants $r_1, r_2, \dots, r_k \in C$, the field of complex numbers, and positive integers $m_1, m_2, \dots, m_k$ such that

$$\sum_{i=1}^k m_i = n.$$

$P(x)$ can be written

$$P(x) = a_n (x - r_1)^{m_1} (x - r_2)^{m_2} \dots (x - r_k)^{m_k}.$$

r_i is a zero of multiplicity m_i . $(x - r_i)^{m_i}$ is a factor of $P(x)$.

A polynomial of n th degree has exactly n zeros (not necessarily distinct) over C .

THEOREM 4

If P and Q are polynomials of degree at most n , and $r_1, r_2, \dots, r_i, i > n$ are distinct numbers with $P(r_i) = Q(r_i) \forall i$, then $P(x) \equiv Q(x)$.

THEOREM 5

If $z = a + bi$ is a complex zero of multiplicity m of the polynomial P , then $z = a - bi$ is also a zero of multiplicity m of P . $(x^2 - 2ax + a^2 + b^2)^m$ is a factor of P .

Horner's Method

Evaluate $P(x_0) = a_0 + a_1x_0 + \dots + a_n x_0^n$.

If

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

and

$$b_n = a_n, \text{ and if } b_i = a_i + b_{i+1}x_0, i = n-1, n-2, \dots, 1, 0,$$

then $b_0 = P(x_0)$.

To reduce $P(x)$:

If

$$Q(x) = b_n x^{n-1} + b_{n-1} x^{n-2} + \dots + b_2 x + b_1,$$

then

$$P(x) = (x - x_0)Q(x) + b_0.$$

Horner's Method of Synthetic Division

Dividing $P(x)$ by $(x - x_0)$ is represented by the scheme:

- 1) Assign the value 0 to T .
- 2) For $i = n$ down to 0 (Step 1) assign the value " $T * x_0 + a_i$ " to T . (T corresponds to b_i .)

THEOREM 6

If Horner's method gives

$$P(x) = (x - x_0)Q(x) + b_0,$$

then

$$P'(x_0) = Q(x_0).$$

ROOTS OF POLYNOMIALS

If P is a polynomial of degree n , and $P(r_i) = 0$, $i = 1, \dots, n$, r_i is not an integer, and if r_i and r_{i+1} are so close that none of the previous root-finding methods work, then we have to find another method.

If $y \in (0, 1]$, then $\frac{1}{y} \in [1, \infty)$.

If $y \in [-1, 0)$, then $\frac{1}{y} \in (-\infty, -1]$.

For $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = 0$

Let

$$\begin{aligned} P\left(\frac{1}{y}\right) &= a_n \left(\frac{1}{y}\right)^n + a_{n-1} \left(\frac{1}{y}\right)^{n-1} + \dots + a_1 \left(\frac{1}{y}\right) + a_0 \\ &= \frac{1}{y^n} (a_n + a_{n-1}y + \dots + a_1 y^{n-1} + a_0 y^n) \end{aligned}$$

Let $\hat{P}(y) = a_0 y^n + a_1 y^{n-1} + \dots + a_{n-1} y + a_n$

Then $\hat{y} \neq 0$ is a zero of $\hat{P}$, if and only if $\frac{1}{\hat{y}}$ is a zero of P .

We then investigate the intervals $[-1, 0]$, $[0, 1]$ to determine sign changes of P and $\hat{P}$.

If $P(x_i)P(x_j) < 0$, then P has a zero between x_i and x_j .

If $\hat{P}(x_i)\hat{P}(x_j) < 0$, then $\hat{P}$ has a zero between x_i and x_j and P has a zero between $\frac{1}{x_i}$ and $\frac{1}{x_j}$. Choose N to be even, and define x_0 to be -1 .

$$x_i = x_{i-1} + \frac{2}{N}, \quad i = 1, \dots, N, \quad \text{giving } N + 1 \text{ points.}$$

Upon locating the intervals, the Newton-Raphson method can be used to refine approximations.

TAYLOR POLYNOMIALS

The algebraic polynomial functions of the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0,$$

n a non-negative integer, $a_0, \dots, a_n$, real constants, map R onto itself.

THEOREM 7 — Weierstrass

If f is defined and continuous on $[a, b]$, with $\varepsilon > 0$, there exists a polynomial P defined on $[a, b]$ $\ni |f(x) - P(x)| < \varepsilon \forall x \in [a, b]$. Any function satisfying the requirements of the Weierstrass theorem can be approximated by a polynomial as closely as desired. The theorem cannot be used effectively for computational purposes; instead of determining closeness, the problem is often finding a polynomial satisfying specific conditions to be used for some problem under consideration, but simultaneously being close in some respects.

Consider the specifications:

- 1) Find a polynomial of degree n or less.
- 2) The polynomial must approximate a given function f .
- 3) The approximations must be valid $\forall x$ near a given x_0 .

The logical approach is as follows:

1) Assume that the polynomial agrees with f and as much of its derivatives near x_0 .

2) Construct $P(x) = \sum_{i=0}^n a_i x^i$.

3) Find a_i such that

$$P(x_0) = f(x_0),$$

$$P'(x_0) = f'(x_0), \text{ and}$$

$$P^{(n)}(x_0) = f^{(n)}(x_0).$$

We will find that

$$a_0 = f(x_0), \quad a_1 = \frac{f'(x_0)}{1}, \quad a_2 = \frac{f''(x_0)}{2!}$$

$$a_n = \frac{f^{(n)}(x_0)}{n!}.$$

Since the approximations must be valid for some domain $(x - x_0) \subseteq D$, in order for $f(x) \Big|_{x_0} = f(x_0)$, we must introduce powers of $(x - x_0)$. If $x_0 \neq 0$, the resulting polynomial is then

$$P(x) = \sum_{i=0}^n a_i (x - x_0)^i = \sum_{i=0}^n \frac{f^{(i)}(x_0) (x - x_0)^i}{i!}.$$

This procedure gives the Taylor polynomial of degree n about x_0 , which agrees with f on $(x - x_0) \subseteq D$, with error,

$$R_n^{(x)} = f(x) - P(x) = \frac{f^{(n+1)}(\xi(x)) (x - x_0)^{n+1}}{(n+1)!}.$$

If f is difficult to integrate as it is, this method provides a polynomial function that can be easily integrated instead of f . We can also integrate the remainder term R to place a bound on approximation error.

If $(x - x_0) \supset D$, and D is always specified, then the approximations using

Taylor polynomials become very poor; the error grows as we move away from the boundaries of D .

Taylor series can be applied to functions of two real variables where the $n + 1$ partial derivatives of f exist and are continuous in some domain D centered about (x_0, y_0) . The expansion for $f(x, y)$ about $f(x_0 + h, y_0 + k)$ for $(x_0 + h, y_0 + k) \in D$ is

$$\begin{aligned} f(x_0 + h, y_0 + k) = & f(x_0, y_0) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right) f(x_0, y_0) + \\ & \dots + \frac{1}{n!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^n f(x_0, y_0) . \end{aligned}$$

With error,

$$R = \frac{\left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^{n+1} f(x_0 + \alpha h, y_0 + \alpha k)}{(n+1)!}$$

$$0 < \alpha < 1 .$$

BERNSTEIN POLYNOMIALS

Another set of approximating polynomials are Bernstein polynomials. This topic includes some combinatorics; therefore, a little review will be helpful.

THEOREM 8 — Binomial Expansion

If p and q commute in multiplication, then

$$(p + q)^N = \sum_{i=0}^N \binom{N}{i} p^i q^{N-i} \quad \text{where} \quad \binom{N}{i} = \frac{N!}{(N-i)!i!} .$$

Some results of the binomial theorem are the following:

$$1) \sum_{i=0}^N \binom{N}{i} x^i (1-x)^{N-i} = 1.$$

$$2) \sum_{i=0}^N \binom{N}{i} x^i (1-x)^{N-i} (i - Nx) = 0.$$

$$3) \sum_{i=0}^N \binom{N}{i} x^i (1-x)^{N-i} (i - Nx)^2 = Nx(1-x).$$

Given any function f defined on $[0, 1]$, the Bernstein polynomial of degree N for f is given by

$$B_N(x) = \sum_{i=0}^N \binom{N}{i} f\left(\frac{i}{N}\right) x^i (1-x)^{N-i}.$$

For $x > 1$, the approximations are poor; this can be overcome by using a change of argument, transforming $[0, x]$ to $[0, 1]$, and using a Bernstein polynomial of higher degree.

THEOREM 9

If $f(x)$ is continuous $\forall x \in [0, 1]$, then the Bernstein polynomial is given by

$$B_N(x) = \sum_{i=0}^N \binom{N}{i} f\left(\frac{i}{N}\right) x^i (1-x)^{N-i}, \text{ which}$$

converges uniformly to $f(x)$ as $N \rightarrow \infty$.

LAGRANGE POLYNOMIAL

Taylor polynomials sometimes are not of any use, especially far from a single point x .

Given certain points in the plane, the problem of finding approximating polynomials that must pass through these points is one whose solution

involves the technique of interpolation. These polynomials are unique.



Consider the problem of constructing a polynomial of degree $\leq n$ passing through the $n + 1$ points $(x_0, f(x_0)) \dots (x_n, f(x_n))$ such that $P(x_k) = f(x_k)$, $k = 0, 1, \dots, n$. The Lagrange method constructs P with

$$\begin{aligned} P(x) &= f(x_0)L_{n,0}(x) + f(x_1)L_{n,1}(x) + \dots + f(x_n)L_{n,n}(x) \\ &= \sum_{k=0}^n f(x_k)L_{n,k}(x) \end{aligned}$$

with

$$\begin{aligned} L_{n,k}(x) &= \frac{(x-x_0)(x-x_1)\dots(x-x_{k-1})(x-x_{k+1})\dots(x-x_n)}{(x_k-x_0)(x_k-x_1)\dots(x_k-x_{k-1})(x_k-x_{k+1})\dots(x_k-x_n)} \\ &= \prod_{\substack{i=0 \\ i \neq k}}^n \frac{(x-x_i)}{(x_k-x_i)}, \quad \forall k=0, 1, \dots, n, \end{aligned}$$

where
$$L_{n,k}(x_i) = \begin{cases} 0 & \forall i \neq k \\ 1 & \forall i = k \end{cases}.$$

When there is no doubt about the degree, $L_{n,k}(x)$ is written as $L_k(x)$.

The arguments $x_0, \dots, x_n$ do not necessarily have to be equally spaced. Most often they are not.

$L_{n,k}(x)$ is invariant under linear transformation.

THEOREM 10

Let $x_0, x_1, \dots, x_n$ be $n + 1$ distinct numbers $\in [a, b]$. Let $f \in C^{n+1}[a, b]$. Then

$$\forall x \in [a, b] \exists \zeta(x) \in (a, b)$$

$$f(x) - P(x) = \frac{f^{(n+1)}(\zeta(x))}{(n+1)!} \prod_{i=0}^n (x - x_i).$$

$P(x)$ is the n th degree Lagrange interpolating polynomial. The error is proportional to $\prod_{i=0}^n (x - x_i)$; although it is theoretically important, its use should be confined to functions with derivatives of known bounds.

INTEGRATED INTERPOLATION

DEFINITION

Let f be a function defined at $n + 1$ points $x_0, \dots, x_n$, and $m_1, \dots, m_k$ k distinct integers such that $0 \leq m_i \leq n \forall i$.

The Lagrange polynomial of degree less than or equal to $k - 1$ agreeing with f at $x_{m_1}, \dots, x_{m_k}$ is denoted by $P_{m_1}, \dots, P_{m_k}$.

THEOREM 11

Let f be defined at $x_0, x_1, \dots, x_k$, with x_j, x_i any two distinct numbers in the set. If

$$P(x) = \frac{(x - x_j)P_{0, 1, \dots, j-1, j+1, \dots, k}(x) - (x - x_i)P_{0, 1, \dots, i-1, i+1, \dots, k}(x)}{(x_i - x_j)},$$

then P is the Lagrange polynomial of degree less than or equal to k which interpolates f at $x_0, x_1, \dots, x_k$.

The polynomial of degree two for arguments x_0, x_1, x_k is

$$P_{0,1,k}(x) = \frac{1}{x_k - x_1} \begin{vmatrix} P_{0,k}(x) & x - x_k \\ P_{0,1}(x) & x - x_1 \end{vmatrix}.$$

Neville's Method

Neville's method uses extended interpolation with values obtained from the Lagrange polynomial. One form arranges values of $P_{0,1}, \dots, n$ in the tabular scheme,

x_0	P_0				
x_1	P_1	$P_{0,1}$			
x_2	P_2	$P_{1,2}$	$P_{0,1,2}$		
x_3	P_3	$P_{2,3}$	$P_{1,2,3}$	$P_{0,1,2,3}$	
x_4	P_4	$P_{3,4}$	$P_{2,3,4}$	$P_{1,2,3,4}$	$P_{0,1,2,3,4}$

Table 3.1

We only need the previous row to add another row.

Another form of Neville's method, more convenient than the first, is as follows:

Let $H_{i,j}$, $i \geq j$, be the interpolating polynomial of degree j on $j+1$ points $x_{i-j}, \dots, x_i$. Compute $H_{i,j} = P_{i-j, i-j+1, \dots, i-1, i}$ by Neville's method. Let $H_{i, j-1} = P_{i-j+1, \dots, i-1, i}$ and $H_{i-1, j-1} = P_{i-j, i-j+1, \dots, i-1}$ in the expression for P to give

$$H_{i,j}(x) = \frac{(x - x_i)H_{i-1, j-1}(x) - (x - x_{i-j})H_{i, j-1}(x)}{x_{i-j} - x_i}$$

$$j = 1, 2, 3, \dots \quad i = j, j+1, \dots$$

$H_{i,0} = f(x_i)$ for each i . This gives the table:

x_0	$H_{0,0}$				
x_1	$H_{1,0}$	$H_{1,1}$			
x_2	$H_{2,0}$	$H_{2,1}$	$H_{2,2}$		
x_3	$H_{3,0}$	$H_{3,1}$	$H_{3,2}$	$H_{3,3}$	
x_4	$H_{4,0}$	$H_{4,1}$	$H_{4,2}$	$H_{4,3}$	$H_{4,4}$
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.

Table 3.2

This gives the same values as the previous table, but it is easier to set up for a computer.

Aitken's Method

Aitken's method is similar to Neville's method, but is even more convenient in that the tabulated values of $x_0, \dots, x_n$ and $f(x_0), \dots, f(x_n)$ are in such a way that one can compute $P_0, \dots, P_n(x)$ by inspection. The layout of the table for $n = 3$ takes the following form:

x_0	y_0				$x - x_0$
x_1	y_1	$P_{0,1}(x)$			$x - x_1$
x_2	y_2	$P_{0,2}(x)$	$P_{0,1,2}(x)$		$x - x_2$
x_3	y_3	$P_{0,3}(x)$	$P_{0,1,3}(x)$	$P_{0,1,2,3}(x)$	$x - x_3$

Table 3.3

$P_{0,1,3}$ is the determinant

$$\begin{vmatrix} P_{0,1}(x) & x - x_1 \\ P_{0,3}(x) & x - x_3 \end{vmatrix}$$

divided by $x_1 - x_3$, which is the difference between the entries in the right-most column.

Inverse interpolation is the process of finding the value of the argument corresponding to a given value of a function $f(x)$ that is between two tabulated values. Aitken's repeated process provides an easy way of doing this after interchanging the roles of the dependent and independent variables.

CUBIC SPLINE INTERPOLATION

Piecemeal polynomial approximation using cubic polynomials between each successive pair of nodes is called Cubic Spline Interpolation.

The cubic spline does not assume that the derivatives agree with those of the function, not even at nodes.

Given f defined on $[a, b]$ with nodes $a = x_0 < x_1 < \dots < x_n = b$, the cubic spline interpolant C for f is a function that meets the following conditions:

- 1) C is a cubic polynomial C_i on the subinterval

$$[x_i, x_{i+1}] \quad \forall i = 0, \dots, n-1$$

- 2) $C(x_i) = f(x_i) \quad \forall i = 0, \dots, n$

- 3) $C_{i+1}(x_{i+1}) = C_i(x_{i+1}), i = 0, \dots, n-2$

- 4) $C'_{i+1}(x_{i+1}) = C'_i(x_{i+1}), i = 0, \dots, n-2$

- 5) $C''_{i+1}(x_{i+1}) = C''_i(x_{i+1}), i = 0, \dots, n-2$

- 6) One of the following sets of boundary conditions is satisfied:

- a) Free Boundary (Natural Spline):

$$C''(x_0) = C''(x_n) = 0$$

b) Clamped Boundary:

$$C'(x_0) = f'(x_0)$$

$$C'(x_n) = f'(x_n)$$

They can be defined with other conditions.

Clamped boundary conditions give more accurate approximations; they include more information. To construct C for f , we apply the conditions to cubic polynomials.

Let

$$C_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$$

$$i = 0, \dots, n - 1.$$

Applying the conditions to $C_i(x)$ gives

$$h_{i-1}c_{i-1} + 2(h_{i-1} + h_i)c_i + h_ic_{i+1} = \frac{3}{h_i}(a_{i+1} - a_i) -$$

$$\frac{3}{h_{i-1}}(a_i - a_{i-1})$$

$$i = 1, 2, \dots, n - 1.$$

This gives a linear system of equations with $c_i, i = 0, \dots, n$ only.

$h_i, i = 0, \dots, n - 1$, and $a_i, i = 0, \dots, n - 1$ are given by the spacing of x_i 's $i = 0, \dots, n$ ($h_i = x_{i+1} - x_i$) and $f(x_i), i = 0, \dots, n$, respectively.

For the natural spline boundary condition, the linear system can be written in the form $A\vec{c} = \vec{d}$, where

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ h_0 & 2(h_0 + h_1) & h_1 & & & \\ 0 & h_1 & 2(h_1 + h_2) & h_2 & & \\ \ddots & \ddots & \ddots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & h_{n-2} & 2(h_{n-2} + h_{n-1}) & h_{n-1} \\ 0 & \dots & \dots & 0 & 0 & 1 \end{bmatrix}$$

$$\vec{c} = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \quad \vec{d} = \begin{bmatrix} 0 \\ \frac{3}{h_1}(a_2 - a_1) - \frac{3}{h_0}(a_1 - a_0) \\ \vdots \\ \frac{3}{h_{n-1}}(a_n - a_{n-1}) - \frac{3}{h_{n-2}}(a_{n-1} - a_{n-2}) \\ 0 \end{bmatrix}.$$

When the c_i for $i = 0, \dots, n$ are known, then b_i , $i = 0, \dots, n - 1$ can be found using the relationship

$$b_i = \frac{1}{h_i}(a_{i+1} - a_i) - \frac{h_i}{3}(2c_i + c_{i+1})$$

and d_i , $i = 0, \dots, n - 1$, can similarly be determined using the relationship

$$d_i = \frac{c_{i+1} - c_i}{3h_i}, \text{ so that}$$

we can construct the cubic polynomials

$$C_i, \quad i = 0, \dots, n - 1.$$

Note that the form of A indicates the c_i 's are unique. Therefore, the cubic spline is unique.

Other important skills associated with numerical analysis are numerical integration and curve fitting. Numerical integration is especially useful when the antiderivative of a function cannot be explicitly represented with elementary functions. Curve fitting is especially useful in data reduction.

TRAPEZOIDAL RULE AND SIMPSON'S RULE

Trapezoidal Rule

To find $\int_a^b f(x)dx$, choose a step size $h = \frac{b-a}{n}$, n is a positive integer.

Let $x_i = x_0 + ih$, $i = 1, 2, \dots, n$. The method sums the areas A_i of the trapezoids constructed, whose corners are $x_i, x_{i+1}, f_i, f_{i+1}$, so that the smaller h is, the better the approximation is. The integral becomes

$$\int_a^b f(x)dx \approx \sum_{i=1}^n A_i = \frac{h}{2} [f_0 + 2f_1 + 2f_2 + \dots + 2f_{n-1} + f_n]$$

with approximation error

$$|E_n| = \left| \frac{-(x_n - x_0)h^2 f^{(2)}(\zeta)}{12} \right|.$$

$$x_0 < \zeta < x_n$$

If $x_0 = a$, $x_1 = x_n = b$, then

$$\int_a^b f(x)dx = \frac{h}{2} [f_0 + f_1].$$

Simpson's Rule

Simpson's rule uses the points

$$x_i, x_2, x_{i+2}, \dots, i = 0, 1, \dots, n-2$$

and replaces the curve between (x_i, f_i) , (x_{i+1}, f_{i+1}) , (x_{i+2}, f_{i+2}) with a quadratic, then sums the areas A_i of all the segments.

To find $\int_a^b f(x)dx$, let $h = \frac{b-a}{n}$, n is an even integer $x_i = x_0 + ih$, $i = 0, 1, \dots, n$. The sum is

$$\int_a^b f(x)dx \approx \sum A_i = \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \dots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)]$$

with approximation error

$$|E_n| = \left| \frac{(x_n - x_0)h^4 f^{(4)}(\zeta)}{180} \right|$$

$$x_0 < \zeta < x_n.$$

Note: The trapezoidal rule and Simpson's rule can be obtained by integrating the Lagrange interpolating polynomial over the interval $[x_0, x_n]$.

NUMERICAL QUADRATURE

Numerical quadrature methods involve approximating $\int_a^b f(x)dx$ using sums of the form $\sum c_i f(x_i)$, which comes from integrating the Lagrange polynomial P_n over $[a, b]$, giving

$$\int_a^b \sum_{i=0}^n f(x_i) L_i(x) dx = \sum_{i=0}^n c_i f(x_i),$$

$$c_i = \int_a^b L_i(x) dx \text{ for each } i = 0, 1, \dots, n.$$

The precision of any quadrature formula is the positive integer n $\exists$. $E(P_k) = 0$, P_k all polynomials of degree less than or equal to n , but $E(P_{n+1}) \neq 0$, where P_{n+1} is some polynomial of degree $n+1$. The $(n+1)$ point closed Newton-Cotes formula has

$$x_0 = a, \quad x_n = b, \quad h = \frac{b-a}{n}, \quad x_i = x_0 + ih.$$

If n is even, then $f \in C^{n+2}[a, b]$

$$\int_a^b f(x) dx = \sum_{i=0}^n c_i f(x_i) + E_n$$

$$c_i = \int_{x_0}^{x_n} L_i(x) dx = \int_{x_0}^{x_n} \prod_{\substack{i=0 \\ i \neq k}}^n \frac{x - x_i}{x_k - x_i} dx.$$

The error,

$$E_n = \frac{h^{n+3} f^{(n+2)}(\zeta)}{(n+2)!} \int_0^n u^2(u-1)\dots(u-n)du,$$

$$\zeta \in [a, b], \quad u = \frac{x - x_0}{h},$$

is used in the evaluation of c_i .

If n is odd, $f \in C^{n+1}[a, b]$, then

$$\int_a^b f(x)dx = \sum_{i=0}^n c_i f(x_i) + E_n,$$

c_i as before.

$$E_n = \frac{h^{n+2} f^{(n+1)}(\zeta)}{(n+1)!} \int_0^n u(u-1)\dots(u-n)du,$$

$$\zeta \in [a, b].$$

DISCRETE LEAST SQUARES APPROXIMATION

Approximation consists mainly of two types of problems:

- 1) Replace a complicated function by a simpler type, such as a polynomial that can be used to give approximate values with some known error bound.
- 2) Find a function that best fits the given data. Some of these methods were mentioned in previous chapters.

Least squares approximation for a given set of data $(x_1, y_1), \dots, (x_n, y_n)$ finds a linear function that best approximates the set of values, while simultaneously minimizing the squares of the differences between the values of the approximating linear function, and the given values.

Method

Assume that $y_j = cx_j + d$ is the approximating line for the arguments $x_1, \dots, x_n$, and that y_j is the j th given value. The constants c and d are needed such that the sum of the squares of the differences

$$D = \sum_{j=1}^n [y_j - (cx_j + d)]^2 \text{ is a minimum.}$$

D can be seen as a function of two variable c and d . The necessary requirements for a minimum are as follows:

$$1) \quad \frac{\partial}{\partial c}(D) = 0$$

$$2) \quad \frac{\partial}{\partial d}(D) = 0$$

So,

$$\frac{\partial}{\partial c} \left(\sum_{j=1}^n [y_j - (cx_j + d)]^2 \right) = 0$$

$$\frac{\partial}{\partial d} \left(\sum_{j=1}^n [y_j - (cx_j + d)]^2 \right) = 0.$$

This yields

$$\sum_{j=1}^n y_j x_j = c \sum_{j=1}^n x_j^2 + d \sum_{j=1}^n x_j, \text{ and}$$

$$\sum_{j=1}^n y_j = c \sum_{j=1}^n x_j + dn,$$

which uniquely determines c and d with

$$c = \left[n \sum_{j=1}^n x_j^2 - \left(\sum_{j=1}^n x_j \right)^2 \right]^{-1} \left(n \sum_{j=1}^n y_j x_j - \sum_{j=1}^n y_j \sum_{j=1}^n x_j \right)$$

$$d = \left[n \sum_{j=1}^n x_j^2 - \left(\sum_{j=1}^n x_j \right)^2 \right]^{-1} \left(\sum_{j=1}^n x_j^2 \sum_{j=1}^n y_j - \sum_{j=1}^n x_j \sum_{j=1}^n y_j x_j \right).$$

The least square fit is $y = cx + d$.

POLYNOMIAL APPROXIMATION AND EXPONENTIAL LEAST SQUARES

Given a set of data $\{(x_j, y_j) \mid j = 0, \dots, N\}$, we need a polynomial P_n of degree $n < N$ of the form

$$P_n = \sum_{i=0}^n c_i x^i$$

to fit the values. Then we must minimize D , where

$$D = \sum_{j=0}^N (y_j - P(x_j))^2.$$

The method is similar to before.

Carry out the square; replace $P(x_j)$ in the $\sum_{i=0}^n c_i x_j^i$. Then set $\frac{\partial}{\partial c_i}(D) = 0$ for $i = 0, \dots, n$.

This procedure gives the normal equations to be solved for $n + 1$ constants c_i ; they are the $n + 1$ equations:

$$c_0 \sum_{j=0}^N x_j^0 + c_1 \sum_{j=0}^N x_j^1 + \dots + c_n \sum_{j=0}^N x_j^n = \sum_{j=0}^N y_j x_j^0$$

$$c_0 \sum_{j=0}^N x_j^1 + c_1 \sum_{j=0}^N x_j^2 + \dots + c_n \sum_{j=0}^N x_j^{n+1} = \sum_{j=0}^N y_j x_j^1$$

$$c_0 \sum_{j=0}^N x_j^n + c_1 \sum_{j=0}^N x_j^{n+1} + \dots + c_n \sum_{j=0}^N x_j^{2n} = \sum_{j=0}^N y_j x_j^n.$$

This system gives unique constants as long as x_j , $j = 0, \dots, N$, are distinct. The least squares polynomial of degree n that fits the data is

$$P_n = \sum_{i=0}^n c_i x^i.$$

Exponential Least Squares

If the data is suspected to be exponentially related, we can still apply least squares approximation to find a fit.

For the assumption $y = de^{cx}$, rewrite $\ln y = cx + \ln d$.

The values (x_j, y_j) , $j = 1, \dots, n$, to be fitted are known data.

Compute $\ln y_j$, x_j^2 , $x_j \ln y_j$, $j = 1, \dots, n$. We replace these in the previous expressions for c and d , giving:

$$c = \left(n \sum_{j=1}^n x_j \ln y_j - \sum_{j=1}^n x_j \sum_{j=1}^n \ln y_j \right) \left[n \sum_{j=1}^n x_j^2 - \left(\sum_{j=1}^n x_j \right)^2 \right]^{-1}$$

$$\ln d = \left(\sum_{j=1}^n x_j^2 \sum_{j=1}^n \ln y_j - \sum_{j=1}^n x_j \sum_{j=1}^n x_j \ln y_j \right) \left[n \sum_{j=1}^n x_j^2 - \left(\sum_{j=1}^n x_j \right)^2 \right]^{-1}.$$

The exponential or logarithmic fit is

$$y = de^{cx},$$

or

$$\ln y = cx + \ln d.$$

LEAST SQUARES APPROXIMATION AND ORTHOGONAL POLYNOMIALS

Given a function $f \in C[a, b]$ for which we require a polynomial P of degree less than or equal to n such that

$$\int_a^b (f(x) - P_n(x))^2 dx \text{ is a minimum,}$$

the least squares method can be used to find this polynomial.

Write
$$P_n(x) = \sum_{i=0}^n c_i x^i.$$

The difference D with respect to the constants c_i to be minimized is

$$D(c_0, \dots, c_n) = \int_a^b \left(f(x) - \sum_{i=0}^n c_i x^i \right)^2 dx.$$

This is accomplished by first squaring out the expression then finding and setting the partial derivative of D with respect to the c_i 's equal to 0, $i = 0, 1, \dots, n$ to obtain the normal equations; they are the $(n + 1)$ linear equations, in $(n + 1)$ unknowns $c_i, i = 0, \dots, n$, given by

$$\sum_{j=0}^n c_j \int_a^b x^{i+j} dx = \int_a^b x^i f(x) dx, \quad i = 0, 1, \dots, n.$$

These have a unique solution as long as $f \in C[a, b]$, $a \neq b$.

THEOREM 13

If $\{\psi_0, \dots, \psi_n\}$ is an orthogonal set of functions on $[a, b]$ with respect to the weight function w , the least squares approximation to f on $[a, b]$ with respect to w is

$$P(x) = \sum_{i=0}^n c_i \psi_i(x),$$

$$c_i = \frac{\int_a^b w(x) \psi_i(x) f(x) dx}{\int_a^b w(x) [\psi_i(x)]^2 dx}.$$

When $w(x) = 1$, $\psi_i(x) = x^i$, $\frac{\int_a^b w(x) \psi_i(x) f(x) dx}{\int_a^b w(x) [\psi_i(x)]^2 dx}$ $i = 0, 1, \dots, n$, we get the previous form. The sets most commonly used are the Chebyshev polynomials. They keep the error stable.

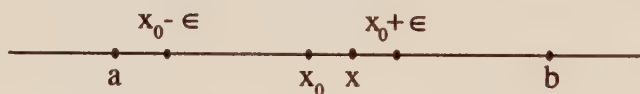
Chapter 4

REAL VARIABLES

REAL LINE

The system of real numbers considered geometrically is merely points on a line. This “real line” is a one-dimensional Euclidean space. The real line is associated with the study of a function of one real variable. Since it can be shown that the real number system is a complete, ordered field, and that the real line is another name for the real number system, then the real line is a complete, ordered field.

If S is a set of points on the real line, S is called an open set if to each $x_0 \in S$ there corresponds some $\epsilon > 0$ such that $x \in S$ whenever $|x - x_0| < \epsilon$. The figure below depicts this case.



Given any number of open sets, the union of all given sets is the open set S .

$$S = S_1 \cup S_2 \text{ or more generally}$$

$$T = \text{union of all } S_n \text{'s } (n = 1, 2, 3, \dots)$$

The empty set, $S = \phi$, is open, as is the entire space.

The intersection of a finite number of open sets is again an open set, but the intersection of an infinite number of open sets may not be open.

A set S is closed if the complement $S^{\sim}$ is open. Let $S \subset R$ (the space); then the complement is $S^{\sim} = R - S$. The set S consisting of a single point is closed; thus, $S^{\sim} = R - S$ is open.

PLANE

The real line denoted by R , being the set of all real numbers, is the basis of the plane which is described by $R \times R$, a "Euclidean Plane." R^2 defines the two-dimensional plane familiar from ordinary geometry.

Suppose R^k exists such that $k > 2$, and there exists an n -dimensional plane termed the hyperplane. The important characteristic of a hyperplane is that the set of all points $x = (\zeta_1, \dots, \zeta_k)$ satisfies a single linear equation of the form

$$\alpha_1 \zeta_1 + \dots + \alpha_k \zeta_k = \beta,$$

Where $\alpha_1 \dots \alpha_k$ and β are fixed and not all α_i 's equal zero. If a specific $\zeta = 0$, such as $\zeta_1 = 0$, the hyperplane is called a coordinate hyperplane.

3 SPACE

Defined as $R \times R \times R$ or R^3 , the element x in this Euclidean three-dimensional space is an ordered triple $x = (\zeta_1, \zeta_2, \zeta_3)$, where the order is significant.

K-SPACE

The real Euclidean space R^k is a collection of points and a certain mathematical structure, which is created by making certain definitions. Each point is an ordered set of k real numbers, $\zeta_1, \zeta_2, \dots, \zeta_k$. A point may be designated as $a = (\alpha_1, \dots, \alpha_k)$ or $x = (\zeta_1, \dots, \zeta_k)$.

The distance between any two points x, y in R^k is

$$D(x, y) = \left[(\zeta_1 - n_1)^2 + \dots + (\zeta_k - n_k)^2 \right]^{\frac{1}{2}}.$$

The following properties exist:

- 1) $D(x, y) = 0$ is equivalent to $x = y$.
- 2) $D(x, y) = D(y, x)$.
- 3) $D(x, y) \leq D(x, a) + D(a, y)$.

The distance function D is called the metric for R^k .

Lastly, the additive and multiplicative algebraic laws govern the R^k vector space.

RIEMANN INTEGRAL

Riemann's theory of integration applied to a bounded, continuous function $f: [a, b] \rightarrow R$, where $[a, b]$ is a compact interval that does not satisfy integration needs for discontinuous functions.

Thus, for a continuous function $F_n(x)$ that converges to $f(x)$ uniformly on $[a, b]$, the integral is

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx ,$$

and

$$\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$$

when

$$f: [a, c] \rightarrow R \text{ and } a < b < c .$$

LEBESGUE INTEGRAL

Before defining the Lebesgue integral, a discussion of measurability is necessary. The exterior Lebesgue measure, $m_e S$, of any bounded linear point set is the greatest lower bound of the combined length of a set of intervals covering S . The interior Lebesgue measure $m_i S$ of S is the difference between the length $b - a$ of any bounded interval (a, b) containing S and the exterior measure of the complement of S with respect to (a, b) .

S is a measurable set if $m_e S = m_i S = mS$. An unbounded set is measurable if $(-x, x) \cap S$ is measurable for all $x > 0$. Every bounded open set is measurable. A function $f(x)$ is measurable on (a, b) if the set of points x in (a, b) exists such that $f(x) \leq C$ is measurable for every real value of C .

Given a function $y = f(x)$ which is measurable and bounded on (a, b) , for

an unbounded function of the Lebesgue integral

$\int_a^b f(x)dx$ is defined by

$$\int_a^b f(x)dx = \lim_{A \rightarrow \infty} \int_a^b f_A(x)dx + \lim_{B \rightarrow \infty} \int_a^b f_B(x)dx ,$$

with

$$f_A(x) = \begin{cases} 0 & \text{if } f(x) \leq 0 \\ f(x) & \text{if } 0 < f(x) \leq A \\ A & \text{if } f(x) \geq A \end{cases}$$

$$f_B(x) = \begin{cases} 0 & \text{if } f(x) \geq 0 \\ f(x) & \text{if } 0 > f(x) \geq B \\ B & \text{if } f(x) < B \end{cases} .$$

However, $\int_a^b f(x)dx$ must exist.

For unbounded intervals,

$\int_a^\infty f(x)dx$ for $x > a$ is

$$\int_a^\infty f(x)dx = \lim_{x_1 \rightarrow \infty} \int_a^{x_1} f_1(x)dx + \lim_{x_2 \rightarrow \infty} \int_a^{x_2} f_2(x)dx ,$$

with

$$f_1(x) = \begin{cases} 0 & \text{if } f(x) \leq 0 \\ f(x) & \text{if } f(x) > 0 \end{cases}$$

$$f_2(x) = \begin{cases} 0 & \text{if } f(x) \geq 0 \\ f(x) & \text{if } f(x) < 0 \end{cases} .$$

Chapter 5

COMPLEX VARIABLES

BASIC CONCEPTS

A complex variable is a variable taking on complex values.

A variable w is a function of the variable z if there is a rule assigning one or more values of w to each value of z . This is written as

$$w = f(z), \quad (5.1)$$

where f is the “name” of the function, and $f(z)$ is the value of w corresponding to the value of z .

Example: For $w = f(z) = z^2$,

$$f(1) = 1, \quad f(2 - i) = (2 - i)^2 = 3 - 4i.$$

A single-valued function $w = f(z)$ is one assigning a unique value w to each value z .

Example: $f(z) = z^2$ is single-valued since, for every value z , the square z^2 is uniquely defined.

A multi-valued function $w = f(z)$ is one that is not single-valued.

For a multi-valued function $w = f(z)$, there are values z for which more than one value of w is defined.

Example: $f(z) = z^{\frac{1}{2}}$ is multi-valued since, for each value of $z \neq 0$, there are two values of w whose square is z ; thus, $f(1) = \pm 1$, while $f(i) = \pm \frac{1+i}{\sqrt{2}}$.

If w is a function of f of z , $w = f(z)$, then z is the independent variable of f , and w is the dependent variable.

The domain of definition or domain of a function $w = f(z)$ is the set of values of the independent variable z for which f is defined.

The domain of a function is determined by its definition. If we define f by the statement

$$“f(z) = z^2 \text{ for all } z,”$$

then the domain of f is the set of all z in the complex plane. On the other hand, a statement of the form

$$“f(z) = \frac{1}{z}”$$

implies that the domain of f is the set of all nonzero z , since we cannot divide by zero. If we wish to define this function at $z = 0$ as well, then we must specify its value, i.e.,

$$“f(z) = \frac{1}{z} \text{ for } z \neq 0”$$

$$“f(0) = 5i,”$$

in which case f is now defined for all z : if $z = 0$, then

$$f(z) = \frac{1}{z}; \quad \text{if } z = 0, \text{ then } f(z) = 5i.$$

The range of a function $w = f(z)$ is the set of values attained by its dependent variable w .

The real and imaginary parts of $w = f(z)$ are real functions of the independent variables $x = \text{Re } z$, $y = \text{Im } z$,

$$u(x, y) = \text{Re } w, \quad v(x, y) = \text{Im } w. \quad (5.2)$$

Example:

$$\text{For } w = f(z) = z^2, u = x^2 - y^2, \quad v = 2xy;$$

$$\text{for } w = \frac{1}{z}, \quad u = \frac{x}{x^2 + y^2}, \quad v = \frac{-y}{x^2 + y^2};$$

$$\text{for } w = \text{mod } z, \quad u = \sqrt{x^2 + y^2}, \quad v = 0;$$

$$\text{for } w = \bar{z}, \quad u = x, \quad v = -y.$$

A real-valued function is one for which $v \equiv 0$.

Example: Some real-valued functions are listed below:

$$w = f(z) = \text{Re } z$$

$$w = g(z) = \text{mod } z$$

$$w = h(z) = (\text{Re } z) + (i \text{Im } z) = z$$

SOME TYPES OF FUNCTIONS

A polynomial function or polynomial is a function of the form

$$\begin{aligned} P(z) &= \sum_{j=0}^n a_j z^j \\ &= a_0 + a_1 z + a_2 z^2 + \dots + a_n z^n; \end{aligned}$$

n is the order, or degree, of the polynomial.

A rational function is a ratio of polynomials

$$R(z) = \frac{P(z)}{Q(z)}; \quad (5.3)$$

its domain is the set of all z for which $Q(z)$ does not vanish.

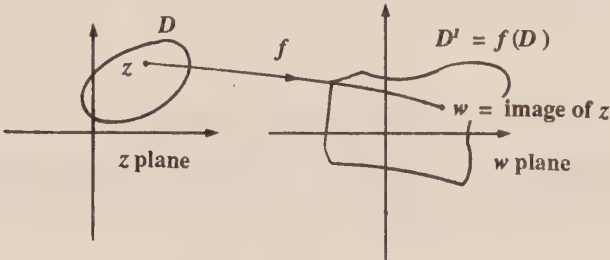
Example: The function

$$w = \frac{az + b}{cz + d}$$

is a rational function defined for $z \neq -\frac{d}{c}$. Special cases include the following

GEOMETRY OF FUNCTIONS

A function $w = f(z)$ defines a mapping or transformation f of points z of a z -plane onto points $w = f(z)$ of a w -plane; a function $w = f(z)$ can also be defined between two points of the same plane.



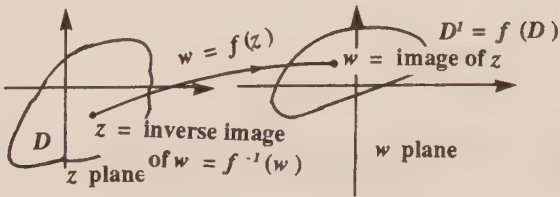
The mapping $w = f(z)$.

The image of a point z of the z -plane, under the mapping f , is the point $w = f(z)$.

The image of a set D in the z -plane, under the mapping f , is the set $D' = f(D)$ of points $w = f(z)$ for z in D . f is a mapping or transformation of D onto its image set $f(D)$.

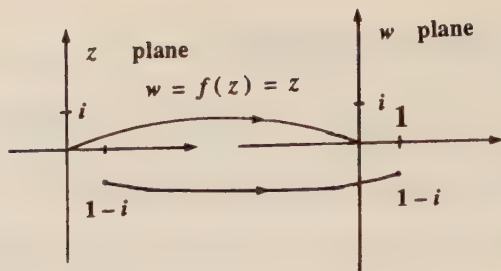
f is a transformation of its domain onto its range.

The inverse image f^{-1} of a point w in the range of f is the set of all points z of its domain, for which $f(z) = w$.



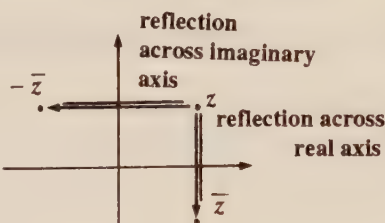
The inverse image of w .

Example: The function $w = f(z) = z$ maps every point z onto itself. The image of $z = 1 - i$ is $w = 1 - i$; the image of $z = 0$ is $w = 0$; the inverse image of $w = 3 + \pi i$ is $z = 3 + \pi i$.



The mapping $w = f(z) = z$.

The mappings $w = \bar{z}$ and $w = -\bar{z}$ are reflections across the real and imaginary axes, respectively.

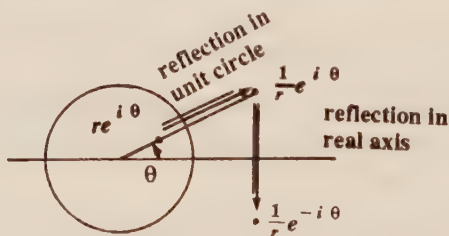


Reflections across the real and imaginary axes.

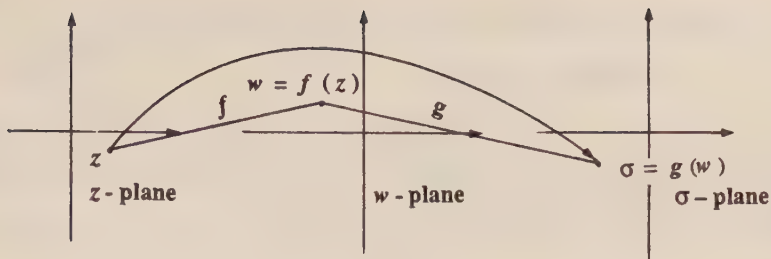
Example: The function $w = f(z) = \frac{1}{z}$ is defined for all $z \neq 0$. For $z = re^{i\theta}$,

$$f(z) = \left(\frac{1}{r}\right)e^{-i\theta}.$$

Thus, $f(z)$ performs two successive mappings: a reflection with respect to the unit circle and a reflection with respect to the real axis.

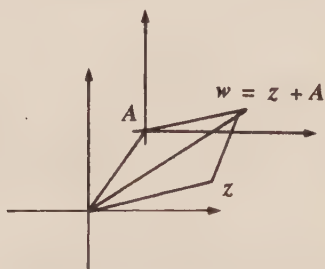


The mapping $w = f(z) = \frac{1}{z}$.



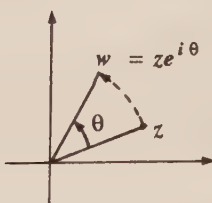
The mapping
 $\sigma = F(z) = g(f(z))$

A translation is a mapping of the form $w = z + A$, with A a complex number.



Translation: $w = z + A$.

A rotation is a mapping of the form $w = ze^{i\phi}$, with ϕ the angle of rotation.



Rotation: $w = ze^{i\phi}$.

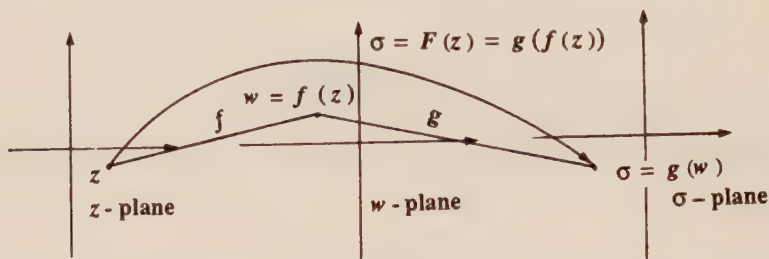
Example:

$w = f(z) = \frac{(1+i)z}{\sqrt{2}}$ rotates the point z through 45° since $\frac{(1+i)}{\sqrt{2}} = e^{\frac{i\pi}{4}}$.

A compound function $\delta = F(z)$ is of the form

$$\delta = F(z) = g(f(z)), \quad (5.4)$$

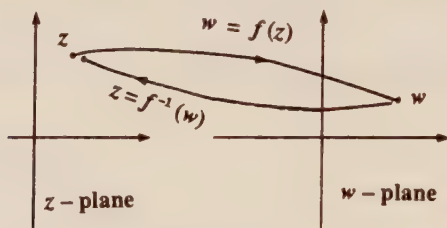
where f is a mapping from its domain D of definition onto its range R , which in turn is a subset of the domain D' of definition of g . We say that F is a composite function of composition or g and f .



A compound function $\sigma = g(f(z))$.

The inverse function $f^{-1}(w)$ of $w = f(z)$ is that function satisfying the relations.

$$f^{-1}(f(z)) = z, \quad f(f^{-1}(w)) = w \quad (5.5)$$



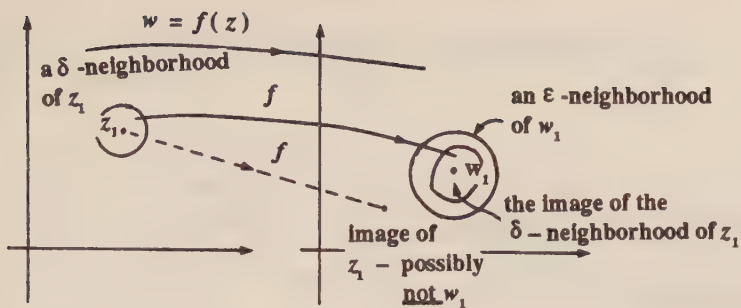
The inverse function: $z = f^{-1}(f(z))$, $w = z$, $f(f^{-1}(w))$.

LIMITS OF FUNCTION

w_1 is the limit of $f(z)$ as z tends to z_1 ,

$$w_1 = \lim_{z \rightarrow z_1} f(z), \quad (5.6)$$

if any ε -neighborhood of w_1 contains the image of some δ -neighborhood of z_1 , with the possible exception of z_1 itself.



$w_1 = \lim_{z \rightarrow z_1} f(z)$: there is a δ -neighborhood of z_1 mapped into a given ϵ -neighborhood of w .

We see that w_1 is a value to which $f(z)$ becomes arbitrarily near as z approaches z_1

Example: $\lim_{z \rightarrow 1} f(z) = 2$ for f defined as

$$f(z) = \begin{cases} z^2 + 1, & \text{for } z \neq 1 \\ -7i, & \text{for } z = 1 \end{cases}$$

Note that $f(1)$ does not coincide with the limiting value.

Example: For g defined as

$$g(z) = \begin{cases} \frac{1}{z}, & \text{for } z \neq 0 \\ 5, & \text{for } z = 0 \end{cases}$$

the limit of g as $z \rightarrow 0$ does not exist, although g is defined for $z = 0$.

Example: For g defined as $g(z) = z^3$, $\lim_{z \rightarrow i} g(z) = -i$ coincides with the value $g(i)$.

If $w_1 = u_1 + iv_1$, $z_1 = x_1 + iy_1$ and $f(z) = u + iv$, then

$$w_1 = \lim_{z \rightarrow z_1} f(z)$$

is equivalent to the limit statements for the real functions $u(x, y)$, $v(x, y)$,

$$\lim_{\substack{x \rightarrow x_1 \\ y \rightarrow y_1}} u(x, y) = u_1, \quad \lim_{\substack{x \rightarrow x_1 \\ y \rightarrow y_1}} v(x, y) = v_1. \quad (5.7)$$

PROPERTIES OF LIMITS

Suppose that

$$\lim_{z \rightarrow z_1} f(z) = w_1, \quad \lim_{z \rightarrow z_1} g(z) = w_2;$$

then

$$\lim_{z \rightarrow z_1} [f(z) + g(z)] = \lim_{z \rightarrow z_1} f(z) + \lim_{z \rightarrow z_1} g(z) = w_1 + w_2 \quad (5.8)$$

$$\lim_{z \rightarrow z_1} [f(z)g(z)] = \left[\lim_{z \rightarrow z_1} f(z) \right] \left[\lim_{z \rightarrow z_1} g(z) \right] = w_1 w_2, \quad (5.9)$$

while, for $w_2 \neq 0$,

$$\lim_{z \rightarrow z_1} \left[\frac{f(z)}{g(z)} \right] = \left[\lim_{z \rightarrow z_1} f(z) \right] \div \left[\lim_{z \rightarrow z_1} g(z) \right] = \frac{w_1}{w_2}. \quad (5.10)$$

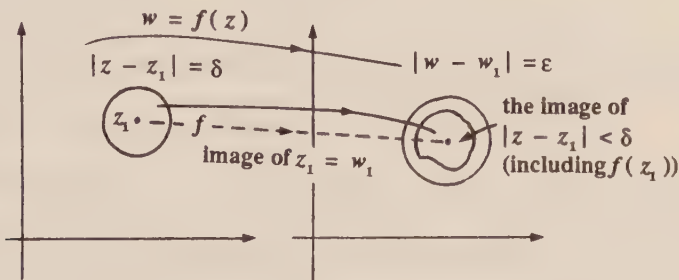
CONTINUITY

$f(z)$ is continuous at $z = z_1$ if:

- 1) $\lim_{z \rightarrow z_1} f(z)$ exists;
- 2) $f(z_1)$ has been defined; and
- 3) $\lim_{z \rightarrow z_1} f(z) = f(z_1)$.

Thus, $f(z)$ is continuous at z_1 , if z_1 is in its domain and the value assigned to it is its limit as z approaches z_1 from any direction. The $\epsilon - \delta$ definition of continuity states that $f(z)$ is continuous at z_1 if $f(z_1)$ is defined, and if any ϵ -neighborhood of $f(z_1)$ contains the image of some δ -neighborhood of z_1 , or,

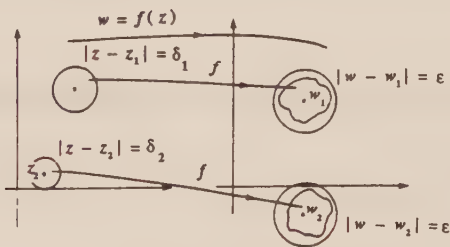
given any $\epsilon > 0$, there exists some $\delta > 0$ such that for $|z - z_1| < \delta$, we have $|f(z) - f(z_1)| < \epsilon$.



f is continuous at z_1 : for a given ϵ there is a δ -neighborhood of z_1 mapped into $|w - w_1| < \epsilon$.

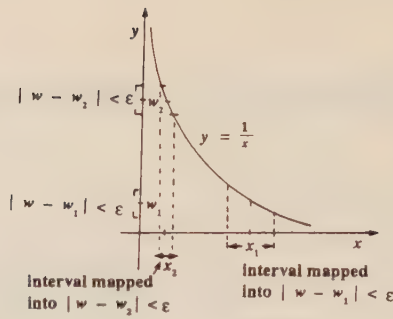
A continuous function is a function that is continuous at every point of its domain of definition.

The δ of our definition of continuity generally depends on the point z_1 . If it is independent of z_1 for all points z_1 of a set S , then f is uniformly continuous on S .



For the same ϵ -value, the δ of $z_2(\delta_2)$ must be smaller than that of $z_1(\delta_1)$ so that $|z - z_1| < \delta_2$ can be mapped into the ϵ -neighborhood of $w_2 = f(z_2)$.

Uniform continuity is vividly exhibited by the behavior of the real function $\frac{1}{x}$ near the origin, where the δ -interval required to restrict the variation (ϵ) in the function becomes sharply smaller as x nears the origin.



$\frac{1}{x}$ is not uniformly continuous for $x > 0$.

PROPERTIES OF CONTINUITY

If f and g are continuous at a point z_1 , then their sum, difference, and product are continuous at z_1 .

If f and g are continuous at z_1 and $g(z_1) \neq 0$, then $\frac{f}{g}$ is also continuous at z_1 .

If $w = f(z)$ is continuous at z_1 and $\sigma = g(w)$ is continuous at $w_1 = f(z_1)$, then the composition

$$\sigma = F(z) = g(f(z))$$

is also continuous.

A function that is continuous on a closed and bounded set is uniformly bounded in absolute value on that set: for some $M > 0$,

$$|f(z)| < M. \quad (5.11)$$

A function that is continuous on a closed and bounded set is uniformly continuous on that set.

DERIVATIVES OF COMPLEX FUNCTIONS

Let $w = f(z)$ be defined in a neighborhood of the point z_1 . The derivative $f'(z_1)$ of f at z_1 is

$$f'(z_1) = \lim_{z \rightarrow z_1} \frac{f(z) - f(z_1)}{z - z_1} \quad (5.12)$$

when this limit exists. We use the notation $\frac{d}{dz}$ or $\frac{dw}{dz}$ to denote the derivative, as in the case of real functions.

$f(z)$ is differentiable at z_1 if its derivative, $f'(z_1)$, exists.

If f is differentiable at a point, then it is continuous at that point. However, the converse may not be true.

PROPERTIES OF THE DERIVATIVE

If $f(z)$, $g(z)$ are differentiable at z , then their sum $f + g$, difference $f - g$, and product fg are differentiable at this point, with their derivatives given by

$$\frac{d}{dz}[f(z) + g(z)] = f'(z) + g'(z) \quad (5.13)$$

$$\frac{d}{dz}[f(z) - g(z)] = f'(z) - g'(z) \quad (5.14)$$

$$\frac{d}{dz}[f(z)g(z)] = f'(z)g(z) + f(z)g'(z) \quad (5.15)$$

If $g(z) \neq 0$, then the quotient $\frac{f(z)}{g(z)}$ of two functions differentiable at a point z is differentiable at z , with the derivative given by

$$\frac{d}{dz}\left[\frac{f(z)}{g(z)}\right] = \frac{f'(z)g(z) - f(z)g'(z)}{g(z)^2}. \quad (5.16)$$

DIFFERENTIABILITY AND THE CAUCHY-RIEMANN EQUATIONS

$f(z)$ is differentiable if it is differentiable at every point of its domain of definition.

Let $f(z) = u(x, y) + iv(x, y)$. If f is differentiable at $z_1 = x_1 + iy_1$, then the first-order partial derivatives u_x, u_y, v_x, v_y of u and v with respect to x and y exist at (x_1, y_1) and satisfy the Cauchy-Riemann equations

$$\begin{aligned}u_x(x_1, y_1) &= v_y(x_1, y_1), \\u_y(x_1, y_1) &= -v_x(x_1, y_1).\end{aligned}\tag{5.17}$$

If u_x, u_y, v_x, v_y exist in a neighborhood of the point (x_1, y_1) , are continuous, and satisfy the Cauchy-Riemann equations at that point, then $f = u + iv$ is differentiable at $z_1 = x_1 + iy_1$.

If $f(z) = u(x, y) + iv(x, y)$ is differentiable at z , then its derivative is given by

$$\begin{aligned}f'(z) &= u_x(x, y) + iv_x(x, y) \\&= v_y(x, y) - iu_y(x, y).\end{aligned}\tag{5.18}$$

Example: The exponential function is defined as

$$\begin{aligned}w = f(z) &= e^z = e^{x+iy} = e^x e^{iy} \\&= e^x (\cos y + i \sin y) \\&= e^x \cos y + ie^x \sin y.\end{aligned}$$

We easily see that the real and imaginary parts of w ,

$$u(x, y) = e^x \cos y, \quad v(x, y) = e^x \sin y,$$

obey the Cauchy-Riemann equations, giving us the differentiability of the function for all z . Moreover,

$$f'(z) = u_x + iv_x = u + iv = f(z),$$

as in the case of the real exponential function.

The Cauchy-Riemann equations in polar coordinates are

$$u_r = \frac{1}{r} v_\theta, \quad \frac{1}{r} u_\theta = -v_r.\tag{5.19}$$

We have noted that a function that is differentiable at a point is continuous there. The converse is not true, as we see in the case of the function $f(z) = x$.

ANALYTICITY

f is analytic or holomorphic at $z = z_1$ if it is differentiable throughout some neighborhood of z_1 (including the point z_1 itself).

Example: $w = \frac{1}{z}$ is analytic at all $z \neq 0$.

f is entire if it is analytic at every point of the complex plane.

Example: $w = e^z$ is entire, since it is analytic at every point of the complex plane; $\frac{1}{z}$ is not entire, since analyticity fails at the origin $z = 0$.

f is analytic in a region R if it is analytic at every point of R .

Example: $\frac{1}{z}$ is analytic in the annulus $1 < |z| < 2$.

z_1 is a singular point or singularity of f if f is not analytic at z_1 but is analytic at some point in every neighborhood of z_1 .

Example: $z = 0$ is a singular point of $\frac{1}{z}$.

The sum, difference, and product of functions analytic in a region R are analytic in R .

If f and g are analytic in R , and g does not vanish anywhere in R , then $\frac{f}{g}$ is analytic in R .

The composition F of analytic functions of f and g , $F(z) = g(f(z))$, is analytic.

HARMONIC FUNCTIONS

A real function $A(x, y)$ is harmonic in a domain D of the x - y plane if all of its partial derivatives of first and second order A_x, A_y, A_{xx}, A_{xy} , and A_{yy} are continuous in D and satisfy the Laplace equation

$$A_{xx}(x, y) + A_{yy}(x, y) = 0 \quad (5.20)$$

throughout D .

The Laplacian operator, or Laplacian, is defined by

$$\Delta A(x, y) = A_{xx}(x, y) + A_{yy}(x, y). \quad (5.21)$$

The Laplace equation may be written as

$$\Delta A(x, y) = 0. \quad (5.22)$$

In polar coordinates, the Laplace equation takes the form

$$r^2 u_{rr}(r, \theta) + r u_r(r, \theta) + u_{\theta\theta}(r, \theta) = 0. \quad (5.23)$$

Example: $A(x, y) = e^x \cos y$ is harmonic, for

$$\begin{aligned} A_x &= e^x \cos y, & A_{xx} &= e^x \cos y \\ A_y &= -e^x \sin y, & A_{yy} &= -e^x \cos y; \end{aligned}$$

therefore,

$$\Delta A = A_{xx} + A_{yy} = 0.$$

If $u(x, y)$, $v(x, y)$ have continuous first-, and second-order partial derivatives with respect to x and y , satisfying the Cauchy-Riemann equations in a region R , then they are both harmonic in R .

The real and imaginary parts of an analytic function in a domain D are harmonic in D .

$u(x, y)$ and $v(x, y)$ are harmonic conjugates to each other in a region R if they are harmonic and satisfy the Cauchy-Riemann equations in R .

Example: The functions $u = e^x \cos y$, $v = e^x \sin y$ are harmonic conjugates to each other everywhere in the complex plane, since they obey the Cauchy-Riemann equations and are harmonic.

A function f is analytic in a domain D if and only if its real and imaginary parts are harmonic conjugates to each other in D .

The level curves of a real function $A(x, y)$ are the curves in the x - y plane along which A is constant:

$$A(x, y) = \text{const.} \quad (5.24)$$

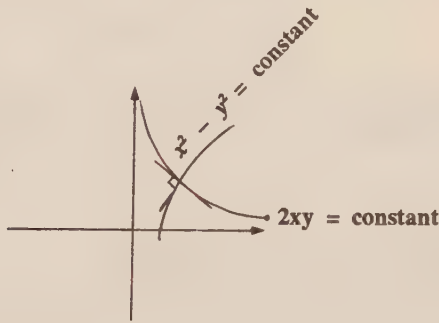
The level curves of the real and imaginary parts of an analytic function are orthogonal.

Example: The function $w = z^2$ is analytic everywhere. Its real and imaginary parts are

$$u = x^2 - y^2, \quad v = 2xy.$$

The level curves of u, v are orthogonal families of hyperbolas

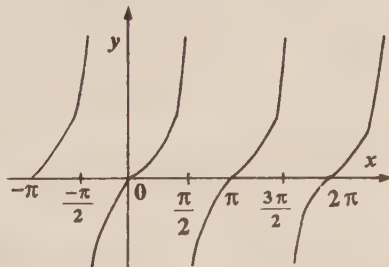
$$x^2 - y^2 = \text{const.}, \quad 2xy = \text{const.}$$



Orthogonality of level curves of $\text{Re}z^2$ and $\text{Im}z^2$.

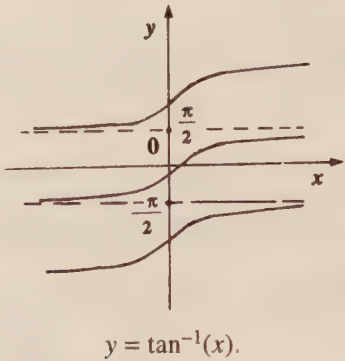
REMARKS ON THE REAL FUNCTION $y = \tan^{-1}(x)$

The graph of $y = \tan(x)$ on the interval $-\frac{\pi}{2} < x < \frac{\pi}{2}$ is duplicated on the intervals $\frac{\pi}{2} < x < \frac{3\pi}{2}$, ..., etc.



$$y = \tan(x).$$

Therefore, the graph of the inverse function $y = \tan^{-1}(x)$ consists of infinitely many alternative pieces, each displaced by (figure below).

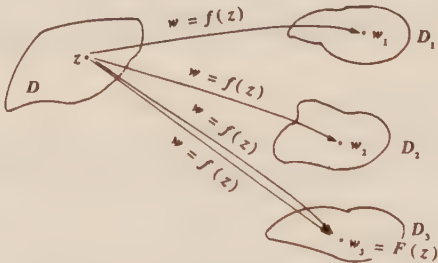


When working with the arctangent, we must specify with which piece or branch we are working.

MULTI-VALUED FUNCTIONS

The zeros of a function $w = f(z)$ are the points z for which the function vanishes, $f(z) = 0$.

A branch of a multi-valued function $w = f(z)$ is any single-valued function $w = F(z)$, analytic in some domain D , at each of whose points it takes on one of the values of f .



$w = F(z)$ maps each z onto one of the values of $f(z)$ and is analytic:
 $w = F(z)$ maps D onto D_3 .

A branch cut is a portion of a line or curve marking the domain D of a branch of a multi-valued function.

The requirement that a branch be analytic demands, in turn, that the

branch be continuous in D . The branch cut is used to remove all possible discontinuities by removing points of discontinuity from D .

A branch point is a singular point belonging to all possible branch cuts of a multi-valued function.

Example: For $z = Re^{i\theta}$, $z^{\frac{1}{2}}$ may take either of the values $\sqrt{Re^{\frac{i\theta}{2}}}$ or $-\sqrt{Re^{\frac{i\theta}{2}}}$, the latter arising since

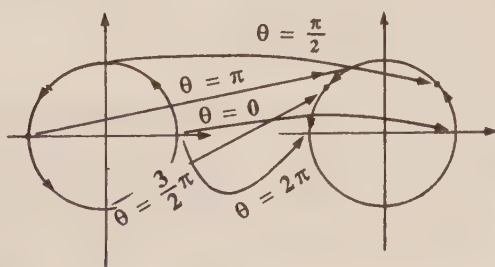
$$e^{i\theta} = e^{i(\theta + 2\pi)},$$

so that

$$(e^{i\theta})^{\frac{1}{2}} = e^{i\left[\frac{\theta}{2} + \pi\right]} = -e^{\frac{i\theta}{2}}.$$

Thus, $F(z) = \sqrt{Re^{\frac{i\theta}{2}}}$ and $G(z) = -\sqrt{Re^{\frac{i\theta}{2}}}$ are candidate branches of $\sqrt{z}$, defined for $0 < \theta < 2\pi$.

Neither F nor G is analytic in the entire plane, since neither is periodic with period 2π in θ . This is seen by tracking a point z that is moving along the unit circle $|z| = 1$ from $\theta = 0$ to $\theta = 2\pi$ in the counterclockwise direction ; the image $w = F(z)$ of z moves along only the upper half of the unit circle in the w -plane, reaching $w = -1$ for $\theta = 2\pi$.



$$\text{Points } F(e^{i\theta}) = e^{\frac{i\theta}{2}}, \quad 0 < \theta < 2\pi$$

$$F(e^0) = e^0$$

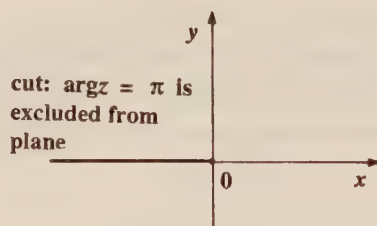
$$F\left(e^{i\frac{\pi}{2}}\right) = e^{i\frac{\pi}{4}} = \frac{1+i}{\sqrt{2}}$$

$$F\left(e^{i\pi}\right) = e^{i\frac{\pi}{2}} = i$$

$$F\left(e^{i\frac{3\pi}{2}}\right) = e^{i\frac{3\pi}{4}} = \frac{1-i}{\sqrt{2}}$$

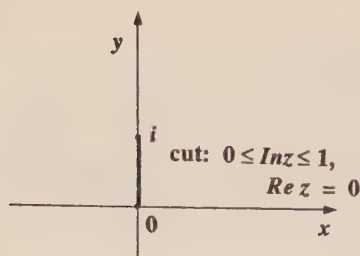
$$F\left(e^{i2\pi}\right) = e^{i\pi} = -1$$

If we do not allow a complete circuit around $z = 0$, then such a discontinuity will not occur. We therefore cut the z -plane by any line (or curve) C and define F or G for all z not on C , resulting in a function analytic in this cut region. C is the branch cut for F , G , a candidate cut being any ray $\theta = \alpha$. Since rotation about the origin must be excluded, $z = 0$ must belong to all branch curves, and is therefore a branch point.



A cut for $F(z) = \sqrt{z}$.

Example: For $f(z) = [z(z-1)]^{\frac{1}{2}}$ to be uniquely defined, we must select one of the branches F or G as above. Whichever choice we make, we must still introduce cuts in the z -plane that will exclude a complete circuit about both $z = 0$ and $z = i$. One such possible cut is shown in the figure below.



A cut for $F(z) = \sqrt{z(z-i)}$.

$w = e^z$ AND ITS KEY PROPERTIES

For $z = x + iy$, the exponential function $w = e^z$ is

$$w = e^{x+iy} = e^x (\cos y + i \sin y). \quad (5.25)$$

$z = iy$ (5.25) reduces to Euler's formula

$$e^{iy} = \cos y + i \sin y.$$

$w = e^z$ is an entire function equal to its derivative,

$$\frac{d}{dz} e^z = e^z. \quad (5.26)$$

$w = e^z$ is periodic with period $2i$,

$$\begin{aligned} e^{z+2\pi i} &= e^{x+iy+2\pi i} = e^x [\cos(y+2\pi i) + i \sin(y+2\pi i)] \\ &= e^x [\cos y + i \sin y] = e^{x+iy} = e^z. \end{aligned} \quad (5.27)$$

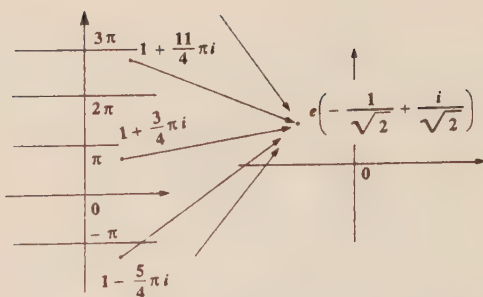
$w = e^z$ never vanishes, and has norm

$$|e^z| = |e^{x+iy}| = e^x |\cos y + i \sin y| = e^x. \quad (5.28)$$

Under the mapping $w = e^z$, the infinitely many points

$$z = \log(R) + i(\phi \pm 2n), \quad n = 0, 1, 2, \dots$$

are mapped onto the same point $w = Re^{i\phi}$.



$w = e^z$ maps infinitely many points into one point: the image of

$$1 + \frac{3}{4}\pi i, 1 + \frac{11}{4}\pi i, \dots, \text{ is } e\left(-\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right).$$

$$\begin{aligned} e^{\log(R) + i(\phi \pm 2n\pi)} &= e^{\log(R)} e^{i(\phi \pm 2n\pi)} \\ &= R[\cos(\phi \pm 2n\pi) + i \sin(\phi \pm 2n\pi)] \\ &= R[\cos \phi + i \sin \phi] = Re^{i\phi}; \end{aligned} \quad (5.29)$$

here, $\log(R)$ is the natural logarithm to the base e .

$w = \log(z)$ AND ITS KEY PROPERTIES

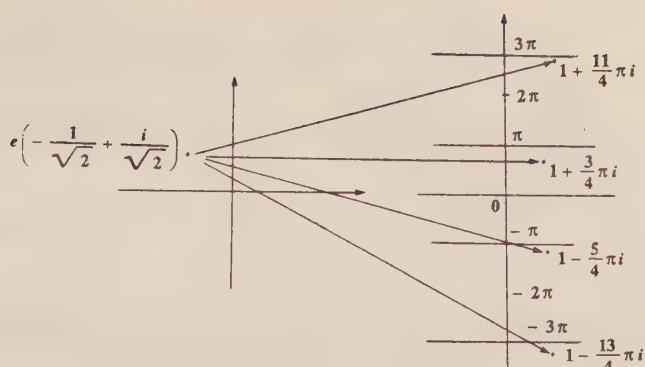
$w = \log(z)$ is defined by

$$e^{\log(z)} = z. \quad (5.30)$$

Interchanging the roles of z and w in (5.29) shows that $\log(z)$ is the multi-valued function taking on the infinitely many values

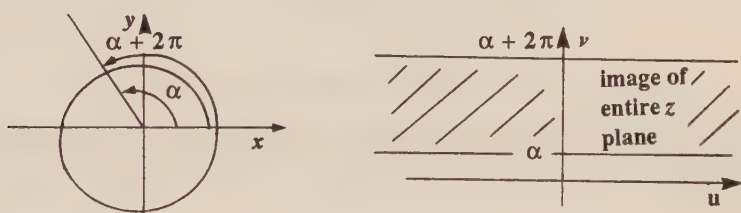
$$w = \log(r) + i(\theta \pm 2n\pi), \quad n = 0, 1, 2, \dots \quad (5.31)$$

that are mapped onto $z = re^{i\theta}$ by $z = e^w$.



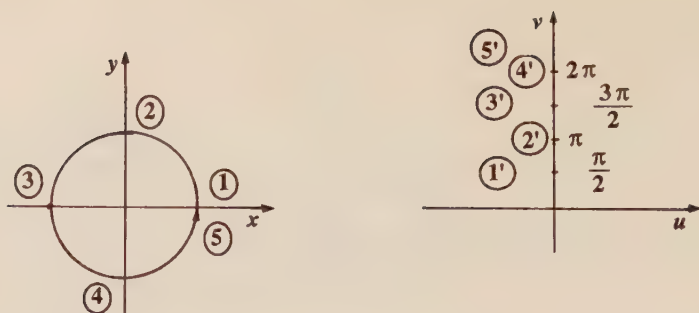
The multi-valuedness of $w = \log(z)$.

A candidate branch or single-valued replica of $\log(z)$ is a mapping of the z -plane onto any strip of height 2π in the w -plane.

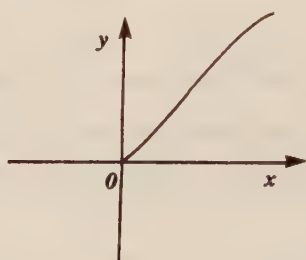


The entire z -plane is mapped onto a strip $-\infty < u < \infty$, $\alpha \leq v < \alpha + 2\pi$ of the $w = u + iv$ plane under $\log(z)$.

As in the case of $z^{\frac{1}{2}}$, any circuit around the origin $z = 0$ will introduce a discontinuity at 2π . For the candidate branch to be analytic, we must cut from the plane a line or curve from $z = 0$ outward. All such cuts pass through the branch point $z = 0$.

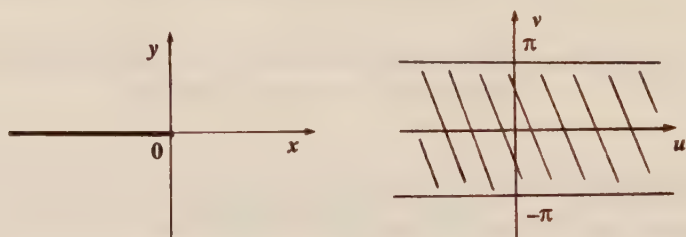


The image $\log e^{i\theta}$, $0 \leq \theta \leq 2\pi$: $1'$ is the image of 1, $2'$ the image of 2,
There is a discontinuity at $z = 1$.



A cut plane for $w = \log(z)$.

The principal branch $\log(z)$ of $\log(z)$ is that branch mapping the z -plane onto the strip $-\pi < v < \pi$ with the cut $\theta = \pi$ in the z -plane.



The cut plane for the principal branch $\log(z)$ of $\log(z)$ and its image.

For any branch of $\log(z)$, $z_1, z_2 \neq 0$,

$$\log(z_1 z_2) = \log(z_1) + \log(z_2) \quad (5.32)$$

$$\log\left(\frac{z_1}{z_2}\right) = \log(z_1) - \log(z_2) \quad (5.33)$$

$$\log(z^\alpha) = \alpha \log(z) \quad (5.34)$$

$$\log(z) \text{ is analytic and } \frac{d}{dz} \log(z) = \frac{1}{z} . \quad (5.35)$$

For any complex number α and $z \neq 0$, we define the α -th power z^α of z by

$$z^\alpha = e^{\alpha \log(z)} . \quad (5.36)$$

For real z , $\log(z)$ is the ordinary log function to the base e .

$\log(z)$ is defined for $-\pi < \arg(z) < \pi$; it is not defined for $z = 0$ and it cannot be continuously defined on the negative real axis, since its limiting values from above and below this axis differ by 2π .

No matter what branch of $\log(z)$ is chosen, it is always true that

$$e^{\log(z)} = z, \quad (5.37)$$

since the exponential function maps every image of z under the log back to itself. On the other hand, it is not necessarily true that

$$z = \log(e^z), \quad (5.38)$$

because of the possible ambiguity in the argument of z . If we limit the variable $z = x + iy$ to the strip $-\pi < y \leq \pi$ and consider only the principal value $\log(z)$, then (5.38) holds, i.e.,

$$z = e^{\log(z)} = \log(e^z) . \quad (5.39)$$

The principal value of the power z^∞ is

$$z^\infty = e^{\alpha \log(z)} . \quad (5.40)$$

TRIGONOMETRIC FUNCTIONS

For any z , the trigonometric functions $\sin(z)$, $\cos(z)$ are defined by

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i} \quad (5.41)$$

$$\cos(z) = \frac{e^{iz} + e^{-iz}}{2} . \quad (5.42)$$

$\sin(z)$ and $\cos(z)$ are entire functions with derivatives

$$\frac{d}{dz}\sin(z) = \cos(z), \quad \frac{d}{dz}\cos(z) = -\sin(z). \quad (5.43)$$

$\sin(z)$, $\cos(z)$ obey the familiar trigonometric relations:

Pythagorean relation:

$$\sin^2(z) + \cos^2(z) = 1 \quad (5.44)$$

periodic with period 2π :

$$\sin(z + 2\pi) = \sin(z), \quad \cos(z + 2\pi) = \cos(z) \quad (5.45)$$

evenness/oddness:

$$\sin(-z) = -\sin(z), \quad \cos(-z) = \cos(z) \quad (5.46)$$

sums of angles-formulas:

$$\sin(z_1 + z_2) = \sin(z_1)\cos(z_2) + \cos(z_1)\sin(z_2) \quad (5.47)$$

$$\cos(z_1 + z_2) = \cos(z_1)\cos(z_2) - \sin(z_1)\sin(z_2) \quad (5.48)$$

double-angle formulas:

$$\sin(2z) = 2\sin(z)\cos(z) \quad (5.49)$$

$$\cos(2z) = \cos^2(z) - \sin^2(z) \quad (5.50)$$

triangle relation:

$$\sin\left(z + \frac{\pi}{2}\right) = \cos(z) \quad (5.51)$$

The $\sin$ and $\cos$ may be written in terms of their real and imaginary parts. For $\sin(z)$,

$$\begin{aligned} \sin(z) &= \frac{e^{iz} - e^{-iz}}{2i} \\ &= \frac{e^{i(x+iy)} - e^{-i(x+iy)}}{2i} \end{aligned}$$

$$\begin{aligned}
&= \frac{e^{ix}e^{-y} - e^{ix}e^y}{2i} \\
&= \frac{e^{-y}(\cos(x) + i \sin(x)) - e^y(\cos(x) + i \sin(x))}{2i} \\
&= \sin(x)\cosh(y) + i \cos(x)\sinh(y) ,
\end{aligned} \tag{5.52}$$

where

$$\sinh(y) = \frac{e^y - e^{-y}}{2} \tag{5.53}$$

$$\cosh(y) = \frac{e^y + e^{-y}}{2} \tag{5.54}$$

are the hyperbolic and cosine respectively, and

$$\cos(z) = \cos(x)\cosh(y) - i \sin(x)\sinh(y) . \tag{5.55}$$

Unlike the real-valued trigonometric functions, the complex sin and cos are no longer bounded. For $z = x + iy$,

$$|\sin(z)|^2 = \sin^2(x) + \sinh^2(y) \tag{5.56}$$

$$|\cos(z)|^2 = \cos^2(x) + \sinh^2(y) . \tag{5.57}$$

To prove these relations, we make use of the relation

$$\cosh^2(y) = 1 + \sinh^2(y) . \tag{5.58}$$

For $y \rightarrow \infty$, $\sinh(y)$ grows exponentially as $\frac{e^y}{2}$, and so the moduli $|\sin(z)|$, $|\cos(z)|$ grow exponentially as well.

Cos and sin vanish at the zeros of $\cos(x)$, $\sin(x)$:

$$\cos(z) = 0 \text{ for } z = \frac{\pi}{2} \pm n\pi, \quad n = 0, 1, 2, \dots \tag{5.59}$$

$$\sin(z) = 0 \text{ for } z = \pm n\pi, \quad n = 0, 1, 2, \dots . \tag{5.60}$$

The complex tangent function is

$$\tan(z) = \frac{\sin(z)}{\cos(z)}. \quad (5.61)$$

$\tan(z)$ is defined and analytic in the z -plane except at the zeros of $\cos(z)$,

$$z = \frac{\pi}{2} \pm n\pi, \quad n = 0, 1, 2, \dots \quad (5.62)$$

Its derivative is given by

$$\frac{d}{dz} \tan(z) = \frac{1}{\cos^2(z)}. \quad (5.63)$$

In addition,

$$1 + \tan^2(z) = \frac{1}{\cos^2(z)}. \quad (5.64)$$

As in the real case the tangent is periodic with period π ,

$$\tan(z + \pi) = \tan(z). \quad (5.65)$$

For all points of the z -plane except for the zeros of cosine, the secant of z is defined by

$$w = \sec(z) = \frac{1}{\cos(z)}. \quad (5.66)$$

For all z except at the zeros of $\sin(z)$, the cosecant is

$$w = \csc(z) = \frac{1}{\sin(z)}, \quad (5.67)$$

while the cotangent is

$$w = \cot(z) = \frac{1}{\tan(z)}. \quad (5.68)$$

The secant, cosecant, and cotangent are analytic for all z where they are defined, with the derivatives

$$\frac{d}{dz} \sec(z) = \sec(z) \tan(z), \quad (5.69)$$

$$\frac{d}{dz} \csc(z) = -\csc(z) \cot(z), \text{ and} \quad (5.70)$$

$$\frac{d}{dz} \cot(z) = -\csc^2(z). \quad (5.71)$$

HYPERBOLIC FUNCTIONS AND THEIR PROPERTIES

The hyperbolic sine and cosine, $\sinh(z)$, $\cosh(z)$ are

$$\sinh(z) = \frac{1}{2} [e^z - e^{-z}] \quad (5.72)$$

$$\cosh(z) = \frac{1}{2} [e^z + e^{-z}] \quad (5.73)$$

The hyperbolic cosine and sine are entire functions of z with derivatives

$$\frac{d}{dz} \sinh(z) = \cosh(z), \quad (5.74)$$

$$\frac{d}{dz} \cosh(z) = \sinh(z). \quad (5.75)$$

The hyperbolic and trigonometric functions are related by

$$\sin(z) = -\sinh(iz), \quad \cos(z) = \cosh(iz), \quad (5.76)$$

$$\sinh(z) = -i \sin(iz), \quad \cosh(z) = \cos(iz). \quad (5.77)$$

The hyperbolic functions have the following properties:

differences of squares:

$$\cosh^2(z) - \sinh^2(z) = 1 \quad (5.78)$$

evenness/oddness:

$$\sinh(-z) = -\sinh(z), \quad \cosh(-z) = \cosh(z) \quad (5.79)$$

sum of arguments formula:

$$\sinh(z_1 + z_2) = \sinh(z_1)\cosh(z_2) + \cosh(z_1)\sinh(z_2) \quad (5.80)$$

$$\cosh(z_1 + z_2) = \cosh(z_1)\cosh(z_2) + \sinh(z_1)\sinh(z_2) \quad (5.81)$$

periodicity: Like the exponential function, the hyperbolic sine and cosine are periodic with period $2\pi i$.

zeros:

$$\text{The zeros of } \cosh(z) \text{ are } z = \left(\frac{1}{2} \pm n\right)\pi i, \quad (5.82)$$

$$n = 0, 1, 2, \dots$$

$$\text{The zeros of } \sinh(z) \text{ are } z = n\pi i, \quad n = 0, 1, 2, \dots \quad (5.83)$$

real and imaginary parts:

$$\text{For } z = x + iy,$$

$$\sinh(z) = \sinh(x)\cos(y) + i \cosh(x)\sin(y), \quad (5.84)$$

$$\cosh(z) = \cosh(x)\cos(y) + i \sinh(x)\sin(y), \quad (5.85)$$

moduli:

$$|\sinh(z)|^2 = \sinh^2(x) + \sin^2(y) \quad (5.86)$$

$$|\cosh(z)|^2 = \cosh^2(x) + \cos^2(y) \quad (5.87)$$

The hyperbolic tangent is defined as

$$\tanh(z) = \frac{\sinh(z)}{\cosh(z)}. \quad (5.88)$$

$\tanh(z)$ is analytic everywhere but at the zeros of $\cosh(z)$, with the derivative

$$\frac{d}{dz} \tanh(z) = \frac{1}{\cosh^2(z)}. \quad (5.89)$$

The hyperbolic, secant, cosecant, and cotangent are, respectively

$$\operatorname{sech}(z) = \frac{1}{\cosh(z)}, \operatorname{csch}(z) = \frac{1}{\sinh(z)}, \quad (5.90)$$

$$\coth(z) = \frac{1}{\tanh(z)}. \quad (5.91)$$

The hyperbolic, secant, cosecant, and cotangent are analytic everywhere but at the zeros of $\cosh(z)$, $\sinh(z)$, and $\tanh(z)$, respectively, with the derivatives

$$\frac{d}{dz} \operatorname{sech}(z) = -\operatorname{sech}(z)\tanh(z), \quad (5.92)$$

$$\frac{d}{dz} \operatorname{csch}(z) = -\operatorname{csch}(z)\coth(z), \text{ and} \quad (5.93)$$

$$\frac{d}{dz} \coth(z) = -\operatorname{csch}^2(z). \quad (5.94)$$

Chapter 6

SET THEORY

WHAT ARE SETS?

A set is a collection of objects. These may be numbers, people, or objects of any kind that are related by the fact that they belong to the set. The objects of a set are said to belong to the set, or to be members or elements of the set.

Notation of Sets and Their Elements

We denote sets and the objects in them by letters. If an object a belongs to a set S , then we write

$$a \in S.$$

On the other hand, if a is not an element of S , then we write

$$a \notin S.$$

The Empty Set ϕ

The empty set ϕ is a set containing no elements.

Finite and Infinite Sets

A set is a finite set if the process of counting the elements of the set eventually ends. In other words, the set would be able to be put into one-to-one correspondence with a part of the sequence of natural numbers.

A set is an infinite set if the process of counting the elements of the set will never end—in other words, if it is not finite.

Example: The set of all natural numbers, denoted by N , is an infinite set

since the process of counting its elements will never end. We denote this by means of three dots: 1, 2, 3, ...

Example: The set of all real numbers, denoted by R , is an infinite set containing the natural numbers

$$1, 2, 3, \dots,$$

zero and the negative integers,

$$0, -1, -2, -3, \dots,$$

the rational numbers,

$$\frac{1}{2}, \frac{-5}{8}, \frac{56}{32}, \dots,$$

and the irrational numbers,

$$\pi, \sqrt{2}, e, \dots$$

An irrational number can be expressed as a *nonterminating decimal*. It is for this reason that the irrational numbers cannot be counted by means of the natural numbers, and, therefore, that the set of real numbers is an infinite set.

Notation for a Set in Terms of Its Elements

If a set S has the elements $a, b, c, \dots$, then we can write

$$S = \{a, b, c, \dots\}.$$

Example: The set S of roots of the equation

$$x^2 - 1 = 0$$

is a set of real numbers and is written:

$$S = \{-1, +1\}.$$

Example: The set S of roots of the equation

$$x^2 + 1 = 0$$

that are real numbers is empty, since there are no real roots to this equation (its roots are pure imaginary).

Point Sets in 1, 2, and 3 Dimensions

A point set S on the real line

$$-\infty < x < \infty ,$$

the x, y plane

$$-\infty < x, y < \infty ,$$

or three-dimensional space

$$-\infty < x, y, z < \infty .$$

is a set of points in the respective spaces.

Examples: The point sets S, T, U, V of the figure below are defined respectively as:

S = set of points x with $-4 < x < 2, x = 5, 6.5$.

T = set of points (x, y) with

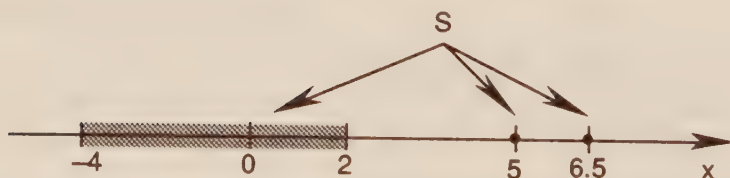
$$x^2 + y^2 = 1 .$$

U = set of points (x, y) with

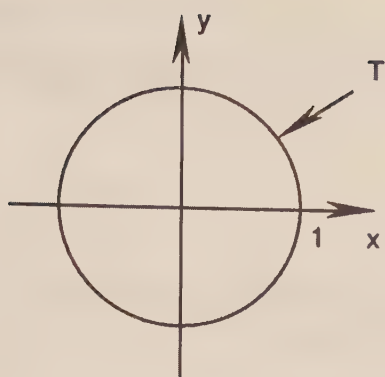
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 .$$

V = set of points (x, y) with

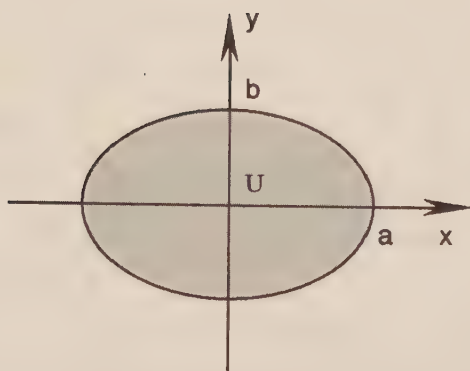
$$|x + y| \geq 1 \text{ or } x = y = 0.$$



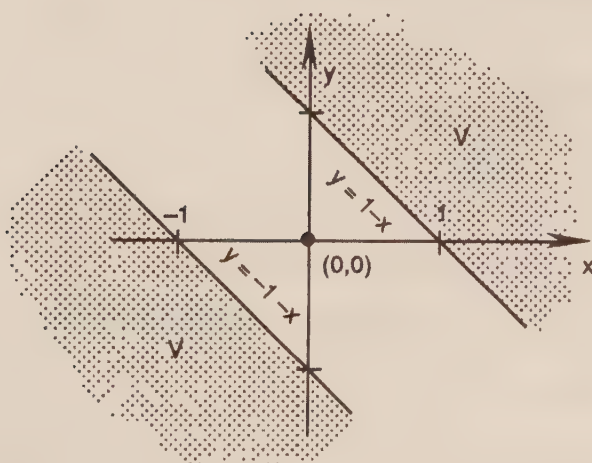
The set S .



The set T .



The set U .

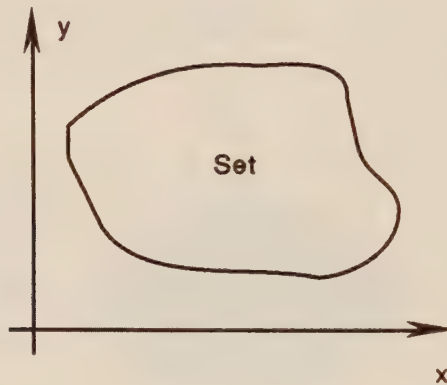


The set V .

Note that S consists of an interval of the real line together with the points 5, 6.5; T is the arc of the unit circle; U is the arc of an ellipse together with its interior, while V consists of the two infinite regions of Figure 6.1d together with the origin.

Venn Diagrams

The representation of a set by point sets in the x, y plane, as in the figure below, is called a Venn diagram.



Venn diagram.

RELATIONSHIPS BETWEEN SETS

Equality of Sets

Two sets A, B are equal,

$$A = B,$$

if they consist of the same collection of objects.

Examples: If $A = \{1, 2, 3\}$, $B = \{3, 1, 2\}$, and $C = \{1, 2, 5\}$, then

$$A = B, A \neq C.$$

Subset of a Set

A set A is a subset of a set B , denoted by

$$A \subseteq B,$$

if every element of A is an element of B .

Examples: If $A = \{1, 2, 3\}$, $B = \{3, 1, 2\}$, $C = \{1, 2, 3, 4\}$, and $D = \{1, 2, 4, 5\}$, then $A = B$ and A is a subset of C . Furthermore, A and B are subsets of each other since they are equal. On the other hand, A is not equal to C , since the number 4 belongs to C but does not belong to A . Note that A is not a subset of D , since the element 3 of A does not belong to D .

If A is not a subset of B , then write

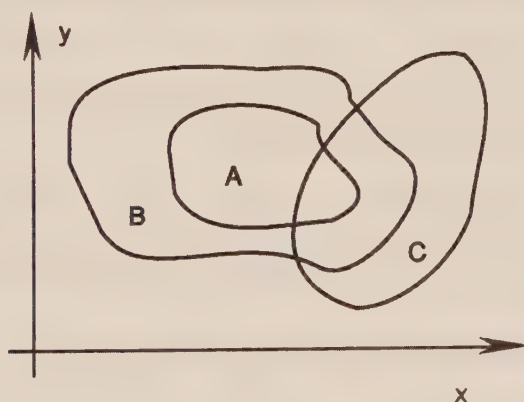
$$A \not\subseteq B.$$

Proper Subsets

The set A is a proper subset of set B if A is a subset of B but is not equal to B , i.e.,

$$A \subseteq B, A \neq B.$$

Clearly, A is a proper subset of B if every element of A is an element of B , while B contains at least one element that does not belong to A .



A is a subset of B but not of C .

Example: Referring to the figure above, A is a proper subset of B , while A is a non-subset of C .

Disjoint Sets

Sets A and B are disjoint if they have no elements in common.

Example: Define the sets of people:

$$A = \{\text{Judy, Henry, Arthur, George}\}$$

$$B = \{\text{Judy, Henry, George}\}$$

$$C = \{\text{Judy, George, Henry}\}$$

$$D = \{\text{Mary, Peter, James}\}$$

Then $B = C$, B and C are proper subsets of A , and A and D are disjoint sets.

Inclusion of Sets

The relation $\subseteq$, defined by the statement

$$A \subseteq B,$$

is called the inclusion relation.

THEOREM 1 — Inclusion Relation Properties

The inclusion relation has the following properties:

- 1) Reflexivity: A set S is a subset of itself:

$$S \subseteq S. \quad (6.1)$$

- 2) Transitivity: If A is a subset of B and B a subset of C , then A is a subset of C (Figure 6.4):

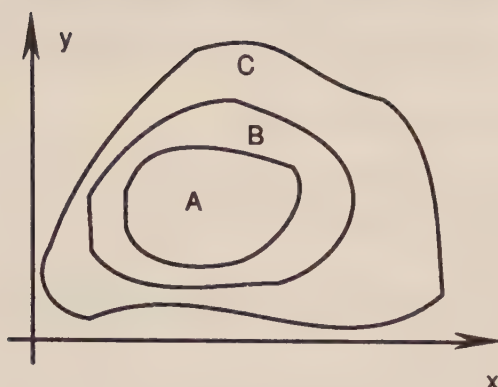
$$A \subseteq B \text{ and } B \subseteq C \text{ implies } A \subseteq C. \quad (6.2)$$

- 3) Equality: If A is a subset of B and B a subset of A , then A is equal to B :

$$A = B \text{ if and only if } A \subseteq B \text{ and } B \subseteq A. \quad (6.3)$$

- 4) Empty set: The empty set is a subset of every other set:

$$\phi \subseteq A, \text{ for all sets } A.$$



Transitivity of inclusion.

Included in the relation $A \subseteq B$ is the possibility that $A = B$. If this possibility is excluded, then we speak of the proper or strong inclusion relation.

Proper Inclusion

The proper or strong inclusion relation

$$A \subset B$$

holds if A is a proper subset of B , that is, A is a subset of B but $A \neq B$.

Example: If $A = \{\text{Dick, Jane, Harvey}\}$, $B = \{\text{Dick, Harvey, Jane}\}$, and $C = \{\text{Jane, Harvey, Dick, Edward}\}$, then

$$A = B, \quad A \subset C, \quad B \subset C.$$

Universal Set

The universal set or universe Ω is the set of all possible objects of interest.

Examples: Some possible universal sets are the following:

- 1) The set H of all human beings; and
- 2) The set R of all real numbers;

3) The set A of all integers between -10 and 7 .

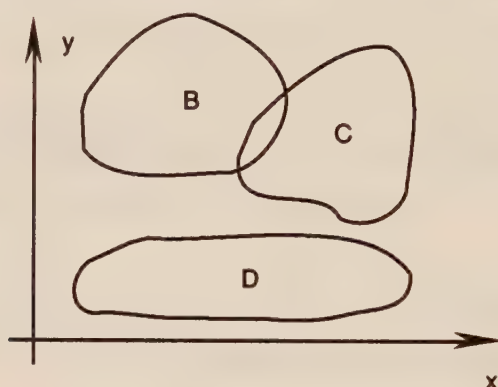
Subsets of these universal sets include the following:

$$X = \{\text{Dick, Jane, Tom}\} \subset H.$$

$$Y = \{1, 5, e, -\pi\} \subset R.$$

$$Z = \{-5, 0, 2\} \subset A.$$

Example: Let the universal set A be the set of all points in the x, y plane. Subsets of A include the point sets B, C , and D , shown in the figure below.



Subsets of the universal set A ; $-\infty < x, y < \infty$.

Complements of a Set

The complement $\bar{A}$ of a set A is the set of all points in the universal set that are not points of A .

Example: For the universal set

$$\Omega = \{\text{Dick, Jane, Henry, Tom}\},$$

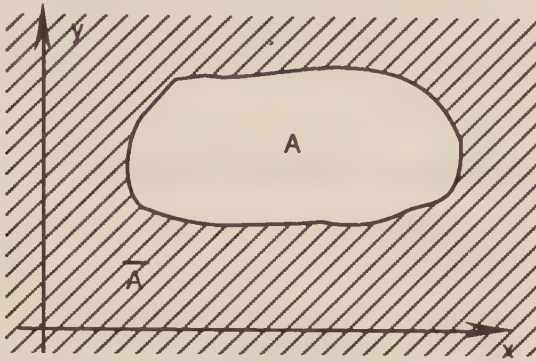
if A is defined as

$$A = \{\text{Jane, Tom}\},$$

then its complement is

$$\bar{A} = \{\text{Dick, Henry}\}.$$

Example: For the subset A of points in the universal set Ω of all points in the x, y plane, the complement of A is the set of all points not in A , as seen in the figure below.



The complement $\bar{A}$ = all points not in A .

Power Sets

The power set $P(\Omega)$ of a universal Ω is the collection of all subsets of Ω . Elements $P(\Omega)$ include the universal set Ω and the empty set ϕ .

Example: If Ω is the set of people

$$\Omega = \{\text{Dick, Jane, Henry}\},$$

then its power set $P(\Omega)$ is the set consisting of the following sets:

$$A_1 = \phi \text{ (the empty set);}$$

$$A_2 = \{\text{Dick}\}, A_3 = \{\text{Jane}\}, A_4 = \{\text{Henry}\};$$

$$A_5 = \{\text{Dick, Jane}\}, A_6 = \{\text{Dick, Henry}\}; \text{ and}$$

$$A_7 = \{\text{Henry, Jane}\}, A_8 = \Omega = \{\text{Dick, Jane, Henry}\}.$$

Partial Ordering

A partial ordering of the subsets of a universal set Ω is a relation, denoted by $\leq$, obeying the transitivity and identity conditions.

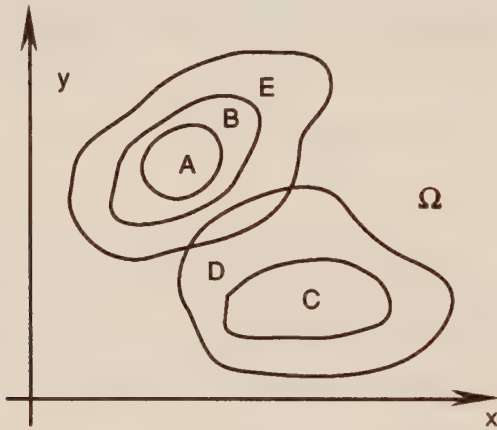
A) Transitivity: If $A \leq B$ and $B \leq C$, then $A \leq C$.

B) Identity: $A = B$ if and only if $A \leq B$ and $B \leq A$. (See Theorem 1, 3.).

Partial ordering defines an arrangement of the sets according to which some sets are “smaller” or “larger” than others, in the same sense that it applies to real numbers.

THEOREM 2 — Inclusion Is an Order Relation

The inclusion and proper inclusion relations define a partial ordering of the subsets of a universal set Ω .



Ordered sets of the following example.

Example: Consider the sets A , B , C , D , and E shown in the Venn diagram of the figure above. With inclusion as an order relation, we see that

$$\phi \leq A \leq B \leq E \leq \Omega, \text{ and}$$

$$\phi \leq C \leq D \leq \Omega.$$

THE UNION AND INTERSECTION OF SETS

Union of Sets

The union of two sets A and B , denoted by $A \cup B$, is the set C consisting

of all the objects belonging to A , to B , or to both sets.

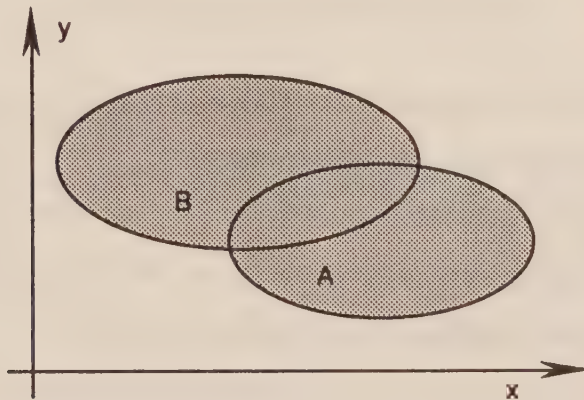
Example: If A and B are sets of people defined as

$$A = \{\text{Barry, Debby, David}\}, \text{ and } B = \{\text{Judy, David}\},$$

then $A \cup B$ is the set of people defined as

$$A \cup B = \{\text{Barry, Debby, David, Judy}\}.$$

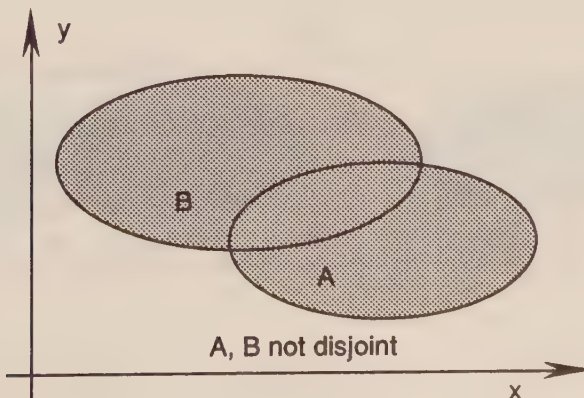
Example: If A and B are the point sets of the x, y plane of the figure below, then $A \cup B$ is the set of all points belonging to either A , B , or both.

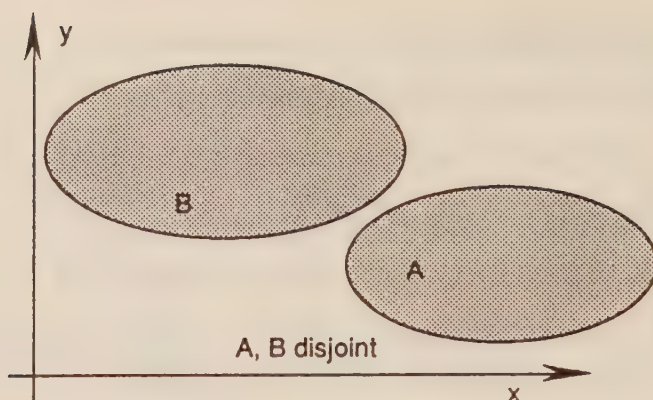


Venn diagram of $A \cup B$ (shaded region).

Venn Diagram for the Union of Sets

The Venn diagram representations of the union of two sets, when the sets are disjoint and when they are not, are shown in the figure below.





$A \cup B$ for A, B not disjoint and for A, B disjoint.

THEOREM 3 — Properties of the Union of Sets

The union obeys the following conditions:

1) Idempotence: $A \cup A = A$. (6.4a)

2) Commutativity: $A \cup B = B \cup A$. (6.4b)

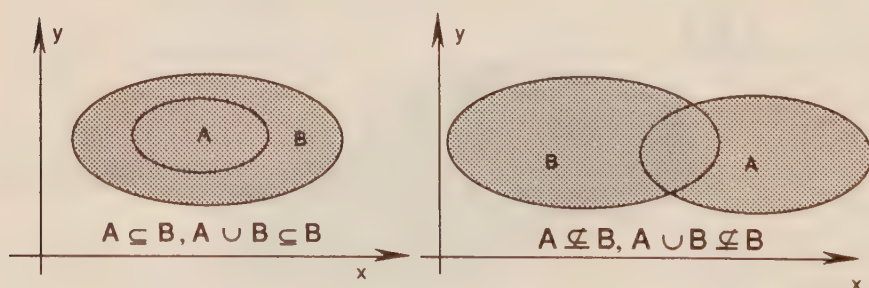
3) Zero property of ϕ : $A \cup \phi = A$. (6.4c)

4) Associativity: $(A \cup B) \cup C = A \cup (B \cup C)$. (6.4d)

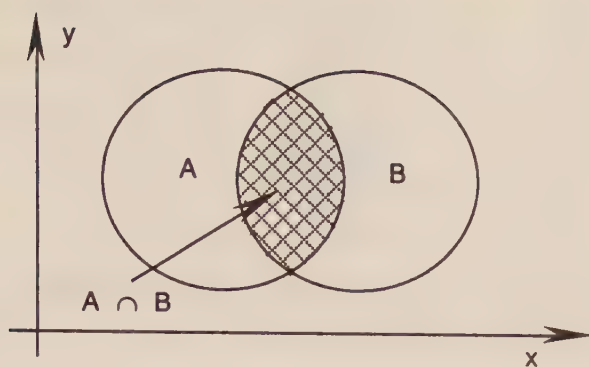
5) Inclusion: $A \cup B \subseteq B$ if and only if $A \subseteq B$. (6.4e)

6) Ordering: $A \subseteq A \cup B$, $B \subseteq A \cup B$. (6.4f)

The validation of 6.4e and 6.4f are seen in the Venn diagrams of the figure below. The remaining properties are obvious.



Condition 6.4e.



Condition 6.4f.

Intersection of Sets

The intersection of two sets A and B is the set, denoted by $A \cap B$, consisting of all those objects belonging to A and to B .

Example: If A and B are the sets of people defined as

$$A = \{\text{Ken, Barbie, Max}\} \text{ and } B = \{\text{Max, Jill, Judy}\},$$

then their intersection is the singleton

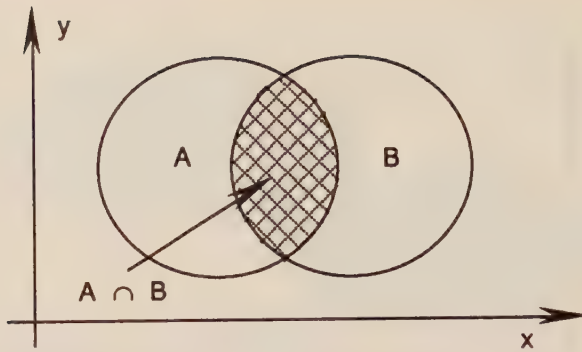
$$A \cap B = \{\text{Max}\}.$$

Venn Diagram of Intersection

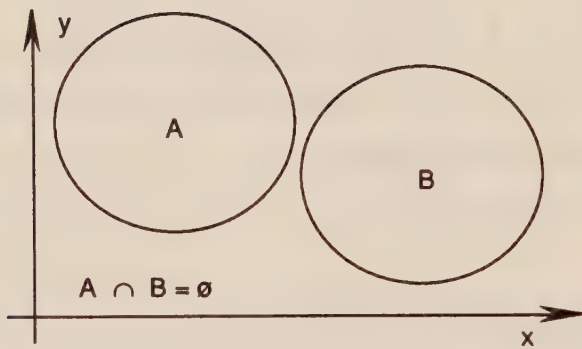
The intersection $A \cap B$ is pictured in the Venn diagram of the figure below.

Empty Set Intersection

The intersection of A and B is the empty set if and only if the sets are disjoint.



Intersection of A, B (shaded region).



$A \cap B$ is empty if and only if A, B are disjoint:
 a) A, B not disjoint; b) A, B disjoint.

THEOREM 4 — Properties of the Intersection of Sets

The intersection obeys the following conditions:

- 1) Idempotence: $A \cap A = A.$ (6.5a)

$$\text{B) Commutativity: } A \cap B = B \cap A. \quad (6.5b)$$

$$\text{C) Zero set: } A \cap \phi = \phi. \quad (6.5c)$$

$$\text{D) Associativity: } (A \cap B) \cap C = A \cap (B \cap C). \quad (6.4d)$$

$$A \cap B = A \text{ if and only if } A \subseteq B. \quad (6.5e)$$

$$A \cap B \subseteq A \text{ and } A \cap B \subseteq B. \quad (6.5f)$$

The definitions of union and intersection can be applied directly to any finite number of sets. Consider the collection $\{B_i\}$ of N sets $B_1, B_2, \dots, B_N$. Then their union,

$$\bigcup_{i=1}^N B_i,$$

is the collection of objects belonging to at least one of the sets, while their intersection,

$$\bigcap_{i=1}^N B_i,$$

is the set of their objects belonging to every one of the sets.

THE ALGEBRA OF SETS

The union and intersection provide us with an algebra for sets that mimics the algebra of numbers.

Recall that the complement of set A is the collection of objects in the universal set that do not belong to A .

THEOREM 5 — Properties of the Complement

The complement obeys the following conditions:

- 1) The complement of the empty set is the universal set:

$$\bar{\phi} = \Omega. \quad (6.6a)$$

2) The complement of the universal set is the empty set:

$$\overline{\Omega} = \phi. \quad (6.6b)$$

3) The complement of the complement is the set itself:

$$\overline{\overline{A}} = A. \quad (6.6c)$$

4) The union of a set A and its complement $\overline{A}$ is the universal set:

$$A \cup \overline{A} = \Omega. \quad (6.6d)$$

5) A set and its complement are disjoint:

$$A \cap \overline{A} = \phi. \quad (6.6e)$$

A central theorem in the algebra of sets A and B is DeMorgan's theorem, relating the union, intersection, and complement:

THEOREM 6 — DeMorgan's Theorem

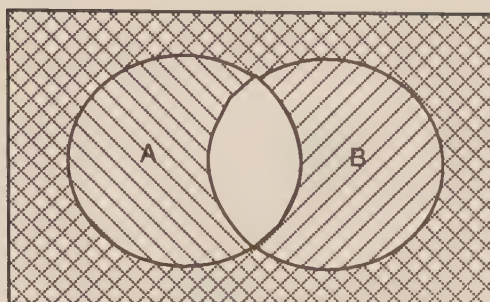
The complement of the union of two sets is the intersection of their complements:

$$\overline{A \cup B} = \overline{A} \cap \overline{B}. \quad (6.7)$$

Similarly, the complement of the intersection of two sets A and B is the union of their complements:

$$\overline{A \cap B} = \overline{A} \cup \overline{B}. \quad (6.8)$$

The Venn diagrams representing DeMorgan's theorem are seen in the figure on the following page.

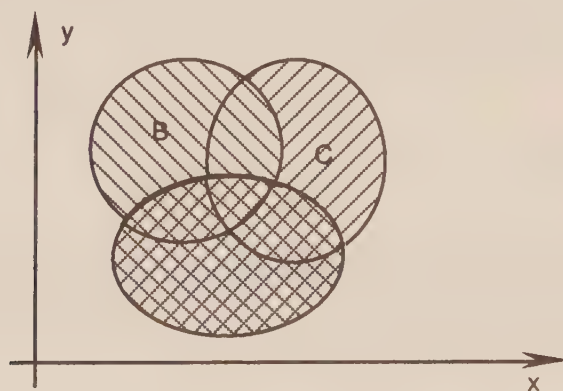


THEOREM 7 — Distributive Laws

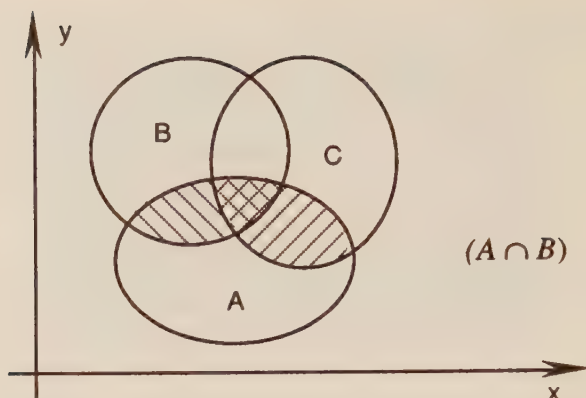
The union and intersection obey the distributive laws:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C); \quad (6.9a)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C), \quad (6.9b)$$



Distributive law 6.9a.



Distributive law 6.9b.

The Venn diagrams representing these relations are seen in the figures above.

Let $\{B_i\}$ be a sequence of N sets. We define their union, denoted by

$$\bigcup_{i=1}^N B_i ,$$

as the set containing all objects belonging to at least one of the sets.

Let $\{B_i\}$ be a sequence of N sets. We define their intersection, denoted by

$$\bigcap_{i=1}^N B_i ,$$

as the set containing all objects belonging to each of the sets.

Relations (6.9a, 6.9b) are valid for unions and intersections of arbitrarily many sets. Thus, for a sequence of sets $\{B_i\}$,

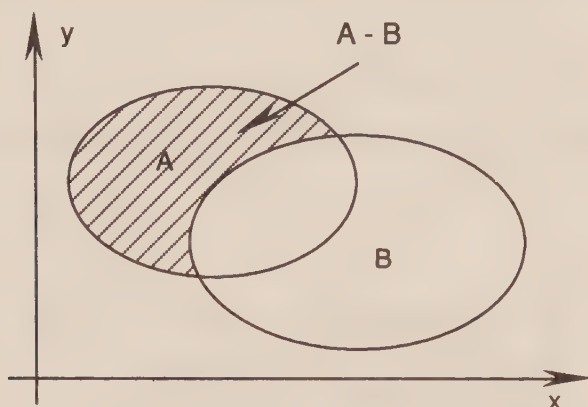
$$A \cup \left(\bigcap_{i=1}^N B_i \right) = \bigcap_{i=1}^N A \cup B_i , \text{ and} \quad (6.10a)$$

$$A \cap \left(\bigcup_{i=1}^N B_i \right) = \bigcup_{i=1}^N A \cap B_i . \quad (6.10b)$$

Difference of Sets

The difference of two sets, A and B , denoted by $A - B$, is the set of all elements of A that are not in B .

The meaning of $A - B$ is seen in the Venn diagram of the figure below.



The Venn diagram for $A - B$.

Example: For A , B , the sets of objects

$$A = \{\text{chair, table, pen}\}, B = \{\text{pen, computer, wheel}\},$$

we find that

$$A - B = \{\text{chair, table}\} \text{ and}$$

$$B - A = \{\text{computer, wheel}\}.$$

THEOREM 8 — Relation for the Difference of Sets

The difference $A - B$ can be given in terms of intersection and complement as

$$A - B = A \cap \bar{B}.$$

Symmetric Difference of Sets

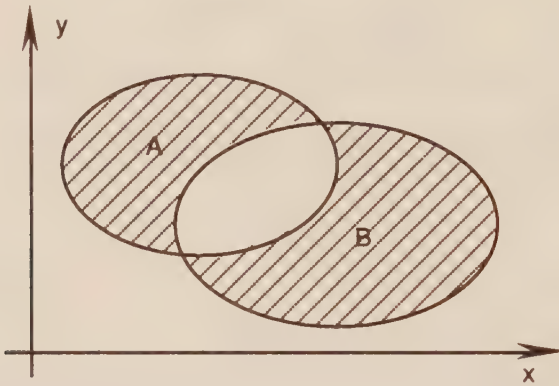
We define the symmetric difference

$$A \Delta B$$

of sets A and B as the set of points belonging either to A or to B , but not to both, i.e.,

$$A \Delta B = (A \cup B) - (A \cap B) .$$

The meaning of the symmetric difference is made clear in the Venn diagram of the figure below.



The Venn diagram for $A \sim B$.

MEASURING THE SIZE OF A SET

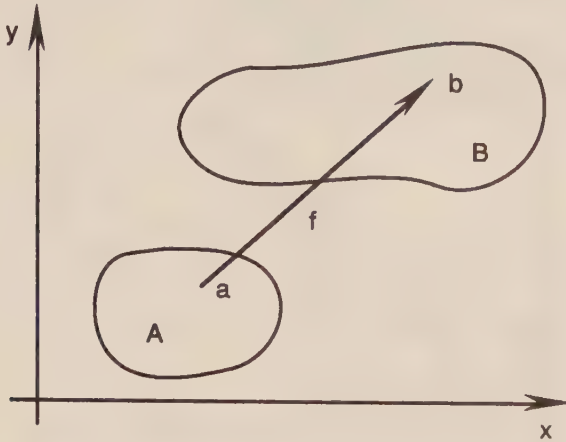
The size of a set N elements is obviously N . However, if the set has infinitely many elements, describing its size is more difficult, requiring a yardstick for comparison with the set of numbers. This yardstick may be based on the idea of mapping or transformation from one set to a second set.

Transformation from Set to Set

Let A and B be two sets. A mapping or transformation from A to B is a rule, which we will usually represent by a letter such as f or g , assigning a unique element b of B to every element a of A . The rule may be represented schematically as seen in the figure below.

Functions

A mapping or transformation f is commonly referred to as a function.



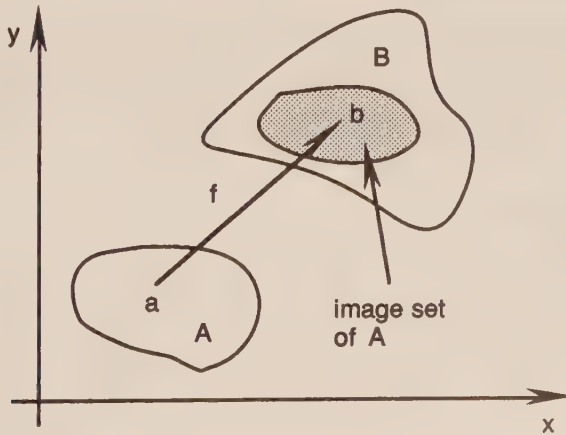
A transformation f from A to B .

Image under a Mapping

If a mapping f assigns an object $b \in B$ to an object $a \in A$, we usually write

$$b = f(a)$$

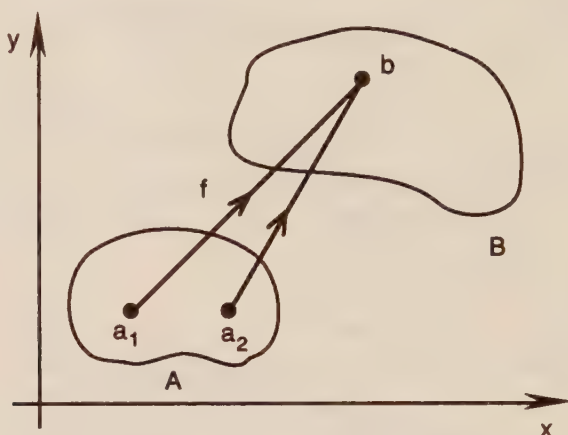
and refer to b as the image of a under the mapping f . We also say that f defines a mapping of A into B , since not every element of B is necessarily the image of some element of A .



f maps A into a subset of B .

Many Points Mapped Onto One Point

The definition of a mapping allows for the possibility that a point in the image set B of a function f is the image of more than one object of A .



Under the mapping f , two points, a_1 , and a_2 , may be mapped onto the same point b .

Mapping Onto

If every point b of B is the image of some point a of A under the mapping f , then we say that f maps the set A onto the set B .

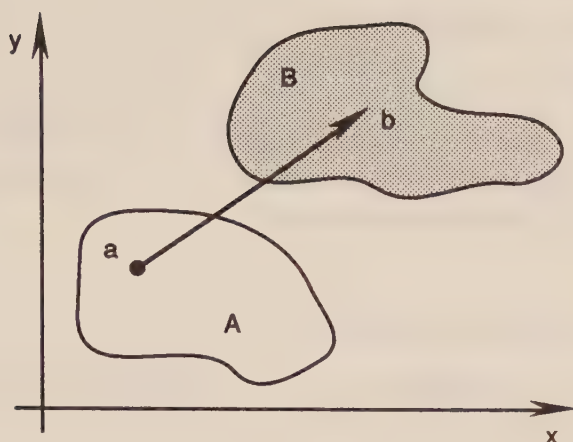
One-to-One Mapping

A mapping f maps the set A onto B in a one-to-one fashion (or f is a one-to-one mapping) if every point b of B is the image of a unique element a of A . In this case, we may define the inverse mapping f^{-1} of f , by the statement that if

$$b = f(a),$$

then

$$a = f^{-1}(b).$$

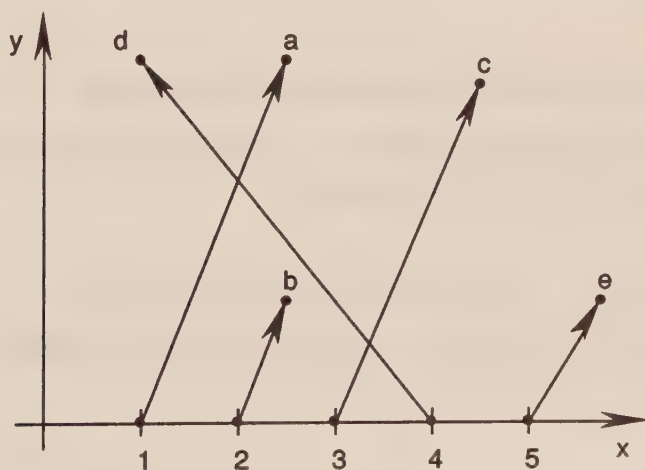


f maps A into a subset of B .

One-to-One Correspondence

A one-to-one correspondence between sets A and B is a one-to-one mapping from either of the sets onto the other.

Example: Let A be a set consisting of the five elements a, b, c, d, e . Then the act of counting the elements constitutes the assignment of a one-to-one correspondence between A and the set of five natural numbers, 1, 2, 3, 4, 5.



A one-to-one correspondence between $\{1, 2, 3, 4, 5\}$ and $\{a, b, c, d, e\}$.

Natural Numbers Are an Infinite Set

The set of all natural numbers,

$$N = \{1, 2, 3, 4, \dots\},$$

is an infinite set since it contains infinitely many elements. Its size, or cardinality, is a measure of “how infinite” it is. To define this size, we employ the idea of a one-to-one correspondence.

Countable (Denumerable) Set

A countable or denumerable set A is one for which a one-to-one correspondence may be defined between it and the set N of all natural numbers.

Example: The set of even natural numbers,

$$E = \{2, 4, 6, 8, \dots\},$$

is countable. To see this, we define the function

$$f(n) = \frac{n}{2},$$

mapping E in a one-to-one manner onto N .

It is remarkable that the set of all rational numbers is countable. Specifically:

THEOREM 9 — Rational Numbers Are Countable

The set R of all rational numbers is countable; thus, the rational numbers may be arranged in an infinite sequence.

THEOREM 10 — Real Numbers Are Not Countable

The set of all real numbers is not countable, but is infinite to a “higher order” than the set of rational numbers.

Cardinality of Sets

We say that A and B have the same cardinality if there is a one-to-one correspondence between them. Conversely, if their cardinalities differ while A has the same cardinality as a subset of B , then we say that A has a smaller cardinality than B .

Chapter 7

TOPOLOGY

TOPOLOGICAL SPACES

Topological Space

Let X be a set and T be a family of subsets of X (called a topology) that satisfies the following:

- 1) ϕ and X belong to T .
- 2) Each union of members of T is also a member of T .
- 3) The intersection of any two sets in T belongs to T .

A couple (X, T) consisting of a set X and a topology T in X is called a topological space.

Elements of topological spaces are called points. The subsets of T are called the open subsets of the topological space (X, T) .

Neighborhood

Let (X, T) be a space and $x \in X$ be a point of X . Any open set U (i.e., $U \in T$) containing x ,

$$x \in U ,$$

is called a neighborhood (an open neighborhood) of x .

Sometimes to denote that “ U is a neighborhood of x ” we write

$$nbd\ U(x) .$$

Example: Let X be a set and

$$T = \{ \phi, X \} .$$

This topology consisting of only two sets is called the indiscrete topology.

Example: Let X be a set and $D = P(X)$ be the family of all subsets of X . D is called the discrete topology; here, every set is an open set.

Example: The set $X = \{0, 1\}$ with the discrete topology (i.e., $D = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$) is denoted by Z .

Example: The set $X = \{0, 1\}$ with the topology

$$T = \{\emptyset, \{0\}, \{0, 1\}\}$$

is called the Sierpinski space S .

A set X may have many topologies; each topology yields a distinct topological space.

Each topology is a subset of $P(X)$. Thus, topologies in X are partially ordered by inclusion. Topology T_1 is finer than T_0 whenever

$$T_0 \subset T_1.$$

THEOREM 1

Let $\{T_\omega \mid \omega \in \Omega\}$ be any family of topologies in X . Then

$$\bigcap_{\omega} T_\omega = \{\cup \mid \forall \omega \in \Omega, \cup \in T_\omega\}$$

is also a topology in X , while $\bigcup_{\omega} T_\omega$ need not be a topology.

Real Line

By R we denote the set of all real numbers. Let A be a set of real numbers. A point $a \in A$ is an interior point of A if and only if a belongs to some open interval I_a which is contained in A .

$$a \in I_a \subset A.$$

The set A is open if each of its points is an interior point.

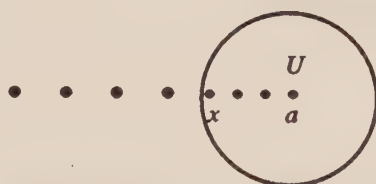
Example: An open interval (a, b) is an open set and the real line R is an open set.

Accumulation Point

Let A be a subset of R . A point

$$a \in R$$

is an accumulation point (or limit point) of A if every open set U that contains a point of A different from a ; i.e.,



$$\forall U (U \text{ open}) : U \ni a$$

$$A \cap (U - \{a\}) \neq \emptyset .$$

The set of accumulation points of A , denoted by A^1 , is called the derived set of A .

Example: $A = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\}$. The point 0 is an accumulation point of A ,

$$A^1 = \{0\} .$$

Example: Let Q be the set of rational numbers. Every real number $x \in R$ is an accumulation point of Q ,

$$Q^1 = R .$$

THEOREM 2 — Bolzano-Weierstrass

Let X be a bounded, infinite set of real numbers; then X has at least one accumulation point.

Euclidean Topology in R

Let R be the set of real numbers. A subset $G \subset R$ is called open if for each $x \in G$ there is an $r > 0$ such that the open interval $B(x, r)$,

$$B(x, r) = (x - r, x + r),$$

is a subset of G . We define the Euclidean topology of R^1 , denoted T , as the family of all open sets. The topological space (R^1, T) is called the Euclidean 1-space.

ELEMENTARY CONCEPTS

Closed Set

Let (X, T) be a topological space. A subset A of X is a closed set if its complement A^C is an open set.

Remember that $A^C = X - A$.

Example: In R^1 , a closed interval $[a, b]$ is a closed set.

Example: In any space, X and $\emptyset$ are both open and closed sets.

THEOREM 3

The intersection of any family of closed sets is a closed set. The union of finitely many closed sets is a closed set.

Observe that the union of infinitely many closed sets does not have to be a closed set. For example, in R^1

$$\bigcup_1^\infty \left[0, 1 - \frac{1}{n}\right] =]0, 1[.$$

We denote closed interval by $[a, b]$ and open interval by $]a, b[$ (or (a, b)).

Closure

Let $A \subset X$. A point $x \in X$ is adherent to A if each neighborhood of x contains at least one point of A .

The closure of A , denoted by $\bar{A}$, is defined by

$$\bar{A} = \{x \in X \mid \forall U(x) : U(x) \cap A \neq \emptyset\}.$$

Example: In R^1 the closure of $A =]0, 1]$ is $\bar{A} = [0, 1]$. The closure of $B = \left\{\frac{1}{n}, n = 1, 2, 3, \dots\right\}$ is $\bar{B} = B \cup \{0\}$.

THEOREM 4

For every set A ,

$$A \subset \bar{A}.$$

A is closed if and only if

$$A = \bar{A}.$$

The next theorem lists some other properties of closed sets.

THEOREM 5

$$A) \quad A \subset B \Rightarrow \bar{A} \subset \bar{B}.$$

$$B) \quad \overline{\bar{A}} = \bar{A}.$$

$$C) \quad \overline{A \cup B} = \bar{A} \cup \bar{B}.$$

$$D) \quad \bar{\emptyset} = \emptyset.$$

THEOREM 6

$$\bar{A} = \bigcap \{F \mid (F \text{ is closed}) \wedge (A \subset F)\}$$

i.e., $\bar{A}$ is the smallest closed set A .

Observe that

$$\overline{A \cap B} \stackrel{?}{=} \bar{A} \cap \bar{B}$$

is not true in general. For example, if $A = [0, 1[$ and $B =]1, 2]$, then $\overline{A \cap B} = \bar{\emptyset} = \emptyset \neq \{1\}$.

The general rule for the closure of intersection is

$$\overline{\bigcap_{\omega} A_{\omega}} \subset \bigcap_{\omega} \bar{A}_{\omega}.$$

Similarly to the definition of accumulation points in R^1 , we can give the definition of accumulation points (sometimes called cluster points) in any topological space.

Accumulation (Cluster) Point

Let $A \subset (X, T)$. A point $x \in X$ is called an accumulation (cluster) point of A if each neighborhood $U(x)$ of x contains at least one point of A distinct from x . The derived set of A , denoted by A^1 , is defined by

$$A^1 = \{x \in X \mid \forall U(x) : U(x) \cap (A - x) \neq \emptyset\},$$

i.e., A^1 consists of all cluster points of A .

Example:

$$A =]0, 1[$$

$$A^1 = [0, 1]$$

If $A = \left\{\frac{1}{n}\right\}$, $n = 1, 2, 3, \dots$, then $A^1 = \{0\}$.

THEOREM 7

A) $\bar{A} = A^1 \cup A.$

B) A is closed if and only if $A^1 \subset A.$

Definition of an Interior

Let $A \subset (X, T)$. The interior $\text{Int}(A)$ of A is the largest open set contained in A ,

$$\text{Int}(A) = \bigcup \{ U \mid (U \text{ open}) \wedge (U \subset A) \}.$$

Example: Let $A =]0, 1] \subset \mathbb{R}$. Then

$$\text{Int}(A) =]0, 1[.$$

Let $A = \left\{ \frac{1}{n}, \mid n = 1, 2, 3, \dots \right\}$. Then

$$\text{Int}(A) = \emptyset.$$

Example: A is open if and only if

$$A = \text{Int}(A).$$

Example: For any set A ,

$$\text{Int}(A) = \left(\overline{A^c} \right)^c.$$

Boundary

Let $A \subset (X, T)$. The boundary $\text{Fr}(A)$ of A is

$$\text{Fr}(A) = \overline{A} \cap \overline{A^c}.$$

Note that the boundary of A is a closed set.

THEOREM 8

Let $A \subset (X, T)$. Then:

$$\text{A) } \overline{A} = \text{Int}(A) \cup \text{Fr}(A).$$

$$\text{B) } \text{Fr}(A) \cap \text{Int}(A) = \emptyset.$$

$$\text{C) } \text{Fr}(A) = \overline{A} - \text{Int}(A).$$

Dense Set

A set $D \subset X$ is said to be dense in X if

$$\overline{D} = X .$$

Example: X is dense in X , and X is the only closed set dense in X .

Example: The set of rational numbers Q is dense in R .

THEOREM 9

The following three conditions are equivalent:

- A) D is dense in X .
- B) The complement of D has an empty interior.
- C) If F is any closed set in X and $D \subset F$, then $F = X$.

NEIGHBORHOOD SYSTEMS AND SEQUENCES

Let x be a point in a topological space X . A subset N of X is a neighborhood of x if N contains an open set U containing x :

$$x \in U \subset N : U \text{ is open.}$$

Neighborhood System

The class of all neighborhoods of $x \in X$, denoted by N_x , is called the neighborhood system of x .

We shall list the fundamental properties of the neighborhood system N_x .

THEOREM 10

- A) $N_x \neq \phi$.
- B) x belongs to each member of N_x .
- C) The intersection of any two members of N_x belongs to N_x .
- D) Every set containing a member N_x belongs to N_x .

- E) Each member $N \in N_x$ contains $G \in N_x$, where G is a neighborhood of each of its points.

The above properties of the neighborhood system may be used to define a topology on X .

Convergent Sequence

A sequence $\{a_1, a_2, \dots\}$ of points in a topological space X converges to a point $a \in X$ if, for each open set U containing a , there exists a positive integer $n_0 \in N$ such that

$$n > n_0 \Rightarrow a_n \in U.$$

Point a is called the limit of the sequence (a_n) , denoted by

$$a_n \rightarrow a \quad \text{or} \quad \lim a_n = a.$$

Let $\{T_\omega \mid \omega \in \Omega\}$ be a family of topologies on a non-empty set X . For any T_1 and $T_2 \in \{T_\omega\}$, if $T_1 \subset T_2$, we say that T_1 is coarser than T_2 and T_2 is finer than T_1 . Two topologies are not comparable if neither is finer than the other.

Subspace, Relative Topology

Let A be a non-empty subset of a topological space (X, T) . The class T_A of all intersections of A with T -open subsets of X is a topology on A .

T_A is called the relative topology on A or the relativization of T to A . The topological space (A, T_A) is called a subspace of (X, T) . It is easy to show that T_A is actually a topology on A .

THEOREM 11

A subspace of a subspace is a subspace of the entire space.

THEOREM 12

Let (X, T) be a space and (Y, T_Y) its subspace. Then:

- A) Let $A \subset Y$. Then A is T_Y -closed if and only if

$$A = F \cap Y$$

where F is T -closed. Thus, the closed sets in Y are the intersections of Y and sets closed in X .

$$B) \overline{A_Y} = Y \cap \overline{A}$$

$$A_Y^1 = Y \cap A^1$$

$$Y \cap \text{Int}(A) \subset \text{Int}_Y(A)$$

$$\text{Fr}_Y(A) \subset Y \cap \text{Fr}(A) .$$

THEOREM 13

Let Y be a subspace of X . If $A \subset Y$ is closed (open) in Y and Y is closed (open) in X , then A is closed (open) in X .

BASIS FOR A GIVEN TOPOLOGY

Instead of listing all open sets to specify a topology, it is easier to give only enough open sets to generate all the open sets.

Basis

Let (X, T) be a topological space.

A family $B \subset T$ is called a basis for T if each open set, that is, each member of T , is the union of members of B .

Example: Let D be the discrete topology on X . Then

$$B = \{ \{x\} \mid x \in X \}$$

is a basis for D .

THEOREM 14

Let B be any subset of T . Then A) is equivalent to B).

A) B is a basis for T .

B) For each $G \in T$ and each $x \in G$, there is a $U \in B$ such that
 $x \in U \subset G$.

THEOREM 15

Let $B \subset T$ be a basis for T . Then A is open ($A \in T$) if and only if for each $x \in A$ there is a $U \in B$ such that

$$x \in U \subset A.$$

Example: Consider space R^n . If

$$x = (x_1, x_2, \dots, x_n) \in R^n \quad \text{and} \quad y = (y_1, y_2, \dots, y_n) \in R^n,$$

then

$$x + y = (x_1 + y_1, \dots, x_n + y_n) \in R^n.$$

We shall write

$$|x| = \sqrt{\sum_{i=1}^n x_i^2}.$$

Hence, the distance between x and y is $|x - y|$. We define a ball of center x and radius r

$$B(x, r) = \{y \mid |x - y| < r\}.$$

We introduce the Euclidean topology in R^n . A set $A \subset R^n$ is called "open" if for each $x \in A$ there is some $r > 0$ such that

$$B(x, r) \subset A.$$

This condition describes a topology in R^n . Topology T of R^n consists of all open sets. Each $B(x, r)$ belongs to T .

With this definition, R^n is called Euclidean n -space and is also denoted by E^n . In each R^n , $n \geq 1$, the family of balls

$$\{B(x, r) \mid x \in R^n, r > 0\}$$

is a basis for the Euclidean topology.

Furthermore, we can show that R^n has a countable basis. The family

$$B_q = \{B(x_q, r_q) \mid x_q \text{ has all coordinates rational,} \\ \text{and } r_q > 0 \text{ is rational}\}.$$

Obviously, B_q is countable. Let G be any open set and $x \in G$; then there is a $B(x, r) \subset G$, where $r > 0$ is rational. We can find a point x_q , with all coordinates rational, such that

$$|x - x_q| < \frac{r}{3}$$

and

$$x \in B\left(x_q, \frac{r}{2}\right) \subset B(x, r) \subset G.$$

THEOREM 16

Let B be a family of subsets of (X, T) . Family B is a base for some topology on X if and only if it has the following two properties:

$$A) X = U\{A \mid A \in B\},$$

B) For any $A_1, A_2 \in B$, $A_1 \cap A_2$ is the union of members of B .

Note that property B) is equivalent to the following:

For any $A_1, A_2 \in B$, if $x \in A_1 \cap A_2$ then $\exists A_x \in B$, such that

$$x \in A_x \subset A_1 \cap A_2.$$

Subbase

Let (X, T) be a topological space. A class S of open subsets of X , i.e., $S \subset T$, is a subbase for the topology T on X if finite intersections of members of S form a base for T .

Example: Let $]a, b[\subset R^1$ be an open interval. Then

$$]a, b[=]-\infty, b[\cap]a, \infty[.$$

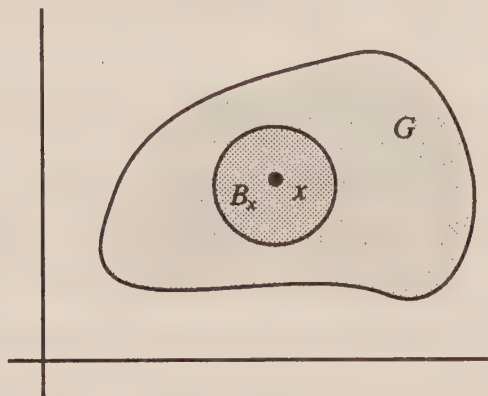
Open intervals form a base for the usual topology on R^1 . Hence, the class of all infinite open intervals is a subbase for R^1 .

Local Base

Let $x \in (X, T)$. A class B_x of open sets containing x is called a local base at x if for each open set G containing x , there exists a set $G_x \in B_x$ such that

$$x \in G_x \subset G.$$

Example: Consider the Euclidean topology in R^2 . Let $x \in R^2$. Then the family of all open balls with centers at x is a local base at x . Indeed, any open set G containing x also contains an open ball B_x with center at x .



THEOREM 17

Let B be a base for a topology T on X . If $x \in X$, then the members of the base B which contain x form a local base at the point x .

THEOREM 18

A sequence $(a_1, a_2, \dots)$ of points in a topological space (X, T) converges to $a \in X$ if and only if each member of some local base B_x at x contains all but a finite number of terms of the sequence.

THEOREM 19

Let B be a base for a topology T on X . Then:

- A) A sequence of points in $X (a_1, a_2, \dots)$ converges to $a \in X$ if and only if each element B containing a contains all but the finite number of terms of the sequence.
- B) $a \in X$ is an accumulation point of $A \subset X$ if and only if each element of B containing a contains a point of A different from a .

CONTINUOUS FUNCTIONS

Let (X, T_X) and (Y, T_Y) be two topological spaces. A function

$$f : X \rightarrow Y$$

maps elements of X onto elements of Y and also induces two functions:

$$f : P(X) \rightarrow P(Y) \text{ and}$$

$$f^{-1} : P(Y) \rightarrow P(X).$$

Note that f^{-1} preserves the Boolean operations:

$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$$

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$$

$$f^{-1}(A - B) = f^{-1}(A) - f^{-1}(B)$$

Continuous Function

Let (X, T_X) and (Y, T_Y) be spaces. A function $f : X \rightarrow Y$ is called continuous if the inverse image of each open set in Y is open in X .

Observe that if f is a continuous function, then $f^{-1} : T_Y \rightarrow T_X$.

Example: A constant function $f : X \rightarrow Y$ is always continuous.

THEOREM 20

- A) If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then

$$g \circ f : X \rightarrow Z$$

is also continuous.

- B) If $f : X \rightarrow Y$ is continuous and $A \subset X$ is taken with the subspace topology, then

$$f|A : A \rightarrow Y$$

is continuous.

- C) If $f : X \rightarrow Y$ is continuous and $f(X)$ is taken with the subspace topology, then

$$f : X \rightarrow f(X)$$

is continuous.

The next theorem lists the basic properties of continuous functions.

THEOREM 21

Let X and Y be topological spaces and let $f : X \rightarrow Y$ be a function. The following conditions are equivalent:

- A) f is continuous.
- B) The inverse image of each closed set in Y is closed in X .
- C) The inverse image of each member of a basis (subbasis) for Y is open in X .
- D) For every $A \subset X$,

$$f(\overline{A}) \subset \overline{f(A)}.$$

- E) $\overline{f^{-1}(B)} \subset f^{-1}(\overline{B})$ for every

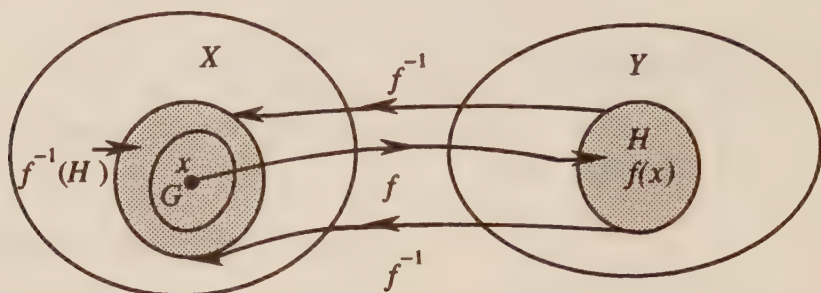
$$B \subset Y.$$

- F) For each $x \in X$ and each neighborhood $U(f(x))$ in Y , there exists a neighborhood of x , $V(x)$ in X such that

$$f(V(x)) \subset U(f(x)).$$

Continuity at a Point

A function $f : X \rightarrow Y$ is continuous at $x \in X$ if the inverse image $f^{-1}(H)$ of every open set $H \subset Y$ that contains $f(x)$ contains an open set $G \subset X$ that contains x .



In other words, $f : X \rightarrow Y$ is continuous at $x \in X$ if the inverse image of every neighborhood of $f(x)$ is a neighborhood of x .

THEOREM 22

Let X and Y be topological spaces. A function $f : X \rightarrow Y$ is continuous if and only if f is continuous at every point of X .

Sequential Continuity at a Point

A function $f : X \rightarrow Y$ is sequentially continuous at $a \in X$ if for every sequence (a_n) in X converging to a , the sequence $(f(a_n))$ converges to $f(a)$; that is:

$$(a_n \rightarrow a) \Rightarrow (f(a_n) \rightarrow f(a)).$$

If a function is continuous at a point, then it is sequentially continuous at this point. Note that the converse of this statement is not true.

Example: Let $f : R^n \rightarrow R^1$; that is, f is a real-valued function of n real variables. Then f is continuous at $x_0 \in R^n$ if, for each open interval $]f(x_0) - \varepsilon, f(x_0) + \varepsilon[$ (where $\varepsilon > 0$), there is a ball $B(x_0, \delta)$ mapped onto this interval by f . We can rewrite this statement as follows:

$$\forall \varepsilon > 0 \exists \delta > 0 : |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon .$$

Open and Closed Functions

If f is a continuous function, then the inverse image of every open set is open and the inverse image of every closed set is closed.

A function $f : X \rightarrow Y$ is called an open function if the image of every open set is open.

A function $f : X \rightarrow Y$ is called a closed function if the image of every closed set is closed.

An open function does not have to be continuous and a continuous function does not have to be open.

THEOREM 23

Let $f : X \rightarrow Y$ be an open function. Given any subset $B \subset Y$, and any closed A such that $A \subset f^{-1}(B)$, there exists a closed $D \subset B$ such that

$$f^{-1}(D) \subset A .$$

HOMEOMORPHISM

Homeomorphism

A continuous bijective (i.e., one-to-one) function $f : X \rightarrow Y$, such that $f^{-1} : Y \rightarrow X$ is also continuous, is called a homeomorphism and is denoted by

$$f : X \cong Y .$$

Two spaces X, Y are homeomorphic, written as $X \cong Y$, if there is a homeomorphism

$$f : X \cong Y .$$

Example: The function

$$f(x) = \frac{x}{1+|x|}$$

is a homeomorphism of $\mathbb{R}$ and $] -1, +1[$.

Homeomorphism is an important tool of topology because a homeomorphism is also an open map. Hence, a function $f : X \rightarrow Y$ which is a homeomorphism is a bijection from X onto Y and also a bijection

$$f \mid T(X) : T(X) \rightarrow T(Y).$$

We conclude that every property of X expressed in terms of set operations and open sets is also possessed by each space homeomorphic to X .

Such properties are called topological properties. In general, any property of topological spaces is called a topological invariant if, whenever this property is true for one space X , it is also true for every space homeomorphic to X .

Thus, topology can be described as the study of topological invariants.

THEOREM 24

Homeomorphism is an equivalence relation in the class in all topological spaces.

Indeed:

- A) $X \cong X$.
- B) $(f : X \cong Y) \Rightarrow (f^{-1} : Y \rightarrow X)$.
- C) $(f : X \cong Y) \wedge (g : Y \cong Z) \Rightarrow (g \circ f : X \cong Z)$.

The relation of homeomorphism divides the class of all topological spaces into mutually disjoint classes. Such classes are called homeomorphism types.

THEOREM 25

Let $f : X \rightarrow Y$ be one-to-one and onto. The following properties are equivalent:

- A) f is a homeomorphism.
- B) f is continuous and open.
- C) f is continuous and closed.
- D) $f(\overline{A}) = \overline{f(A)}$ for each $A \subset X$.

THEOREM 26

Let $f : X \cong Y$ and $A \subset X$. Then $f|A : A \cong f(A)$ and $f|X-A : X-A \cong Y-f(A)$.

Example: The closed interval $[a, b] \subset R$ is homeomorphic to the closed unit interval $[0, 1]$.

The linear function

$$f : [0, 1] \rightarrow [a, b],$$

defined by

$$f(x) = (b - a)x + a.$$

is one-to-one, onto and bicontinuous. Hence, f is a homeomorphism.

Note that a function f is called bicontinuous if f is continuous and open.

Chapter 8

LOGIC

FUNDAMENTAL CONCEPTS

Sentential calculus is the “calculus of sentences,” a field in which the truth or falseness of assertions is examined by using algebraic tools.

Sentence

A sentence is any expression that can be labeled either true or false.

Examples: Expressions to which the terms “true” or “false” can be assigned include the following:

- 1) “It is raining where I am standing.”
- 2) “My name is George.”
- 3) “ $1 + 2 = 3$ ”

Examples: Expressions to which the terms “true” or “false” cannot be assigned include the following:

- 1) “I will probably be healthier if I exercise.”
- 2) “It will rain on this day, one year from now.”
- 3) “What I am saying at this instant is a lie.”

Sentential Calculus

Sentential calculus is the “calculus of sentences,” in which tools of algebra are used to examine whether sentences are true or false (when possible). Sentential calculus is also referred to as propositional calculus or algebra of logic.

Sentences can be combined to form new sentences using the connectives, AND, OR, NOT, and IF-THEN.

Examples: The sentences

- 1) "John is tired."
- 2) "Mary is cooking."

can be combined to form

- 1) "John is tired AND Mary is cooking."
- 2) "John is tired OR Mary is cooking."
- 3) "John is NOT tired."
- 4) "IF John is tired THEN Mary is cooking."

Literals

In any context, there is a collection of basic or fundamental sentences which, when linked together via the connectives NOT, AND, OR, and IF-THEN, can serve to express the content of any other sentence. These sentences, together with their negations, are referred to as the fundamental sentences, literals, or atoms of sentential calculus.

In the following we will use capital letters X , Y , Z , ... to represent sentences, and develop algebraic tools to represent new sentences formed by linking them with the above connectives. Our connectives may be regarded as operations transforming one or more sentences into a new sentence. To describe them in greater detail, we introduce symbols to represent them. You will find that different symbols representing the same idea may appear in different references.

Logical Value

The logical value of a sentence X is TRUE (or T) if X is true, and FALSE (or F) if X is false.

TRUTH TABLES AND BASIC LOGICAL OPERATIONS

Truth Table

The truth table for a sentence X is the exhaustive list of possible logical values of X .

Negation

If X is a sentence, then $\sim X$ represents the negation, the opposite, or the contradiction of X . Thus, the logical values of $\sim X$ are as shown in the following truth table:

X	$\sim X$
T	F
F	T

Table 8.1 — Truth table for negation.

$\sim$ is called the negation operation on sentences.

Example: For

$X = \text{“Jane is eating an apple.”}$

$\sim X = \text{“Jane is NOT eating an apple.”}$

Negation Is a “Unary” Operation

The negation operation is called unary, since it transforms a sentence into a unique image sentence.

IFF

We use the word IFF to represent the expression “if and only if.”

AND

For sentences X and Y , “ X AND Y ,” represented by $X \wedge Y$, is the sentence which is true IFF both X and Y are true. The truth table for $\wedge$ is as follows:

X	Y	$X \& Y$
T	T	T
T	F	F
F	T	F
F	F	F

Table 8.2 — Truth Table for AND.

$\&$ is called the conjunction operator.

Conjunction Is a Binary Operation

The conjunction $\&$ is a binary operation since it transforms a pair of sentences into a unique image sentence.

Example: For

$X =$ “Jane is eating an apple,”

and

$Y =$ “All apples are sweet,”

$X \& Y =$ “Jane is eating an apple AND all apples are sweet.”

OR

For sentences X and Y , “ X OR Y ,” represented by $X \vee Y$ (sometimes referred to as the “inclusive “OR” or “AND/OR”), denotes the sentence that is true if either or both X and Y are true. The truth table for $\vee$ is as follows:

X	Y	$X \vee Y$
T	T	T
T	F	T
F	T	T
F	F	F

Table 8.3 — Truth table for OR.

$\vee$ is called the disjunction operator.

Disjunction Is a Binary Operation

As with the conjunction operator, the disjunction is a binary operation, transforming the pair of sentences X , Y into a unique image sentence $X \vee Y$.

Example: For

$X = \text{"Jane is eating the apple,"}$

and

$Y = \text{"Marvin is running,"}$

$X \vee Y = \text{"Jane is eating the apple OR Marvin is running or both."}$

IF-THEN-

For sentences X and Y , $X \rightarrow Y$ represents the statement "IF X THEN Y ." $X \rightarrow Y$ is false IFF X is true and Y is false; otherwise, it is true. The truth table for $\rightarrow$ is as follows:

X	Y	$X \rightarrow Y$
T	T	T
T	F	F
F	T	T
F	F	T

Table 8.4 — Truth Table for IF-THEN-.

$\rightarrow$ is referred to as the implication operator.

Implication $\rightarrow$ Is a Binary Operation

$\rightarrow$ is a binary operator on pairs of sentences X, Y .

X IFF Y

For sentences X and Y , $X \leftrightarrow Y$ is true IFF X and Y have the same truth value; otherwise, it is false. The truth table for $\leftrightarrow$ is as follows:

X	Y	$X \leftrightarrow Y$
T	T	T
T	F	F
F	T	F
F	F	T

Table 8.5 — Truth table for equivalence.

$\leftrightarrow$ represents logical equivalence, “IFF.”

Example: For

X = “Jane eats apples,”

and

$Y = \text{"apples are sweet,"}$

$X \leftrightarrow Y = \text{"Jane is eats apples IFF apples are sweet."}$

Equivalence $\leftrightarrow$ Is a Binary Operation

$\leftrightarrow$ is again a binary operation on pairs of sentences.

LOGICAL EQUIVALENCE

Logically Equivalent Sentences

Two sentences, X , Y for which $X \leftrightarrow Y$ are said to be logically equivalent.

Logical Equivalence and Meaning the Same

Logical equivalence $\leftrightarrow$ is not the same as equivalence of meanings. Thus, if Jane is eating an apple and Barbara is frightened of mice, then for $X = \text{"Jane is eating an apple"}$ and $Y = \text{"Barbara is frightened of mice,"}$ X and Y are logically equivalent, since both are correct. However, they do not have the same meaning. Statements having the same meaning are, for example, the double negative $\sim\sim X$ (not-not) and X itself.

If Sentences Have the Same Meaning

If X AND Y have the same meaning, then we write

$$X \text{ EQ } Y.$$

There is no truth table for EQ, since having the same meaning is far stronger than simply being simultaneously true or false.

Sentential Calculus and Equivalence

Sentential calculus is concerned only with logical equivalence, and ignores the meaning of the sentences considered.

THEOREM 1 — Double Negation Equals Identity

For any sentence X ,

$$\sim\sim X \leftrightarrow X .$$

FUNDAMENTAL PROPERTIES OF OPERATIONS

THEOREM 2 — Properties of Conjunction Operation

For any sentences X, Y, Z , the following properties hold:

- 1) Commutativity: $X \& Y \leftrightarrow Y \& X .$
- 2) Associativity: $X \& (Y \& Z) \leftrightarrow (X \& Y) \& Z .$

THEOREM 3 — Properties of Disjunction Operation

For any sentences X, Y, Z , the following properties hold:

- 1) Commutativity: $X \vee Y \leftrightarrow Y \vee X .$
- 2) Associativity: $X \vee (Y \vee Z) \leftrightarrow (X \vee Y) \vee Z .$

THEOREM 4 — Distributive Laws

For any sentences X, Y, Z , the following laws hold:

- 1) $X \vee (Y \& Z) \leftrightarrow (X \vee Y) \& (X \vee Z) .$
- 2) $X \& (Y \vee Z) \leftrightarrow (X \& Y) \vee (X \& Z) .$

THEOREM 5 — DeMorgan's Laws for Sentences

For any sentences X, Y , the following laws hold:

- 1) $\sim (X \& Y) \leftrightarrow (\sim X) \vee (\sim Y) .$
- 2) $\sim (X \vee Y) \leftrightarrow (\sim X) \& (\sim Y) .$

Proof of A): Assertion A) can be proved by developing a truth table over all possible combinations of X and Y and observing that all values assumed by

the sentences are the same. To this end, we first evaluate the expression $\sim (X \& Y)$:

	X	Y	$X \& Y$	$\sim (X \& Y)$
	T	T	T	F
	T	F	F	T
	F	T	F	T
	F	F	F	T

Table 8.6a — Truth table for negation of conjunction.

Now we evaluate $(\sim X) \vee (\sim Y)$:

	X	Y	$\sim X$	$\sim Y$	$(\sim X) \vee (\sim Y)$
	T	T	F	F	F
	T	F	F	T	T
	F	T	T	F	T
	F	F	T	T	T

Table 8.6b — Truth table for disjunction of negation.

The last columns of the truth tables coincide, proving our assertion.

THEOREM 6 — Two Logical Identities

For any sentences X , Y , the sentences X and $(X \& Y) \vee (X \& \sim Y)$ are logically equivalent, i.e.,

$$(X \& Y) \vee (X \& \sim Y) \leftrightarrow X.$$

This is proven in the following truth table:

X	Y	$\sim Y$	$X \& Y$	$X \& \sim Y$	$(X \& Y) \vee (X \& \sim Y)$
T	T	F	T	F	T
T	F	T	F	T	T
F	T	F	F	F	F
F	F	T	F	F	F

Table 8.7a — Truth table for $(X \& Y) \vee (X \& \sim Y) \leftrightarrow X$.

For any sentences X , Y , the sentences X and $X \vee (Y \& \sim Y)$ are logically equivalent, i.e.,

$$X \vee (Y \& \sim Y) \leftrightarrow X.$$

This is proven in the following truth table:

X	Y	$\sim Y$	$Y \& \sim Y$	$X \vee (Y \& \sim Y)$
T	T	F	F	T
T	F	T	F	T
F	T	F	F	F
F	F	T	F	F

Table 8.7b — Truth table for $X \vee (Y \& \sim Y) \leftrightarrow X$.

THEOREM 7 — Proof by Contradiction

For any sentences X , Y , the following holds:

$$X \rightarrow Y \leftrightarrow \sim Y \rightarrow \sim X.$$

To prove this, we consider the following truth table:

X	Y	$X \rightarrow Y$	$\sim Y$	$\sim X$	$\sim Y \rightarrow \sim X$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Table 8.8 — Truth table for proof by contradiction.

SENTENCES, LITERALS, AND FUNDAMENTAL CONJUNCTIONS

We have seen that logically equivalent sentences may be expressed in different ways, the simplest examples being that a sentence is equal to its double negation,

$$\sim\sim X \leftrightarrow X,$$

while by DeMorgan's theorem,

$$X \vee Y \leftrightarrow \sim(\sim X \& \sim Y).$$

The significance of sentential calculus and the algebra of logic is that it provides us with a method of producing a "standard" form for representing a statement in terms of the literals. This is indeed unique and, while not usually the simplest representation, it does serve as a standard form for comparison and evaluation of sentences. This standard form is the disjunctive normal form introduced in the following lines.

Subsentences

Suppose that a sentence X is formed as the result of operating on sentences $A, B, C, \dots$, by means of the operators $\&, \vee, \sim, \rightarrow$, and $\leftrightarrow$, for example, $X = (A \vee B) \& ((\sim C) \rightarrow D)$. Then the sentences $A, B, C, \dots$, are referred to as subsentences of X .

Fundamental Conjunction

A sentence is a fundamental Conjunction if it is either a literal or conjunction $X \& Y$ of at least two literals in which each literal only appears once.

Examples: If $A, B, C, \dots$, are literals, then the sentences

$$X = A, X = A \& B, X = A \& B \& D$$

are fundamental conjunctions, while the sentence

$$X = (A \& B) \& A$$

is not.

Subjunction

A fundamental subjunction X is a subjunction of the fundamental conjunction Y if every literal appearing in X is a literal of Y .

Example: The fundamental conjunction

$$X = A \& B$$

is a subjunction of the fundamental conjunction

$$Y = A \& C \& B.$$

Disjunctive Normal Form

A sentence X is in disjunctive normal form (DNF) if it is a fundamental conjunction or it is a disjunction of at least two disjuncts, or fundamental conjunctions, none of which is a subconjunction of any of the others.

Examples: Let $A, B, C, D, \dots$ be literals. Then the following sentences are in disjunctive normal form:

$$X = A, \quad Y = A \& B, \quad Z = A \vee (A \& C);$$

$$X = (A \& B \& C) \vee (D \& C) \vee (A \& D).$$

The following sentences are not in disjunctive normal form:

$$X = A \& (B \vee C), \quad Y = (A \& B) \vee (C \& A \& B) .$$

THEOREM 8 — Fundamental Theorem of Sentences

Every sentence which is not a contradiction is logically equivalent to a sentence which is in disjunctive normal form.

Examples: Consider the sentence

$$X = \{ [(A \vee \sim B) \vee C] \& [A \vee (\sim \sim B)] \} \& \sim A .$$

We can reduce this sentence to disjunctive normal form as follows. Firstly, $\sim \sim B = B$. Similarly, the parentheses around the disjunction $A \vee \sim B$ may be removed using the associative law for disjunction. Thus, we find

$$X = [A \vee \sim B \vee C] \& [A \vee B] \& \sim A .$$

Now by distributivity and Theorem 6,

$$\begin{aligned} [A \vee \sim B] \& \sim A &= (A \& \sim A) \vee (B \& \sim A) \\ &= B \& \sim A, \end{aligned}$$

so that

$$X = (A \vee \sim B \vee C) \& (B \& \sim A) .$$

By distributivity, this becomes

$$X = C \& B \& \sim A,$$

which is the disjunctive normal form consisting only of a single fundamental conjunction.

Full Disjunctive Normal Form

A sentence X is in full disjunctive normal form with respect to the set of literals $A_1, A_2, A_3, \dots, A_N$ if the only literals appearing in X are chosen from among the A_2 and any disjunct in X contains all of the literals $A_1, A_2, A_3, \dots, A_N$.

Example: In the system with literals A and B , B may be represented in the full disjunctive normal form as

$$B \Leftrightarrow (A \& B) \vee (\sim A \& B) .$$

ADEQUATE SYSTEMS OF CONNECTIVES AND DUALITY

Are the connectives in our logic system independent, or are some of them redundant, that is, expressible in terms of the others? For example, the implication connective $\rightarrow$ may be expressed in terms of the negation and disjunction connectives

$$X \rightarrow Y \Leftrightarrow Y \vee \sim X,$$

as we see in Table 8.9:

X	Y	$X \rightarrow Y$	$\sim X$	$Y \vee \sim X$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Table 8.9 — Truth table for $X \rightarrow Y \Leftrightarrow Y \vee \sim X$.

Adequate System of Connectives

An adequate system of connectives is a set of connectives that can be used to represent every logical sentence in terms of their operation on its literals.

THEOREM 9 — Some Adequate Systems of Connectives

Each of the following systems of connectives constitutes an adequate system:

$$\{\sim, \&, \vee\}, \{\sim, \&\}, \{\sim, \vee\}, \{\sim, \rightarrow\} .$$

Dependence of Connectives

The operation $X \& Y$ may be presented in terms of $\sim$ and $\vee$ by the negation of DeMorgan's theorem

$$X \& Y \leftrightarrow \sim(\sim X \vee \sim Y) .$$

Similarly,

$$X \vee Y \leftrightarrow \sim(\sim X \& \sim Y) .$$

Sheffer Stroke

The Sheffer stroke $|$ is a connective defined by the truth table:

X	Y	$X Y$
T	T	F
T	F	T
F	T	T
F	F	T

Table 8.10 — Definition of Sheffer stroke.

Joint Denial

We define the joint denial connective JD by the truth table:

X	Y	$X \text{ JD } Y$
T	T	F
T	F	F
F	T	F
F	F	T

Table 8.11 — Truth table for joint denial operation.

THEOREM 10 — Adequate Systems of One Connective

The Sheffer stroke and joint denial operations are each adequate systems of connectives consisting of exactly one operation.

Proof: For the Sheffer stroke and any sentences X, Y ,

$$\sim X \leftrightarrow X \mid X$$

$$X \vee Y \leftrightarrow (X \mid X) \mid (Y \mid Y) ;$$

similarly, $X \& Y$ can be expressed via DeMorgan's theorem in terms of $\sim$ and $\vee$, and so, in terms of $\mid$.

THEOREM 11 — Other Adequate Systems Do Not Exist

The joint denial and Sheffer stroke operations are the only adequate connective systems consisting of a single connective.

We have seen earlier that

$$\sim(X \& Y) \Leftrightarrow \sim X \vee \sim Y,$$

while

$$\sim(X \vee Y) \Leftrightarrow \sim X \& \sim Y.$$

These duality results can be extended to the general principle of duality for sentences.

THEOREM 12 — Duality Principle

Given a sentence formed from literals and their negations by means of conjunctions and disjunctions only, the negation of the sentence is found by interchanging the symbols $\&$ and $\vee$ of conjunction and disjunction, and replacing every literal by its negation.

LOGICAL EQUIVALENCE CLASSES

It is 12 noon and Mary is eating sweets. The two statements

$$X = \text{"It is 12 noon."}$$

$$Y = \text{"Mary is eating sweets."}$$

are both true, and hence are logically equivalent:

$$X \leftrightarrow Y .$$

However, they do not have the same meaning. Therefore, from the viewpoint of logical equivalence, these two totally different and unrelated statements are the same: in the collection of all literals and sentences of our universe, X and Y , while not saying the same thing, are equivalent. We may make their equivalence more rigorous by introducing the idea of equivalence classes over the sentences of our logical universe.

Statement Bundle

Let X be any sentence. Then $\{X\}$ denotes the collection of all sentences which are logically equivalent to X , that is, the set of all sentences Y for which

$$Y \leftrightarrow X .$$

The collection $\{X\}$ is referred to as the equivalence class or statement bundle corresponding to X .

Null Logical Equivalence Class

The null logical equivalence class $\emptyset$ is the equivalence class of the false sentence $X \& \sim X$ for any sentence X :

$$\emptyset = \{X \ \& \ \sim X\} .$$

Null Equivalence Class and False Sentences

The null logical equivalence class $\emptyset$ is the class of all false sentences.

Unit Logical Equivalence Class

The unit logical equivalence class 1 is the equivalence of the true sentence $X \vee \sim X$ for any sentence X :

$$1 = \{X \vee \sim X\} .$$

Unit Equivalence Class

The unit logical equivalence class 1 is the class of all true sentences.

Chapter 9

ALGORITHMIC PROCESSES

DEFINITION

An algorithm is the sequence of logical steps required to perform a specific task, such as solving a problem. The algorithm is a map that leads to the solution when completed.

SPECIFIC ALGORITHMIC PROPERTIES

- 1) Each step must be deterministic, with nothing left to chance. The final answer cannot depend on the person executing the instructions.
- 2) The process must end after a finite number of steps. An open-ended algorithm will not solve a problem.
- 3) The algorithm must be general enough to handle unknowns or contingencies that arise.

TOP-DOWN APPROACH

Start with a simple, broad-based solution and add specificity as time and problem understanding allow. Take the large problem, and break it down into small pieces; then put the solution together. Since a program language is ordinarily part of the process, the following steps outline the process:

- 1) Flesh out the problem – Ask questions that will help formulate direction.

- 2) Understand the problem language structure – This will aid in ensuring that your questions from part 1) are applicable to the solution.
 - 3) Determine the tools needed to organize – Information maps, for example, may be helpful.
 - 4) Sequential vs. control statements – Control statements tend to branch the program.
 - 5) Final organization – This helps to answer the question/solve the problem.
 - 6) Execution and checks – These are necessary to test the algorithm. The best algorithm often fails to answer the real problem. Testing is necessary to validate the algorithm.
-

KEY DEFINITIONS AND SYMBOLS FOR GRE MATH

$\in$: “a member of” or “in”

$\ni$: “such that”

$\exists$: “there exists”

$\forall$: “for all”

$\subset$: “is a subset of”

$\cup$: indicates union of two sets

$\cap$: indicates intersection of two sets

$\emptyset$: indicates “empty or null set”

The Graduate Record Examination in
MATHEMATICS

Practice Test I

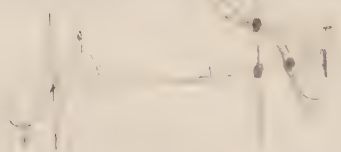
The Graduate Record Examination in **MATHEMATICS** Answer Sheet Test I

- | | | |
|---|---|---|
| 1. ☐ A ☐ B ☐ C ☐ D ☐ E | 23. ☐ A ☐ B ☐ C ☐ D ☐ E | 45. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 2. ☐ A ☐ B ☐ C ☐ D ☐ E | 24. ☐ A ☐ B ☐ C ☐ D ☐ E | 46. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 3. ☐ A ☐ B ☐ C ☐ D ☐ E | 25. ☐ A ☐ B ☐ C ☐ D ☐ E | 47. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 4. ☐ A ☐ B ☐ C ☐ D ☐ E | 26. ☐ A ☐ B ☐ C ☐ D ☐ E | 48. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 5. ☐ A ☐ B ☐ C ☐ D ☐ E | 27. ☐ A ☐ B ☐ C ☐ D ☐ E | 49. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 6. ☐ A ☐ B ☐ C ☐ D ☐ E | 28. ☐ A ☐ B ☐ C ☐ D ☐ E | 50. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 7. ☐ A ☐ B ☐ C ☐ D ☐ E | 29. ☐ A ☐ B ☐ C ☐ D ☐ E | 51. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 8. ☐ A ☐ B ☐ C ☐ D ☐ E | 30. ☐ A ☐ B ☐ C ☐ D ☐ E | 52. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 9. ☐ A ☐ B ☐ C ☐ D ☐ E | 31. ☐ A ☐ B ☐ C ☐ D ☐ E | 53. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 10. ☐ A ☐ B ☐ C ☐ D ☐ E | 32. ☐ A ☐ B ☐ C ☐ D ☐ E | 54. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 11. ☐ A ☐ B ☐ C ☐ D ☐ E | 33. ☐ A ☐ B ☐ C ☐ D ☐ E | 55. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 12. ☐ A ☐ B ☐ C ☐ D ☐ E | 34. ☐ A ☐ B ☐ C ☐ D ☐ E | 56. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 13. ☐ A ☐ B ☐ C ☐ D ☐ E | 35. ☐ A ☐ B ☐ C ☐ D ☐ E | 57. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 14. ☐ A ☐ B ☐ C ☐ D ☐ E | 36. ☐ A ☐ B ☐ C ☐ D ☐ E | 58. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 15. ☐ A ☐ B ☐ C ☐ D ☐ E | 37. ☐ A ☐ B ☐ C ☐ D ☐ E | 59. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 16. ☐ A ☐ B ☐ C ☐ D ☐ E | 38. ☐ A ☐ B ☐ C ☐ D ☐ E | 60. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 17. ☐ A ☐ B ☐ C ☐ D ☐ E | 39. ☐ A ☐ B ☐ C ☐ D ☐ E | 61. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 18. ☐ A ☐ B ☐ C ☐ D ☐ E | 40. ☐ A ☐ B ☐ C ☐ D ☐ E | 62. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 19. ☐ A ☐ B ☐ C ☐ D ☐ E | 41. ☐ A ☐ B ☐ C ☐ D ☐ E | 63. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 20. ☐ A ☐ B ☐ C ☐ D ☐ E | 42. ☐ A ☐ B ☐ C ☐ D ☐ E | 64. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 21. ☐ A ☐ B ☐ C ☐ D ☐ E | 43. ☐ A ☐ B ☐ C ☐ D ☐ E | 65. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 22. ☐ A ☐ B ☐ C ☐ D ☐ E | 44. ☐ A ☐ B ☐ C ☐ D ☐ E | 66. ☐ A ☐ B ☐ C ☐ D ☐ E |

GRE MATHEMATICS TEST I

TIME: 2 hours and 50 minutes
66 Questions

DIRECTIONS: Choose the best answer for each question and mark the letter of your selection on the corresponding answer sheet.



1. The graph of the arccosine function is the graph of the arcsine function
- (A) translated horizontally $\pi / 2$ units to the right
 - (B) first reflected in the horizontal axis and then translated vertically $\pi / 2$ units upward
 - (C) first translated horizontally $\pi / 2$ units to the left and then reflected in the horizontal axis
 - (D) first translated vertically $\pi / 2$ units downward and then reflected in the vertical axis
 - (E) translated horizontally $\pi / 2$ units to the left
2. If $f(x) = e^x - e^{-x}$, then $[f'(x)]^2 - [f(x)]^2$ equals
- (A) 4
 - (B) $4e^{-2x}$

(C) $2e^{-x}$

(E) $2e^x$

(D) 2

3. The domain of $f(x) = \frac{\sqrt[3]{x+2}}{x-6}$ is given by

(A) $(6, +\infty)$

(D) $[-2, +\infty) \setminus \{6\}$

(B) $[-2, +\infty)$

(E) $\mathbb{R} \setminus \{6\}$

(C) $\mathbb{R} \setminus \{-2, 6\}$

4. Let $M = \begin{bmatrix} 1 & 2 \\ 3 & 9 \end{bmatrix}$. The determinant of the adjoint of M is

(A) 9

(D) 18

(B) 6

(E) 3

(C) 27

5. The number of generators of a cyclic group of order 8 is

(A) 6

(D) 2

(B) 4

(E) 1

(C) 3

6. An integrating factor for the ordinary differential equation

$$\frac{-2y}{x} dx + (x^2 y \cos y + 1) dy = 0 \text{ is}$$

- (A) 1 (D) $-2x$
 (B) $\frac{-2}{x}$ (E) x^2
 (C) $\frac{1}{x^2}$

7. Assuming convergence, find $x = \sqrt{3 + \sqrt{3 + \sqrt{3 + \dots}}}$

- (A) $\frac{1}{2} (\sqrt{5} + 1)$
 (B) $\frac{1}{2} (\sqrt{13} - 1)$
 (C) $\frac{1}{2} (\sqrt{5} - 1)$
 (D) $\frac{1}{2} (\sqrt{13} + 1)$
 (E) $\frac{1}{2} (\sqrt{13} - \sqrt{5})$

$$\begin{aligned}
 x &= \sqrt{3 + x} \\
 x^2 &= 3 + x \\
 x^2 - x - 3 &= 0 \\
 x &= \frac{1 \pm \sqrt{1 + 12}}{2} \\
 x &= \frac{1 \pm \sqrt{13}}{2}
 \end{aligned}$$

8. Let x be a random variable possessing the probability density function

$$f(x) = \begin{cases} cx & x \in [0, 10] \\ 0 & \text{otherwise} \end{cases}$$

where $c \in R$. The probability that x is an element of $[1, 2]$ is

- (A) $\frac{1}{100}$ (B) $\frac{3}{100}$

(C) $\frac{5}{100}$

(E) $\frac{9}{100}$

(D) $\frac{7}{100}$

9. Let the random variable X have the probability density function

$$f(x) = \begin{cases} 1 - \frac{x}{2} & x \in (0, 2) \\ 0 & \text{otherwise} \end{cases}$$

The expected value of the random variable X^2 is

(A) $\frac{1}{3}$

(D) $\frac{1}{6}$

(B) $\frac{5}{6}$

(E) $\frac{2}{3}$

(C) $\frac{1}{2}$

10. Find the number of solutions of the set of all algebraic equations of height two.

(A) 0

(D) 3

(B) 1

(E) 4

(C) 2

11. Define a metric on $R^2 = R \times R$ by $d[(x_1, y_1); (x_2, y_2)] = |x_2 - x_1| + |y_2 - y_1|$. The unit ball $d[(0, 0); (x, y)] < 1$ is

- (A) the interior of a circle with center $(0, 0)$ and radius 1
- (B) $(0, 0)$
- (C) the interior of a square with vertices $(-1, 1), (1, 1), (1, -1)$ and $(-1, -1)$
- (D) the interior of a square with vertices $(-1, 0), (0, 1), (1, 0)$ and $(0, -1)$
- (E) the interior of a triangle with vertices $(-1, -1), (0, \sqrt{3}),$ and $(1, -1)$

12. Which of the following is not equal to $f(x) = \frac{x+1}{x-1}$ when both are defined?

- (A) $-f(x^{-1})$
- (B) $[f(-x)]^{-1}$
- (C) $f^{-1}(x)$
- (D) $f^{-1}(x^{-1})$
- (E) $\frac{1}{2} [f^{-1}(x) - f(x^{-1})]$

13. Let $M = \begin{bmatrix} 6 & 10 \\ -2 & -3 \end{bmatrix}$. The trace of M^5 equals

- (A) 27
- (B) 3^5

(C) 5^3

(E) 33

(D) $6^5 + (-3)^5$

14. The degree of the minimum polynomial satisfied by a nonscalar, 8 by 8, idempotent matrix M is

(A) 4

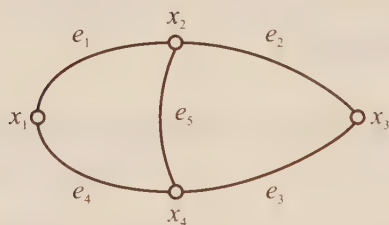
(D) 3

(B) 2

(E) 6

(C) 8

15. What is the incidence matrix for the graph shown below?



(A)
$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

(B)
$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$(C) \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \quad (D) \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$(E) \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

16. The fixed point(s) of the Mobius transformation

$$w(z) = \frac{z-2}{z-1} \text{ is (are)}$$

(A) $1 \pm \sqrt{3}$

(D) $1 \pm i$

(B) $1 \pm 2i$

(E) $-1 \pm \sqrt{2}i$

(C) $2i$

17. The sum of the 9th roots of unity is

(A) 0

(D) 10

(B) 1

(E) $1 + i$

(C) 9

18. Let x, y, z represent Boolean variables. Which of the following is not a Boolean function?

(A) $f(x, y) = x\sqrt{y}$

(B) $f(x, y, z) = \max \{x, y, z\}$

(C) $f(x, y) = x^2 + y - xy$

(D) $f(x, y, z) = x + y + z - xy - yz$

(E) $f(x, y, z) = xyz$

19. What is the maximum perimeter of all rectangles that can be inscribed in $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$?

(A) $4\sqrt{a^2 + b^2}$

(D) $a^2 + b^2$

(B) $\frac{8}{\sqrt{a^2 + b^2}}$

(E) $2(a^2 + b^2)$

(C) $2\sqrt{a^2 + b^2}$

20. Which of the following is a topological property?

(A) boundedness

(B) being a Cauchy sequence

(C) completeness

(D) being an accumulation (limit) point

(E) length

21. The value of $I = \oint_C \frac{\cos z}{z(z - \pi)} dz$ where C is the circle $|z - 1| = 2$ is

(A) 0

(D) $-4i$

(B) $2i$

(E) $4i$

(C) $-2i$

22. Let $p(x)$, $q(x)$, and $r(x)$ be open statements relative to the set S . Then $\sim (\exists x \in S) [(p(x) \vee q(x)) \wedge r(x)]$ is equivalent to

(A) $(\forall x \in S) \{[(\sim p(x)) \vee (\sim q(x))] \vee [\sim r(x)]\}$

(B) $(\forall x \in S) \{[(\sim p(x)) \wedge (\sim q(x))] \vee [\sim r(x)]\}$

(C) $(\forall x \in S) \{[(\sim p(x)) \wedge (\sim q(x))] \vee [r(x)]\}$

(D) $(\forall x \in S) \{[(\sim p(x)) \vee (\sim q(x))] \wedge [\sim r(x)]\}$

(E) $(\forall x \in S) \{[(\sim p(x)) \vee (\sim q(x))] \wedge [r(x)]\}$

23. Let $x_n = \frac{n^n}{n!}$ for $n = 1, 2, 3, \dots$. Then $\lim_{n \rightarrow +\infty} \frac{x_{n+1}}{x_n}$ equals

(A) $\sqrt{e}$

(D) e^2

(B) e

(E) e^{-1}

(C) $\sqrt{e^3}$

24. The derivative of $f(x) = \int_x^0 \frac{\cos xt}{t} dt$ is

(A) $\frac{1}{x} [1 - 2 \cos x^2]$

(D) $-\frac{\sin 2x^2}{x}$

(B) $-\frac{\cos 2x^2}{x}$

(E) $\frac{\cos x^2}{x}$

(C) $\frac{1}{x} [1 + 2 \sin x^2]$

25. The maximum value of the directional derivative on the surface $z = f(x, y) = xe^{xy} + y \cos x$ at $P(0, 1)$ is

(A) 1

(D) $\sqrt{4}$

(B) $\sqrt{2}$

(E) $\sqrt{5}$

(C) $\sqrt{3}$

26. The number of ordered partitions of the positive integer 5 is

(A) 20 (D) 14

(B) 18 (E) 12

(C) 16

27. The Wronskian of $f_1(x) = x^2 \sin x$ and $f_2(x) = x^2 \cos x$ is

(A) x^2 (D) $-x^4$

(B) $-x^2$ (E) $2x^4$

(C) x^4

28. Given the linear second-order difference equation

$$y_{k+2} - y_{k+1} - 2y_k = 0 ; k = 0, 1, 2, \dots$$

$$y_0 = 9 ; y_1 = -12$$

find y_6 .

(A) -54 (D) 27

(B) 64 (E) 54

(C) -32

29. Which of the following numbers is divisible by 9?

(A) 7224466

(D) 5224466

(B) 9224466

(E) 1224466

(C) 3224466

30. The inflection point for $f(x) = \frac{\ln x}{x}$ occurs at $x =$

(A) $\sqrt{e}$

(D) e^{-1}

(B) e

(E) $\sqrt{e^{-1}}$

(C) $\sqrt{e^3}$

31. The dimension of the null space of

$$M = \begin{bmatrix} 1 & 2 & -1 & 0 \\ 3 & 2 & 0 & 1 \\ 1 & 2 & 0 & 2 \\ -1 & 0 & 1 & 3 \end{bmatrix}$$

is:

(A) 2

(D) 3

(B) 1

(E) 0

(C) 4

32. What is the general solution of $3xy^3dx - (x + 2)dy = 0$?
(where c is a constant)

(A) $3x - 4 \ln|x + 2| + \frac{1}{y^2} = c$

(B) $3x - \ln|x + 2| + \frac{1}{3y^2} = c$

(C) $3x - 5 \ln|x + 2| + \frac{3}{y^2} = c$

(D) $3x - 6 \ln|x + 2| + \frac{1}{2y^2} = c$

(E) $3x - 2 \ln|x + 2| + \frac{2}{y^2} = c$

33. Which of the following is equivalent to $\sin^3 x \cos^2 x$?

(A) $\frac{1}{16} [2 \sin x - \sin 3x - 2 \sin 5x]$

(B) $\frac{1}{16} [\sin x - 2 \sin 3x - \sin 5x]$

(C) $\frac{1}{16} [2 \sin x - \sin 3x - \sin 5x]$

(D) $\frac{1}{16} [2 \sin x + \sin 3x - \sin 5x]$

(E) $\frac{1}{16} [\sin x + \sin 3x - \sin 5x]$

34. The absolute maximum of $f(x) = \cos 2x - 2 \cos x$ on $[0, 2\pi]$ occurs at $x =$

(A) $\frac{\pi}{3}$

(D) $\frac{5\pi}{3}$

(B) $\frac{\pi}{2}$

(E) $\frac{3\pi}{4}$

(C) π

35. The eigenvalue which corresponds to the eigenvector

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix} \text{ for } M = \begin{bmatrix} 1 & -3 \\ -2 & 2 \end{bmatrix} \text{ is}$$

- (A) 1 (D) -4
(B) 4 (E) 2
(C) -1

36. Evaluate the sum $\sum_{n=1}^m \arctan\left(\frac{1}{n^2 + n + 1}\right)$

- (A) $m^2 + 1$
(B) $\frac{1}{m^2 + m}$
(C) $\cot(m + 1) - \frac{1}{m^2 + 1}$
(D) $\arctan(m + 1) - \frac{\pi}{4}$
(E) $(-1)^m \sin(m + 1) + \tan m$

37. The conjugates of an element are the other roots of the irreducible polynomial of which the given element is a root. The conjugates of $\sqrt{\sqrt{3} + 1}$ over the field of rational numbers are

- (A) $\sqrt{\sqrt{3} + 1}, \sqrt{\sqrt{3} - 1}$

(B) $\sqrt{\sqrt{3} + 1}, -\sqrt{\sqrt{3} + 1}$

(C) $\pm \sqrt{1 + \sqrt{3}}, \pm \sqrt{1 - \sqrt{3}}$

(D) $\pm \sqrt{\sqrt{3} + 1}, \pm \sqrt{\sqrt{3} - 1}$

(E) $\sqrt{\sqrt{3} + 1}, -\sqrt{\sqrt{3} - 1}$

38. The smallest positive integer n for which the inequality $2^n > n^2$ is true for $\{n, n + 1, \dots\}$ is

(A) 1

(D) 4

(B) 2

(E) 5

(C) 3

39. Consider the two player (P_1, P_2) game G with payoff matrix

$$P_2 \begin{matrix} \\ P_1 \end{matrix} \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}.$$

The minimax value of G is

(A) $\frac{5}{3}$

(D) 5

(B) 1

(E) 0

(C) $\frac{3}{2}$

40. The first Newton approximation x_1 for a zero of $f(x) = x^3 - 2x$ with initial approximation $x_0 = 2$ is

(A) $\frac{12}{5}$ (D) $\frac{6}{5}$

(B) 2 (E) $\frac{7}{5}$

(C) $\frac{8}{5}$

41. The value of $\int_1^4 |x - 2| dx$ is

(A) 3 (D) $\frac{3}{2}$

(B) $\frac{5}{2}$ (E) $\frac{7}{2}$

(C) 2

42. A Sylow 3-subgroup of a group of order 72 has order

(A) 3 (D) 27

(B) 9 (E) 36

(C) 18

43. Which of the following sets, together with the given binary operation $*$, does not form a group?

Note: Z = integers
 Q = rationals
 R = reals
 C = complex numbers

- (A) $G = \{a + b\sqrt{2} \in R \setminus \{0\} \mid a, b \in Q\}$
*: usual multiplication of real numbers
- (B) $G = \{a + bi\sqrt{2} \in C \setminus \{0\} \mid a, b \in Q\}$
*: usual multiplication of complex numbers
- (C) $G = \{\sqrt[3]{a} \in R \mid a \in Z\}$
*: for $a, b \in G$, $\sqrt[3]{a} * \sqrt[3]{b} = \sqrt[3]{a + b}$
- (D) $G = R \setminus \{0\}$
*: for $a, b \in G$, $a * b = |a| |b|$
- (E) $G = \{z \in C \mid |z| = 1\}$
*: usual multiplication of complex numbers

44. Let $M = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$. Then $M^6 = kM$ for $k =$

- (A) 2^6 (D) 2^{12}
(B) 2^8 (E) 2^{14}
(C) 2^{10}

45. From a group of 15 mathematics graduate school applicants, 10 are selected at random. Let P be the probability that 4 of the 5 applicants who would make the best graduate students are included in the 10 selected. Which of the following statements is true?

(A) $0 \leq P \leq \frac{1}{5}$

(D) $\frac{3}{5} < P \leq \frac{4}{5}$

(B) $\frac{1}{5} < P \leq \frac{2}{5}$

(E) $\frac{4}{5} < P \leq 1$

(C) $\frac{2}{5} < P \leq \frac{3}{5}$

46. The equation $r = 2 \sin \theta - \cos \theta$ in rectangular coordinates is given by

(A) $x^2 + y^2 + x - 2y = 0$

(B) $x^2 - x + 2y = 0$

(C) $x^2 + y^2 + 2x - y = 0$

(D) $x^2 - y^2 - x + 2y = 0$

(E) $y^2 - x^2 - x + 2y = 0$

47. The decimal $2.0259\overline{259}$ is equivalent to which of the following?

(A) $\frac{20237}{9990}$

(B) $\frac{547}{270}$

(C) $\frac{20239}{9999}$

(D) $\frac{747}{370}$

(E) $\frac{737}{380}$

48. The general term of the Maclaurin series for xe^{-x^2} is

(A) $\frac{(-1)^n x^{2n}}{(n+1)!}$

(D) $\frac{(-1)^n x^{2n+1}}{n!}$

(B) $\frac{(-1)^{n+1} x^{2n+1}}{n!}$

(E) $\frac{(-1)^{n+1} x^{2n}}{n!}$

(C) $\frac{(-1)^n x^{2n+1}}{(n+1)!}$

49. The solution set for the inequality $x - \frac{3}{x} > 2$ is given by

(A) $(0, +\infty)$

(D) $(-\infty, 0) \cup (3, +\infty)$

(B) $(3, +\infty)$

(E) $(-\infty, 3)$

(C) $(-1, 0) \cup (3, +\infty)$

50. The volume (in cubic units) generated by rotating the region defined by the curves

$$y = x$$

$$y = 2\sqrt{x}$$

around the x -axis is

(A) $\frac{16\pi}{5}$

(B) $\frac{32\pi}{15}$

(C) $\frac{16\pi}{3}$

(D) $\frac{32\pi}{3}$

(E) π

51. Let A and B be subsets of U and denote the complement of subset X of U by X^c . Find $[(A \cap (A \cap B^c)) \cap B]^c$.

(A) B^c

(D) U

(B) A^c

(E) $\emptyset$

(C) $A \cup B^c$

52. The cross product $\vec{u} \times \vec{v}$ of the vectors

$$\vec{u} = 2\vec{i} - \vec{j} + 3\vec{k}$$

$$\vec{v} = \vec{i} + 2\vec{j} - \vec{k}$$

is given by

(A) $7\vec{i} + \vec{j} + 3\vec{k}$

(D) $-5\vec{i} + 5\vec{j} + 5\vec{k}$

(B) $5\vec{i} + 5\vec{j} + 3\vec{k}$

(E) $7\vec{i} - \vec{j}$

(C) -3

53. Find the number of left cosets of the cyclic subgroup generated by $(1, 1)$ of $Z_2 \times Z_4$ where Z_n denotes the cyclic group of $\{0, 1, 2, \dots, n-1\}$ under addition modulo n .

(A) 1

(D) 6

(B) 2

(E) 8

(C) 4

54. Up to isomorphism, how many abelian groups are there of order 36?

(A) 1

(D) 12

(B) 4

(E) 18

(C) 9

55. If $i = \sqrt{-1}$, then $\sum_{j=0}^{10} (-i)^j$ is

(A) i

(D) $1 + i$

(B) -1

(E) $1 - i$

(C) $-i$

56. The set of gaussian integers, $R = \{a + ib \mid a, b \in \mathbb{Z}(\text{integers})\}$, is a commutative subring of the complex numbers. An element $u = e + id$ in R is a unit of R if there exists $V \in R$ such that $uv = 1$. The unit(s) of R is (are)

- (A) ± 1 (D) $\pm 1, \pm i$
(B) $\pm i$ (E) 1
(C) $1, i$

57. The number of solutions (equivalence classes) of the congruence $3x + 11 \equiv 20 \pmod{12}$ is:

- (A) no solutions (D) 4
(B) 1 (E) 6
(C) 3

58. Let R be the region defined by

$$y = x - 1; \quad x = 1; \quad y = -x + 3$$

Find the maximum value of $f(x, y) = -2x + 3y$ on R .

- (A) -2 (D) 4
(B) 1 (E) -1
(C) 2

59. If $|x|$ is large, then $f(x) = \frac{x^5 - x^4 + x^3 + x}{x^3 - 1}$ is approximately

(A) $x^2 + x$

(D) $x^2 + 1$

(B) $x^2 - x + 1$

(E) $x^2 - x$

(C) x^2

60. The number of vertices of an ordinary polyhedron with 12 faces and 17 edges is

(A) 7

(D) 9

(B) 5

(E) 13

(C) 11

61. Let T be a linear transformation of the plane such that $T(1, 1) = (-1, 1)$ and $T(2, 3) = (1, 2)$. Then $T(2, 4)$ equals

(A) $(4, 2)$

(D) $(2, 4)$

(B) $(2, -4)$

(E) $(-3, 2)$

(C) $(3, -2)$

62. The function $f(z) = \sin x \cosh y + v(x, y)i$ is analytic for $v(x, y)$ equal to

- (A) $\cos x \cosh y$ (D) $\sin x \sinh y$
(B) $\cos x \sinh y$ (E) $\sin y \cosh x$
(C) $-\sin y \cosh x$

63. Let T represent a nonsingular linear transformation from E^n into E^n . Which of the following is not true?

- (A) Null space of $T = \{0\}$
(B) T is one-to-one
(C) Dimension of null space is zero: $\text{Dim } N(T) = 0$
(D) Dimension of range space is n : $\text{Dim } R(T) = n$
(E) $\text{Dim } N(T^{-1}) = \text{Dim } R(T)$

64. Find the value of the sum: $1 + \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \dots$

- (A) $\frac{11}{3}$ (D) 3
(B) $\frac{5}{2}$ (E) ∞
(C) 2

65. The circles

$$c_1: x^2 + y^2 + 2ax + 2by + c = 0$$

$$c_2: x^2 + y^2 + 2a'x + 2b'y + c' = 0$$

are orthogonal if

(A) $2aa' + 2bb' = c + c'$

(B) $a + a' + b + b' = cc'$

(C) $aa' - bb' = c - c'$

(D) $2aa' - 2bb' = c - c'$

(E) $a + b + c = a' + b' + c'$

66. The inverse of the matrix $M = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \\ 4 & 3 & 1 \end{bmatrix}$ is the matrix

$$M^{-1} = \frac{1}{6} \begin{bmatrix} -7 & 8 & a \\ 8 & -10 & -4 \\ 4 & b & -2 \end{bmatrix} \text{ where}$$

(A) $a = 5; b = -2$ (D) $a = 2; b = -3$

(B) $a = 3; b = 2$ (E) $a = 2; b = 3$

(C) $a = 1; b = -3$

GRE MATHEMATICS

TEST I

ANSWER KEY

1.	B	23.	B	45.	B
2.	A	24.	A	46.	A
3.	E	25.	B	47.	B
4.	E	26.	C	48.	D
5.	B	27.	D	49.	C
6.	C	28.	A	50.	D
7.	D	29.	C	51.	D
8.	B	30.	C	52.	D
9.	E	31.	E	53.	B
10.	B	32.	D	54.	B
11.	D	33.	D	55.	C
12.	D	34.	C	56.	D
13.	E	35.	C	57.	C
14.	B	36.	D	58.	D
15.	C	37.	C	59.	B
16.	D	38.	E	60.	A
17.	A	39.	A	61.	A
18.	D	40.	C	62.	B
19.	A	41.	B	63.	E
20.	D	42.	B	64.	D
21.	C	43.	D	65.	A
22.	B	44.	C	66.	A

GRE MATHEMATICS TEST I

DETAILED EXPLANATIONS OF ANSWERS

1. (B)

Whereas the fundamental identity for the trigonometric functions is $\sin^2 x + \cos^2 x = 1$, the fundamental identity for the inverse trigonometric functions is $\arcsin x + \arccos x = \pi/2$. Thus $\arccos x = \pi/2 - \arcsin x$. The curve of $\arcsin x$ reflected in the horizontal axis will represent the curve of $-\arcsin x$. Adding $\pi/2$ is geometrically equivalent to translating the curve vertically $\pi/2$ units upward.

2. (A)

Since $f'(x) = e^x + e^{-x}$, we have

$$\begin{aligned}[f'(x)]^2 - [f(x)]^2 &= [e^x + e^{-x}]^2 - [e^x - e^{-x}]^2 \\ &= e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x} \\ &= 4\end{aligned}$$

Using the identities

$$f(x) = 2 \sinh x, \quad f'(x) = 2 \cosh x, \quad \text{and} \quad \cosh^2 x - \sinh^2 x = 1,$$

we also have

$$\begin{aligned}[f'(x)]^2 - [f(x)]^2 &= [2 \cosh x]^2 - [2 \sinh x]^2 \\ &= 4.\end{aligned}$$

3. (E)

The numerator is defined for all real numbers since it contains a cube root. The denominator is defined and nonzero except for $x = 6$. So, the domain is $\mathbb{R} \setminus \{6\}$.

4. (E)

The determinant of M , denoted $|M|$, is given by

$$|M| = \begin{vmatrix} 1 & 2 \\ 3 & 9 \end{vmatrix} = 3.$$

For an $n \times n$ nonsingular matrix M , we have $M^{-1} = |M|^{-1} \text{adj}(M)$ and $|M^{-1}| = |M|^{-1}$ so that

$$|M^{-1}| = | |M|^{-1} \text{adj}(M) |$$

$$|M|^{-1} = (|M|^{-1})^n |\text{adj}(M)|$$

$$|\text{adj}(M)| = |M|^{n-1}$$

Since $|M| = 3$, $|\text{adj}(M)| = |M|^{2-1} = 3$.

5. (B)

Let g be any generator of a cyclic group G of order 8. The generators of G are of the form g^r where r is relatively prime to 8 (that is, the greatest common divisor of r and 8 is 1). The positive integers less than 8 and relatively prime to 8 are 1, 3, 5, 7. Therefore the four generators of G are g^1 , g^3 , g^5 , and g^7 .

6. (C)

An integrating factor for an ordinary differential equation of the form $M dx + N dy = 0$ is a function of the form $J(x, y)$ such that $JM dx + JN dy = 0$ is exact, that is,

$$\frac{\partial(JM)}{\partial y} = \frac{\partial(JN)}{\partial x}.$$

The given equation is not exact since

$$\begin{aligned}\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} &= \frac{-2}{x} - 2xy \cos y \\ &= \frac{-2}{x} (1 + x^2 y \cos y) \\ &\neq 0\end{aligned}$$

but since $r(x) = \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{-2}{x}$ is a function of x , an integrating factor is given by

$$\begin{aligned}J(x) &= e^{\int r(x) dx} \\ &= e^{-\int \frac{2}{x} dx} \\ &= e^{-2 \ln x} \\ &= \frac{1}{x^2}.\end{aligned}$$

7. (D)

We have $x = \sqrt{3+x}$ or $x^2 = 3+x$ so that $x^2 - x - 3 = 0$.

Using the quadratic formula, we obtain $x = \frac{1 \pm \sqrt{13}}{2}$.

We choose the positive square root, obtaining $x = \frac{1 + \sqrt{13}}{2}$ since x is positive.

8. (B)

In order for $f(x)$ to be a valid probability density function

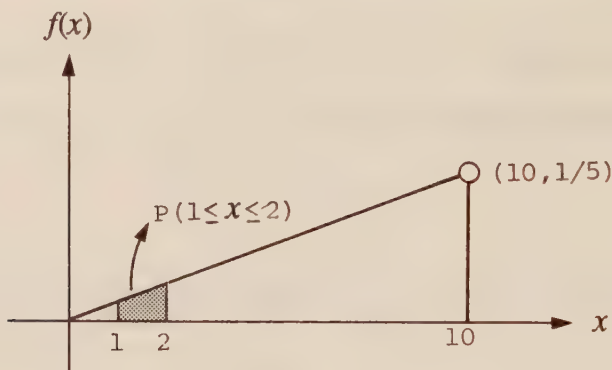
$$\int_{-\infty}^{+\infty} f(x) dx = 1.$$

Thus, $1 = \int_0^{10} cx dx = 50c$ so that $c = \frac{1}{50}$.

The probability that x is in $[1, 2]$, denoted $P(1 \leq x \leq 2)$, is given by

$$P(1 \leq x \leq 2) = \int_1^2 \frac{x}{50} dx = \frac{1}{50} \left[\frac{x^2}{2} \right]_1^2 = \frac{3}{100}.$$

The graph of the probability density function is shown below.



9. (E)

The expected value of a function $g(x)$ of a random variable X , denoted $E[g(X)]$, is defined by

$$E[g(X)] = \int_{-\infty}^{+\infty} g(x) f(x) dx$$

where $f(x)$ is the probability function for X . We have

$$\begin{aligned} E[X^2] &= \int_{-\infty}^{\infty} X^2 f(x) dx = \int_0^2 X^2 \left(1 - \frac{x}{2}\right) dx \\ &= \left[\frac{x^3}{3} - \frac{x^4}{8} \right]_0^2 \\ &= \frac{2}{3}. \end{aligned}$$

10. (B)

An algebraic equation is of the form

$$a_N x^N + a_{N-1} x^{N-1} + \dots + a_1 x + a_0 = 0$$

where N is a positive integer and $a_N, \dots, a_0$ are integers. ($a_N \neq 0$). The height h is defined by $h = N + |a_N| + \dots + |a_0|$. For $h = 2$, we have $N = 1, |a_1| = 1$, and $|a_0| = 0$. This implies that $a_1 = \pm 1$ so that $x = 0$. Note that there are no other possibilities except for $N = 1$. We see that if $N = 0$ then the equation is of degree 0 and is of the form $a_0 = 0$, which is unacceptable, as we require $a_N \neq 0$. If $N = 2$, on the other hand, we obtain $h = 2 = N + 0 + 0 + \dots + 0$, so $a_2 = 0 = a_1 = a_0$ which is again unacceptable.

11. (D)

A point (x, y) is in the unit ball if and only if $|x| + |y| < 1$. The boundary consists of the following lines:

a) $x \geq 0; y \geq 0: x + y = 1$

b) $x \geq 0; y \leq 0: x - y = 1$

c) $x \leq 0; y \geq 0: -x + y = 1$

d) $x \leq 0; y \leq 0: -x - y = 1$

These lines intersect at $(-1, 0)$, $(0, 1)$, $(1, 0)$, and $(0, -1)$.

12. (D)

Trying the given solutions:

$$-f(x^{-1}) = -\left(\frac{\frac{1}{x} + 1}{\frac{1}{x} - 1}\right) = \frac{x + 1}{x - 1} = f(x)$$

$$\text{and } [f(-x)]^{-1} = \left[\frac{-x+1}{-x-1} \right]^{-1} = \frac{-x-1}{-x+1} = \frac{x+1}{x-1} = f(x).$$

But if we solve $y = f(x) = \frac{x+1}{x-1}$ for x :

$$y(x-1) = x+1$$

$$xy - x = y + 1$$

$$x = \frac{y+1}{y-1}$$

Thus $f^{-1}(x) = f(x)$. However,

$$\begin{aligned} f^{-1}(x^{-1}) &= \frac{x^{-1}+1}{x^{-1}-1} \\ &= \frac{1+x}{1-x} \\ &= -f(x) \end{aligned}$$

13. (E)

Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of an $n \times n$ matrix M . The trace of M , denoted $tr(M)$, equals $\sum_{k=1}^n \lambda_k$. If p is a positive integer, then $tr(M^p)$ equals $\sum_{k=1}^n \lambda_k^p$. The eigenvalues of M are given by

$$\begin{vmatrix} 6-\lambda & 10 \\ -2 & -3-\lambda \end{vmatrix} = 0$$

which implies

$$-18 - 6\lambda + 3\lambda + \lambda^2 + 20 = 0$$

$$(\lambda - 1)(\lambda - 2) = 0$$

$$\lambda = 1, 2$$

Thus $tr(M^5) = 1^5 + 2^5 = 33$.

14. (B)

The Cayley-Hamilton theorem guarantees that M will satisfy a polynomial of degree 8. However, since M is idempotent $M^2 = M$ or $M^2 - M = 0$. Thus M satisfies $p(x) = x^2 - x$. Assuming that M satisfies a linear polynomial (degree 1) equation $\hat{p}(x) = a_1x + a_0 = 0$ implies that M is a scalar matrix, i.e.,

$$a_1 M + a_0 I_8 = 0$$

$$M = -\frac{a_0}{a_1} I_8,$$

which contradicts the fact that M is nonscalar.

15. (C)

The correct matrix must have 4 rows and 5 columns. The number of rows corresponds to the number of vertices, denoted by x_1, x_2, x_3 , and x_4 . The number of columns corresponds to the number of edges, denoted by e_1, e_2, e_3, e_4 , and e_5 . Each of the $(4)(5) = 20$ entries must be 0 or 1. The number 1 is used if the indicated vertex and edge are connected. Otherwise, the number 0 is used. The actual headings will appear as follows:

$$\begin{array}{c} x_1 \\ x_2 \\ x_3 \\ x_4 \end{array} \begin{bmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

Consider vertex x_1 . We note that it is connected to edges e_1 and e_4 . Thus, the number 1 is placed in each of the first and fourth columns. The number 0 is placed in the other three columns. Consider vertex x_4 . We note that it is connected to edges e_3, e_4 , and e_5 . Thus, the number 1 is placed in each of the third, fourth, and fifth columns. The number 0 is placed in the other two columns. Each of the rows containing x_2 and x_3 are filled in the same way.

16. (D)

A fixed point z must satisfy $z = w(z) = \frac{z-2}{z-1}$ which implies $z^2 - z = z - 2$. The solutions of $z^2 - 2z + 2 = 0$ are given by

$$z = \frac{2 \pm \sqrt{4 - 4(1)(2)}}{2} = 1 \pm i.$$

17. (A)

The n^{th} roots of unity are solutions of the polynomial equation $x^n - 1 = 0$. For $n > 1$ and a leading coefficient of 1, the sum of the roots of a polynomial is equal to the negative of the coefficient of the x^{n-1} term. Hence the sum of the n^{th} roots of unity ($n > 1$) is zero. For another demonstration of this fact, let $1, \alpha, \dots, \alpha^{n-1}$ denote the n^{th} roots of unity ($n > 1$). Then

$$1 + \alpha + \dots + \alpha^{n-1} = \frac{1 - \alpha^n}{1 - \alpha} = 0 \text{ since } \alpha^n = 1.$$

18. (D)

A Boolean variable is a variable whose value can be either 0 or 1. A Boolean function is a function whose variables (both independent and dependent) are Boolean variables. The function $f(x, y, z) = x + y + z - xy - yz$ is not a Boolean function since $f(1, 0, 1) = 2$.

19. (A)

The perimeter $P(x, y)$ for the rectangle shown in the figure below is $P(x, y) = 4x + 4y$. We want to maximize this quantity subject to the constraint

$$\varnothing(x, y) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0.$$

Introducing a Lagrange multiplier λ , we must maximize $L(x, y) = P(x, y) + \lambda \varnothing(x, y)$. Taking the first order partial derivatives of $L(x, y)$ and setting them equal to zero yields

$$\frac{\partial L}{\partial x} = 4 + \lambda \frac{2x}{a^2} = 0$$

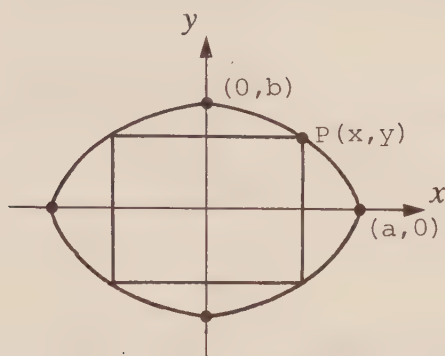
$$\frac{\partial L}{\partial y} = 4 + \lambda \frac{2y}{b^2} = 0.$$

Thus $\frac{2x}{a^2} = \frac{2y}{b^2}$. Substituting $y = \frac{b^2 x}{a^2}$ into $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

yields $\frac{x^2}{a^2} + \frac{b^4 x^2}{a^4 b^2} = 1$, so that $x = \frac{a^2}{\sqrt{a^2 + b^2}}$ and

$$y = \frac{b^2}{\sqrt{a^2 + b^2}}.$$

The maximum perimeter is $4\sqrt{a^2 + b^2}$.



20. (D)

A one-to-one bicontinuous function h from a topological space (X, τ) onto a topological space (X', τ') is called a homeomorphism. A property P of sets of a topological space (X, τ) is called topological if it is invariant under homeomorphisms. To see that the property of being an accumulation point is topological, let x be an accumulation point of a set $S \subseteq X$ and consider $h(x) \in h(S) \subseteq X'$. Let $D_{x'}$ be an open set in X' containing $h(x)$. Then $D_x = h^{-1}(D_{x'})$ is an open set in X containing x . Since x is an accumulation point of S , there exists a $\bar{x} \in S \cap D_x$ such that $x \neq \bar{x}$. Thus $h(\bar{x}) \in h(S) \cap D_{x'}$, distinct from $h(x)$, which implies that $h(x)$ is an accumulation point of $h(S)$.

21. (C)

According to Cauchy's integral formula we have

$$f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-a} dz,$$

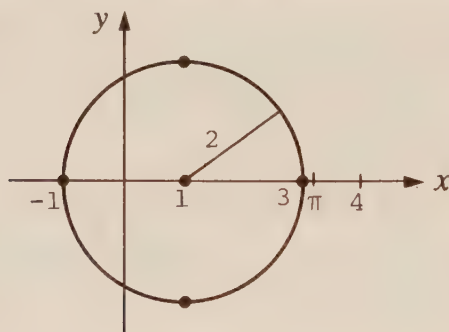
where C is a closed contour which includes a , and f is analytic within and on C . Now, if we let

$$f(z) = \frac{(\cos z)}{(z - \pi)},$$

and $a = 0$, we will have

$$\begin{aligned} f(a) = f(0) &= \frac{1}{2\pi i} \oint_C \frac{f(z)}{z} dz \\ &= \frac{1}{2\pi i} \oint_C \frac{\cos z}{z - \pi} dz \\ &= \frac{\cos 0}{0 - \pi} \\ &= \frac{1}{-\pi} \\ \Rightarrow \oint_C \frac{\cos z}{z - \pi} dz &= \frac{-1}{\pi} (2\pi i) \\ &= -2i \end{aligned}$$

Note that we can use the Cauchy integral formula for the circle $C : |z - 1| = 2$ since it does not include the point $z = \pi$.



22. (B)

Propositional algebra is similar to set algebra with union ($\cup$) replaced with disjunction ($\vee$), intersection ($\cap$) replaced with conjunction ($\wedge$), complementation (c) replaced with negation ($\sim$), the universal set (Ω) replaced with tautology (1), and the empty set ($\emptyset$) replaced with absurdity (0). Also,

$$\sim(\forall x \in S)w(x) \Leftrightarrow (\exists x \in S) [\sim w(x)] \text{ and}$$

$$\sim(\exists x \in S)w(x) \Leftrightarrow (\forall x \in S) [\sim w(x)]$$

for an open statement $w(x)$ relative to S . We have

$$\sim(\exists x \in S) [(p(x) \vee q(x)) \wedge r(x)]$$

$$\Leftrightarrow (\forall x \in S) \{ \sim [(p(x) \vee q(x)) \wedge r(x)] \}$$

$$\Leftrightarrow (\forall x \in S) \{ [\sim(p(x) \vee q(x))] \vee [\sim r(x)] \}$$

$$\Leftrightarrow (\forall x \in S) \{ [(\sim p(x)) \wedge (\sim q(x))] \vee [\sim r(x)] \}$$

23. (B)

We have

$$\begin{aligned} \lim_{n \rightarrow +\infty} \frac{x_{n+1}}{x_n} &= \lim_{n \rightarrow +\infty} \frac{(n+1)^{n+1} n!}{(n+1)! n^n} \\ &= \lim_{n \rightarrow +\infty} \frac{(n+1)^n (n+1) n!}{n^n (n+1)!} \\ &= \lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n \\ &= e \end{aligned}$$

24. (A)

If

$$f(x) = \int_x^0 \frac{\cos xt}{t} dt = - \int_0^x \frac{\cos xt}{t} dt$$

Then

$$f'(x) = -\frac{\cos x^2}{x} - \int_0^x -\frac{t \sin xt}{t} dt$$

$$\begin{aligned}
&= -\frac{\cos x^2}{x} - \left[\frac{\cos x^2}{x} - \frac{1}{x} \right] \\
&= -\frac{\cos x^2}{x} - \frac{\cos x^2}{x} + \frac{1}{x} \\
&= \frac{-2\cos x^2}{x} + \frac{1}{x} \\
&= \frac{1}{x} [1 - 2\cos x^2]
\end{aligned}$$

25. (B)

The directional derivative of the function $P(x, y)$ along the path through (x, y) and parallel to the unit vector $\vec{u}$ is equal to $\frac{\partial f}{\partial u} = \nabla P \cdot \vec{u}$. Therefore, since the values $|\nabla P|$ and $|\vec{u}| = 1$ are constant, the directional derivative of $\frac{\partial f}{\partial u}$ is maximized when the angle between ∇P and $\vec{u}$ is zero. Therefore, the maximum value of the directional derivative at $P(x, y)$ equals the magnitude of the gradient

$$\nabla f(x, y) = \frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j$$

at $P(x, y)$. We have

$$\frac{\partial f}{\partial x} = xye^{xy} + e^{xy} - y \sin x \quad ; \quad \frac{\partial f(0,1)}{\partial x} = 1$$

$$\frac{\partial f}{\partial y} = x^2 e^{xy} + \cos x \quad ; \quad \frac{\partial f(0,1)}{\partial y} = 1$$

Thus $\nabla f(0,1) = \vec{i} + \vec{j}$ so that $|\nabla f(0,1)| = \sqrt{2}$.

26. (C)

Let n and k represent positive integers with k satisfying $1 \leq k \leq n$. An ordered partition of n into k parts is a decomposition of n into the sum of k positive integers:

$$n = x_1 + x_2 + x_3 + \dots + x_k.$$

For a fixed k , the number of ordered partitions of n into k parts is the number of distinct ways of placing $\bar{k} = k - 1$ identical markers in the $\bar{n} = n - 1$ spaces between a row of n ones. The first marker can be placed in $\bar{n}$ ways, the second in $\bar{n} - 1$ ways, ..., and the last marker in $\bar{n} - [\bar{k} - 1]$ ways. The number of ways is therefore $\bar{n}(\bar{n} - 1) \dots (\bar{n} - \bar{k} + 1)$.

Since the markers are identical, in each way of placing them we can switch the markers around without changing the chosen spaces and still have the same way of placing. Therefore, each group of $k!$ ways of placing the k markers, as long as the spaces chosen are still the same, are actually identical. Hence we must divide the above result by $k!$ to arrive at the correct answer. That is, the number of distinct ways is:

$$\begin{aligned} \frac{\bar{n}(\bar{n} - 1) \dots (\bar{n} - [\bar{k} - 1])}{k!} &= \frac{\bar{n}!}{k!(\bar{n} - \bar{k})!} \\ &= \binom{\bar{n}}{k}, \end{aligned}$$

which is the binomial coefficient. Since we can have ordered partitions of 5 into 1, 2, 3, 4 or 5 parts (that is, we can decompose 5 as a sum of 1, 2, 3, 4 or 5 positive integers), the number of ordered partitions of 5 is

$$\sum_{k=0}^4 \left[\frac{4}{k} \right] = (1+1)^4 = 2^4 = 16.$$

27. (D)

The Wronskian is given by

$$\begin{aligned} W(x) &= \begin{vmatrix} f_1(x) & f_2(x) \\ f_1'(x) & f_2'(x) \end{vmatrix} \\ &= \begin{vmatrix} x^2 \sin x & x^2 \cos x \\ 2x \sin x + x^2 \cos x & 2x \cos x - x^2 \sin x \end{vmatrix} \\ &= 2x^3 \sin x \cos x - x^4 \sin^2 x - 2x^3 \sin x \cos x - x^4 \cos^2 x \\ &= -x^4. \end{aligned}$$

28. (A)

Assuming a solution of the form $y_k = r^k$, we obtain

$$r^{k+2} - r^{k+1} - 2r^k = 0$$

$$r^k(r-2)(r+1) = 0$$

so that $r = -1, 2$. Hence

$$y_k = c_1(-1)^k + c_2 2^k$$

and since $y_0 = 9$ and $y_1 = -12$, we have

$$c_1 + c_2 = 9$$

$$-c_1 + 2c_2 = -12$$

This implies $c_1 = 10$ and $c_2 = -1$ which implies $y_k = 10(-1)^k - 2^k$. For $k=6$, we obtain $y_6 = -54$. Note that in the general form of a second order homogeneous recursion equation, we have $y_{k+2} + Ay_{k+1} + By_k = 0$. The general form of the solution of this equation is $c_1 r_1^k + c_2 r_2^k = y_k$. To evaluate r_1 and r_2 , we can insert the special

solution r^k directly into the equation and get:

$$\begin{aligned} r^{k+2} + Ar^{k+1} + Br^k &= 0 \\ \Rightarrow r^k(r^2 + Ar + B) &= 0 \\ \Rightarrow r^2 + Ar + B &= 0. \end{aligned}$$

We observe that if this quadratic equation has 2 distinct roots, r_1 and r_2 , then any expression of the form $c_1 r_1^k + c_2 r_2^k$ will be a solution to the equation, where c_1, c_2 are arbitrary constants, because of the linearity of the equation.

29. (C)

Let N represent a positive integer and write N as

$$N = u_0 + 10u_1 + 10^2 u_2 + 10^3 u_3 + \dots + 10^n u_n.$$

Recalling that $a \equiv b \pmod{c}$ means that $a - b$ is divisible by c , we have that $u_j 10^j \equiv u_j \pmod{9}$ for $0 \leq j \leq n$. Thus $N \equiv u_0 + u_1 + \dots + u_n \pmod{9}$ which implies $N - (\text{sum of digits of } N)$ is divisible by 9. Therefore N is divisible by 9 if and only if the sum of its digits is divisible by 9. Since $(3 + 2 + 2 + 4 + 4 + 6 + 6)/9 = 3$, 3224466 is divisible by 9.

30. (C)

The possible inflection points of f occur where $f''(x) = 0$ or where $f''(x)$ does not exist. We have

$$\begin{aligned} f'(x) &= \frac{x \left(\frac{1}{x} \right) - \ln x}{x^2} = \frac{1 - \ln x}{x^2} \\ f''(x) &= \frac{x^2 \left(\frac{-1}{x} \right) - 2x(1 - \ln x)}{x^4} = \frac{2 \ln x - 3}{x^3} \end{aligned}$$

Setting $f''(x) = 0$ implies that $x = e^{3/2}$. There are no elements in the domain of $f(x)$ such that $f''(x)$ does not exist. The function f is concave downward on $(0, e^{3/2}]$ and concave upward on $[e^{3/2}, +\infty)$. Thus, there is an inflection point at $x = e^{3/2}$.

31. (E)

The dimension of the null space of

$$M = \begin{bmatrix} 1 & 2 & -1 & 0 \\ 3 & 2 & 0 & 1 \\ 1 & 2 & 0 & 2 \\ -1 & 0 & 1 & 3 \end{bmatrix}$$

(E) 0 not (B) 1.

Using elementary row operations:

$$\begin{aligned} M &\xrightarrow{-3R_1 + R_2} \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 4 & 3 & 1 \\ 1 & 2 & 0 & 2 \\ -1 & 0 & 1 & 3 \end{bmatrix} \xrightarrow{R_1 + R_4} \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & -4 & 3 & 1 \\ 1 & 2 & 0 & 2 \\ 0 & 2 & 0 & 3 \end{bmatrix} = \\ &\xrightarrow{-1R_1 + R_3} \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & -4 & 3 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 2 & 0 & 3 \end{bmatrix} \xrightarrow{\frac{1}{2}R_2} \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & -2 & \frac{3}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 2 \\ 0 & 2 & 0 & 3 \end{bmatrix} = \\ &\xrightarrow{\begin{matrix} R_2 + R \\ R_2 + R_{41} \end{matrix}} \begin{bmatrix} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -2 & \frac{3}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & \frac{3}{2} & \frac{7}{2} \end{bmatrix} \xrightarrow{2R_4} \begin{bmatrix} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -2 & \frac{3}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 7 \end{bmatrix} = \\ &\xrightarrow{13R_3} \begin{bmatrix} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -2 & \frac{3}{2} & \frac{1}{2} \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \Rightarrow \end{aligned}$$

$$\Rightarrow \text{Rank}(M) = 4 \Rightarrow \text{Null}(M) = \phi$$

Dimension null space is zero.

32. (D)

By separating variables,

$$\frac{3x \, dx}{x+2} - \frac{dy}{y^3} = 0$$

$$\int \left(\frac{3x \, dx}{x+2} - \frac{dy}{y^3} \right) = \int 0 = c$$

$$\int \left(3 - \frac{6}{x+2} \right) dx - \int \frac{1}{y^3} dy = c$$

$$3x - 6 \ln|x+2| + \frac{1}{2y^2} = c$$

33. (D)

Using the identities $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$ and $\cos x = \frac{e^{ix} + e^{-ix}}{2}$, we obtain

$$\begin{aligned} \sin^3 x \cos^2 x &= \left[\frac{e^{ix} - e^{-ix}}{2i} \right]^3 \left[\frac{e^{ix} + e^{-ix}}{2} \right]^2 \\ &= \frac{1}{-32i} \left[e^{2xi} - e^{-2xi} \right]^2 \left[e^{xi} - e^{-xi} \right] \\ &= \frac{1}{-32i} \left[e^{4xi} - 2 + e^{-4xi} \right] \left[e^{xi} - e^{-xi} \right] \\ &= \frac{1}{-32i} \left[e^{5xi} - 2e^{xi} + e^{-3xi} - e^{3xi} + 2e^{-xi} - e^{-5xi} \right] \\ &= \frac{1}{16} [2 \sin x + \sin 3x - \sin 5x] \end{aligned}$$

34. (C)

Since f is continuous on $[0, 2\pi]$, an absolute maximum exists. It must occur at the endpoints $[(0, -1); (2\pi, -1)]$ or at an interior point

where $f'(x) = 0$. Setting $f'(x) = -2 \sin 2x + 2 \sin x$ equal to zero implies $\sin x (1 - 2 \cos x) = 0$, so that $x = \pi/3, \pi, 5\pi/3$. Since $f(\pi/3) = -3/2, f(\pi) = 3, f(5\pi/3) = -3/2, f(0) = -1$ and $f(2\pi) = -1$, the absolute maximum of f occurs at $x = \pi$.

35. (C)

A number λ is called an eigenvalue for a matrix M if there exists a nonzero vector X such that $MX = \lambda X$. This implies that the determinant $|\lambda I - M| = 0$. Thus

$$\begin{vmatrix} \lambda - 1 & 3 \\ 2 & \lambda - 2 \end{vmatrix} = 0$$

$$\lambda^2 - 3\lambda - 4 = 0; \lambda = -1, 4$$

For $\lambda = -1$, $[\lambda I - M]X = 0$ yields

$$\begin{bmatrix} -2x_1 & + & 3x_2 \\ 2x_1 & - & 3x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

so that $2x_1 - 3x_2 = 0$. Therefore $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ is an eigenvector for $\lambda = -1$.

The solution also follows from the fact that $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ is an eigenvector:

$$\lambda \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 & -3 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = - \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

which implies $\lambda = -1$.

36. (D)

We can rearrange each term in the sum as follows:

$$\frac{1}{n^2 + n + 1} = \frac{(n+1) - n}{1 + n(n+1)}.$$

Now if we define $\tan a_n = n$, then we will have

$$\frac{1}{n^2 + n + 1} = \frac{\tan a_{n+1} - \tan a_n}{1 + \tan a_n \tan a_{n+1}} = \tan (a_{n+1} - a_n).$$

$$\text{So } \arctan \left(\frac{1}{n^2 + n + 1} \right) = \arctan (\tan (a_{n+1} - a_n)) = a_{n+1} - a_n$$

$$\begin{aligned} \text{So } \sum_{n=1}^m \arctan \left(\frac{1}{n^2 + n + 1} \right) &= \sum_{n=1}^m (a_{n+1} - a_n) \\ &= (a_2 - a_1) + (a_3 - a_2) + \dots + (a_{m+1} - a_m) \\ &= a_{m+1} - a_1 \\ &= \arctan (m + 1) - \arctan 1 \\ &= \arctan (m + 1) - \frac{\pi}{4} \end{aligned}$$

37. (C)

The conjugates of $\sqrt{\sqrt{3} + 1}$ consist of the set of all zeros of the irreducible polynomial of $\sqrt{\sqrt{3} + 1}$ over the rational numbers. We first determine a polynomial over the rationals for which $\sqrt{\sqrt{3} + 1}$ is a zero:

$$\begin{aligned} x &= \sqrt{\sqrt{3} + 1} \\ x^2 &= \sqrt{3} + 1 \\ p(x) &= x^4 - 2x^2 - 2 \end{aligned}$$

We now use the Eisenstein test to establish the irreducibility of $p(x) = x^4 + 2x^2 - 2$ over the rational numbers. Firstly, $p(x)$ is an element of the set of polynomials with integer coefficients. Secondly, $a_2 = 1 \not\equiv 0 \pmod{2}$, $a_1 = -2 \equiv 0 \pmod{2}$, $a_0 = -2 \not\equiv 0 \pmod{2^2}$. Therefore $p(x)$ is irreducible over the rational numbers. The conjugates of

$\sqrt{\sqrt{3} + 1}$ are the zeros of $p(x)$:

$$x^4 - 2x^2 - 2 = 0$$

$$x^2 = \frac{2 \pm \sqrt{4+8}}{2} = 1 \pm \sqrt{3}$$

$$x = \pm \sqrt{1 + \sqrt{3}}, \pm \sqrt{1 - \sqrt{3}}.$$

38. (E)

We have

$$2^1 > 1^2 \text{ is true}$$

$$2^2 > 2^2 \text{ is false}$$

$$2^3 > 3^2 \text{ is false}$$

$$2^4 > 4^2 \text{ is false}$$

$$2^5 > 5^2 \text{ is true}$$

The inequality is true for $n = 5$. Assume $2^n > n^2$ for $n = k$. We will show that $2^{k+1} > (k+1)^2$. Consider $f(x) = 2^x - 2x - 1$ so that $f'(x) = 2^x \ln 2 - 2$. We have $f(5) > 0$ and $f'(x) > 0$ for $x \in [5, +\infty)$ which implies $2^k > 2k + 1$ for $k \geq 5$. Since $2^k > k^2$, we have

$$2^k + 2^k > k^2 + 2k + 1$$

$$2^{k+1} > (k+1)^2.$$

39. (A)

For a nonstrictly determined two player (P_1, P_2) game G with payoff matrix M

$$P_2 \begin{matrix} \\ P_1 \end{matrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},$$

the minimax value v is given by $v = \frac{|M|}{d}$ where $d = (a_{11} + a_{22}) - (a_{12} + a_{21})$.

Thus $v = \frac{5}{(1+3) - (-1+2)} = \frac{5}{3}$. This game favors player P_1 to the extent that it will, on the average, pay him $5/3$ units/game.

40. (C)

The first Newton approximation can be obtained from the formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

with $n = 0$. Since $f'(x) = 3x^2 - 2$, we have

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2 - \frac{4}{10} = \frac{8}{5}.$$

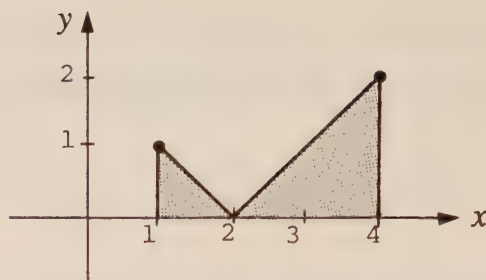
41. (B)

$$\text{Note that } |x - 2| = \begin{cases} -(x - 2) & \text{if } x \leq 2 \\ (x - 2) & \text{if } x > 2 \end{cases}$$

Therefore

$$\begin{aligned} \int_1^4 |x - 2| dx &= - \int_1^2 (x - 2) dx + \int_2^4 (x - 2) dx \\ &= - \left[\frac{x^2}{2} - 2x \right]_1^2 + \left[\frac{x^2}{2} - 2x \right]_2^4 \\ &= - \left[(2 - 4) - \left(\frac{1}{2} - 2 \right) \right] + [(8 - 8) - (2 - 4)] \\ &= \frac{5}{2}. \end{aligned}$$

The graph of $|x - 2|$ on $[1, 4]$ is shown following.



42. (B)

For a group of order n , a Sylow p -subgroup has order p^k where k is the largest positive integer such that p^k divides n . Since $72 = 2^3 \cdot 3^2$, a Sylow 3-subgroup has 9 elements.

43. (D)

A set G , together with a binary operation $*$ is called a group, denoted $(G, *)$, if

- 1) the binary operation $*$ is associative; $f, g, h \in G$ implies $(f * g) * h = f * (g * h)$,
- 2) G contains an identity element: there exists $e \in G$ such that $e * g = g * e = g$ for all $g \in G$.
- 3) Each element of G has an inverse: if $g \in G$, there exists $g' \in G$ such that $g * g' = g' * g = e$.

The set $G = \mathbb{R} \setminus \{0\}$ together with the binary operation $a * b = |a|b$ does not form a group. The number 1 is a “right” identity element, but it is not a “left” identity element:

$$1) \quad 1 * b = b \text{ for all } b \in G$$

$$2) \quad a * 1 \neq a \text{ for } a \in G \text{ when } a < 0$$

44. (C)

We have

$$M^2 = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}^2 = \begin{bmatrix} 8 & 16 \\ 4 & 8 \end{bmatrix} = 4M$$

$$\text{so that } M^6 = (4M)^3 = 4^3 M^2 M = 4^4 M M = 4^5 M.$$

45. (B)

The total number of selections possible is the number of ways of selecting 10 graduate students from 15 applicants which is $\begin{bmatrix} 15 \\ 10 \end{bmatrix}$. Since the selection process was random, the probability of any selection is $\frac{1}{\begin{bmatrix} 15 \\ 10 \end{bmatrix}}$. We must determine the number of selections

which include 4 of the 5 "best students." Firstly, 4 of the possible 10 people selected must be selected from the 5 "best students." This can be accomplished in $\begin{bmatrix} 5 \\ 4 \end{bmatrix}$ ways. The other $10 - 4 = 6$ people must come from the other $15 - 5 = 10$ applicants. This can be accomplished in $\begin{bmatrix} 10 \\ 6 \end{bmatrix}$ ways. Thus, there are $\begin{bmatrix} 5 \\ 4 \end{bmatrix} \begin{bmatrix} 10 \\ 6 \end{bmatrix}$ ways of selecting 4 of the 5 "best students." Hence, the probability of selecting 4 of the 5 "best students" is

$$\frac{\begin{bmatrix} 5 \\ 4 \end{bmatrix} \begin{bmatrix} 10 \\ 6 \end{bmatrix}}{\begin{bmatrix} 15 \\ 10 \end{bmatrix}} = \frac{5!}{4! 1!} \frac{10!}{6! 4!} \frac{10! 5!}{15!} = \frac{50}{143}$$

46. (A)

Since $r^2 = x^2 + y^2$, $\sin \theta = y/r$, and $\cos \theta = x/r$, we have

$$r = 2y/r - x/r$$

$$r^2 = 2y - x$$

$$x^2 + y^2 + x - 2y = 0$$

47. (B)

Set $x = 0.0259\overline{259}$. Then $9990x = 10000x - 10x = 259.\overline{259} - 0.\overline{259}$ so that

$$x = \frac{259}{9990} = \frac{7}{270}. \text{ Thus } 2.0259\overline{259} = \frac{547}{270}.$$

48. (D)

The Maclaurin series for e^x is $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots$

Therefore

$$xe^{-x^2} = x \left[1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \frac{x^8}{4!} + \dots + \frac{(-1)^n x^{2n}}{n!} + \dots \right],$$

so that the general term is given by $\frac{(-1)^n x^{2n+1}}{n!}$.

49. (C)

The solution set for the inequality is equivalent to finding where the function

$$f(x) = x - \frac{3}{x} - 2 = \frac{(x-3)(x+1)}{x}$$

is positive. The figure below shows the x -axis subdivided into regions where f is continuous and never zero; we always omit endpoints. Thus f has the same sign throughout each subinterval; the signs are shown

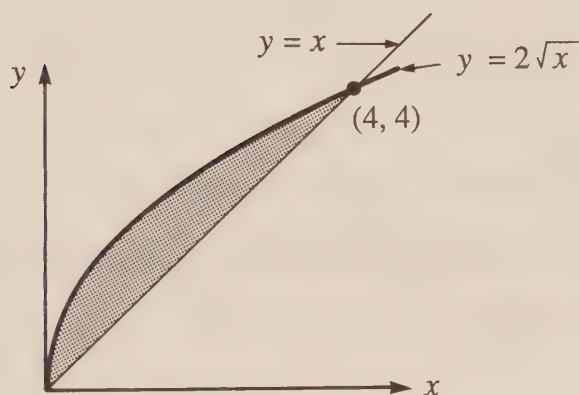
below. The solution of the inequality is $(-1, 0) \cup (3, +\infty)$.



50. (D)

The x -coordinates of the points of intersection of the curves are solutions of $x = 2\sqrt{x}$ which implies $x = 0, 4$. The region is shown below. The volume generated is given by

$$\begin{aligned} \text{Volume} &= \pi \int_0^4 [(2\sqrt{x})^2 - (x)^2] dx \\ &= \pi \left[2x^2 - \frac{x^3}{3} \right]_0^4 \\ &= \frac{32\pi}{3}. \end{aligned}$$



51. (D)

Using the associative law for intersection, we can write:

$$\begin{aligned} [A \cap (A \cap B^C)] \cap B]^C &= [A \cap A \cap B \cap B^C]^C \\ &= [(A \cap A) \cap (B \cap B^C)]^C \\ &= [A \cap (\emptyset)]^C \\ &= \emptyset^C \\ &= U. \end{aligned}$$

52. (D)

The cross product of $\vec{u} \times \vec{v}$ is equal to the determinant

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 3 \\ 1 & 2 & -1 \end{vmatrix} = (\vec{i} + 4\vec{k} + 3\vec{j}) - (-\vec{k} + 6\vec{i} - 2\vec{j}) \\ = -5\vec{i} + 5\vec{j} + 5\vec{k}.$$

53. (B)

The group $Z_2 \times Z_4$ contains eight elements

$$Z_2 \times Z_4 = \{(0,0), (0,1), (0,2), (0,3), (1,0), (1,1), (1,2), (1,3)\}$$

The cyclic subgroup $\langle (1,1) \rangle$ is given by

$$\langle (1,1) \rangle = \{(1,1), (0,2), (1,3), (0,0)\}$$

Since

$$(0,1) + \langle (1,1) \rangle = \{(1,2), (0,3), (1,0), (0,1)\}$$

there are two left cosets.

54. (B)

Let G be an abelian group with order n . Then G is isomorphic to the products of the form

$$Z_{(p_1^{n_1})} \times Z_{(p_2^{n_2})} \times \dots \times Z_{(p_k^{n_k})},$$

Where the p_j 's, not necessarily distinct, are the primes in the factorization of n and $(p_1^{n_1}) (p_2^{n_2}) \dots (p_k^{n_k}) = n$. Here Z_n denotes the cyclic group of $\{0, 1, 2, 3, 4, 5, 6, 7\}$ under addition modulo n . For $n = 36 = 2^2 3^2$, we have G isomorphic to

$$Z_4 \times Z_9 = Z_{(2^2)} \times Z_{(3^2)}$$

$$Z_2 \times Z_2 \times Z_9 = Z_{(2^1)} \times Z_{(2^1)} \times Z_{(3^2)}$$

$$Z_4 \times Z_3 \times Z_3 = Z_{(2^2)} \times Z_{(3^1)} \times Z_{(3^1)}$$

$$Z_2 \times Z_2 \times Z_3 \times Z_3 = Z_{(2^1)} \times Z_{(2^1)} \times Z_{(3^1)} \times Z_{(3^1)}.$$

55. (C)

The sum of $a + ar + ar^2 + \dots + ar^{n-1}$... is $\frac{a(1 - r^n)}{1 - r}$.
We have

$$\sum_{j=0}^{10} (-i)^j = 1 - i + i^2 - i^3 + \dots + i^{10}$$

so that $a = 1$, $r = -i$, and $n = 11$. Thus

$$\begin{aligned} \sum_{j=0}^{10} (-i)^j &= \frac{1 - (-i)^{11}}{1 - (-i)} = \frac{1 + i}{1 - i} \frac{1 - i}{1 - i} = -i \\ &= \frac{1 - i}{1 + i} \frac{1 - i}{1 - i} = -i. \end{aligned}$$

56. (D)

The number 1 is the identity element R . An element $u = a + ib$ in R is a unit if there exists $v = c + id$ in R such that $uv = 1$. If u is a unit,

then $\bar{u} = a - ib$ is also a unit since $\overline{uv} = 1$. We have

$$\begin{aligned} 1 = uv = \overline{uv} &\Rightarrow 1 = uv(\overline{uv}) = (u\bar{u})(v\bar{v}) \\ &= |u|^2 |v|^2 = (a^2 + b^2)(c^2 + d^2) \end{aligned}$$

Since a, b, c, d are integers and $a^2 + b^2 \neq 0$, we know that $a^2 + b^2 = 1$. The solutions are $a = 0, b = \pm 1$ and $a = \pm 1, b = 0$ which implies that the units are $\pm 1, \pm i$.

57. (C)

We have

$$3x + 11 \equiv 20 \pmod{12}$$

$$3x \equiv 9 \pmod{12}$$

$$x \equiv 3 \pmod{4}$$

The numbers in the set $\{\dots, -9, -5, -1, 3, 7, 11, 15, \dots\}$ satisfy $3x + 11 \equiv 20 \pmod{12}$. Each of these numbers is in one of the following equivalence classes:

$$\langle 3 \rangle = \{\dots, -9, 3, 15, 27, \dots\}$$

$$\langle 7 \rangle = \{\dots, -5, 7, 19, 31, \dots\}$$

$$\langle 11 \rangle = \{\dots, -1, 11, 23, 35, \dots\}$$

58. (D)

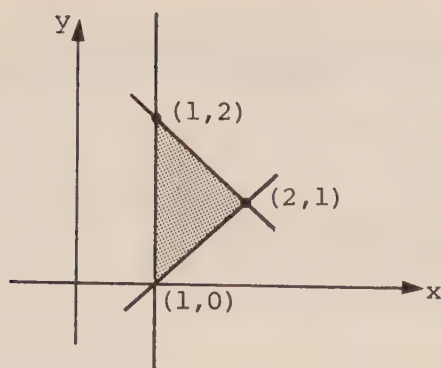
The region R is shown below. Since R is convex and $f(x, y)$ is linear, the maximum of f occurs at a corner point. We have

$$f(1,0) = -2$$

$$f(2,1) = -1$$

$$f(1,2) = 4.$$

Therefore the maximum is 4.



59. (B)

We have

$$\begin{array}{r}
 x^2 - x + 1 \\
 x^3 - 1 \overline{) x^5 - x^4 + x^3 + 0x^2 + x} \\
 \underline{x^5} \\
 -x^4 + x^3 + x^2 + x \\
 \underline{-x^4} \\
 x^3 + x^2 + 0x \\
 \underline{x^3} \\
 x^2 + 1
 \end{array}$$

so that $\frac{x^5 - x^4 + x^3 + x}{x^3 - 1} = x^2 - x + 1 + \frac{x^2 + 1}{x^3 - 1}.$

As $|x| \rightarrow +\infty, \frac{x^2 + 1}{x^3 - 1} \rightarrow 0$ so that $f(x) \approx x^2 - x + 1.$

60. (A)

Let F represent the number of faces, E the number of edges, and V the number of vertices of an ordinary polyhedron. Euler's theorem states that $F - E + V = 2$. Thus $12 - 17 + V = 2$ so that $V = 7$.

61. (A)

We have $(2, 4) = a(1, 1) + b(2, 3)$ so that

$$2 = a + 2b$$

$$4 = a + 3b$$

Thus $b = 2$ and $a = -2$ which implies

$$T(2, 4) = -2T(1, 1) + 2T(2, 3) = -2(-1, 1) + 2(1, 2) = (4, 2).$$

62. (B)

The function $f(z)$ is analytic if and only if the Cauchy-Riemann conditions are satisfied:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Thus

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = \cos x \cosh y$$

and

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = -\sin x \sinh y.$$

We have

$$v(x, y) = \int \frac{\partial u}{\partial x} dy = \int \cos x \cosh y dy = \cos x \sinh y + g(x).$$

Also,

$$-\sin x \sinh y = \frac{\partial v}{\partial x} = -\sin x \sinh y + g'(x),$$

so that $g(x) = \text{constant}$. Hence $v(x, y) = \cos x \sinh y + \text{constant}$.

63. (E)

A linear transformation T from E^n into E^n is called nonsingular if there exists a linear transformation T^{-1} from E^n into E^n such that TT^{-1}

$= T^{-1}T = I_n$, I_n being the identity transformation. The transformation T^{-1} is called the inverse of T and is denoted T^{-1} . Let $x, y \in E^n$ and assume $Tx = Ty$. Then $x = T^{-1}Tx = T^{-1}Ty = y$ so that T is one-to-one. If $Tx = 0$, then $x = 0$ since $T0 = 0$ and T is one-to-one. Thus the null space of $T = \{0\}$ and $\text{Dim } N(T) = 0$. The equation $n = \text{Dim } N(T) + \text{Dim } R(T)$ shows that $n = \text{Dim } R(T)$. Reversing the roles of T and T^{-1} above shows that T^{-1} is one-to-one; $N(T^{-1}) = \{0\}$, and $\text{Dim } N(T^{-1}) = 0$. Thus $\text{Dim } N(T^{-1}) \neq \text{Dim } R(T)$.

64. (D)

The given series is geometric, so its sum is $\frac{1}{1 - \frac{2}{3}} = 3$.

65. (A)

Two circles C_1 and C_2 are said to be orthogonal if they intersect at right angles. This means that at a point of intersection $P(\bar{x}, \bar{y})$ of C_1 and C_2 , the radius r_1 of C_1 is tangent to C_2 at P and the radius of r_2 of C_2 is tangent to C_1 at P . Note that the centers of C_1 and C_2 are $(-a, -b)$ and $(-a', -b')$, respectively. The slope of the tangent line at the point of intersection $P(\bar{x}, \bar{y})$ is equal to the negative reciprocal of the slope through $(-a, -b)$ and $P(\bar{x}, \bar{y})$. It is also equal to the slope through the points $P(\bar{x}, \bar{y})$ and $(-a', -b')$. Thus

$$\frac{-1}{\frac{\bar{y} + b}{\bar{x} + a}} = \frac{\bar{y} + b'}{\bar{x} + a'}$$

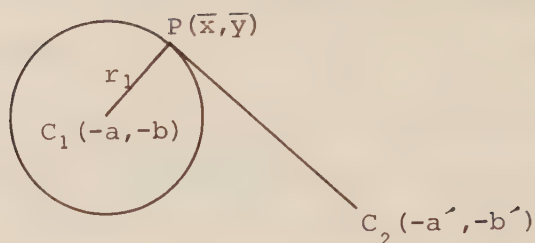
$$-[\bar{x}^2 + (a + a')\bar{x} + aa'] = \bar{y}^2 + (b + b')\bar{y} + bb'.$$

Multiplying by 2 and rearranging terms,

$$\Rightarrow -[(\bar{x}^2 + 2a\bar{x} + \bar{y}^2 + 2b\bar{y}) + (\bar{x}^2 + 2a'\bar{x} + \bar{y}^2 + 2b'\bar{y})]$$

$$= 2aa' + 2bb'$$

$$\Rightarrow c + c' = 2aa' + 2bb'$$



66. (A)

Using elementary row operations, we transform $[M \mid I]$ into $[I \mid M^{-1}]$:

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 3 & 1 & 0 & 0 \\ 0 & -1 & 2 & 0 & 1 & 0 \\ 4 & 3 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 3 & 1 & 0 & 0 \\ 4 & 3 & 1 & 0 & 0 & 1 \\ 0 & -1 & 2 & 0 & 1 & 0 \end{array} \right]$$

$$R_2 \Leftrightarrow R_3$$

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -5 & -2 & 0 & 1 \\ 0 & -1 & 2 & 0 & 1 & 0 \end{array} \right]$$

$$-2R_1 + R_2 \Rightarrow R_2$$

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -5 & -2 & 0 & 1 \\ 0 & 0 & -3 & -2 & 1 & 1 \end{array} \right]$$

$$R_2 + R_3 \Rightarrow R_3$$

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & -1 & 1 & 1 \\ 0 & 1 & -5 & -2 & 0 & 1 \\ 0 & 0 & -3 & -2 & 1 & 1 \end{array} \right]$$

$$R_3 + R_1 \Rightarrow R_1$$

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & -1 & 1 & 1 \\ 0 & 1 & 0 & 4/3 & -5/3 & -2/3 \\ 0 & 0 & -3 & -2 & 1 & 1 \end{array} \right] \quad -5/3R_3 + R_2 \Rightarrow R_2$$

$$\left[\begin{array}{ccc|ccc} 2 & 0 & 0 & -7/3 & 8/3 & 5/3 \\ 0 & 1 & 0 & 4/3 & -5/3 & -2/3 \\ 0 & 0 & -3 & -2 & 1 & 1 \end{array} \right] \quad R_1 - R_2 \Rightarrow R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -7/6 & 8/6 & 5/6 \\ 0 & 1 & 0 & 4/3 & -5/3 & -2/3 \\ 0 & 0 & 1 & 2/3 & -1/3 & -1/3 \end{array} \right] \quad \begin{array}{l} 1/2R_1 \Rightarrow R_1 \\ -1/3R_3 \Rightarrow R_3 \end{array}$$

The Graduate Record Examination in
MATHEMATICS

Practice Test 2

The Graduate Record Examination in **MATHEMATICS** Answer Sheet Test 2

- | | | |
|-------------------------|-------------------------|-------------------------|
| 1. (A) (B) (C) (D) (E) | 23. (A) (B) (C) (D) (E) | 45. (A) (B) (C) (D) (E) |
| 2. (A) (B) (C) (D) (E) | 24. (A) (B) (C) (D) (E) | 46. (A) (B) (C) (D) (E) |
| 3. (A) (B) (C) (D) (E) | 25. (A) (B) (C) (D) (E) | 47. (A) (B) (C) (D) (E) |
| 4. (A) (B) (C) (D) (E) | 26. (A) (B) (C) (D) (E) | 48. (A) (B) (C) (D) (E) |
| 5. (A) (B) (C) (D) (E) | 27. (A) (B) (C) (D) (E) | 49. (A) (B) (C) (D) (E) |
| 6. (A) (B) (C) (D) (E) | 28. (A) (B) (C) (D) (E) | 50. (A) (B) (C) (D) (E) |
| 7. (A) (B) (C) (D) (E) | 29. (A) (B) (C) (D) (E) | 51. (A) (B) (C) (D) (E) |
| 8. (A) (B) (C) (D) (E) | 30. (A) (B) (C) (D) (E) | 52. (A) (B) (C) (D) (E) |
| 9. (A) (B) (C) (D) (E) | 31. (A) (B) (C) (D) (E) | 53. (A) (B) (C) (D) (E) |
| 10. (A) (B) (C) (D) (E) | 32. (A) (B) (C) (D) (E) | 54. (A) (B) (C) (D) (E) |
| 11. (A) (B) (C) (D) (E) | 33. (A) (B) (C) (D) (E) | 55. (A) (B) (C) (D) (E) |
| 12. (A) (B) (C) (D) (E) | 34. (A) (B) (C) (D) (E) | 56. (A) (B) (C) (D) (E) |
| 13. (A) (B) (C) (D) (E) | 35. (A) (B) (C) (D) (E) | 57. (A) (B) (C) (D) (E) |
| 14. (A) (B) (C) (D) (E) | 36. (A) (B) (C) (D) (E) | 58. (A) (B) (C) (D) (E) |
| 15. (A) (B) (C) (D) (E) | 37. (A) (B) (C) (D) (E) | 59. (A) (B) (C) (D) (E) |
| 16. (A) (B) (C) (D) (E) | 38. (A) (B) (C) (D) (E) | 60. (A) (B) (C) (D) (E) |
| 17. (A) (B) (C) (D) (E) | 39. (A) (B) (C) (D) (E) | 61. (A) (B) (C) (D) (E) |
| 18. (A) (B) (C) (D) (E) | 40. (A) (B) (C) (D) (E) | 62. (A) (B) (C) (D) (E) |
| 19. (A) (B) (C) (D) (E) | 41. (A) (B) (C) (D) (E) | 63. (A) (B) (C) (D) (E) |
| 20. (A) (B) (C) (D) (E) | 42. (A) (B) (C) (D) (E) | 64. (A) (B) (C) (D) (E) |
| 21. (A) (B) (C) (D) (E) | 43. (A) (B) (C) (D) (E) | 65. (A) (B) (C) (D) (E) |
| 22. (A) (B) (C) (D) (E) | 44. (A) (B) (C) (D) (E) | 66. (A) (B) (C) (D) (E) |

GRE MATHEMATICS TEST II

TIME: 2 hours and 50 minutes
66 Questions

DIRECTIONS: Choose the best answer for each question and mark the letter of your selection on the corresponding answer sheet.

1. Find the sum of the series $\sum_{n=1}^{\infty} \frac{n}{4^{n+1}}$.

(A) $\frac{1}{12}$

(D) $\frac{4}{3}$

(B) $\frac{1}{9}$

(E) $\frac{1}{3}$

(C) $\frac{1}{6}$

2. A rectangle has dimensions a units by b units with $a > b$. A diagonal divides the rectangle into two triangles. A square, with sides parallel to those of the rectangle, is inscribed in each triangle. Find the distance between the vertices (of the squares) that lie in the interior of the rectangle.

(A) $\frac{(a-b)\sqrt{a^2+b^2}}{a+b}$

(D) $\frac{a^2-b^2}{\sqrt{a^2+b^2}}$

(B) $\sqrt{a^2-b^2}$

(E) $\frac{(a-b)\sqrt{a^2-b^2}}{a+b}$

(C) $\frac{a^2-b^2}{\sqrt{ab}}$

(C) $(n-1, n)$

(D) $(n+1, n)$

(E) $(n, n+1)$

6. Let the set S be infinite and let the set T be countably infinite. Let $\bar{S}$ denote the complement of S . If S and T are both subsets of the real numbers, which of the following pairs of sets must be of the same cardinality?

(A) T and $S \cap T$

(D) Both (A) and (B)

(B) S and $S \cup T$

(E) Both (A) and (C)

(C) T and $\bar{S} \cup T$

7. Find the maximum value of $f(x) = 5 \sin 7x + 12 \cos 7x$.

(A) 12

(D) 17

(B) 5

(E) 13

(C) 7

8. In a computer program, separate loops with distinct indices produce M and N operations respectively. If these reside internally in a loop with an independent index producing P

operations, find the total number of operations represented by the three loops.

(A) P^{M+N}

(D) $P^M + P^N$

(B) $(M + N)^P$

(E) $M^P + N^P$

(C) $P(M + N)$

9. Given that the Taylor series for $g(x)$ is

$$\sum_{n=0}^{\infty} a_n x^n \text{ and } f(x) = [g(x)]^3, \text{ find } f''(0).$$

(A) $3a_0(2a_1^2 + a_0a_2)$

(D) $6a_0(a_1 + a_0a_1)$

(B) $6a_0(a_1^2 + a_0a_2)$

(E) None of these

(C) $3a_0(2a_1 + a_0a_2)$

10. If $f(x) = x/(1 - x)$, find $f(f(f(x)))$.

(A) $\frac{x^3}{1 - x^3}$

(D) $\frac{3x}{1 - 3x}$

(B) $\frac{3x}{3 - x}$

(E) $\frac{x}{1 - 3x}$

(C) $\frac{x}{3 - x}$

11. Define $S_k = \{\frac{k}{j} \mid j = k, k+1, \dots\}$ for $k=1, 2, \dots$. Find the intersection $\bigcap_{k=1}^{\infty} S_k$.

- (A) S_1 (D) the interval $(0, 1)$
 (B) the empty set (E) $\{\frac{k}{j} \mid 0 < \frac{k}{j} \leq 1\}$
 (C) the interval $[0, 1]$

12. Evaluate $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n(1 + \frac{k}{n})}$.

- (A) 1 (D) $+\infty$
 (B) $\ln 2$ (E) Does not exist
 (C) 0

13. Evaluate the definite integral $\int_0^a \frac{x^2 + b^2}{x^2 + a^2} dx$.

- (A) $\frac{((4 - \pi)a + \pi b^2)}{4a}$ (D) $a + (b^2 - a^2) \ln 2$
 (B) $4a^2 + \pi(b^2 - a^2)$ (E) $(4 - \pi) \frac{a}{4} + \frac{\pi b^2}{4a}$
 (C) $\frac{((2 - \pi)a + \pi b^2)}{2a}$

14. Let $f(x, y) = x^3 - axy + y^2 - x$. Find the greatest lower bound for a so that $f(x, y)$ has a relative minimum point.

(A) 0 (D) 6
(B) $\sqrt{48}$ (E) Does not exist
(C) 12

15. Given that S and T are subspaces of a vector space, which of the following is also a subspace?

(A) $S \cap T$ (D) Both (A) and (C)
(B) $S \cup T$ (E) Both (B) and (C)
(C) $2S$

16. Find the point $x + iy$ at which the complex function $f(x + iy) = x^2 + y^2 + 2xi$ is differentiable.

(A) 0 (D) $1 + i$
(B) i (E) $-i$
(C) $1 - i$

17. Define a function, $f(x)$, to be differentiably redundant of order n if the n -th derivative $f^{(n)}(x) = f(x)$ but $f^{(k)}(x) \neq f(x)$ when $k < n$. For easy examples, in this context, e^x is of order 1, e^{-x} is of order 2, and $\cos x$ is of order 4. Which of the following functions is differentiably redundant of order 6?

(A) $e^{-x} + e^{\frac{-x}{2}} \cos\left(\frac{\sqrt{3}x}{2}\right)$

(B) $e^{-x} + \cos x$

(C) $e^{\frac{x}{2}} \sin\left(\frac{\sqrt{3}x}{2}\right)$

(D) Both (A) and (C)

(E) (A), (B) and (C)

18. In a homogeneous system of 5 linear equations in 7 unknowns, the rank of the coefficient matrix is 4. The maximum number of independent solution vectors is

(A) 5

(D) 1

(B) 2

(E) 3

(C) 4

19. Let X, Y, Z represent statements. Then the compound statement $(X \vee Y) \wedge (X \wedge \sim Z)$ is true under which one of the following conditions?

(A) $X = \text{true}, Y = \text{false}, Z = \text{false}$

(B) $X = \text{true}, Y = \text{false}, Z = \text{true}$

(C) $X = \text{false}, Y = \text{true}, Z = \text{true}$

- (D) $X = \text{true}, Y = \text{true}, Z = \text{true}$
- (E) $X = \text{false}, Y = \text{true}, Z = \text{false}$
20. From the set of integers $\{1, 2, \dots, 9\}$, how many nonempty subsets sum to an even integer?
- (A) 512
- (D) $\frac{9!}{2!}$
- (B) 255
- (E) None of these
- (C) $\frac{9!}{2!7!}$
21. How many topologies are possible on a set of 2 points?
- (A) 5
- (D) 2
- (B) 4
- (E) 1
- (C) 3
22. In the (ϵ, δ) definition of the limit, $\lim_{x \rightarrow c} f(x) = L$, let $f(x) = x^3 + 3x^2 - x + 1$ and let $c = 2$. Find the least upper bound on δ so that $f(x)$ is bounded within ϵ of L for all sufficiently small $\epsilon > 0$.
- (A) $\frac{\epsilon}{8}$
- (D) $\frac{(\epsilon^2)}{16}$
- (B) $\frac{\epsilon}{23}$
- (E) $\frac{\epsilon}{19}$
- (C) $\frac{(\epsilon^3)}{4}$

23. Find the remainder on dividing 3^{20} by 7.

(A) 1

(D) 4

(B) 2

(E) 5

(C) 3

24. In the partial fractions expansion of $\frac{s^2 + 1}{(s^2 - 2)(s^2 + 3)}$, the numerator of the fraction with denominator $s^2 + 3$ is

(A) 3

(D) $2s + 1$

(B) $\frac{3}{5}$

(E) $1 - 2s$

(C) $\frac{2}{5}$

25. For how many distinct real coefficients, a , will the system of equations $y = ax^2 + 2$ and $x = ay^2 + 2$ admit a solution with $y = 1$?

(A) 0

(D) 3

(B) 1

(E) 4

(C) 2

26. A subgroup H in group G is invariant if $gH = Hg$ for every g in G . If H and K are both invariant subgroups of G , which of the following is also an invariant subgroup?

(A) $H \cap K$ (D) Both (A) and (B)
(B) HK (E) Both (B) and (C)
(C) $H \cup K$

27. Let R be a ring such that $x^2 = x$ for each $x \in R$. Which of the following must be true?

(A) $x = -x$ for all $x \in R$
(B) R is commutative
(C) $xy + yx = 0$ for all $x, y \in R$
(D) Both (A) and (C)
(E) (A), (B), and (C)

28. In the integral domain $D = \{r + s\sqrt{17} \mid r, s \text{ integers}\}$, which of the following is irreducible?

(A) $8 + 2\sqrt{17}$ (D) $7 + \sqrt{17}$
(B) $3 - \sqrt{17}$ (E) $13 + \sqrt{17}$
(C) $9 - 2\sqrt{17}$

29. For which value of k is x^k a solution for the differential equation $x^2y'' - 3xy' - 4y = 0$?

(A) 4

(D) 1

(B) 3

(E) None of these

(C) 2

30. Which of the following is a factor of $x^4 + 1$?

(A) $x + 1$

(D) $x^2 + \sqrt{2}x - 1$

(B) $x^2 + 1$

(E) None of these

(C) $x^2 - \sqrt{2}x + 1$

31. The surface given by $z = x^2 - y^2$ is cut by the plane given by $y = 3x$, producing a curve in the plane. Find the slope of this curve at the point $(1, 3, -8)$.

(A) 3

(D) 0

(B) -16

(E) $\frac{18}{\sqrt{10}}$

(C) $-8\sqrt{\frac{2}{5}}$

32. If A is an $n \times n$ matrix with diagonal entries, a , and other entries, b , then one eigenvalue of A is $a - b$. Find another eigenvalue of A .

(A) $b - a$ (D) 0
(B) $nb + a - b$ (E) None of these
(C) $nb - a + b$

33. In ancient Egypt, the formula $A = \left(\frac{8d}{9}\right)^2$ was used for the area of a circle of diameter d . Using the correct multiple, relating the volume of a sphere to the area of a circle, what should the Egyptian formula be for the volume of the sphere of diameter d ?

(A) $\frac{\pi d^3}{3}$ (D) $\frac{2^8 \pi d^3}{3^5}$
(B) $\left(\frac{8d}{9}\right)^3$ (E) $\frac{2^7 d^3}{3^5}$
(C) $4\left(\frac{8}{9}d\right)^3$

34. Find k so that the matrix

$$A = \begin{pmatrix} k & 1 & 2 \\ 1 & 2 & k \\ 1 & 2 & 3 \end{pmatrix}$$

has eigenvalue $\lambda = 1$.

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$

(C) 0

(D) 1

(E) -1

35. Express z^{14} in the form $a + bi$ if $z = \frac{(1+i)}{\sqrt{2}}$.

(A) $-i$

(D) i

(B) -1

(E) $\frac{(1+i)}{128}$

(C) 1

36. Which number is nearest a solution of $e^{\frac{-x}{100}} - e^{-x} = e^{-1}$?

(A) $1 - \frac{1}{2} \ln 4$

(D) -1

(B) $1 - \frac{1}{2} \ln 3$

(E) 0

(C) $1 - \frac{1}{2} \ln 2$

37. In the Laurent series for $f(z) = \frac{1}{(z-4)}$ centered at $z = 1$, the coefficient of $(z-1)^{-2}$ is

(A) 9

(D) 3

(B) -9

(E) -1

(C) -3

38. If $f(x) = \int_1^{x^2} \frac{dt}{1+t^3}$ then $f'(2)$ is

(A) $\frac{4}{65}$ (D) $\ln\left(\frac{9}{2}\right)$

(B) $\frac{1}{9}$ (E) 0.23

(C) $\ln\left(\frac{65}{2}\right)$

39. The series $\sum_{n=2}^{\infty} \frac{1}{n \cdot 3^n}$ must

(A) converge to a value greater than $1/4$

(B) converge to a value greater than $1/9$

(C) converge to a value less than $1/18$

(D) converge to a value less than $1/12$

(E) diverge

40. If A is a square matrix of order $n \geq 4$ and $a_{ij} = i + j$ represents the entry in row i and column j , then the rank of A is always

(A) 1 (D) $n - 1$

(B) 2 (E) None of these

(C) $n - 2$

41. In the finite field, Z_{17} , the multiplicative inverse of 10 is

(A) 13

(D) 9

(B) 12

(E) 7

(C) 11

42. The system of equations $x^2 - y = a$ and $y - x = b$ has exactly one value of x in its solution(s). This value of x must be

(A) 0

(D) $-\frac{1}{2}$

(B) 1

(E) $\frac{1}{2}$

(C) $\frac{3}{2}$

43. $\log_4 64$ is identical to

(A) $\log_7 343$

(D) Both (A) and (B)

(B) $\frac{\log_{10} 64}{\log_{10} 4}$

(E) Both (A) and (C)

(C) $\log_8 256$

44. Let $S = \{\tan(k) \mid k = 1, 2, \dots\}$. Find the set of limit points of S on the real line.

(A) $(-\infty, \infty)$ (D) $(-\infty, 0]$
(B) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (E) The empty set
(C) $[0, \infty)$

45. Given the sequence $a_n = \sin n$, $n \geq 1$, find $\liminf_{n \rightarrow \infty} \{a_n\} - \limsup_{n \rightarrow \infty} \{a_n\}$.

(A) 2 (D) -1
(B) 1 (E) -2
(C) 0

46. Find the length of a diagonal of a regular pentagon of side length 1.

(A) $2 \cos \frac{\pi}{5}$ (D) $\sqrt{2}$
(B) $\sqrt{2} \left(1 + \cos \left(\frac{\pi}{5}\right)\right)$ (E) $\sqrt{2} + \sqrt{2 \cos \left(\frac{2\pi}{5}\right)}$
(C) $\sqrt{2} \left(1 + \cos \left(\frac{2\pi}{5}\right)\right)$

47. Which of the following is a solution to the differential equation $y'' = 2y' + 8y$?

(A) $e^{2x} - e^{-4x}$

(D) $e^{4x} - e^{-2x}$

(B) xe^{4x}

(E) xe^{-2x}

(C) e^{2x}

48. Let $f(x)$ be defined and strictly increasing on $(a, b]$. Find the maximum value of $g(x) = -f(x)$ on $(a, b]$.

(A) $g(b)$

(D) $f(a)$

(B) $f(b)$

(E) Does not exist

(C) $g(a)$

49. Find the number of distinguishable permutations of six colored blocks if one is red, two are yellow, and three are blue.

(A) 360

(D) 120

(B) 60

(E) 240

(C) 720

50. Let $f(x) = 4x^3 - x + 2$ on the closed interval $[-3, 3]$. Find a value of c in $[-3, 3]$ such that the slope of $f(x)$ at $x = c$ is equal to the slope of the secant line on $[-3, 3]$.

(A) 0 (D) $\sqrt{3}$

(B) 1 (E) $\sqrt{6}$

(C) $\sqrt{2}$

51. The volume, V , of the region in space bounded above by the surface $x^2 + y^2 + z^2 = 4$ and below by $z = -\sqrt{x^2 + y^2}$ is represented by a triple integral in spherical coordinates as

$$\int \int \int_V \rho^2 \sin \phi d\rho d\phi d\theta.$$

V

find the upper limit of integration for ϕ .

(A) π (D) $\frac{\pi}{4}$

(B) $\frac{3\pi}{4}$ (E) $\frac{\pi}{6}$

(C) $\frac{\pi}{2}$

52. The reliability of component C is R . A system is designed as a series of n subcomponents each of which is doubly redundant using two components, C . Find the reliability of the system.

(A) $R^n(1 - R)^n$ (B) $R^n(2 - R)^n$

(C) R^{2n} (D) $\frac{R^n}{2^n}$

(E) $1 - (1 - R)^{2n}$

53. A family of curves is represented by the differential equation $ydx - xdy = 0$. Which of the following best describes the family of orthogonal trajectories to this given family?

(A) all parabolas with vertex at $(0, 0)$

(B) all hyperbolas with centers at $(0, 0)$

(C) all lines through $(0, 0)$

(D) all circles with centers at $(0, 0)$

(E) all lines parallel to the y -axis

54. Which of the following sequences is a solution for the difference equation $x_{n+2} + 36n = x_{n+1} + 6x_n$?

(A) $3^n + 6n$

(D) $3^n + 6 + n$

(B) $2^n + 6n + 1$

(E) $(-1)^n 2^{n+1} + 6n + 1$

(C) $6^n + 6n$

55. Let f be a mapping from a topological space $\overline{X}$ onto itself. Which of the following is true for continuous f ?

(A) Every open set in $\overline{X}$ is the image of an open set in $\overline{X}$

(B) $f^{-1}(B)$ is bounded for each bounded set B in $\overline{X}$

(C) f is one-to-one

(D) Both (A) and (B)

(E) Both (A) and (C)

56. At the point $(2, -1, 2)$ on the surface $z = xy^2$, find a direction vector for the greatest rate of decrease of z .

(A) $\hat{i} - 2\hat{j}$

(D) $-\hat{i} + 4\hat{j}$

(B) $\hat{i} - 4\hat{j}$

(E) $\hat{i} + \hat{j}$

(C) $\frac{(\hat{i} - 4\hat{j})}{\sqrt{17}}$

57. Let G be a polyhedron with 27 vertices and 40 edges. Find the number of faces on G .

(A) 12

(D) 15

(B) 13

(E) Cannot be decided
based on the given
information

(C) 14

58. Find the coefficient of x^2y^3 in the binomial expansion of $(x - 2y)^5$.

(A) -160

(D) 8

(B) 80

(E) -8

(C) -80

59. In the evaluation of the integral $\int \frac{dx}{x(2 + 3x)^2}$, the coefficient of $\ln |2 + 3x|$ is

(A) $-\frac{3}{4}$

(D) $\frac{1}{4}$

(B) $-\frac{1}{4}$

(E) $-\frac{1}{36}$

(C) $\frac{3}{4}$

60. A biased coin is tossed repeatedly until the first "tail" occurs. The expected number of tosses required to produce the first tail is estimated as T . Assuming this is true, find the probability of at least two tails in $3T$ tosses.

(A) $\frac{T^{3T} - (T - 1)^{3T-1}(4T)}{T^{3T}}$

(B) $\frac{T^{3T} - (T - 1)^{3T-1}(3T)}{T^{3T}}$

(C) $\frac{T^{3T} - (T - 1)^{3T-1}(3T - 1)}{T^{3T}}$

$$(D) \frac{T^{3T} - (T-1)^{3T-1}(4T-1)}{T^{3T}}$$

(E) None of these

61. The vertices of a quadrilateral are $(0, 0)$, $(1, 4)$, $(3, 2)$, and $(5, 5)$. Find the first coordinate of the centroid of the region.

(A) $\frac{9}{4}$ (D) $\frac{17}{6}$

(B) $\frac{13}{6}$ (E) $\frac{8}{3}$

(C) $\frac{7}{3}$

62. Let R be a ring and let $x \neq 0$ be a fixed element in R . Which of the following is a subring of R ?

(A) $\{ r \mid xr = 0 \}$

(B) $\{ x \mid x^{-1} \text{ exists in } R \}$

(C) $\{ x^n \mid n = 1, 2, \dots \}$

(D) $\{ nx \mid n \text{ is an integer} \}$

(E) Both (A) and (D)

63. If $f(x) = \ln x$, find

$$\lim_{m \rightarrow 0} \left[\lim_{n \rightarrow 0} \frac{f(2+m+n) - f(2+m) - f(2+n) + f(2)}{mn} \right]$$

(A) $\frac{1}{4}$

(D) $\frac{1}{2}$

(B) 1

(E) $-\frac{1}{4}$

(C) -1

64. Let $S = \{x_1, x_2, \dots, x_n, \dots\}$ be a topological space where the open sets are $U_n = \{x_1, \dots, x_n\}$, $n = 1, 2, \dots$. Let $E = \{x_2, x_4, \dots, x_{2k}, \dots\}$. Find the set of cluster points of E .

(A) $S - \{x_1, x_2\}$

(D) $E - \{x_2\}$

(B) $\{x_1\}$

(E) $S - E$

(C) $\{x_2\}$

65. A stiff beam on two supports that are 20ft. apart is loaded by two uniform blocks with dimensions and weights as shown below.



How much of the total weight is supported at the left support?

(A) 150

(D) 100

(B) 120

(E) None of these

(C) 110

66. Given that $\sum_{n=1}^{\infty} a_n$ converges to L , which conclusion is valid

for $\sum_{n=1}^{\infty} a_n^2$?

(A) It may diverge

(B) It converges absolutely

(C) It converges to $M < L$

(D) It converges to $M > L$

(E) It converges to L^2

GRE MATHEMATICS

TEST II

ANSWER KEY

1.	B	23.	B	45.	E
2.	A	24.	C	46.	A
3.	A	25.	D	47.	D
4.	E	26.	D	48.	E
5.	D	27.	E	49.	B
6.	B	28.	C	50.	D
7.	E	29.	E	51.	B
8.	C	30.	C	52.	B
9	B	31.	C	53.	D
10.	E	32.	B	54.	E
11.	A	33.	E	55.	A
12.	B	34.	A	56.	D
13.	E	35.	A	57.	D
14.	E	36.	B	58.	C
15.	D	37.	D	59.	B
16.	E	38.	A	60.	D
17.	D	39.	D	61.	B
18.	E	40.	B	62.	A
19.	A	41.	B	63.	E
20.	B	42.	E	64.	A
21.	B	43.	D	65.	C
22.	B	44.	A	66.	A

GRE MATHEMATICS TEST II

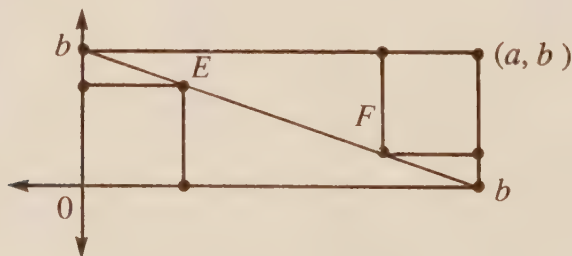
DETAILED EXPLANATIONS OF ANSWERS

1. (B)

It is well-known that $(1 - x)^{-1} = \sum_{n=0}^{\infty} x^n$, so $(1 - x)^{-2} = \sum_{n=1}^{\infty} nx^{n-1}$ by differentiation. Of course $|x| < 1$ is required for convergence. The given series is $\frac{1}{16} \sum_{n=1}^{\infty} n \left(\frac{1}{4}\right)^{n-1} = \frac{1}{16} \left(1 - \frac{1}{4}\right)^{-2} = \frac{1}{9}$.

2. (A)

From the diagram shown, vertex E must lie on the lines $y = x$ and $y = -\frac{b}{a}x + b$, so it has coordinates $\left(\frac{ab}{a+b}, \frac{ab}{a+b}\right)$, found by solving these equations simultaneously. By symmetry we must have coordinates for vertex F as $\left(\frac{a^2}{a+b}, \frac{b^2}{a+b}\right)$.



Applying the distance formula,

$$d = \frac{\sqrt{a^2(a-b)^2 + b^2(b-a)^2}}{(a+b)}$$

which reduces to (A).

3. (A)

If $\lambda = (1\ 3\ 2)(4)(5)$, then $\lambda^2 = (1\ 2\ 3)(4)(5)$ and $\lambda^3 = (1)(2)(3)(4)(5)$, the identity. Hence the cyclic subgroup generated by λ is of order 3. A_5 is of order $5!/2 = 60$ and so the required index is $60/3 = 20$.

4. (E)

The equation can be solved by separation of variables $\frac{dM}{M} = -\frac{dt}{10}$ and integration, yielding $\ln M = -\frac{t}{10} + C$. The value $M = M_0$ when $t = 0$ implies that $C = \ln M_0$. Taking the exponential of both sides of the solution yields

$$M = M_0 e^{-\frac{t}{10}} \text{ so } M(20) = M_0 e^{-2} = \frac{M_0}{e^2}.$$

5. (D)

First, we must have $f(1) = 1 - a + b = 0$. Then since $f'(1) = n - 1 + 2 - a = 0$ is also required, we must have $a = n + 1$ so $b = n$.

6. (B)

If S is countable, then $S \cup T$ is the union of countable sets and hence countable. If S is uncountable, then $S \cup T$ is uncountable as its cardinality is at least that of S . Counterexamples for the other cases:

(A) $S \cap T$ is countable but it may be finite.

For example, $S = \{0, 1, 2, \dots\}$, $T = \{0, -1, -2, \dots\}$.

(C) $\bar{S}$ may be uncountable and then $\bar{S} \cup T$ is also.

7. (E)

The amplitude of any wave in the form $a \sin kx + b \cos kx$ is $\sqrt{a^2 + b^2}$. This follows since factoring $\sqrt{a^2 + b^2}$ from the expression leaves coefficients $\frac{a}{\sqrt{a^2 + b^2}}$ and $\frac{b}{\sqrt{a^2 + b^2}}$ which have sum of squares equal to one. Hence we can rewrite the wave as

$$\sqrt{a^2 + b^2} \sin(kx \pm \theta) \text{ where } \theta = \sin^{-1}\left(\frac{b}{\sqrt{a^2 + b^2}}\right).$$

As $\sin(kx \pm \theta)$ can be at most 1, the maximum is at

$$\sqrt{a^2 + b^2} = \sqrt{5^2 + 12^2} = \sqrt{169} = 13.$$

A calculus approach is much more difficult..

8. (C)

Each of the P passes through the external loop creates $M + N$ operations in the internal loops. Basic counting technique implies the product $P(M + N)$, gives the total number of operations.

9. (B)

Directly, using the chain rule, $f'(x) = 3 [g(x)]^2 \cdot g'(x)$ and then $f''(x) = 6 g(x)[g'(x)]^2 + 3 [g(x)]^2 g''(x)$. Now the coefficients of the

series are related to $g(x)$ in general as $g^{(n)}(0) = n!a_n$. So $g(0) = a_0$, $g'(0) = a_1$ and $g''(0) = 2a_2$. Hence $f''(0) = 6a_0a_1^2 + 3a_0^2 \cdot 2a_2 = 6a_0(a_1^2 + a_0a_2)$.

10. (E)

$$f(f(x)) = f\left(\frac{x}{1-x}\right) = \frac{\frac{x}{1-x}}{\left(1 - \frac{x}{1-x}\right)} = \frac{x}{1-x-x} = \frac{x}{1-2x}$$

Likewise,

$$f(f(f(x))) = f\left(\frac{x}{1-2x}\right) = \frac{\frac{x}{1-2x}}{\left(1 - \frac{x}{1-2x}\right)} = \frac{x}{1-3x}$$

11. (A)

S_1 contains 1 and all unit fractions $1/n$. In all other S_k , only the numerators of the fractions change. Since the unit fraction $\frac{1}{n} = \frac{k}{nk}$ and this is in S_k , we find that

$$S_1 \subseteq S_k \text{ for all } k, \text{ so } \bigcap_{k=1}^{\infty} S_k = S_1.$$

12. (B)

It is simplest to view the limit as a definite integral in the form

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n f(a + k \Delta x) \Delta x.$$

In this case, $\Delta x = 1/n$ and $a = 1$ so $f(x) = 1/x$. As usual, we can use $\Delta x = \frac{b-a}{n}$ for interval $[a, b]$ to conclude that $b = 2$. Thus, the limit is

$$\int_1^2 \frac{dx}{x} = \ln x \Big|_1^2 = \ln 2 - \ln 1 = \ln 2.$$

13. (E)

$$\begin{aligned}
 \int_0^a \frac{x^2 + b^2}{x^2 + a^2} dx &= \int_0^a \left(\frac{x^2 + a^2}{x^2 + a^2} + \frac{b^2 - a^2}{x^2 + a^2} \right) dx \\
 &= \int_0^a \left(1 + \frac{b^2 - a^2}{x^2 + a^2} \right) dx \\
 &= \left(x + \frac{b^2 - a^2}{a} \tan^{-1} \left(\frac{x}{a} \right) \right) \Big|_0^a \\
 &= a + \frac{(b^2 - a^2)}{a} \frac{\pi}{4} \\
 &= \frac{4a^2}{4a} + \frac{(b^2 - a^2)\pi}{4a} \\
 &= \frac{4a^2 + b^2\pi - a^2\pi}{4a} \\
 &= \frac{(4 - \pi)a^2 + b^2\pi}{4a} \\
 &= (4 - \pi) \frac{a}{4} + \frac{\pi b^2}{4a} .
 \end{aligned}$$

14. (E)

For a relative minimum, we must simultaneously have $f_x = 3x^2 - ay - 1 = 0$, $f_y = -ax + 2y = 0$, and $D = f_{xx} \cdot f_{yy} - f_{xy}^2 = 12x - a^2 > 0$ since $f_{yy} = 2 > 0$ independent of x, y . The first two conditions imply $y = ax/2$ and $3x^2 - \frac{a^2}{2}x - 1 = 0$. Solving the quadratic always yields two roots,

$$x = \frac{\left(\frac{a^2}{2} \pm \sqrt{\frac{a^4}{4} + 12} \right)}{6}$$

where the + sign will yield the necessary positive root for $D > 0$. Then

$D = a^2 + \sqrt{a^4 + 48} - a^2 = \sqrt{a^4 + 48} > 0$ independent of the choice of a . No lower bound exists.

15. (D)

The result (A) is a standard theorem in vector spaces. For (C), we need only note that subspaces are closed under scalar multiplication so $2S \subseteq S$. But, for every x in S , $1/2 x$ is in S and $2(1/2 x) = x$ so $S \subseteq 2S$, and we can conclude that $S = 2S$, so $2S$ is a subspace.

To show that (B) is not necessarily true, let the vector space be R^2 , $S = \{ (a, 0) \mid a \in R \}$, $T = \{ (a, b) \mid a + b = 0, a, b \in R \}$. Then S and T are subspaces of R^2 (this is easily checked), and $S \cup T = \{ (a, b) \mid b = 0 \text{ or } a + b = 0, a, b \in R \}$. We see that $S \cup T$ is not a subspace of R^2 because it is not closed under addition. For example, if we take $s, t \in S \cup T$ where $s = (1, 0)$ and $t = (1, -1)$, then $s + t = (2, -1) \notin S \cup T$.

16. (E)

The Cauchy-Riemann conditions to be satisfied are

$$\frac{\partial(x^2 + y^2)}{\partial x} = \frac{\partial(2x)}{\partial y} \quad \text{and} \quad \frac{\partial(x^2 + y^2)}{\partial y} = -\frac{\partial(2x)}{\partial x},$$

which imply $x = 0$ and $y = -1$.

17. (D)

If $y = f(x)$ is differentially redundant of order n then $y^{(n)} - y = 0$ and solutions are related to the n th roots of unity. Clearly, the sum of two functions, one of order m and the other of order n , yields a function of order $\text{lcm}(m, n)$. This eliminates (B) which must be of order 4. (A) is of order 6 since $-1/2 \pm i\sqrt{3}/2$ are 3rd roots of unity. (C) is of order

6 since $1/2 \pm i\sqrt{3}/2$ are 6th roots of unity. In each of these latter cases we have primitive roots and this is necessary. For instance -1 is also a 6th root of unity but e^{-x} is not of order 6.

18. (E)

The dimension of the solution space of a matrix always is given by the number of columns (7) or unknowns less the rank of the matrix (4).

19. (A)

By substitution,

$$\begin{aligned} (\text{true} \vee \text{false}) \wedge (\text{true} \wedge \sim \text{false}) \\ &= \text{true} \wedge (\text{true} \wedge \text{true}) \\ &= (\text{true} \wedge \text{true}) \\ &= \text{true} \end{aligned}$$

20. (B)

It may simply be well known that half the sums will be even (including sum zero for the empty set) and the number of subsets is $2^9 = 512$, so $1/2(512) - 1 = 255$ (for the empty set). A more direct analysis follows in that an even sum results in each of the $2^4 = 16$ subsets containing only the even integers, 2, 4, 6, 8. These may be combined with either 0, 2, or 4 odd numbers and the number of ways to choose these is

$$\binom{5}{0} = 1, \binom{5}{2} = 10, \text{ or } \binom{5}{4} = 5, \text{ respectively.}$$

Hence the total $16(1 + 10 + 5) = 16^2 = 256$ which includes the empty set.

21. (B)

Given $\overline{X} = \{x, y\}$, every topology must contain the open sets $\overline{X}$ and ϕ , the empty set. When these are the only open sets, we have one (trivial) topology. The other three possible topologies are $\{\{x\}\}$, $\{\{y\}\}$, and $\{\{x\}, \{y\}\}$.

22. (B)

This is very tedious by direct means, but generally the bound is $\epsilon/|f'(c)|$ when f is differentiable and $f'(c) \neq 0$, on an interval containing c . In this case $f'(2) = 23$.

23. (B)

Since $3^3 = 27 \equiv (-1) \pmod{7}$ we find $3^{18} \equiv (-1)^6 \equiv 1 \pmod{7}$, so $3^{20} \equiv 3^2 \pmod{7} \equiv 2 \pmod{7}$.

24. (C)

The Heaviside method yields the coefficient

$$\left. \frac{s^2 + 1}{s^2 - 2} \right|_{s^2 = -3} = \frac{-2}{-5} = \frac{2}{5}.$$

25. (D)

By substitution $y = a(ay^2 + 2)^2 + 2$. Then $y = 1$ implies $a(a + 2)^2 + 1 = 0$ or $a^3 + 4a^2 + 4a + 1 = 0$ with obvious solution $a = -1$ yielding the factor $a^2 + 3a + 1$ with two real (irrational) roots.

26. (D)

$H \cap K$ is always a subgroup and invariance follows easily. HK is a subgroup since invariance guarantees that $HK = KH$. Finally, HK is invariant since $gHK = HgK = HKg$, for all $g \in G$.

27. (E)

Since $x = x^2$ we must have $x + y = (x + y)^2 = x^2 + y^2 + xy + yx = x + y + xy + yx$. So $xy + yx = 0$. When $x = y$ we get $2x^2 = 2x = 0$ so $x = -x$. Combining we find $xy = -yx = yx$ and R is commutative.

28. (C)

For $a = r + s\sqrt{17} \in D$ let $\bar{a} = r - s\sqrt{17} \in D$. The norm of a is $N(a) = a\bar{a} = r^2 - 17s^2$. It is well known that $N(a)N(b) = N(ab)$ and this implies that whenever $N(a)$ is prime then a is irreducible. The norms of the choices show only one prime, $N(9 - 2\sqrt{17}) = 81 - 68 = 13$.

29. (E)

Direct substitution yields

$$k(k-1)x^k - 3kx^k - 4x^k = 0,$$

and since x is not always 0, we can claim that $k^2 - 4k - 4 = 0$ and solve for $k = 2 \pm 2\sqrt{2}$.

30. (C)

Since $(x^2 + 1)^2 = x^4 + 2x^2 + 1$ we can find that

$$x^4 + 1 = (x^2 + 1 + \sqrt{2}x)(x^2 + 1 - \sqrt{2}x).$$

31. (C)

This slope is the directional derivative of z at $(1, 3)$ in the direction given by $\hat{i} + 3\hat{j} = \bar{v}$ (a vector of slope 3). To compute this, we need the gradient vector $\nabla z = z_x \hat{i} + z_y \hat{j}$ or $2x\hat{i} - 2y\hat{j} = 2\hat{i} - 6\hat{j}$. Then the directional derivative is

$$\nabla z \cdot \frac{\bar{v}}{|\bar{v}|} = \frac{(2 - 18)}{\sqrt{10}} = -8\sqrt{\frac{2}{5}}.$$

32. (B)

Since the row sums are all equal to $a + (n - 1)b$ this must be an eigenvalue corresponding to the eigenvector, $\bar{x} = (1, 1, \dots, 1)$.

33. (E)

The correct relationship is

$$\frac{\text{VOLUME}}{\text{AREA}} = \frac{(4\pi r^3)}{\pi r^2} = \frac{4}{3}r.$$

The formula should be

$$\frac{4}{3} \frac{d}{2} \left(\frac{8}{9} d \right)^2 = \frac{2^7 d^3}{3^5}.$$

34. (A)

To have eigenvalue $\lambda = 1$ we must have $|A - I| = 0$.

$$\Rightarrow \begin{vmatrix} k-1 & 1 & 2 \\ 1 & 1 & k \\ 1 & 2 & 2 \end{vmatrix} = 2k^2 - 5k + 2 = (2k-1)(k-2) = 0,$$

so $k = 1/2$ or $k = 2$ will suffice.

35. (A)

In polar form, $z = e^{i\pi/4}$ and $z^{14} = e^{i14\pi/4}$. The periodic quality of e^z means $z^{14} = e^{-i\pi/2} = -i$.

36. (B)

For a solution of relatively small magnitude, as proposed, we may approximate $e^{-x/100}$ by $e^0 = 1$. Then solving $1 - e^{-x} = e^{-1}$ implies $e^{-x} = 1 - e^{-1}$ or $-x = \ln(1 - e^{-1})$ or

$$x = \ln\left(\frac{e}{e-1}\right) = 1 - \ln(e-1) = 1 - \ln(1.718 +).$$

Since $\sqrt{3} = 1.732 +$ the best choice is $1 - \ln \sqrt{3} = 1 - \frac{1}{2} \ln 3$.

37. (D)

The necessary algebra is

$$\frac{1}{z-4} = \frac{1}{z-1-3} = \frac{\frac{1}{(z-1)}}{1 - \left[\frac{3}{(z-1)}\right]} = \frac{1}{z-1} \sum_{n=0}^{\infty} \left(\frac{3}{z-1}\right)^n,$$

by the formula for the sum of a geometric series. Then the coefficient corresponds to $n = 1$ in the form

$$\sum_{n=0}^{\infty} 3^n (z-1)^{-n-1}.$$

38. (A)

The Fundamental Theorem of Calculus states that

$$\frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x).$$

Then by the chain rule we must have

$$\frac{d}{dx} \left[\int_a^{x^2} f(t) dt \right] = 2x f(x^2) = \frac{2x}{(1+x^6)}$$

in this case. Evaluation at $x = 2$ yields $\frac{4}{(1+64)} = \frac{4}{65}$.

39. (D)

By comparison methods the series has partial sums

$$\begin{aligned} \frac{1}{2 \cdot 3^2} + \frac{1}{3 \cdot 3^3} + \frac{1}{4 \cdot 3^4} + \dots + \frac{1}{n \cdot 3^n} &< \frac{1}{2 \cdot 3^2} + \frac{1}{2 \cdot 3^3} \\ &+ \frac{1}{2 \cdot 3^4} + \dots + \frac{1}{2 \cdot 3^n}. \end{aligned}$$

The latter sum is geometric and its corresponding series sums to

$$\frac{1}{18} \left(\frac{1}{1 - \frac{1}{3}} \right) = \frac{1}{12}.$$

40. (B)

Since the entries in each row are one unit apart, the given matrix is row equivalent to the matrix shown where the rank of 2 is easily visible. Actually $n \geq 2$ could be used without changing the result.

$$\begin{pmatrix} 2 & 3 & 4 & \dots & (n+1) \\ 1 & 1 & 1 & \dots & 1 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix}$$

41. (B)

The inverse x , must satisfy the equation $10x \equiv 1 \pmod{17}$ in modular arithmetic or $10x = 17y + 1$. Apparently $17y$ ends in the digit 9, and so $y = 7$ must be the case. Then $10x = 119 + 1$ implies $x = 12$.

42. (E)

By substitution of $y = x + b$ from the latter equation we get $x^2 - x - b = a$ or $x^2 - x - (a + b) = 0$ in the former. The quadratic formula implies two distinct solutions for x unless $(a + b) = -1/4$, and then $x = 1/2$ follows from $(x - 1/2)^2 = 0$.

43. (D)

By basic properties, $\log_4 64 = \log_4 4^3 = 3$. Clearly, $\log_7 343 = \log_7 7^3 = 3$. Also a standard identity for change of base is $\log_b x = \frac{\log_a x}{\log_a b}$, so (B) is also correct.

44. (A)

The points on the unit circle corresponding to arcs of integral measurement (whole radians) are dense in the circle. We know that y is a limit point of S if every neighborhood of y contains $x \neq y$ such that $x \in S$. Since the whole radians are dense in the circle, the values of $\tan k$, $k = 1, 2, 3, \dots$ are dense in the real numbers. Hence if y is any real number, then any neighborhood of y will also contain a value of $\tan(k) \neq y$, $k = 1, 2, 3, \dots$, so the limit points of S consist of all the points on the real line. This means that the entire range $(-\infty, \infty)$ of the real function $f(x) = \tan x$ is the set of limit points.

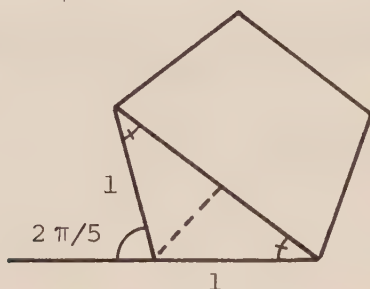
45. (E)

As in problem 44, the points on the unit circle corresponding to whole radians are dense in the circle. Hence the values of $\sin(n)$, $n = 1, 2, 3, \dots$ are dense in the range of $\sin n$, namely $[-1, 1]$. Therefore,

$\liminf_{n \rightarrow \infty} \{\sin n\} = \inf\{-1, 1\} = -1$ and $\limsup_{n \rightarrow \infty} \{\sin n\} = \sup\{-1, 1\} = 1$ so the difference is $-1 - (-1) = 0$.

46. (A)

In the diagram, the diagonal is the base of an isosceles triangle with external angle $2\pi/5$. The base angles must be $\pi/5$. Therefore, the base is $2 \cos \pi/5$.

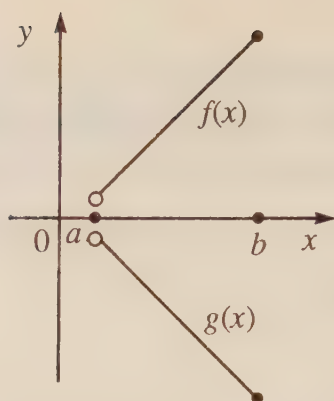


47. (D)

The characteristic polynomial of the equation is $m^2 - 2m - 8 = (m - 4)(m + 2)$, implying that the general form of all solutions is $y = c_1 e^{4x} + c_2 e^{-2x}$. Only (D) is in this form.

48. (E)

If $f(x)$ is strictly increasing then $g(x) = -f(x)$ must be strictly decreasing and hence has no maximum since $g(a)$ is undefined.



49. (B)

Permutations of $n = n_1 + n_2 + n_3$ objects, when groups of n_1 , n_2 , and n_3 are not distinguishable, total $\frac{n!}{n_1!n_2!n_3!}$. In this case $n = 6$, $n_1 = 1$, $n_2 = 2$, $n_3 = 3$. So we get $\frac{6!}{1!2!3!} = \frac{720}{12} = 60$.

50. (D)

The slope of the secant line

$$= \frac{f(3) - f(-3)}{3 - (-3)} = \frac{107 - (-103)}{6} = 35.$$

Slope of $f(x)$ at $x = c$ is $f'(c) = 12c^2 - 1$.

Then $12c^2 - 1 = 35$, so $c^2 = 3$ and $c = \sqrt{3}$.

51. (B)

In spherical coordinates, ϕ represents the angle from the positive z -axis to the line generating a cone with z -axis of symmetry and center at the origin. In spherical coordinates $\rho^2 = x^2 + y^2 + z^2 = 4$, so $\rho = 2$.

$$\text{Also } 2 \cos \Phi = z = -\sqrt{x^2 + y^2}$$

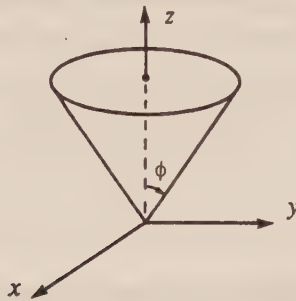
$$= -\sqrt{\rho^2 \sin^2 \Phi \cos^2 \theta + \rho^2 \sin^2 \Phi \sin^2 \theta}$$

$$\Rightarrow 2 \cos \Phi = -\sqrt{\rho^2 \sin^2 \Phi} = -\rho \sin \Phi = -2 \sin \theta$$

$$\Rightarrow \frac{\sin \theta}{\cos \theta} = -1$$

$$\Rightarrow \theta = \frac{3\pi}{4}.$$

So in this case, the surfaces, in spherical coordinates, have equations $\rho = 2$, (a sphere) and $\Phi = 3\pi/4$ (a lower half-cone).



52. (B)

Each subcomponent has reliability $1 - (1-R)^2 = R(2-R)$. Then the series $0 \leq n$ subcomponents must have reliability $R^n(2-R)^n$.

53. (D)

For the given family, $y' = y/x$; thus, the orthogonal family satisfies the differential equation $y' = -x/y$. In separated form, we have $x dx + y dy = 0$, which can be integrated directly to yield $\frac{x^2}{2} + \frac{y^2}{2} = C$, where C is arbitrary, so $x^2 + y^2 = 2C = r^2$, giving circles centered at $(0, 0)$.

54. (E)

In standard form the linear difference equation is $x_{n+2} - x_{n+1} - 6x_n = -36n$, which implies characteristic roots 3 and -2 and a particular solution in the form $An + B$. Substitution leads to matching of coefficients via $-6An + A - 6B = -36n$, so $-6A = -36$ and $A = 6$. Then $6 - 6B = 0$ and $B = 1$. All solutions must be in the form $x_n = c_1 3^n + c_2 (-2)^n + 6n + 1$ where c_1, c_2 are arbitrary constants. (E) is in this form with $c_1 = 0$ and $c_2 = 2$.

55. (A)

Continuity implies $f^{-1}(U)$ must be open in $\overline{X}$ for every open set U in $\overline{X}$. But then $f(f^{-1}(U)) = U$ shows U is the image under f of an open set. Since f is a surjection, $f^{-1}(U)$ is well-defined for every open set. A counterexample for both (B) and (C) is $f(x) = e^x \sin x$ on the reals.

56. (D)

The negative of the gradient vector always supplies this direction. Here

$$\begin{aligned} -\nabla z &= -y^2 \hat{i} + 2xy \hat{j} \Big|_{(2, -1)} \\ &= -\hat{i} + 4\hat{j}. \end{aligned}$$

57. (D)

Euler's formula states that $V - E + F = 2$. Here $V = 27$ and $E = 40$ so $F = 15$.

58. (C)

The term in the expansion is given combinatorially by

$$\binom{5}{3} x^2 (-2y)^3 = -80 x^2 y^3 .$$

59. (B)

The partial fractions expansion multiplied by x gives

$$x \left[\frac{A}{x} + \frac{B}{2+3x} + \frac{C}{(2+3x)^2} \right] = \frac{x}{x(2+3x)^2}$$

Taking limits as $x \rightarrow +\infty$ yields $A + B/3 = 0$. $A = 1/4$ by the following calculations:

$$\frac{A}{x} + \frac{B}{2+3x} + \frac{Cx+D}{(2+3x)^2} = \frac{1}{x(2+3x)^2}$$

$$\Rightarrow A(2+3x)^2 + Bx(2+3x) + (Cx+D)x = 1$$

$$\Rightarrow 4A = 1, \text{ since all other terms will contain a factor of } x,$$

$$\Rightarrow A = \frac{1}{4}.$$

So $B = -3/4$ and the integral of this term is $\left(-\frac{3}{4}\right)\left(\frac{1}{3}\right) \ln |2+3x|$.

60. (D)

It is well known that in the waiting time model the probability of a tail will be $1/T$. Then the probability of at least two tails is the complement of zero or one tail. As usual, this is

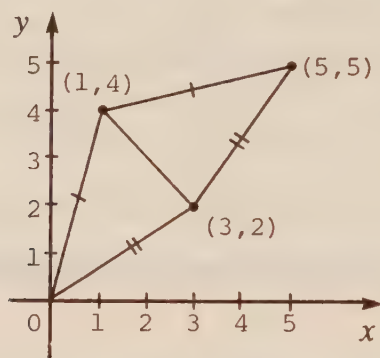
$$1 - \left(1 - \frac{1}{T}\right)^{3T} - 3T \left(1 - \frac{1}{T}\right)^{3T-1} \left(\frac{1}{T}\right).$$

Using T^{3T} as a common denominator yields the numerator

$$\begin{aligned} T^{3T} - (T - 1)^{3T} - 3T(T - 1)^{3T-1} \\ = T^{3T} - (T - 1)^{3T-1}(T - 1 + 3T). \end{aligned}$$

61. (B)

The quadrilateral can be split into two triangles by the diagonal joining $(1, 4)$ and $(3, 2)$ as shown. The triangles have simple centroidal first coordinates $4/3$ and $9/3$ as the average of those for their vertices. The sides of the triangles are identical, so the average of these is the centroid coordinate $4/6 + 9/6 = 13/6$.



62. (A)

All that is needed is to show closure under the operations. For (A), if $xr = 0$ and $xs = 0$ then $xr + xs = 0 + 0 = 0$ and $(xr)(xs) = 0 \cdot 0 = 0$.

(B) will not contain 0 and this is a must for a subring.

(C) may not contain 0 if x is not nilpotent.

(D) fails to guarantee that $(mx)(nx) = mnx^2$ is a multiple of x .
For example, let R be the ring of 2×2 matrices.

63. (E)

The limit can be simplified as

$$\begin{aligned} & \lim_{m \rightarrow 0} \frac{1}{m} \left[\lim_{n \rightarrow 0} \frac{f(2+m+n) - f(2+m)}{n} - \lim_{n \rightarrow 0} \frac{f(2+n) - f(2)}{n} \right] \\ &= \lim_{m \rightarrow 0} \left[\frac{f'(2+m) - f'(2)}{m} \right] = f''(2) = \frac{-1}{2^2} = -\frac{1}{4} \end{aligned}$$

by the definition of derivative since f is twice differentiable at $x = 2$.

64. (A)

Every open set containing a cluster point of E must contain some other point in E . Since $\{x_1\}$ is open, x_1 can never be a cluster point. Likewise $\{x_1, x_2\}$ is open and contains only x_2 in E so x_2 is not a cluster point. For $x_n, n \geq 3$, every open set containing x_n also contains x_2 in E , so these are all cluster points.

65. (C)

The weights can be thought to act as their centers. Further, the proportion of the weight supported at the left is the ratio of the distance to the right support and the total distance between supports. In this case, we get $100(16/20) + 200(3/20) = 80 + 30 = 110$.

66. (A)

The series may diverge if a_n is not a series of terms all of the same sign. For example, let $a_n = \frac{(-1)^n}{\sqrt{n}}$. Then $\sum_{n=1}^{\infty} a_n$ converges by the alternating series test. However, $\sum_{n=1}^{\infty} a_n^2$ is the well-known divergent harmonic series.

The Graduate Record Examination in
MATHEMATICS

Practice Test 3

The Graduate Record Examination in **MATHEMATICS** Answer Sheet Test 3

1. (A) (B) (C) (D) (E)

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66. (A) (B) (C) (D) (E)

GRE MATHEMATICS TEST III

TIME: 2 hours and 50 minutes
66 Questions

DIRECTIONS: Choose the best answer for each question and mark the letter of your selection on the corresponding answer sheet.

1. The generating function $f(x)$ for the Fibonacci numbers 1, 1, 2, 3, 5, 8, 13, 21, ... is given by

(A) $(1 - x + x^2 - x^3)^{1/2}$ (D) $(1 + x + x^2)^{-1/2}$

(B) $(1 - x - x^2)^{-1}$ (E) $(1 - x^2 + x^3)^{-1}$

(C) $(1 - x - x^2 - x^3)^{-1}$

2. The number of solutions of $p(x) = x^2 + 3x + 2$ in Z_6 is

(A) 0 (D) 3

(B) 1 (E) 4

(C) 2

3. Which of the following matrices is normal? ($i = \sqrt{-1}$)

(A) $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} i & 1 \\ -1 & 0 \end{bmatrix}$

(B) $\begin{bmatrix} 0 & i \\ -1 & 1 \end{bmatrix}$

(E) $\begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}$

4. Find the locus of all points (x, y) , such that the sum of those distances from $(0, 1)$ and $(1, 0)$ is 2.

(A) $x^2 + xy + y^2 - 2x - 2y + 2 = 0$

(B) $3x^2 - 2xy + 3y^2 - 4x + 4y - 2 = 0$

(C) $4x^2 - 2xy + 4y^2 - 2x - 2y = 0$

(D) $3x^2 + 2xy + 3y^2 - 4x - 4y = 0$

(E) $x^2 - 2xy + y^2 + 2x - 2y + 4 = 0$

5. Find $\prod_{k=2}^{+\infty} \left(1 - \frac{1}{k^2}\right)$

(A) $\frac{1}{2}$

(D) $\frac{3}{4}$

(B) $\frac{1}{4}$

(E) $\frac{1}{8}$

(C) 0

6. The cross ratio of the following set of lines is

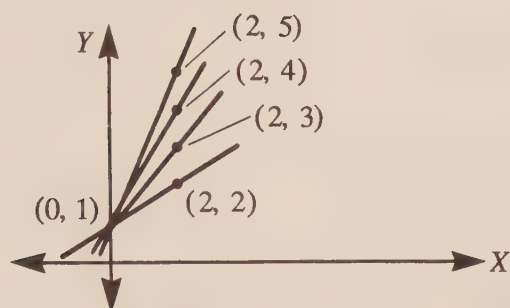
(A) $\frac{4}{3}$

(D) $-\frac{5}{6}$

(B) $\frac{3}{2}$

(E) 1

(C) $\frac{1}{5}$



7. Find the characteristic of the ring $Z_2 + Z_3$.

(A) 0

(D) 4

(B) 6

(E) 2

(C) 3

8. Find $\lim_{n \rightarrow +\infty} (\sqrt{n^4 + i n^2} - n^2)$

(A) $\frac{i}{2}$

(D) $-\frac{1}{2}$

(B) 0

(E) $\sqrt{i}$

(C) $+\infty$

9. Which of the following is a solution of

$$u(x) = x + \int_0^x (t - x) u(t) dt ?$$

- (A) $\sin x$ (D) $x e^{-x}$
(B) $x \cos x$ (E) $x e^x$
(C) $\ln(x + 1)$

10. Find $\int_0^1 \left(\ln \frac{1}{x} \right)^5 dx$

- (A) 120 (D) 720
(B) $+\infty$ (E) 24
(C) 1

11. Find the Laplace transform of

$$f(x) = \begin{cases} 0 & \text{if } x \in (-\infty, 1) \\ 1 & \text{if } x \in (1, +\infty) \end{cases}$$

- (A) e^{-p} (D) $p e^p$
(B) $\frac{1}{p}$ (E) $\frac{1}{p e^p}$
(C) $\frac{1}{e^{p-1}}$

12. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$T(x, y) = \begin{bmatrix} 2x - y \\ x + 3y \end{bmatrix}$$

Find the adjoint T^* of T .

(A) $\begin{bmatrix} 2x + y \\ -x + 3y \end{bmatrix}$

(D) $\begin{bmatrix} \frac{x}{2} - y \\ -x + \frac{y}{3} \end{bmatrix}$

(B) $\begin{bmatrix} x + 2y \\ x - 3y \end{bmatrix}$

(C) $\begin{bmatrix} 2x + y \\ x - 3y \end{bmatrix}$

(E) $\begin{bmatrix} 3x - y \\ x + 2y \end{bmatrix}$

13. The value of $I = \int_0^{\frac{\pi}{2}} \frac{\cos x}{\cos x + \sin x} dx$ is

(A) 1

(D) $\frac{\pi}{4}$

(B) $\frac{\pi}{2}$

(E) π

(C) 0

14. Find the discriminant of the ternary quadratic form $x^2 - y^2 + z^2 - 2xy + 4yz - 6xz$

(A) 25

(D) 15

(B) 13

(E) 19

(C) 0

15. The radius of curvature of $f(x) = x + \frac{1}{x}$ at $P(1, 2)$ is

(A) 1

(D) 2

(B) $\sqrt{2}$

(E) $\frac{1}{2}$

(C) 4

16. If $\Gamma(p)$ represents the gamma function, then $\int_0^{+\infty} e^{-x^2} dx$ is equal to $\frac{1}{2} \Gamma(p)$ when p is equal to

(A) -1

(D) $\frac{1}{4}$

(B) $\frac{1}{2}$

(E) 2

(C) 1

17. The factor group $\frac{(Z_2 \times Z_3)}{\langle (1, 0) \rangle}$ has order

(A) 2

(D) 1

(B) 3

(E) 6

(C) 4

18. Let $R[0, 1]$ denote the set of Riemann integrable functions defined on $[0, 1]$. Which of the following is not satisfied by the function d defined on $R[0, 1]$ by

$$d(f, g) = \int_0^1 |f(x) - g(x)| dx ?$$

- (A) $d(f, f) = 0$
- (B) $d(f, g) \geq 0$
- (C) $d(f, g) > 0$ if $f \neq g$
- (D) $d(f, g) = d(g, f)$
- (E) $d(f, g) \leq d(f, h) + d(h, g)$

19. Find the simple continued fraction for $\frac{13}{42}$.

(A) $\frac{1}{2 + \frac{1}{3 + \frac{1}{3}}}$

(D) $\frac{1}{4 + \frac{1}{4 + \frac{1}{3}}}$

(B) $\frac{1}{3 + \frac{1}{3 + \frac{1}{4}}}$

(E) $\frac{1}{3 + \frac{1}{4 + \frac{1}{3}}}$

(C) $\frac{1}{3 + \frac{1}{2 + \frac{1}{4}}}$

20. Which of the following polynomials satisfies an Eisenstein criterion for irreducibility over the rationals?

(A) $x^5 + 3x^4 + 18x^2 + 15x + 9$

(B) $2x^5 + 9x^4 + 15x^2 + 3x + 18$

(C) $3x^5 + 18x^4 + 15x^2 + 9x + 3$

(D) $4x^5 + 15x^4 + 9x^2 + 3x + 18$

(E) $5x^5 + 9x^4 + 18x^2 + 3x + 15$

21. The number of degrees that the conic, defined by

$$x^2 - y^2 + 2\sqrt{3}xy = 2$$

must be rotated in order to eliminate the xy term is

(A) 15

(D) 60

(B) 30

(E) 75

(C) 45

22. For the initial value problem $y'' + 6y' + 9y = 0$; $y(0) = 3$; $y'(0) = -11$ find $\mathcal{L}(y)$, the Laplace transform of y .

(A) $\frac{1}{p+3} + \frac{2}{(p+3)^2}$

(B) $\frac{2}{p+3} - \frac{3}{(p+3)^2}$

(C) $\frac{3}{(p+3)^2}$

(D) $\frac{3}{p+3} - \frac{2}{(p+3)^2}$

(E) $\frac{1}{p+3} - \frac{2}{(p+3)^2}$

23. Find the join of the subgroups $\langle 4 \rangle$ and $\langle 6 \rangle$ of Z_{12} .

(A) $\langle 0 \rangle$

(D) $\langle 4 \rangle$

(B) $\langle 2 \rangle$

(E) $\langle 5 \rangle$

(C) $\langle 3 \rangle$

24. For which n is the regular n -gon not constructible with a straightedge and compass?

(A) 6

(D) 17

(B) 9

(E) 20

(C) 15

25. Given $T = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, the sum of the elements in T^n is

(A) $3n$

(B) $n+3$

(C) n

(D) $2n$

(E) $n + 2$

26. Let $b: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ be a bilinear form defined by

$$b(X, Y) = x_1 y_1 - 2x_1 y_2 + x_2 y_1 + 3x_2 y_2$$

where $X = (x_1, x_2)$ and $Y = (y_1, y_2)$. Find the 2×2 matrix B of b relative to the basis $U = \{u_1, u_2\}$ where $u_1 = (0, 1)$ and $u_2 = (1, 1)$.

(A) $\begin{bmatrix} 5 & -3 \\ 0 & 2 \end{bmatrix}$

(D) $\begin{bmatrix} 3 & 4 \\ 1 & 3 \end{bmatrix}$

(B) $\begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$

(E) $\begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}$

(C) $\begin{bmatrix} -1 & 4 \\ 2 & 3 \end{bmatrix}$

27. Let $\mathcal{X} = \{a, b, c\}$. Which of the following classes of subsets of $\mathcal{X}$ does not form a topology on $\mathcal{X}$?

(A) $\{\mathcal{X}, \emptyset\}$

(B) $\{\mathcal{X}, \emptyset, \{a\}\}$

(C) $\{\mathcal{X}, \emptyset, \{a\}, \{b\}, \{a, b\}\}$

(D) $\{\mathcal{X}, \emptyset, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$

(E) $\mathcal{P}(\mathcal{X})$, the power set of $\mathcal{X}$

28. The eigenvalues for the initial value–eigenvalue problem

$$y'' + \lambda y = 0$$

$$y(0) = 0; \quad y(\pi) = 0$$

are given by

- (A) 1, 2, 3, 4, ... (D) 0, ± 1 , ± 4 , ± 9 , ± 16 , ...
(B) 1, 4, 9, 16, ... (E) ..., -3, -2, -1
(C) 0, ± 1 , ± 2 , ± 3 , ± 4 , ...

29. For switching functions f , g , and h , the expression $(f \vee g)(\bar{f} \vee h)$ is equivalent to

- (A) $g\bar{f} \wedge gh$ (D) $g\bar{f} \vee fh \vee gh$
(B) $g \vee h$ (E) $g\bar{f} \wedge fh \vee gh$
(C) $g \wedge fh$

30. For the inner product $\langle A, B \rangle = \text{trace}(B' A)$ defined on the vector space of 2 by 2 matrices on $\mathbb{R}$, find the square of the norm of

$$T = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}.$$

- (A) 5 (B) 10

(C) 15

(D) 20

(E) 25

31. Find the limit of the series

$$x^4 + \frac{x^4}{1+x^2} + \frac{x^4}{(1+x^2)^2} + \frac{x^4}{(1+x^2)^3} + \dots$$

(A) $x^6 + x^4$

(D) $x^4 + \frac{x^6}{1+x^2}$

(B) $\frac{x^6}{1+x^2}$

(E) $x^4 + x^2$

(C) x^6

32. The domain of $f(x) = \int (x + 2x^2 + 3x^3 + \dots) dx$ is

(A) $(-1, 1)$

(D) $\left[\frac{1}{2}, \frac{1}{2}\right)$

(B) $[-1, 1)$

(E) $(-1, 1]$

(C) $\left[-\frac{1}{2}, \frac{1}{2}\right]$

33. Determine the number of homomorphisms from the group Z_8 onto the group Z_4 .

(A) 0

(B) 1

(C) 2

(D) 3

(E) 4

34. The solution of $x^2 y'' + 6xy' + 6y = 0$, for $x > 0$, is given by

(A) $c_1 x^3 - c_2 x^2$

(D) $c_1 e^{-2x} + c_2 e^{-3x}$

(B) $c_1 x + c_2 \ln x$

(E) $c_1 x \ln x + c_2 \ln x$

(C) $\frac{c_1}{x^3} + \frac{c_2}{x^2}$

35. The remainder of 5^{34} when divided by 17 is

(A) 0

(D) 6

(B) 2

(E) 8

(C) 4

36. Find the curl of $\vec{u} = xyz \vec{i} + xy^2 \vec{j} + yz \vec{k}$

(A) $xz \vec{i} + (x - yz) \vec{j} + 2y \vec{k}$

(B) $(x - z) \vec{i} - yz \vec{j} + xyz \vec{k}$

(C) $z \vec{i} + xy \vec{j} + (y^2 - xz) \vec{k}$

(D) $xy \vec{i} - (z - y) \vec{j} + (xy - yz) \vec{k}$

(E) $(xy - yz) \vec{i} - yz \vec{j} + x \vec{k}$

37. The inverse of the function $f(x) = \frac{x}{x-1}$ is

(A) $\frac{x}{x+1}$

(D) $1 - \frac{1}{x}$

(B) $\frac{x}{x-1}$

(E) $\frac{x-1}{x+1}$

(C) $1 + \frac{1}{x}$

38. Find the slope of the tangent line to the ellipse

$$2x^2 + y^2 + 30 = 8y - 12x$$

at (x_0, y_0) , where $x_0 = -2$ and $y_0 > 4$.

(A) $\frac{1}{\sqrt{2}}$

(D) $\frac{1}{2}$

(B) $-\sqrt{2}$

(E) -2

(C) 2

39. For matrices

$$A = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix},$$

$$\text{and } D = \begin{bmatrix} 3 & 3 \\ 0 & -2 \end{bmatrix},$$

the matrix D is a linear combination ($aA + bB + cC$) of A, B, C for a, b, c , given by

(A) $1, 1, -1$

(D) $1, -2, 1$

(B) $2, 1, -1$

(E) $-1, 1, -2$

(C) $2, 2, -2$

40. Given $p(x) = \sum_{k=1}^{+\infty} \frac{(x-2)^k}{k^2}$, find the interval in which $p'(x)$ converges.

(A) $\{2\}$

(D) $(1, 3]$

(B) $[1, 3)$

(E) $\mathbb{R} \rightarrow \mathbb{R}$

(C) $[1, 3]$

41. Find the Laplace transform of $\int_0^x \sin 2t \, dt$.

(A) $\frac{1}{p^2 + 4}$

(D) $\frac{4}{p^4 + 16}$

(B) $\frac{2p}{p^2 + 4}$

(E) $\frac{1}{p^2 + 2p}$

(C) $\frac{2}{p^3 + 4p}$

42. If $f'(x_0) = \sqrt{3}$, then the tangent line to the graph of f at x_0 makes an angle of β degrees with the positive x -axis. Find β .

(A) 0 (D) 60

(B) 30 (E) 90

(C) 45

43. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $T(x, y, z) = (x + y, x - y + z, y + 2z)$. Find the trace of T .

(A) 5 (D) 7

(B) -1 (E) 2

(C) 0

44. Find the sum of $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots$.

(A) $\frac{3}{4}$ (D) $\frac{3}{2}$

(B) 1 (E) $\frac{7}{4}$

(C) $\frac{5}{4}$

45. Which of the following is not a proper ideal of the ring Z_{12} ?

(A) $\langle 5 \rangle$

(D) $\langle 3 \rangle$

(B) $\langle 8 \rangle$

(E) $\langle 4 \rangle$

(C) $\langle 2 \rangle$

46. Assuming that a person selects an answer to each of the first ten questions on this examination at random and that the selections are independent, what is the probability that he/she will guess exactly five answers correct?

(A) $\frac{(63)4^6}{5^{10}}$

(D) $\frac{(61)4^6}{5^{10}}$

(B) $\frac{(65)4^6}{5^{10}}$

(E) $\frac{(67)4^6}{5^{10}}$

(C) $\frac{4^9}{5^{10}}$

47. Find the Jacobian of the transformation from the xy -plane to the uv -plane defined by

$$u = f(x, y) = xe^{xy}$$

$$v = g(x, y) = ye^{xy}$$

(A) $2xye^{xy}$

(D) $(2xy + 1)e^{2xy}$

(B) $(1 - x^2y^2)e^{2xy}$

(E) 0

(C) $2e^{2xy}$

48. On average, a baseball player gets a hit in one out of three attempts. Assuming that the attempts are independent, what is the probability that he gets exactly three hits in six attempts?

(A) $\frac{160}{3^6}$

(D) $\frac{80}{3^6}$

(B) $\frac{160}{3^5}$

(E) $\frac{40}{3^6}$

(C) $\frac{1}{2}$

49. Find the number of units in the ring Z_5 .

(A) 0

(D) 3

(B) 1

(E) 4

(C) 2

50. Define $f(x) = x$ for $x \in (0, 1)$. Find the coefficient of the third term in the half range Fourier sine series.

(A) $\frac{2}{3\pi}$

(D) $\frac{1}{3\pi}$

(B) $\frac{4}{9\pi^2}$

(E) $\frac{1}{9\pi^2}$

(C) $\frac{2}{9\pi^2}$

51. Let V be the vector space of functions $f: \mathbb{R} \rightarrow \mathbb{R}$. Let S be the subspace generated by $\{e^x, e^{2x}, e^{-2x}\}$. Define D_x to be the derivative operator on S . Find the determinant of D_x .

(A) 2

(D) 1

(B) 0

(E) -1

(C) -4

52. Find Green's function for $y'' + 5y' + 6y = \sin x$

(A) $2e^{2(t-x)} + 3e^{3(t-x)}$

(D) $2e^{(t-x)} - 3e^{(t-x)}$

(B) $e^{2(t-x)} - e^{3(t-x)}$

(E) $e^{3(x-t)} - e^{2(x-t)}$

(C) $e^{(t+x)} - e^{(t-x)}$

53. The Maclaurin series for xe^{-x^2} is given by

(A) $x - x^3 + \frac{x^5}{2!} - \frac{x^7}{3!} + \dots$

(B) $x^3 - \frac{x^5}{2!} + \frac{x^7}{3!} - \frac{x^9}{4!}$

(C) $x - \frac{x^3}{2!} + \frac{x^5}{3!} - \frac{x^7}{5!} + \dots$

(D) $x^3 + \frac{x^5}{2!} - \frac{x^7}{3!} + \frac{x^9}{4!} - \dots$

(E) $x + x^3 - \frac{x^5}{2!} + \frac{x^7}{3!} - \dots$

54. The symmetric difference of the sets $S = \{1, 2, 3, 4, 5\}$ and $T = \{4, 5, 6, 7, 8\}$ is

- (A) $\{4, 5\}$ (D) $\{3, 4, 5, 6\}$
(B) $\emptyset$ (E) $\{1, 2, 3, 6, 7, 8\}$
(C) $\{1, 2, 3, 4, 5, 6, 7, 8\}$

55. Given

$$x_{n+2} + 6x_{n+1} + 9x_n = 0 \quad (n = 0, 1, 2, \dots)$$

$$x_0 = 1 ; x_1 = 0 ,$$

then $x_5 =$

- (A) 576 (D) 972
(B) -834 (E) 774
(C) 1068

56. Let V be the vector space of real polynomials with inner product

$$(f, g) = \int_0^1 f(x) g(x) dx$$

where $f, g \in V$. Find the cosine of the angle between $f(x) = 2$ and $g(x) = x$.

- (A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$

(C) 0

(D) 1

(E) $\frac{\sqrt{2}}{2}$

57. The integral $\int_{-24}^4 \frac{dx}{\sqrt[3]{(x-3)^2}}$

(A) converges to 6

(D) converges to 12

(B) diverges to $+\infty$

(E) diverges to $-\infty$

(C) converges to 9

58. Let n be a positive integer greater than 3. Then $n^3 + (n+1)^3 + (n+2)^3$ is divisible by

(A) 9

(D) 6

(B) 4

(E) 15

(C) 12

59. If F is a finite field, then which of the following numbers can be the cardinality of F ?

(A) 21

(B) 45

(C) 27

(D) 14

(E) 33

60. Consider the set $S = \{2, 3, 4, 6, 8, 9\}$ ordered by “ s is a multiple of t ”. How many minimal elements does S have?

(A) 0

(D) 3

(B) 1

(E) 4

(C) 2

61. Which of the following is a neighborhood of 0 relative to the usual topology τ for the real numbers?

(A) $(0, 1)$

(D) $[0, 1]$

(B) $[-1, 1]$

(E) $(-1, 0)$

(C) $[-1, 0]$

62. Find the Cauchy number for the permutation

$$\sigma = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 6 & 5 & 1 & 4 & 7 & 2 \end{bmatrix} \in S_7.$$

(A) 1

(B) 2

(C) 3

(D) 4

(E) 5

63. Which of the following ordinary differential equations is exact?

(A) $(x + e^y) dx + (xe^y - 2xye^y - x^2y) dy = 0$

(B) $(ye^{xy} + \cos x) dx + xe^{xy} + 1) dy = 0$

(C) $(\sin x \sin y + y^2) dx + (\cos x \cos y - 2xy) dy = 0$

(D) $(x^2y - y^2) dx + (2 - xy) dy = 0$

(E) $(2x + 3y - 4) dx + (6x + 9y + 2) dy = 0$

64. Let G be a graph with vertices x_1, x_2, x_3, x_4, x_5 . If $\text{val}(x_1) = 2$, $\text{val}(x_2) = 2$, $\text{val}(x_3) = 3$, $\text{val}(x_4) = 3$ and $\text{val}(x_5) = 4$, where the valence of vertex x is denoted $\text{val}(x)$, how many edges does G have?

(A) 4

(D) 8

(B) 9

(E) 5

(C) 7

65. If τ is the discrete topology on the real numbers $\mathbb{R}$, find the closure of (a, b) .

(A) (a, b)

(D) $[a, b]$

(B) $(a, b]$

(E) $\mathbb{R}$

(C) $[a, b)$

66. Define $f: \mathbb{C}^3 \rightarrow \mathbb{C}$ by $f(c) = x - iy + (2 + i)z$ where $c = (x, y, z)$. Find a $\hat{c} \in \mathbb{C}^3$ such that $f(c) = (c, \hat{c})$ for every $\hat{c} \in \mathbb{C}^3$ where $(c, \hat{c})$ is the usual inner product on $\mathbb{C}^3$.

(A) $(1, i, 2 - i)$

(D) $(-1, i, -2 - i)$

(B) $(-1, -i, 2 - i)$

(E) $(-1, i, -2 - i)$

(C) $(1, -i, 2 + i)$

GRE MATHEMATICS

TEST III

ANSWER KEY

1.	B	23.	B	45.	A
2.	E	24.	B	46.	A
3.	D	25.	E	47.	D
4.	D	26.	D	48.	A
5.	A	27.	D	49.	E
6.	A	28.	B	50.	A
7.	B	29.	D	51.	C
8.	A	30.	C	52.	B
9.	A	31.	E	53.	A
10.	A	32.	A	54.	E
11.	E	33.	C	55.	D
12.	A	34.	C	56.	B
13.	D	35.	E	57.	D
14.	D	36.	C	58.	A
15.	E	37.	B	59.	E
16.	B	38.	B	60.	D
17.	B	39.	B	61.	B
18.	C	40.	B	62.	E
19.	E	41.	C	63.	B
20.	E	42.	D	64.	C
21.	B	43.	E	65.	D
22.	D	44.	B	66.	A

GRE MATHEMATICS TEST III

DETAILED EXPLANATIONS OF ANSWERS

1. (B)

Let $a_0, a_1, a_2, \dots, a_n, \dots$ be a sequence. The generating function for this sequence is given by

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n + \dots$$

Since the Fibonacci sequence satisfies $a_n = a_{n-1} + a_{n-2}$ for $n = 2, 3, 4, \dots$, we have

$$\begin{aligned} f(x) &= a_0 + a_1x + (a_0 + a_1)x^2 + (a_1 + a_2)x^3 + \dots \\ &\quad + (a_{n-1} + a_{n-2})x^n + \dots \\ &= a_0 + x(a_1 + a_1x + a_2x^2 + \dots) + x^2(a_0 + a_1x + \dots) \\ &= 1 + xf(x) + x^2f(x) \end{aligned}$$

where use was made of the fact that $a_0 = a_1 = 1$. Hence

$$(1 - x - x^2)f(x) = 1$$

so that

$$f(x) = (1 - x - x^2)^{-1}$$

2. (E)

We have

$$p(0) = 0 + 0 + 2 = 2 \equiv 2 \pmod{6}$$

$$p(1) = 1 + 3 + 2 = 6 \equiv 0 \pmod{6}$$

$$p(2) = 4 + 6 + 2 = 12 \equiv 0 \pmod{6}$$

$$p(3) = 9 + 9 + 2 = 20 \equiv 2 \pmod{6}$$

$$p(4) = 16 + 12 + 2 = 30 \equiv 0 \pmod{6}$$

$$p(5) = 25 + 15 + 2 = 42 \equiv 0 \pmod{6}$$

so that there are four elements of Z_6 satisfying $p(x) = 0$.

3. (D)

A complex matrix M is said to be normal if $MM^* = M^*M$, where M^* is the conjugate transpose of M . If M is real then $M^* = M^T$, where M^T is the transpose of M . We have for

$$M = \begin{bmatrix} i & 1 \\ -1 & 0 \end{bmatrix}, \quad M^* = \begin{bmatrix} -i & -1 \\ 1 & 0 \end{bmatrix}$$

so that

$$MM^* = \begin{bmatrix} i & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} -i & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -i \\ i & 1 \end{bmatrix}$$

and

$$M^*M = \begin{bmatrix} -i & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} i & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -i \\ i & 1 \end{bmatrix},$$

so M is normal.

4. (D)

We have

$$\sqrt{x^2 + (y-1)^2} + \sqrt{(x-1)^2 + y^2} = 2$$

$$\sqrt{x^2 + (y-1)^2} = 2 - \sqrt{(x-1)^2 + y^2}$$

$$4\sqrt{(x-1)^2 + y^2} = 4 - 2x + 2y$$

$$3x^2 + 2xy + 3y^2 - 4x - 4y = 0$$

5. (A)

Let $\{u_k\}_k^{+\infty} = 1$ be a sequence of non-zero numbers and form the sequence $\{p_n\}_n^{+\infty} = 1$ of partial products $p_n = u_1 \cdot u_2 \cdot \dots \cdot u_n = \prod_{k=1}^n u_k$ for $n = 1, 2, \dots$ (actually, this sequence does not have to start at $n = 1$). If p_n converges to p , we say that the infinite product $\prod_{k=1}^{+\infty} u_k$ converges to p . For $u_k = 1 - \frac{1}{k^2}$ we have

$$p_2 = \frac{3}{4} \quad \left(= \frac{2+1}{2 \cdot 2} \right)$$

$$p_3 = \frac{3}{4} \cdot \frac{8}{9} = \frac{2}{3} \quad \left(= \frac{3+1}{2 \cdot 3} \right)$$

$$p_4 = \frac{2}{3} \cdot \frac{15}{16} = \frac{5}{8} \quad \left(= \frac{4+1}{2 \cdot 4} \right)$$

$$p_5 = \frac{5}{8} \cdot \frac{24}{25} = \frac{3}{5} \quad \left(= \frac{5+1}{2 \cdot 5} \right)$$

We claim that $p_n = \frac{n+1}{2n}$. This is true for $p_2 = \frac{2+1}{4} = \frac{3}{4}$.

We assume that it is true for $m = n$ and let $m = n + 1$. Then

$$\begin{aligned} p_{n+1} &= p_n u_{n+1} \\ &= \left[\frac{n+1}{2n} \right] \left[\frac{(n+1)^2 - 1}{(n+1)^2} \right] \\ &= \frac{n^2 + 2n}{2n(n+1)} \\ &= \frac{(n+1)+1}{2(n+1)} \end{aligned}$$

Therefore $p_n = \frac{n+1}{2n}$ for $n = 2, 3, \dots$ so that

$$\begin{aligned}\prod_{k=2}^{+\infty} \left(1 - \frac{1}{k^2}\right) &= \lim_{n \rightarrow +\infty} p_n \\ &= \lim_{n \rightarrow +\infty} \frac{n+1}{2n} \\ &= \frac{1}{2}\end{aligned}$$

6. (A)

The cross ratio R_c of a set of four distinct concurrent lines l_1, l_2, l_3, l_4 , is given by

$$R_c = \frac{(m_3 - m_1)(m_4 - m_2)}{(m_3 - m_2)(m_4 - m_1)}$$

where m_1, m_2, m_3 , and m_4 represent the slopes of l_1, l_2, l_3 and l_4 , respectively. We have $m_1 = 1/2, m_2 = 2/2, m_3 = 3/2$, and $m_4 = 4/2$ so that

$$R_c = \frac{(3-1)(4-2)}{(3-2)(4-1)} = \frac{4}{3}$$

7. (B)

Let R be a ring and n a positive integer such that $n \cdot r = 0$ for all $r \in R$. Then the least positive integer satisfying the equation is called the characteristic of R . If there does not exist a positive integer satisfying the equation, then R is said to have characteristic 0. The ring

$$Z_2 + Z_3 = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2)\}$$

We have

$$1(0,1) \neq (0, 0)$$

$$2(0,1) = (0, 2) \neq (0, 0)$$

$$3(1,1) = (1, 0) \neq (0, 0)$$

$$4(1,1) = (0, 1) \neq (0, 0)$$

$$5(0,1) = (0, 2) \neq (0, 0)$$

However, $6(r, \bar{r}) = (0, 0)$ for all $(r, \bar{r}) \in Z_2 + Z_3$, so the characteristic of $Z_2 + Z_3$ is 6.

8. (A)

We have

$$\begin{aligned} & \lim_{n \rightarrow +\infty} (\sqrt{n^4 + in^2} - n^2) \\ &= \lim_{n \rightarrow +\infty} (\sqrt{n^4 + in^2} - n^2) \frac{\sqrt{n^4 + in^2} + n^2}{\sqrt{n^4 + in^2} + n^2} \\ &= \lim_{n \rightarrow +\infty} \frac{in^2}{\sqrt{n^4 + in^2} + n^2} \\ &= \lim_{n \rightarrow +\infty} \frac{i}{\sqrt{1 + \frac{i}{n^2}} + 1} \\ &= \frac{i}{2} \end{aligned}$$

9. (A)

Successive substitutions of the right hand side of

$$u(x) = x + \int_0^x (t - x) u(t) dt$$

into the $u(t)$ in the integral yields

$$\begin{aligned} u(x) &= x + \int_0^x (t - x) \left[t + \int_0^t (t_1 - t) u(t_1) dt_1 \right] dt \\ &= x + \int_0^x (t^2 - xt) dt + \int_0^x (t - x) \int_0^t (t_1 - t) u(t_1) dt_1 dt \end{aligned}$$

$$\begin{aligned}
&= x - \frac{x^3}{3!} + \int_0^x (t-x) \int_0^t (t_1-t) u(t_1) dt_1 dt \\
&= x - \frac{x^3}{3!} + \int_0^x (t-x) \int_0^t (t_1-t) \left[t_1 + \int_0^{t_1} (t_2-t_1) u(t_2) dt_2 \right] dt_1 dt \\
&= x - \frac{x^3}{3!} + \int_0^x (t-x) \int_0^t (t_1-t) u(t_1) dt_1 dt \\
&\quad + \int_0^x (t-x) \int_0^t (t_1-t) \int_0^{t_1} (t_2-t_1) u(t_2) dt_2 dt_1 dt \\
&= x - \frac{x^3}{3!} + \int_0^x (t-x) \left[-\frac{t^3}{3!} \right] dt \\
&\quad + \int_0^x (t-x) \int_0^t (t_1-t) \int_0^{t_1} (t_2-t_1) u(t_2) dt_2 dt_1 dt \\
&= x - \frac{x^3}{3!} + \frac{x^5}{5!} + \int_0^x (t-x) \int_0^t (t_1-t) \int_0^{t_1} (t_2-t_1) u(t_2) dt_2 dt_1 dt \\
&= \dots = \sin x
\end{aligned}$$

10. (A)

Setting $u = \ln x^{-1}$, we obtain $x = e^{-u}$ and $dx = -e^{-u} du$. Also, $x \rightarrow 0^+$ implies $u \rightarrow +\infty$ and $x \rightarrow 1^-$ implies $u \rightarrow 0^+$. Thus

$$\begin{aligned}
\int_0^1 \left(\ln \frac{1}{x} \right)^5 dx &= \int_{+\infty}^0 u^5 (-e^{-u}) du \\
&= \int_0^{+\infty} u^{6-1} e^{-u} du \\
&= \Gamma(6) = 5! = 120
\end{aligned}$$

11. (E)

Let $f(x)$ be defined on $[0, +\infty)$. The Laplace transform of $f(x)$, denoted $\mathfrak{L}[f(x)](p)$, is the function of p defined by

$$\mathfrak{L}[f(x)](p) = \int_0^{+\infty} e^{-px} f(x) dx$$

The domain of $\mathfrak{L}[f(x)](p)$ is the set of all real numbers p for which the integral converges. We have

$$\begin{aligned} \mathfrak{L}[f(x)](p) &= \int_1^{+\infty} e^{-px} dx \\ &= \lim_{b \rightarrow +\infty} \int_1^b e^{-px} dx \\ &= \lim_{b \rightarrow +\infty} \left[-\frac{e^{-px}}{p} \right]_1^b \\ &= \lim_{b \rightarrow +\infty} \left[\frac{e^{-p}}{p} - \frac{e^{-pb}}{p} \right] \\ &= \frac{e^{-p}}{p} \end{aligned}$$

for $p \in (0, +\infty)$.

12. (A)

The matrix M_T of T relative to $\{(1, 0), (0, 1)\}$ is given by

$$M_T = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$$

The matrix M_T^* for the adjoint T^* of T is

$$M_T^* = M_T^* = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

Thus

$$T^*(x, y) = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x + y \\ -x + 3y \end{bmatrix}$$

13. (D)

Set $x = \pi/2 - u$. Then $dx = -du$, $x = 0$ implies that $u = \pi/2$, and $x = \pi/2$ implies that $u = 0$. Using the identities $\sin(\pi/2 - u) = \cos u$ and $\cos(\pi/2 - u) = \sin u$, we obtain

$$\begin{aligned} I &= \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx = \int_{\pi/2}^0 \frac{\sin u}{\sin u + \cos u} (-du) \\ &= \int_0^{\pi/2} \frac{\sin u}{\sin u + \cos u} du \end{aligned}$$

Hence

$$\begin{aligned} 2I &= \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx + \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx \\ &= \int_0^{\pi/2} 1 dx \\ &= \frac{\pi}{2} \end{aligned}$$

so that $I = \pi/4$.

14. (D)

The discriminant D of the ternary quadratic form

$$a_{11} x^2 + a_{22} y^2 + a_{33} z^2 + 2 a_{12} xy + 2 a_{23} yz + 2 a_{13} xz$$

is the determinant

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$$

Thus

$$\begin{aligned} D &= \det \begin{bmatrix} 1 & -1 & -3 \\ -1 & -1 & 2 \\ -3 & 2 & 1 \end{bmatrix} \\ &= (-1 + 6 + 6) - (-9 + 4 + 1) \\ &= 15 \end{aligned}$$

15. (E)

The curvature $\kappa(x)$ of $f(x)$ at $P(x_0, f(x_0))$ is given by

$$\kappa(x_0) = \frac{f''(x_0)}{\{1 + [f'(x_0)]^2\}^{3/2}}$$

and the radius of curvature $R(x)$ of $f(x)$ at $P(x_0, f(x_0))$ is given by

$$R(x_0) = \frac{1}{|\kappa(x_0)|}.$$

We have

$$f'(x) = 1 - \frac{1}{x^2}; \quad f'(1) = 0$$

$$f''(x) = -\frac{2}{x^3}; \quad f''(1) = -2$$

so that $\kappa(1) = -2$ and $R(1) = 1/2$.

16. (B)

The gamma function $\Gamma(p)$ is defined by

$$\Gamma(p) = \int_0^{+\infty} x^{p-1} e^{-x} dx$$

Setting $x = u^{1/2}$, we obtain $dx = u^{-1/2} du$ so that

$$\begin{aligned} \int_0^{+\infty} e^{-x^2} dx &= \int_0^{+\infty} \frac{1}{2} u^{-1/2} e^{-u} du \\ &= \frac{1}{2} \int_0^{+\infty} u^{1/2-1} e^{-u} du \\ &= \frac{1}{2} \Gamma\left(\frac{1}{2}\right) \end{aligned}$$

17. (B)

We have

$$\langle (1, 0) \rangle = \{(0, 0), (1, 0)\}$$

so that all cosets of $\langle (1, 0) \rangle$ must have two elements. Since $Z_2 \times Z_3$ has six elements, $(Z_2 \times Z_3)/\langle (1, 0) \rangle$ contains three elements:

$$\begin{aligned} (0, 1) + \langle (1, 0) \rangle &= \{(0, 1), (1, 1)\} \\ (0, 0) + \langle (1, 0) \rangle &= \{(0, 0), (1, 0)\} \\ (1, 2) + \langle (1, 0) \rangle &= \{(1, 2), (0, 2)\} \end{aligned}$$

18. (C)

The function d would be called a metric for $R[0, 1]$ if it satisfied all of the conditions (A) – (E). Since condition (C) is not satisfied, the function d is called a pseudometric for $R[0, 1]$. To see this, define

$$f(x) = 1 \text{ if } x \in [0, 1]$$

and

$$g(x) = \begin{cases} 1 & \text{if } x \in [0, 1) \\ 2 & \text{if } x = 1 \end{cases}.$$

Then $f, g \in R[0, 1]$ with $f \neq g$, but $d(f, g) = 0$, as

$$\int_0^1 |f(x) - g(x)| dx = \int_0^1 |1 - 1| dx = 0.$$

19. (E)

Let $m (> 0)$ and n be integers. The Division Algorithm states that there exist unique integers q and r such that

$$\frac{n}{m} = q + \frac{r}{m} \text{ where } 0 \leq \frac{r}{m} < 1.$$

Repeated applications of this algorithm yield

$$\begin{aligned} \frac{13}{42} &= 0 + \frac{1}{\frac{42}{13}} \\ &= \frac{1}{3 + \frac{1}{\frac{13}{3}}} \\ &= \frac{1}{3 + \frac{1}{4 + \frac{1}{3}}}. \end{aligned}$$

20. (E)

Let $p(x) = a_N x^N + a_{N-1} x^{N-1} + \dots + a_1 x + a_0$ be an element of $Z[x]$. The Eisenstein criterion states that for a prime p , if $a_N \not\equiv 0 \pmod{p}$, $a_j \equiv 0 \pmod{p}$ for $j = 0, 1, 2, 3, \dots, N-1$ and $a_0 \not\equiv 0 \pmod{p^2}$, then $p(x)$ is irreducible over the rationals. We have, for $p = 3$,

$$5 \not\equiv 0 \pmod{3}$$

$$9 \equiv 0 \pmod{3}$$

$$0 \equiv 0 \pmod{3}$$

$$18 \equiv 0 \pmod{3}$$

$$3 \equiv 0 \pmod{3}$$

$$15 \equiv 0 \pmod{3}$$

$$15 \not\equiv 0 \pmod{9}$$

Therefore $5x^5 + 9x^4 + 18x^2 + 3x + 15$ satisfies an Eisenstein criterion ($p = 3$).

21. (B)

The angle θ that a conic $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ must be rotated in order to eliminate the xy term must satisfy

$$\cot 2\theta = \frac{A - C}{B}.$$

For the given conic, we must have

$$\cot 2\theta = \frac{1 + 1}{2\sqrt{3}} \text{ or } \tan 2\theta = \sqrt{3}.$$

Therefore $\theta = 30^\circ$.

22. (D)

We have

$$\begin{aligned} 0 &= \mathcal{L}[y'' + 6y' + 9y] \\ &= p^2 \mathcal{L}(y) - [p y(0) + y'(0)] + 6\{p \mathcal{L}(y) - y(0)\} + 9 \mathcal{L}(y) \\ &= [p^2 + 6p + 9] \mathcal{L}(y) = 3p + 7 \end{aligned}$$

Thus

$$\begin{aligned}\mathfrak{f}(y) &= \frac{3p+7}{(p+3)^2} \\ &= \frac{A}{p+3} + \frac{B}{(p+3)^2}\end{aligned}$$

Setting $p = -3$ in $3p+7 = A(p+3) + B$ yields $B = -2$. For $p = 0$, we obtain $A = 3$.

23. (B)

The join of subgroups S_1 and S_2 of G is the smallest subgroup of G containing S_1 and S_2 . Thus the join must contain $\{0, 4, 6, 8\}$. Since $4 + 6 \equiv 10 \pmod{12}$ and $6 + 8 \equiv 2 \pmod{12}$, the join equals $\langle 2 \rangle$.

24. (B)

Since a 60° angle cannot be trisected, a 20° angle cannot be constructed. A regular 9-gon is constructible if and only if the angle $\frac{360}{9} = 40^\circ$ is constructible. Thus, a 9-gon is not constructible because one of the 40° angles formed could then be bisected to construct a 20° angle.

25. (E)

We have

$$T^2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

$$T^3 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^3 = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

and, in general,

$$T^n = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}.$$

Hence, the sum of the elements in T^n is $n + 2$.

26. (D)

Let V be a finite dimensional vector space over a field F . A bilinear form $b(v, \bar{v})$ is a function $b: V \times V \rightarrow F$ which satisfies

- 1) $b(\alpha v_1 + \beta v_2, \bar{v}) = \alpha b(v_1, \bar{v}) + \beta b(v_2, \bar{v})$
- 2) $b(v, \delta \bar{v}_1 + v \bar{v}_2) = \delta b(v, \bar{v}_1) + v b(v, \bar{v}_2)$
- 3) $b(\lambda v, \bar{v}) = \lambda b(v, \bar{v}) = b(v, \lambda \bar{v})$

where $v_1, v_2, \bar{v}_1, \bar{v}_2 \in V$ and $\alpha, \beta, \delta, v \in F$. The matrix $B = (b_{ij})$ is given by $b_{ij} = b(u_i, u_j)$ for $1 \leq i \leq 2$ and $1 \leq j \leq 2$. We have

$$\begin{aligned} b_{11}(u_1, u_1) &= b((0, 1), (0, 1)) = 0 \times 0 - 2(0 \times 1) + 1 \times 0 + 3(1 \times 1) \\ &= 0 - 0 + 0 + 3 = 3 \\ b_{12}(u_1, u_2) &= b((0, 1), (1, 1)) = 0 \times 1 - 2(0 \times 1) + 1 \times 1 + 3(1 \times 1) \\ &= 0 - 0 + 1 + 3 = 4 \\ b_{21}(u_2, u_1) &= b((1, 1), (0, 1)) = 1 \times 0 - 2(1 \times 1) + 1 \times 0 + 3(1 \times 1) \\ &= 0 - 2 + 0 + 3 = 1 \\ b_{22}(u_2, u_2) &= b((1, 1), (1, 1)) = 1 \times 1 - 2(1 \times 1) + 1 \times 1 + 3(1 \times 1) \\ &= 1 - 2 + 1 + 3 = 3 \end{aligned}$$

Thus,

$$B = \begin{bmatrix} 3 & 4 \\ 1 & 3 \end{bmatrix}$$

27. (D)

A class of subsets τ of X is called a topology on X if τ satisfies the following axioms:

$$(1) \quad X \in \tau; \emptyset \in \tau$$

$$(2) \quad S_\alpha \in \tau \text{ for } \alpha \in A \text{ implies } \bigcup_{\alpha \in A} S_\alpha \in \tau$$

$$(3) \quad S_1, S_2 \in \tau \text{ implies } S_1 \cap S_2 \in \tau$$

Consider $\tau = \{X, \emptyset, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$. We have $\{a, b\} \in \tau$ and $\{b, c\} \in \tau$ but $\{b\} = \{a, b\} \cap \{b, c\}$ is not an element of τ . Thus τ does not form a topology on X .

28. (B)

An eigenvalue λ for a differential equation of the form $y'' + \lambda y = 0$ with initial conditions $y(x_1) = y_1; y(x_2) = y_2$ is a number for which the initial value problem has a non-trivial solution. The auxiliary equation of the given equation is $m^2 + \lambda = 0$ so that $m = \pm \sqrt{-\lambda}$. We consider three cases: $\lambda = 0; \lambda < 0; \lambda > 0$. For $\lambda = 0$, the general solution is $y(x) = c_1 + c_2 x$. Since $y(0) = 0$ and $y(\pi) = 0$, we have

$$0 = y(0) = c_1$$

$$0 = y(\pi) = c_2 \pi$$

so that $y(x) \equiv 0$. For $\lambda < 0$, the general solution is

$$y(x) = c_1 e^{\sqrt{-\lambda}x} + c_2 e^{-\sqrt{-\lambda}x}$$

Substitution of the initial conditions into $y(x)$ yields

$$0 = y(0) = c_1 + c_2$$

$$0 = c_1 e^{\sqrt{-\lambda}\pi} + c_2 e^{-\sqrt{-\lambda}\pi}$$

Since

$$\begin{vmatrix} 1 & 1 \\ e^{\sqrt{-\lambda}\pi} & e^{-\sqrt{-\lambda}\pi} \end{vmatrix} \neq 0$$

$c_1 = c_2 = 0$, so that $y(x) \equiv 0$. For $\lambda > 0$, the general solution is

$$y(x) = c_1 \sin \sqrt{\lambda} x + c_2 \cos \sqrt{\lambda} x.$$

Substitution of the initial conditions into $y(x)$ yields

$$0 = y(0) = c_2$$

$$0 = y(\pi) = c_1 \sin \sqrt{\lambda} \pi.$$

To obtain a non-trivial solution, $\sin \sqrt{\lambda} \pi = 0$, that is $\sqrt{\lambda} \pi = k\pi$ for $k = 1, 2, 3, \dots$. Therefore, the eigenvalues are $\lambda = k^2$ for $k = 1, 2, 3, \dots$

29. (D)

Switching algebra is similar to set algebra with union ($\cup$) replaced with disjunction ($\vee$), intersection replaced with conjunction (or juxtaposition) and complementation denoted by ($\bar{}$). We have

$$\begin{aligned} (f \vee g) (\bar{f} \vee h) &= (f \vee g) \bar{f} \vee (f \vee g) h \\ &= f \bar{f} \vee g \bar{f} \vee f h \vee g h \\ &= 0 \vee g \bar{f} \vee f h \vee g h \\ &= g \bar{f} \vee f h \vee g h \end{aligned}$$

30. (C)

Using the identity $||T||^2 = \langle T, T \rangle$, we obtain

$$\begin{aligned} ||T|| &= \text{trace } (T^t T) \\ &= \text{trace} \left[\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \right] \\ &= \text{trace} \begin{bmatrix} 5 & 1 \\ 1 & 10 \end{bmatrix} \\ &= 15 \end{aligned}$$

31. (E)

A geometric series is a series of the form $a + ar + ar^2 + \dots + ar^{n-1} + \dots$. If $|r| < 1$, the series converges to $\frac{a}{1-r}$. For the given series, $r = \frac{1}{1+x^2} < 1$ so that its sum is

$$\frac{x^4}{1 - \frac{1}{1+x^2}} = \frac{x^4}{\frac{1+x^2-1}{1+x^2}} = x^4 + x^2.$$

32. (A)

To within an additive constant,

$$f(x) = \frac{x^2}{2} + \frac{2x^3}{3} + \dots + \frac{n x^{n+1}}{n+1} + \dots$$

Using the ratio test with $u_n = \frac{n x^{n+1}}{n+1}$, we have

$$\begin{aligned} L &= \lim_{n \rightarrow +\infty} \left| \frac{u_{n+1}}{u_n} \right| \\ &= \lim_{n \rightarrow +\infty} \left| \frac{(n+1)(n+1)x^{n+1}}{n(n+2)x^{n+1}} \right| \\ &= |x| \end{aligned}$$

so that the series is absolutely convergent for $|x| < 1$ and divergent for $|x| > 1$. For

$$x = -1, u_n = \frac{(-1)^{n+1} n}{n+1}$$

which does not go to 0 which implies that -1 is not in $\text{Dom}(f)$. For

$$x = 1, u_n = \frac{n}{n+1},$$

which does not approach 0 which implies that 1 is not in $\text{Dom}(f)$. Thus $\text{Dom}(f) = (-1, 1)$.

33. (C)

Let $(G, *)$ and $(G', *')$ be groups. A homomorphism from G into G' is a function ϕ such that $\phi(gh) = \phi(g) \phi(h)$. A homomorphism ϕ must take the identity $e \in G$ onto the identity $\phi(e) \in G'$. When G is cyclic, the homomorphism ϕ is completely determined by $\phi(g)$ where g is a generator of G . Since 1 is a generator of Z_8 , the value $\phi(1) \in \{0, 1, 2, 3\}$ is critical. Because we desire only the "onto" homomorphisms, $\phi(1)$ must be a generator of Z_4 . The generators of Z_4 are the elements of $\{0, 1, 2, 3\}$ relatively prime to 4, namely 1 and 3. The functions "defined by" $\phi_1(1) = 1$ and $\phi_2(1) = 3$ are both homomorphisms from Z_8 onto Z_4 . In particular, note that they both take the identity element in Z_8 onto the identity element in Z_4 .

34. (C)

Setting $x = e^t$, we have

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt} = y' x, \text{ since } \frac{dx}{dt} = \frac{d(e^t)}{dt} = e^t = x$$

and

$$\begin{aligned} \frac{d^2 y}{dt^2} &= \frac{d}{dt} \left(x \frac{dy}{dx} \right) = x \frac{d}{dt} \left(\frac{dy}{dx} \right) + xy' \\ &= x \frac{d}{dx} \left(\frac{dy}{dx} \right) \frac{dx}{dt} + xy' = x^2 y'' + xy' \end{aligned}$$

Thus

$$xy' = \frac{dy}{dt} \text{ and } x^2 y'' = \frac{d^2 y}{dt^2} - \frac{dy}{dt}.$$

We have

$$\begin{aligned} \frac{d^2 y}{dt^2} - \frac{dy}{dt} + 6 \frac{dy}{dt} + 6y &= 0 \\ \frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} + 6y &= 0. \end{aligned}$$

The auxiliary equation is given by $m^2 + 5m + 6 = 0$, so that the general solution is $c_1 e^{-3t} + c_2 e^{-2t}$ which equals $\frac{c_1}{x^3} + \frac{c_2}{x^2}$.

35. (E)

Fermat's "little" theorem states that if t is an integer and p a prime not dividing t , then $t^{p-1} \equiv 1 \pmod{p}$. Thus

$$5^{34} \equiv (5^{16})^2 5^2 \equiv (5^{17-1})^2 5^2 \equiv 1^2 5^2 \equiv 25 \equiv 8 \pmod{17}$$

36. (C)

Let $\vec{u} = u_1 \vec{i} + u_2 \vec{j} + u_3 \vec{k}$. The curl of $\vec{u}$, denoted $\text{curl } (\vec{u})$, is the cross product of the vector operator $\nabla \equiv \vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z}$ and the vector $\vec{u}$:

$$\text{curl}(\vec{u}) = \nabla \times \vec{u} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u_1 & u_2 & u_3 \end{vmatrix}$$

For $\vec{u} = xyz \vec{i} + xy^2 \vec{j} + yz \vec{k}$, we have

$$\text{curl}(\vec{u}) = (z - 0) \vec{i} + (xy - 0) \vec{j} + (y^2 - xz) \vec{k}$$

37. (B)

The inverse f^{-1} of f is found from $y = \frac{x}{x-1}$ by first solving for x in terms of y and then replacing y with x and x with y . Thus

$$y = \frac{x}{x-1}$$

$$xy - y = x$$

$$x(y-1) = y \Rightarrow x = \frac{y}{y-1}$$

$$f^{-1}(x) = \frac{x}{x-1}$$

Thus $f^{-1}(x) = f(x)$.

38. (B)

For $x_0 = -2$, we have $8 + y_0^2 + 30 = 8y_0 + 24$ so that

$$y_0 = \frac{8 \pm \sqrt{64 - 4(14)}}{2} = 4 \pm \sqrt{2}$$

Since $y_0 > 4$, $y_0 = 4 + \sqrt{2}$. Taking the derivative of both sides of $2x^2 + y^2 + 30 = 8y - 12x$ with respect to x yields

$$4x + 2yy' = 8y' - 12$$

$$4x + 12 = y'(8 - 2y)$$

$$y' = \frac{2x + 6}{4 - y}$$

Therefore, the slope of the tangent line at $(-2, 4 + \sqrt{2})$ is

$$y' = \frac{-4 + 6}{4 - (4 + \sqrt{2})} = -\sqrt{2}$$

39. (B)

The equation $D = aA + bB + cC$ implies that

$$\begin{bmatrix} 3 & 3 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} a - c & a + b \\ b + c & -a + b + c \end{bmatrix}$$

Thus,

$$a - c = 3$$

$$a + b = 3$$

$$b + c = 0$$

$$-a + b + c = -2$$

Since $b + c = 0$, $a = 2$ which implies that $b = 1$, $c = -1$.

40. (B)

The derivative of $p(x)$ is given by

$$p'(x) = \sum_{n=1}^{+\infty} \frac{(x-2)^{n-1}}{n}$$

The n th term of $p'(x)$ is $a_n = \frac{(x-2)^{n-1}}{n}$.

$$\begin{aligned}\rho &= \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{(x-2)^n n}{(x-2)^{n-1} (n+1)} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{(x-2) n}{n+1} \right| \\ &= |x-2|\end{aligned}$$

By the Ratio Test, the series converges absolutely if $|x-2| < 1$, i.e., $1 < x < 3$. Now we consider endpoints. When $x = 1$, the series converges by the Alternating Series Test. When $x = 3$, the series is the divergent harmonic series. Therefore, the series converges in $[1, 3)$.

41. (C)

Let $f(x)$ be defined on $[0, +\infty)$. The Laplace transformation of f , denoted $\mathcal{L}[f(x)](p)$, is defined by

$$\mathcal{L}[f(x)](p) = \int_0^{+\infty} e^{-px} f(x) dx$$

for all p for which the integral converges. If, in addition, f is of exponential order and is piecewise continuous on $[0, b]$ for every $b > 0$, then

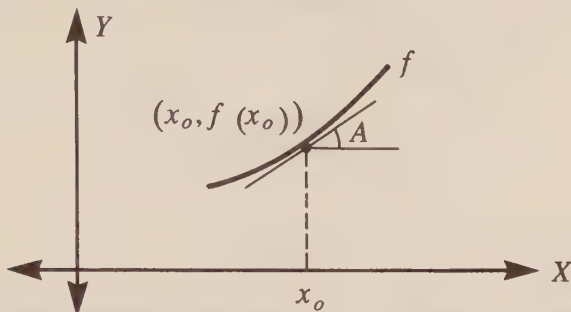
$$\mathcal{L}\left[\int_0^x f(t) dt\right](p) = \frac{1}{p} \mathcal{L}[f(x)](p)$$

Since $\mathcal{L}[\sin ax](p) = \frac{a}{p^2 + a^2}$, we have

$$\mathcal{L}\left[\int_0^x \sin 2t dt\right](p) = \frac{2}{p^3 + 4p}.$$

42. (D)

The slope of the tangent line to the graph of f at x_0 is equal to $f'(x_0)$. It is also equal to the $\tan A$ where A is the angle formed by the tangent line and the positive x -axis. Thus $\tan A = f'(x_0) = \sqrt{3}$ so that $a = 60^\circ$.



43. (E)

The trace of T is the trace of M_T where M_T is a matrix representing T in some fixed basis. For the basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$,

$$M_T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

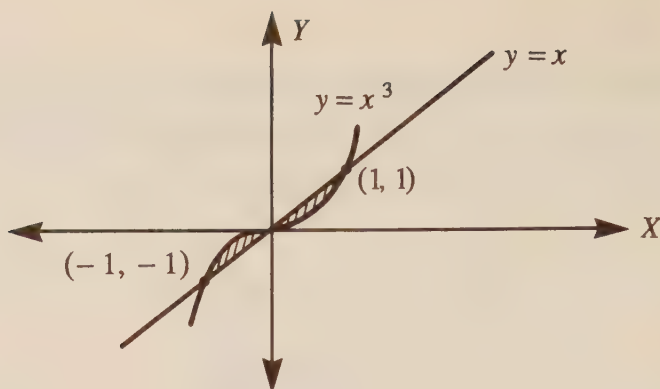
so that the trace of T is 2.

44. (B)

The n 'th partial sum s_n can be written as

$$\begin{aligned} s_n &= \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} \\ &= \frac{2-1}{2!} + \frac{3-1}{3!} + \frac{4-1}{4!} + \dots + \frac{(n+1)-1}{(n+1)!} \\ &= 1 - \frac{1}{2!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{3!} - \frac{1}{4!} + \dots + \frac{1}{n!} - \frac{1}{(n+1)!} \\ &= 1 - \frac{1}{(n+1)!} \rightarrow 1 \text{ as } n \rightarrow +\infty. \end{aligned}$$

Therefore, the sum is 1.



45. (A)

An ideal I of a ring R is a subring of R such that $rI \subseteq I$ and $Ir \subseteq I$, for all $r \in R$. An ideal I is called proper if $I \neq R$ and $I \neq \{0\}$. Since 5 and 12 are relatively prime, 5 is a generator of Z_{12} , that is, $\langle 5 \rangle = Z_{12}$. Thus, $\langle 5 \rangle$ is not a proper ideal of Z_{12} .

46. (A)

A set of answers to the ten questions is a 10-tuple of the form $(a_1, \dots, a_{10})$ where a_j represents an answer to question Q_j , $1 \leq j \leq 10$. An answer is an element of the set $\{r_j, w_{j1}, \dots, w_{j4}\}$ where r_j represents the right answer and $w_{j1}, \dots, w_{j4}$ represent wrong answers. There are 5^{10} possible sets of answers. The number of ways to obtain a 10-tuple with exactly five correct answers is

$$\left[\begin{matrix} 10 \\ 5 \end{matrix} \right] = \frac{10!}{5! 5!} = 252,$$

each with probability

$$\left(\frac{1}{5}\right)^5 \left(\frac{4}{5}\right)^5 = \frac{4^5}{5^{10}}.$$

Hence the probability desired is

$$\frac{252 \cdot 4^5}{5^{10}} = \frac{(63) \cdot 4^6}{5^{10}}.$$

47. (D)

The Jacobian J of a transformation from the xy -plane into the uv -plane defined by

$$u = f(x, y)$$

$$v = g(x, y)$$

is given by

$$J = \begin{vmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{vmatrix}$$

We have

$$\begin{aligned} J &= \begin{vmatrix} xy e^{xy} + e^{xy} & x^2 e^{xy} \\ y^2 e^{xy} & xy e^{xy} + e^{xy} \end{vmatrix} \\ &= e^{2xy} [(xy + 1)^2 - x^2 y^2] \\ &= [2xy + 1] e^{2xy} . \end{aligned}$$

48. (A)

The outcome of six attempts is a 6-tuple of the form $(O_1, \dots, O_6)$, where O_j is either a hit (H) or an out (O). There are $2^6 = 64$ possible outcomes. The number of ways to select a 6-tuple with exactly 3 H 's and 3 O 's is

$$\begin{bmatrix} 6 \\ 3 \end{bmatrix} = \frac{6!}{3! 3!} = 20,$$

each with probability

$$\left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^3 = \frac{8}{3^6}.$$

Hence the desired probability is $\frac{160}{3^6}$.

49. (E)

Let R be a ring with a multiplicative identity e ($re = er = r$, for all $r \in R$). A unit of R is an element u of R that possesses a multiplicative inverse in R . In $Z_5 = \{0, 1, 2, 3, 4\}$, we have that 1 is the multiplicative identity and

$$1 \cdot 1 \equiv 1 \pmod{5}$$

$$2 \cdot 3 \equiv 1 \pmod{5}$$

$$3 \cdot 3 \equiv 1 \pmod{5}$$

$$4 \cdot 4 \equiv 1 \pmod{5}$$

which means that Z_5 contains four units.

50. (A)

Let f be defined on an interval $(0, L)$. Define f in $(-L, L)$ to be odd. The half range Fourier sine series is given by

$$\sum_{k=1}^{+\infty} b_k \sin \frac{k\pi x}{L}$$

where

$$b_k = \frac{2}{L} \int_0^L f(x) \sin \frac{k\pi x}{L} dx$$

for $k = 1, 2, 3, \dots$. We have

$$\begin{aligned} b_k &= 2 \int_0^1 x \sin k\pi x dx \\ &= 2 \left\{ \left[-\frac{x}{k\pi} \cos k\pi x \right]_0^1 + \int_0^1 \frac{1}{k\pi} \cos k\pi x dx \right\} \\ &= 2 \left\{ -\frac{1}{k\pi} \cos k\pi + \frac{1}{k\pi} \left[\frac{1}{k\pi} \sin k\pi x \right]_0^1 \right\} \\ &= -\frac{2}{k\pi} \cos k\pi. \end{aligned}$$

For $k = 3$, $b_3 = -\frac{2}{3\pi} \cos 3\pi = \frac{2}{3\pi}$.

51. (C)

Since $\{e^x, e^{2x}, e^{-2x}\}$ is a linearly independent set, it forms a basis for S . In this basis, we have

$$D_x(e^x) = 1e^x + 0e^{2x} + 0e^{-2x}$$

$$D_x(e^{2x}) = 0e^x + 2e^{2x} + 0e^{-2x}$$

$$D_x(e^{-2x}) = 0e^x + 0e^{2x} - 2e^{-2x}$$

so that the corresponding matrix M_{D_x} of D_x is given by

$$M_{D_x} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

The determinant of D_x is -4 .

52. (B)

Let $a_2(x)$, $a_1(x)$, $a_0(x)$, and $f(x)$ be continuous on an interval I with $a_2(x) \neq 0$ for each $x \in I$. The Green's function for the ordinary differential equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = f(x)$ is given by

$$G(x, t) = -\frac{1}{a_2(t)} \frac{\begin{vmatrix} y_1(x) & y_2(x) \\ y_1(t) & y_2(t) \end{vmatrix}}{\begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix}}$$

where y_1 and y_2 are linearly independent solutions of the corresponding homogeneous equation. Note that

$$\int_{x_0}^x G(x,t) f(t) dt$$

is a particular solution of the non-homogeneous equation. We have $m^2 + 5m + 6 = 0$ so that $y_1(x) = e^{-3x}$ and $y_2(x) = e^{-2x}$. Since

$$\begin{vmatrix} e^{-3x} & e^{-2x} \\ e^{-3t} & e^{-2t} \end{vmatrix} = e^{-(3x+2t)} - e^{-(3t+2x)}$$

and

$$\begin{vmatrix} e^{-3t} & e^{-2t} \\ -3e^{-3t} & -2e^{-2t} \end{vmatrix} = e^{-5t}.$$

Therefore,

$$G(x,t) = -e^{5t}(e^{-(3x+2t)} - e^{-(3t+2x)}) = e^{2(t-x)} - e^{3(t-x)}.$$

53. (A)

The Maclaurin series for e^u is given by

$$e^u = 1 + u + \frac{u^2}{2!} + \dots + \frac{u^n}{n!} + \dots$$

Setting $u = -x^2$ and multiplying the result by x yields the Maclaurin series for xe^{-x^2} :

$$\begin{aligned} xe^{-x^2} &= x \left(1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \dots \right) \\ &= x - x^3 + \frac{x^5}{2!} - \frac{x^7}{3!} + \dots \end{aligned}$$

54. (E)

The symmetric difference is defined by

$$S \Delta T = (S \setminus T) \cup (T \setminus S).$$

We have

$$S \setminus T = \{1, 2, 3\}$$

and

$$T \setminus S = \{6, 7, 8\}$$

so that

$$S \Delta T = \{1, 2, 3, 6, 7, 8\}.$$

55. (D)

Set $x_n = \alpha^n$ for $n = 0, 1, 2, 3, \dots$. Then

$$\alpha^{n+2} + 6\alpha^{n+1} + 9\alpha^n = 0$$

$$\alpha^2 + 6\alpha + 9 = 0$$

$$\alpha = (-3, -3).$$

Thus, the general solution is given by $x_n = a(-3)^n + b n(-3)^n$ where $a, b \in \mathbb{R}$. Since $x_0 = 1$ and $x_1 = 0$, we have

$$1 = x_0 = a$$

$$0 = x_1 = 3a + 3b$$

so that $a = 1$ and $b = -1$. Therefore, $x_5 = (-3)^5 - 5(-3^5) = -4(-3^5) = 972$.

56. (B)

If A represents the angle between f and g , then

$$\cos A = \frac{(f, g)}{\|f\| \|g\|}.$$

We have

$$\|f\|^2 = (f, f) = \int_0^1 2^2 dx = 4 \quad (\|f\| = 2)$$

$$\|g\|^2 = (g, g) = \int_0^1 x^2 dx = \frac{1}{3} \quad (\|g\| = 1/\sqrt{3})$$

$$(f, g) = \int_0^1 2x \, dx = 1.$$

Therefore $\cos A = \sqrt{3}/2$.

57. (D)

The integral is improper since the integrand has a vertical asymptote at $x = 3$. Hence

$$\begin{aligned} \int_{-24}^4 \frac{dx}{\sqrt[3]{(x-3)^2}} &= \int_{-24}^3 \frac{dx}{\sqrt[3]{(x-3)^2}} + \int_3^4 \frac{dx}{\sqrt[3]{(x-3)^2}} \\ &= \lim_{\delta \rightarrow 0^+} \int_{-24}^{3-\delta} \frac{dx}{\sqrt[3]{(x-3)^2}} + \lim_{\epsilon \rightarrow 0^+} \int_{3+\epsilon}^4 \frac{dx}{\sqrt[3]{(x-3)^2}} \\ &= \lim_{\delta \rightarrow 0^+} \left[3\sqrt[3]{x-3} \right]_{-24}^{3-\delta} + \lim_{\epsilon \rightarrow 0^+} \left[3\sqrt[3]{x-3} \right]_{3+\epsilon}^4 \\ &= 3 \lim_{\delta \rightarrow 0^+} \left[\sqrt[3]{-\delta} - \sqrt[3]{-27} \right] + 3 \lim_{\epsilon \rightarrow 0^+} \left[\sqrt[3]{1} - \sqrt[3]{\epsilon} \right] \\ &= 12. \end{aligned}$$

58. (A)

The statement is actually true for all positive integers. For $n = 1$, we have $1^2 + 2^2 + 3^2 = 36$ which is divisible by 9. Assume that $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9 for some positive integer n . We have

$$\begin{aligned} &(n+1)^3 + (n+2)^3 + (n+3)^3 \\ &= (n+1)^3 + (n+2)^3 + n^3 + 9n^2 + 27n + 27 \\ &= \{n^3 + (n+1)^3 + (n+2)^3\} + 9\{n^2 + 3n + 3\} \end{aligned}$$

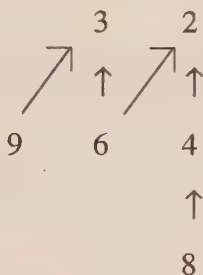
which is divisible by 9. Thus the statement is true for all positive integers.

59. (C)

If F is a finite field, then it must have p^n elements, P a prime and a non-negative integer. Among the given choices, 27 is equivalent to a prime number raised to a non-negative power (3^3). So 27 can be the cardinality of a finite field.

60. (D)

The following diagram represents S with the given ordering:



Therefore, the minimal elements of S are 6, 8, 9.

61. (B)

A subset of a topological space is a neighborhood of a point if it contains an open set containing the point. Since $0 \in (-1, 1) \subseteq [-1, 1]$ and $(-1, 1)$ is open with respect to the usual topology on $\mathbb{R}$, $[-1, 1]$ is a neighborhood of 0.

62. (E)

Let $\sigma \in S_n$, the symmetric group of degree n . If

$$\sigma = \sigma_m \sigma_{m-1} \dots \sigma_i \dots \sigma_2 \sigma_1$$

is the decomposition of σ into disjoint cycles where σ_i is a K_i -cycle, the Cauchy number of σ is given by

$$C(\sigma) = \sum_{i=1}^m (k_i - 1).$$

We have $\sigma = (1 \ 3 \ 5 \ 4) (2 \ 6 \ 7)$. Thus $C(\sigma) = (4 - 1) + (3 - 1) = 5$.

63. (B)

Consider the ordinary differential equation $M(x, y) dx + N(x, y) dy = 0$. If $M(x, y)$, $N(x, y)$, $\frac{\partial M(x, y)}{\partial y}$, and $\frac{\partial N(x, y)}{\partial x}$ are continuous in a region R , then the differential equation is exact if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} . \text{ For}$$

$$(ye^{xy} + \cos x) dx + (xe^{xy} + 1) dy = 0$$

we have

$$\frac{\partial M}{\partial y} = ye^{xy} + e^{xy} = \frac{\partial N}{\partial x} .$$

64. (C)

Let x be a vertex of a graph G . The valence of x , denoted $\text{val}(x)$, is the number of edges of G with x as one end point. The valences of the vertices and the number of vertices are related by $\text{val}(x_1) + \dots + \text{val}(x_5) = 2e$ where e represents the number of edges. Thus $2 + 2 + 3 + 3 + 4 = 2e$ so that $e = 7$.

65. (A)

Since the discrete topology contains all subsets of R , every subset of R is both open and closed. Every subset of R is both closed and open in the discrete topology, and therefore (a, b) is closed. The closure of a set A is the smallest closed set that contains A . If A is closed, then A will be its closure. Therefore, (a, b) is the closure of (a, b) .

66. (A)

For $c, \hat{c} \in C^3$ the usual inner product is given by

$$(c, \hat{c}) = x \bar{\hat{x}} + y \bar{\hat{y}} + z \bar{\hat{z}}$$

where $c = (x, y, z)$ and $\hat{c} = (\hat{x}, \hat{y}, \hat{z})$. Therefore $\hat{c}$ can be chosen so that $\bar{\hat{x}} = 1$, $\bar{\hat{y}} = -i$, and $\bar{\hat{z}} = 2 + i$, that is $\hat{c} = (1, i, 2 - i)$.

The Graduate Record Examination in
MATHEMATICS

Practice Test 4

The Graduate Record Examination in **MATHEMATICS** Answer Sheet Test 4

- | | | |
|---|---|---|
| 1. ☐ A ☐ B ☐ C ☐ D ☐ E | 23. ☐ A ☐ B ☐ C ☐ D ☐ E | 45. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 2. ☐ A ☐ B ☐ C ☐ D ☐ E | 24. ☐ A ☐ B ☐ C ☐ D ☐ E | 46. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 3. ☐ A ☐ B ☐ C ☐ D ☐ E | 25. ☐ A ☐ B ☐ C ☐ D ☐ E | 47. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 4. ☐ A ☐ B ☐ C ☐ D ☐ E | 26. ☐ A ☐ B ☐ C ☐ D ☐ E | 48. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 5. ☐ A ☐ B ☐ C ☐ D ☐ E | 27. ☐ A ☐ B ☐ C ☐ D ☐ E | 49. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 6. ☐ A ☐ B ☐ C ☐ D ☐ E | 28. ☐ A ☐ B ☐ C ☐ D ☐ E | 50. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 7. ☐ A ☐ B ☐ C ☐ D ☐ E | 29. ☐ A ☐ B ☐ C ☐ D ☐ E | 51. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 8. ☐ A ☐ B ☐ C ☐ D ☐ E | 30. ☐ A ☐ B ☐ C ☐ D ☐ E | 52. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 9. ☐ A ☐ B ☐ C ☐ D ☐ E | 31. ☐ A ☐ B ☐ C ☐ D ☐ E | 53. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 10. ☐ A ☐ B ☐ C ☐ D ☐ E | 32. ☐ A ☐ B ☐ C ☐ D ☐ E | 54. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 11. ☐ A ☐ B ☐ C ☐ D ☐ E | 33. ☐ A ☐ B ☐ C ☐ D ☐ E | 55. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 12. ☐ A ☐ B ☐ C ☐ D ☐ E | 34. ☐ A ☐ B ☐ C ☐ D ☐ E | 56. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 13. ☐ A ☐ B ☐ C ☐ D ☐ E | 35. ☐ A ☐ B ☐ C ☐ D ☐ E | 57. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 14. ☐ A ☐ B ☐ C ☐ D ☐ E | 36. ☐ A ☐ B ☐ C ☐ D ☐ E | 58. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 15. ☐ A ☐ B ☐ C ☐ D ☐ E | 37. ☐ A ☐ B ☐ C ☐ D ☐ E | 59. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 16. ☐ A ☐ B ☐ C ☐ D ☐ E | 38. ☐ A ☐ B ☐ C ☐ D ☐ E | 60. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 17. ☐ A ☐ B ☐ C ☐ D ☐ E | 39. ☐ A ☐ B ☐ C ☐ D ☐ E | 61. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 18. ☐ A ☐ B ☐ C ☐ D ☐ E | 40. ☐ A ☐ B ☐ C ☐ D ☐ E | 62. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 19. ☐ A ☐ B ☐ C ☐ D ☐ E | 41. ☐ A ☐ B ☐ C ☐ D ☐ E | 63. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 20. ☐ A ☐ B ☐ C ☐ D ☐ E | 42. ☐ A ☐ B ☐ C ☐ D ☐ E | 64. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 21. ☐ A ☐ B ☐ C ☐ D ☐ E | 43. ☐ A ☐ B ☐ C ☐ D ☐ E | 65. ☐ A ☐ B ☐ C ☐ D ☐ E |
| 22. ☐ A ☐ B ☐ C ☐ D ☐ E | 44. ☐ A ☐ B ☐ C ☐ D ☐ E | 66. ☐ A ☐ B ☐ C ☐ D ☐ E |

GRE MATHEMATICS TEST IV

TIME: 2 hours and 50 minutes
66 Questions

DIRECTIONS: Choose the best answer for each question and mark the letter of your selection on the corresponding answer sheet.

1. Given that $(1, 2, 3)$ is an eigenvector for the matrix

$\begin{pmatrix} 2 & 3 & -1 \\ 3 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix}$, find the corresponding eigenvalue.

(A) 5

(D) 2

(B) 4

(E) None of these

(C) 3

2. A rectangular box resides in 3-space with one vertex at the origin, $(0, 0, 0)$ and three faces in the coordinate planes. If another vertex is $(x, 2, 3)$, $x > 0$, and the angle in radians between the diagonals from $(0, 0, 0)$ to $(x, 2, 3)$ and the other vertex in the xz plane is $\frac{\pi}{6}$, find x .

(A) 1

(D) 3

(B) $\sqrt{2}$

(E) $\sqrt{3}$

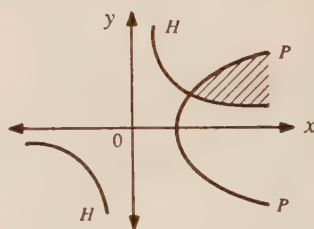
(C) 2

3. The random variable, X , is discrete and uniformly distributed with values 1, 2, 3, 4, 5. The variance of X is

(A) 1 (D) 4
(B) 2 (E) None of these
(C) 3

4. Which collection of inequalities represents the shaded region as shown in the plane? The curves represent $y^2 = x - 1$ and $2xy = 1$.

(A) $y^2 + 1 < x$ and $2x > \frac{1}{y}$
(B) $x < y^2 + 1$ or $2x > \frac{1}{y}$
(C) $2x < \frac{1}{y}$ and $x - 1 > y^2$
(D) $2x < \frac{1}{y}$ or $x - 1 < y^2$
(E) $2xy > 1$ and $y^2 + 1 > x$



5. A whispering gallery is constructed as part of the surface formed on rotation of the ellipse $\frac{x^2}{100} + \frac{y^2}{k} = 1$ with x and y in yards. Each whisperer stands at a focus on the x -axis that is three feet from the nearest vertex. Find k .

(A) 6 (B) 7

(C) 18

(E) 36

(D) 19

6. Which of the following is a divisor of $3^{10} - 1$?

(A) 3

(D) 17

(B) 7

(E) 19

(C) 11

7. In the permutation group, S_5 , find the product of the elements $(21453) \cdot (35214)$, where $(a b c d e)$ means $1 \rightarrow a, 2 \rightarrow b$, etc.

(A) (41352)

(D) (43125)

(B) (43215)

(E) (34251)

(C) (43152)

8. In the algebra of sets, which of the following is identical to $(\overline{S} \cap T)$, where $\overline{S}$ denotes the complement of S ?

(A) $S \cap \overline{T}$

(D) $S \cup \overline{T}$

(B) $\overline{(S \cup \overline{T})}$

(E) None of these

(C) ϕ

9. Find the interval on which the power series

$$\sum_{n=1}^{\infty} \frac{(-1)^n (x-3)^{2n+1}}{n \cdot 4^n} \quad \text{converges.}$$

- (A) $(-1, 7)$ (D) $(1, 5)$
(B) $(-1, 7]$ (E) $[1, 5]$
(C) $(1, 5]$

10. Let f be a function such that the graph of f lies on a parabola through $(0, 0)$ and $(4, 2)$. If function g is the composition of f with itself, which of the following points could lie on the graph g ?

- (A) $(1, 1)$ (D) either (A) or (B)
(B) $(16, 2)$ (E) (A), (B), or (C)
(C) $(8, 8)$

11. When 4% interest is compounded quarterly, what is the effective annual rate to the nearest hundredth of a percent?

- (A) 4.04 (D) 4.12
(B) 4.03 (E) 4.08
(C) 4.06

12. Given that f is a differentiable function with normal line $4x + y = 9$ at $x = 2$ then

$$\lim_{x \rightarrow 0} \frac{f(2-x) - f(2+x)}{x} \text{ is}$$

- (A) $\frac{1}{4}$ (D) $-\frac{1}{2}$
(B) $-\frac{1}{4}$ (E) Undefined
(C) $\frac{1}{2}$

13. Given that a 3×3 matrix A has only one eigenvalue, what is the dimension of the corresponding eigenspace?

- (A) 1 (D) 1 or 2
(B) 2 (E) 1, 2 or 3
(C) 3

14. If $f(e^x) = \sqrt{x}$ for $x \geq 1$, then $f^{-1}(x)$ is

- (A) $(\log x)^2$ (D) $e^{\sqrt{x}}$
(B) e^{x^2} (E) $\sqrt{\log x}$
(C) $2 \log x$

15. Let $*$ be a binary operation defined on the rational numbers by $a * b = a + b - ab$. Then the number of integers with integer $*$ -inverse is

(A) 0 (D) 3
(B) 1 (E) 4
(C) 2

16. Evaluate the limit, $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left[1 + \frac{2k}{n}\right]^2 \left[\frac{2}{n}\right]$.

(A) $\frac{26}{3}$ (D) $\frac{7}{3}$
(B) 6 (E) 2
(C) $\frac{13}{3}$

17. Define a linear transformation from R^2 to R^3 by $(x, y)T = (x + y, x - y, x)$. Define S , a linear transformation from R^3 to R^2 , to be an inverse of T if $((x, y)T)S = (x, y)$. Which of the following represent an inverse of T ?

(A) $(x, y, z)S = \left(z, \frac{x}{2} - \frac{y}{2}\right)$
(B) $(x, y, z)S = \left(\frac{x}{2} + \frac{y}{2}, \frac{x}{2} - \frac{y}{2} - z\right)$
(C) $(x, y, z)S = \left(\frac{x}{3} + \frac{y}{3} + \frac{z}{3}, x - y\right)$
(D) both (A) and (B)
(E) all three, (A), (B), and (C)

18. The mapping $\phi : G \rightarrow G$ given by $g\phi = a^2ga^2$ for fixed $a \in G$ and for each element g in G , is a homomorphism if

- (A) $a^4 = e$ (D) G is abelian
 (B) $a^3 = e$ (E) G is finite
 (C) $aG = Ga$

19. Given the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 1 & 1 & 5 \end{pmatrix}$ and $B = A^{-1}$, find the entry in row 3 and column 2 of B .

- (A) $\frac{1}{4}$ (D) $-\frac{1}{2}$
 (B) $\frac{1}{2}$ (E) $-\frac{1}{4}$
 (C) 1

20. Given the function $f : (A \cup B) \rightarrow C$, which of the following is always true? ($\bar{A}$ denotes the complement of A).

- (A) $f(A \cap B) = f(A) \cap f(B)$
 (B) $f(\bar{A}) = \overline{f(A)}$
 (C) $f(A - B) = f(A) - f(B)$

(D) $f(A \cup B) = f(A) \cup f(B)$

(E) All of these

21. Evaluate $\int_0^{\pi} \cos^2 nx \, dx$, where n is a positive integer.

(A) $\frac{\pi}{2}$

(D) $\frac{\pi}{2n}$

(B) π

(E) $n\pi$

(C) $\frac{n\pi}{2}$

22. Let R be the ring of integers modulo four. Which of the polynomials shown is irreducible over R ?

(A) $x^2 + x + 2$

(D) $x^3 - 3$

(B) $x^2 + x + 3$

(E) $x^4 + 3$

(C) $x^2 + 3x - 2$

23. In a circle of radius $\frac{4}{\pi}$ find the area between an arc of length 2 and its chord.

(A) $\frac{4}{\pi} \left[1 - \frac{2}{\pi} \right]$

(D) $\frac{4}{\pi}$

(B) $\frac{2}{\pi} - \frac{4}{\pi^2}$

(E) $\frac{8}{\pi}$

(C) $\frac{2}{\pi}$

24. Let $f_n(x) = x^n$ on $[0, 1]$ and $f(x) = \lim_{n \rightarrow \infty} f_n(x)$. Which of the following is true?

- (A) f is constant.
- (B) f is monotone.
- (C) $\{f_n\}$ converges uniformly to f .
- (D) Both (A) and (B)
- (E) Both (B) and (C)

25. Find $\lim_{x \rightarrow \infty} \frac{(e^x + e^{-x})}{2 - 3e^x}$.

- (A) 0
- (B) $\frac{1}{2}$
- (C) Undefined
- (D) -2
- (E) $-\frac{1}{3}$

26. The column space of a 5×6 matrix is spanned by the vectors $(1, 0, 0, 0, 0)$, $(0, 0, 1, 0, 0)$, and $(2, 0, 3, 0, 0)$. Find the dimension of the solution space of the matrix.

- (A) 3
- (B) 4
- (C) 6
- (D) 2
- (E) 5

27. For $f(x,y) = x^2 + y^2$, find the rate of change of f at $(1,2,5)$ in the direction $\vec{v} = 3\hat{i} - 4\hat{j}$.

- (A) -2 (D) -13
 (B) -10 (E) None of these
 (C) -5

28. Which pair of the following elements of the symmetric group, S_4 , has a product in the alternating group A_4 ?

$$(i) \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 2 & 4 \end{pmatrix} \quad (ii) \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix} \quad (iii) \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$$

- (A) (i), (ii) (D) both (i), (ii) and (i), (iii)
 (B) (i), (iii) (E) both (i), (ii) and (ii), (iii)
 (C) (ii), (iii)

29. All functions, f , defined on the xy -plane such that

$$\frac{\partial f}{\partial x} = y^2(3x^2 + y^2) \quad \text{and} \quad \frac{\partial f}{\partial y} = 2xy(x^2 + 2y^2)$$

are given by

- (A) $x^2y^3 + xy^4 + C$
 (B) $xy^2(x^2 + y^2) + C$

(C) $x^2y(x^2 + y^2) + C$

(D) $x^4y + x^2y^3 + C$

(E) $xy(x^3 + y^3) + C$

30. The moment with respect to the yz plane of the volume in the first octant bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$ is written as an integral in cylindrical coordinates as $\iiint_V F(z, r, \theta) dz dr d\theta$. Find $F(z, r, \theta)$.

V

(A) $r \cos \theta$

(D) $r^2 \sin \theta$

(B) $r \sin \theta$

(E) $r^2 \cos \theta$

(C) $rz \cos \theta$

31. In the solution of the differential equation $xy^2 dx = dy - x dx$ satisfying the initial condition $y(0) = -1$, find y when $x = \sqrt{\pi}$

(A) 0

(D) $\frac{\pi}{4}$

(B) -1

(E) $-\frac{\pi}{4}$

(C) 1

32. The set of all points in the plane satisfying $y = x \sin \left(\frac{1}{x} \right)$ together with the origin
- (A) is compact but not connected
 - (B) is connected but not compact
 - (C) is compact and connected
 - (D) contains an open set
 - (E) does not contain all of its limit points.
33. A regular deck of 52 cards contains 4 suits of 13 denominations; 2, 3, ..., 10, J, Q, K, A. In how many ways may we select a subset of 5 cards containing exactly 3 of the same denomination? In the answers below, $\cdot$ represents multiplication as usual.
- (A) $24 \cdot 47 \cdot 52$
 - (B) $47 \cdot 48 \cdot 52$
 - (C) $24 \cdot 39 \cdot 47$
 - (D) $39 \cdot 47 \cdot 48$
 - (E) $6 \cdot 47 \cdot 48 \cdot 52$
34. Find the integrating factor for the linear equation $xy' - 2y = x^2$.
- (A) x^{-2}
 - (B) e^{-2x}

(C) e^{2x}

(E) x^2

(D) x

35. If T is a linear transformation mapping vectors $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$ to the vectors $(1,2,3)$, $(2,3,1)$, and $(1,1,-2)$ respectively, which vector is the image of the vector $(3,-2,1)$ under T ?

(A) $(1,1,7)$

(D) $(0,1,9)$

(B) $(1,0,5)$

(E) $(1,7,0)$

(C) $(0,1,5)$

36. If the ninth partial sum of the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ is used to approximate its sum, which statement about E , the sum minus the approximation, is true?

(A) $E > .01$

(D) $-.01 < E < 0$

(B) $0 < E < .01$

(E) $E < -1/81$

(C) $E < -.01$

37. The base ten number, 73, is given in binary as

(A) 10001001

(B) 1001001

(C) 10010001

(E) 101011

(D) 101001

38. For $f(x) = x^3 - 3x^2 + k$ on $[1,3]$, the maximum and minimum values have the same absolute value. Find k .

(A) 4

(D) 1

(B) 3

(E) None of these

(C) 2

39. A regular polygon has ten times as many diagonals as it has vertices. How many sides does it have?

(A) 17

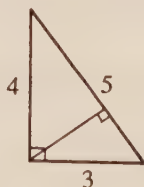
(D) 23

(B) 19

(E) None of these

(C) 21

40. An altitude is drawn to the hypotenuse of the triangle as shown. Find the ratio of the areas of the two smaller triangles.



(A) $\frac{9}{16}$

(D) $\frac{4}{5}$

(B) $\frac{3}{4}$

(E) 1

(C) $\frac{3}{5}$

41. The number of different ways, not counting rotations, to seat 6 different people around a circular table is

(A) 720

(D) 180

(B) 60

(E) 120

(C) 360

42. Find the average value of the function $f(x) = \sqrt{4 - x^2}$ on $[0, 2]$.

(A) 1

(D) $\frac{\pi}{2}$

(B) $\frac{3}{2}$

(E) None of these

(C) $\frac{\pi}{4}$

43. Let $F(x)$ represent the Fourier series of the periodic extension of the function $f(x) = x^2$ on the interval, $(0, 2)$. Find $F(4)$.

(A) 16

(B) 2

(C) 4

(E) Undefined

(D) 8

44. Let f be a function on $[0, \infty)$ such that for each point in its graph, $(x, y) = (y^2, y)$. At how many points must each such f have a limit?

(A) 1

(D) 4

(B) 2

(E) Infinitely many

(C) 3

45. Given that $\{x_n\}$ is a bounded, divergent, infinite sequence of real numbers, which of the following must be true?

(A) $\{x_n\}$ contains infinitely many convergent subsequences

(B) $\{x_n\}$ contains convergent subsequences with different limits.

(C) $\{y_n = \min_{k \leq n} x_k\}$ is convergent.

(D) All of the above

(E) (A) and (C) only

46. Let p , q , r , and s be statements with negations $\bar{p}$, $\bar{q}$, $\bar{r}$, and $\bar{s}$. If p implies q , r implies s , and $\bar{p}$ implies $\bar{q}$ and $\bar{r}$, which statements below must be true?

I. $\bar{p}$ and $\bar{q}$ are equivalent

II. $\bar{r}$ implies $\bar{s}$

III. r implies q

(A) I

(D) I and II

(B) II

(E) I and III

(C) III

47. Given $|x|, |y| \leq 2$ where x, y are real and $|x| \neq |y|$, we state

I. $|x^2 - y^2| \geq 1$ if and only if $|x + y||x - y| \geq 1$,

II. $|x + y||x - y| \geq 1$ if and only if $|x - y| \geq \frac{1}{|x + y|}$

III. $|x - y| \geq \frac{1}{|x + y|}$ if and only if $|x - y| \geq \frac{1}{|x| + |y|}$

IV. $|x - y| \geq \frac{1}{|x| + |y|}$ if and only if $|x - y| \geq \frac{1}{4}$

Find all the general true statements above.

(A) (I)

(D) (I), (II), and (III)

(B) (II)

(E) (I), (II), and (IV)

(C) (I) and (II)

48. In a sequence of consecutive throws of a die, find the probability that six will show before a one or a two.

(A) $\frac{1}{6}$

(D) $\frac{5}{6}$

(B) $\frac{1}{2}$

(E) $\frac{1}{3}$

(C) $\frac{2}{3}$

49. Given $x^2z - 2yz^2 + xy = 0$, find $\frac{\partial x}{\partial z}$ at $(1, 1, 1)$.

(A) 0

(D) 1

(B) $\frac{4}{3}$

(E) None of these

(C) -1

50. All commutators in a group, G , are of the form $aba^{-1}b^{-1}$. The inverse of this commutator is

(A) the identity element

(D) $aba^{-1}b^{-1}$

(B) $a^{-1}b^{-1}ab$

(E) $b^{-1}a^{-1}ba$

(C) $bab^{-1}a^{-1}$

51. If H is a left ideal containing the identity element of a ring R , then

(A) $HR = H$

(D) Both (B) and (C)

(B) $HR = H$

(E) Both (A) and (C)

(C) $RR = H$

52. Let C_n be a sequence of closed, bounded, nonempty intervals in the real line with the usual topology. The intervals are also nested in the sense that $C_{n+1} \subseteq C_n$. Which of the following is true of the intersection

$$S = \bigcap_{k=1}^{\infty} C_k$$

(A) S may be open or closed.

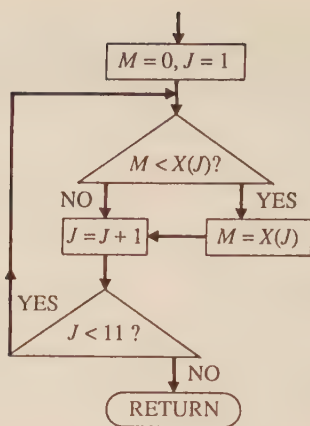
(B) S may be empty.

(C) S must be nonempty and closed.

(D) S must contain an interval.

(E) S must not contain an interval.

53. Find the best statement describing the output that the following subroutine, S , (flow chart) returns to the main program which provides the array $X(I)$.



- (A) S returns the minimum of the first 10 values of any $X(J)$
- (B) S returns the maximum of the first 11 values of any $X(J)$
- (C) S returns the maximum of the first 10 values of any positive $X(J)$
- (D) S returns the minimum of the first 11 values of any $X(J)$
- (E) S returns the maximum of the first 10 values of any $X(J)$ if one of these values is nonnegative.

54. In drawing two balls, from an urn containing 10 balls of each of the colors, red, white, and blue, find the probability of getting two different colors.

- | | |
|---------------------|---------------------|
| (A) $\frac{2}{3}$ | (D) $\frac{20}{29}$ |
| (B) $\frac{1}{3}$ | (E) None of these |
| (C) $\frac{10}{29}$ | |

55. A root of the polynomial $x^3 - 3x^2 + 5x - 3$ is

(A) -1

(D) $1 \pm i\sqrt{2}$

(B) $2 + \sqrt{2i}$

(E) 3

(C) $1 - \sqrt{8i}$

56. Find the fifth derivative, $f^{(5)}(2)$, of the given

$$f(x) = \sum_{n=0}^{\infty} \frac{(x-2)^n}{2^n(n+1)}$$

(A) 0

(D) $\frac{3}{4}$

(B) $\frac{1}{32}$

(E) $\frac{5}{8}$

(C) $\frac{1}{192}$

57. Let a and b be constants. If $f(x) = ae^{-x^2} + bx \tan x - |x|$ and if $f(7) = 5$, then $f(-7)$ is

(A) 5

(C) -2

(B) -5

(D) Undefined

(E) Not uniquely determined by the information given.

58. If the determinants $|A| = 3$ and $|B| = 2$, find $|2(AB)^{-1}|$ for 4×4 matrices A and B .

(A) $\frac{1}{3}$

(D) $\frac{8}{3}$

(B) $\frac{2}{3}$

(E) 12

(C) $\frac{4}{3}$

59. Let x and y be integers such that $7x + 3y$ is divisible by 13. For which value of k must $8x + ky$ be divisible by 13?

(A) 11

(D) 5

(B) 9

(E) 3

(C) 7

60. In Euler's method of tangents for approximating points on the solution curve of $x dy - (x + y) dx = 0$, $y(1) = 1$, find the approximation of $y(1.2)$ using two steps of length 0.1.

(A) $\frac{31}{22}$

(D) 2

(B) $\frac{7}{5}$

(E) $\frac{3}{2}$

(C) $\frac{29}{20}$

61. Find the number of distinct divisors of 1440.
- (A) 48 (D) 24
(B) 36 (E) 10
(C) 30
62. S is a set containing exactly 10 points. C is a collection of subsets of S . Find the maximum number of subsets that C may contain and satisfy the requirement that the intersection of all the subsets in C is nonempty.
- (A) 255 (D) 512
(B) 256 (E) 1023
(C) 511
63. Given that $f(x)$ and $g(x)$ are independent solutions of a linear homogeneous differential equation on (a, b) , which of the following must also be solutions?
- (A) 0 (D) Both (A) and (B)
(B) $2f(x) - 3g(x)$ (E) Both (B) and (C)
(C) $f(x)g(x)$

64. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and let I be an identity matrix. Which matrix polynomial is zero?

(A) $A^2 - 10A + I$ (D) $A^2 + 5A - 2I$

(B) $A^2 - 10A$ (E) $A^2 + 5A + 2I$

(C) $A^2 - 5A - 2I$

65. Which of the following subsets of the plane may be translated or rotated into a proper subset of itself?

I. $Z = \{(x, y) \mid x \text{ and } y \text{ are positive integers}\}$

II. $I = \{(x, y) \mid y \geq x\}$

III. $C = \{(\cos n, \sin n) \mid n = 0, 1, 2, \dots\}$

(A) I and II (D) I, II, and III

(B) I and III (E) None of these

(C) II and III

66. Which of the following is an eigenvalue of the matrix

$$A = \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 1 \end{pmatrix}$$

(A) -1

(D) 2

(B) -2

(E) 0

(C) 1

GRE MATHEMATICS

TEST IV

ANSWER KEY

1.	A	23.	A	45.	D
2.	E	24.	B	46.	E
3.	B	25.	E	47.	C
4.	A	26.	B	48.	E
5.	D	27.	A	49.	D
6.	C	28.	B	50.	C
7.	D	29.	B	51.	E
8.	B	30.	E	52.	C
9.	E	31.	C	53.	E
10.	E	32.	B	54.	D
11.	C	33.	A	55.	D
12.	D	34.	A	56.	E
13.	E	35.	C	57.	A
14.	B	36.	B	58.	D
15.	C	37.	B	59.	B
16.	A	38.	C	60.	A
17.	A	39.	D	61.	B
18.	A	40.	A	62.	D
19.	E	41.	E	63.	D
20.	D	42.	D	64.	C
21.	A	43.	D	65.	D
22.	B	44.	A	66.	B

GRE MATHEMATICS TEST IV

DETAILED EXPLANATIONS OF ANSWERS

1. (A)

When $\bar{x}$ is an eigenvector for matrix, A , then $A\bar{x} = \lambda\bar{x}$ defines the eigenvalue, λ . In this case,

$$A \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 10 \\ 15 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \text{ so } \lambda = 5.$$

2. (E)

The other vertex is $(x, 0, 3)$ and so, using dot products,

$$\begin{aligned} \cos\left(\frac{\pi}{6}\right) &= \sqrt{\frac{3}{4}} \\ &= \frac{x^2 + 9}{\sqrt{x^2 + 13} \sqrt{x^2 + 9}} \\ &= \sqrt{\frac{x^2 + 9}{x^2 + 13}} \end{aligned}$$

Then $4(x^2 + 9) = 3(x^2 + 13)$ implies $x^2 = 3$ and $x = \sqrt{3}$.

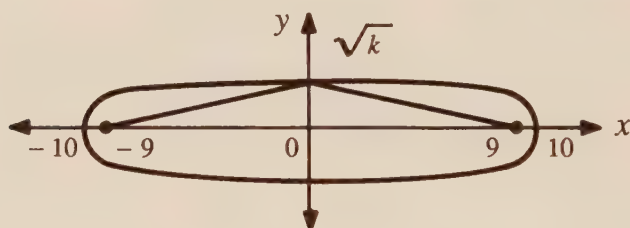
3. (B)

“Uniform” means that each value has equal probability, $\frac{1}{5}$. Then variance of x is

$$\begin{aligned} E(x^2) - (E(x))^2 &= \sum_{x=1}^5 \frac{x^2}{5} - \left(\sum_{x=1}^5 \frac{x}{5} \right)^2 \\ &= 11 - 3^2 \\ &= 2. \end{aligned}$$

4. (A)

The parabola is $y^2 = x - 1$ (P) and the hyperbola is $2xy = 1$ (H). Since the shaded region is above (H) we must have $y > 1/2x$ or $2x > 1/y$. And, since the shaded region is to the right of (P) we must have $x > y^2 + 1$ or $y^2 + 1 < x$. As usual “and” represents intersection.



5. (D)

The vertices from standard form are as shown. The statement implies that the foci must be at $(\pm 9, 0)$. The locus definition of an ellipse implies that $20 = 2\sqrt{k + 9^2}$ so that $100 = k + 9^2$ and $k = 19$.

6. (C)

Fermat's Little Theorem states that $a^{p-1} \equiv 1 \pmod{p}$ for any prime p and any a which is not divisible by p . Here, the form indicates $p = 11$.

7. (D)

Since permutations are functions, the "product" is actually a composition and proceeds from right to left. So, $1 \rightarrow 3 \rightarrow 4$, $2 \rightarrow 5 \rightarrow 3$, $3 \rightarrow 2 \rightarrow 1$, $4 \rightarrow 1 \rightarrow 2$, and $5 \rightarrow 4 \rightarrow 5$. We get $(4\ 3\ 1\ 2\ 5)$.

8. (B)

One can apply DeMorgan's Law which states $\overline{A \cap B} = \overline{A} \cup \overline{B}$.
 $\overline{(\overline{S} \cup \overline{T})} = \overline{\overline{S}} \cap \overline{\overline{T}} = \overline{\overline{S}} \cap T$.

9. (E)

The ratio test, $\rho = \lim_{n \rightarrow \infty} \frac{|x-3|^2 n}{4 \cdot (n+1)} = \left| \frac{x-3}{2} \right|^2 < 1$, implies

$|x-3| < 2$ or $-2 < x-3 < 2$ and so $1 < x < 5$. Finally, checking endpoints directly we find for $x = 1$ that

$$2 \sum_{n=1}^{\infty} \frac{(-1)^{3n+1}}{n}$$

converges by the alternating series (Leibniz) test and for $x = 5$ that

$$2 \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

likewise converges.

10. (E)

There are two possible parabolic arcs that will contain (0, 0) and (4, 2). Either $f(x) = \sqrt{x}$ or $g(x) = \frac{x^2}{8}$ will do. Then, we have

$$g_1(x) = f \circ f = f(f(x)) = \sqrt{\sqrt{x}} = \sqrt[4]{x}$$

or

$$g_2(x) = f \circ f = f(f(x)) = \frac{\left(\frac{x^2}{8}\right)^2}{8} = \frac{x^4}{512}$$

All the given points in (A), (B), and (C) satisfy either $g_1(x)$ or $g_2(x)$. So choice (E) is the most accurate.

11. (C)

The effective rate is $(1 + .01)^4 - 1 = .04060401 \cong 4.06\%$.

12. (D)

The symmetric limit for the derivative at $x = 2$ has the form

$$\lim_{x \rightarrow 0} \frac{f(2+x) - f(2-x)}{2x} = f'(2) = \frac{1}{4}$$

since the tangent line slope is the negative reciprocal of the normal line slope. Since our limit has a reversed numerator and only $\frac{1}{2}$ the denominator, it has value $-\frac{1}{2}$.

13. (E)

A single eigenvalue must then be of multiplicity 3 for a 3×3 matrix. In general the dimension of the eigenspace may be any number greater than zero and less than or equal to the multiplicity.

14. (B)

Since $f(e^x) = \sqrt{x}$ we must have $f(x) = f(e^{\log x}) = \sqrt{\log x}$ and here is where $x \geq 1$ is needed for definition of the radical. Since

$y = \sqrt{\log x}$ may be solved for $x = e^{y^2} = f^{-1}(y)$ we find that $f^{-1}(x) = e^{x^2}$.

15. (C)

The $*$ -identity is 0 since $a*0 = 0*a = a$ for each rational, a . Then the $*$ -inverse of a is found by solving $a*x = a + x - ax = 0$ for $x = \frac{a}{(a-1)}$, $a \neq 1$. Now x is an integer when $a = 0$ and $a = 2$ and in no other case.

16. (A)

In general

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(a + k \Delta x) \Delta x \quad \text{where} \quad \Delta x = \frac{(b-a)}{n}.$$

This may be called the limit of the right end point Riemann sum. In this case we let $f(x) = x^2$, $a = 1$, and $b = 3$ so $\Delta x = \frac{2}{n}$.

It follows by the fundamental theorem of calculus that the limit is

$$\int_1^3 x^2 dx = \left. \frac{x^3}{3} \right|_1^3 = 9 - \frac{1}{3} = \frac{26}{3}.$$

17. (A)

T is representable as a 2×3 matrix $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$.

Then S is an inverse if $TS = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

The given transformation for S are representable as 3×2 matrices

$$\begin{bmatrix} 0 & \frac{1}{2} \\ 0 & -\frac{1}{2} \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \\ 0 & -1 \end{bmatrix}, \text{ and } \begin{bmatrix} \frac{1}{3} & 1 \\ \frac{1}{3} & -1 \\ \frac{1}{3} & 0 \end{bmatrix} \text{ respectively.}$$

Direct multiplications show that only (A) is inverse.

18. (A)

The test for a homomorphism is to show that

$$(gh^{-1})\phi = (g\phi)(h\phi)^{-1} \text{ for arbitrary } g, h \in G.$$

In this case

$$(g\phi)(h\phi)^{-1} = a^2ga^2(a^2ha^2)^{-1} = a^2ga^2a^{-2}h^{-1}a^{-2} = a^2(gh^{-1})a^{-2}$$

which will be $(gh^{-1})\phi$ as desired if $a^2 = a^{-2}$ or $a^4 = e$.

19. (E)

The entry is the cofactor of the entry in row 2 and column 3 divided by the determinant of A . The cofactor is

$$(-1)^5 \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = 1 \text{ and } |A| = -4.$$

20. (D)

It is well-known that $f(A \cup B) = f(A) \cup f(B)$. (A) is contradicted by a constant function with disjoint A and B . Well-chosen constant functions and sets will also contradict (B) and (C).

21. (A)

$$\cos^2(nx) = \frac{1 + \cos 2(nx)}{2} \text{ with integral } \left(\frac{x}{2} + \frac{\sin^2 nx}{4n} \right) \bigg|_0^\pi = \frac{\pi}{2}$$

since $\sin^2 n\pi = \sin^2 0 = 0$ for all n .

22. (B)

If we represent the polynomial by $p(x)$ then in (A) we find $p(1) = 4 \equiv 0 \pmod{4}$ so $p(x)$ is reducible. In (B) we find $p(0) = 3$, $p(1) = 1$, $p(2) = 1$, and $p(3) = 3$ modulo four and so $p(x)$ is irreducible.

23. (A)

The circumference is $2\pi \left(\frac{4}{\pi}\right) = 8$ and so the chord of an arc of length 2 is the hypotenuse of a right triangle of side lengths $\frac{4}{\pi}$ and area $\frac{1}{2} \left(\frac{4}{\pi}\right)^2 = \frac{8}{\pi^2}$. The area of the quarter-circle is $\frac{\pi}{4} \left(\frac{4}{\pi}\right)^2 = \frac{4}{\pi}$ and the desired area is the difference $\frac{4}{\pi} - \frac{8}{\pi^2} = \frac{4}{\pi} \left[1 - \frac{2}{\pi}\right]$.

24. (B)

The limit function $f(x) = 0$ for all x in $[0, 1)$ and $f(1) = 1$. Since the uniform limit of continuous functions is always continuous, (C) is impossible.

25. (E)

Dividing both numerator and denominator by e^x results in a determinate form.

26. (B)

Since the third vector is a linear combination of the two elementary basis vectors, the column rank is two. The dimension of the solution space must be $6 - 2 = 4$.

27. (A)

The rate of change is the directional derivative,

$$(f_x \hat{i} + f_y \hat{j}) \cdot \frac{(3\hat{i} - 4\hat{j})}{5} = \frac{(6x - 8y)}{5} = -\frac{10}{5} = -2$$

at $(1, 2, 5)$.

28. (B)

A_4 is the subgroup of even permutations in S_4 . It is easy to inspect that

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 2 & 4 \end{pmatrix}^3 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix} = e, \text{ so (i) is odd.}$$

Likewise $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}^2 = e$, so (ii) is even

and $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix}^3 = e$, so (iii) is odd. The product of odd and

even is odd and the product of odd and odd is even so on (i), (iii) will have an even product.

29. (B)

By partially integrating the given derivatives we find at first that

$$f(x, y) = \int (3x^2y^2 + y^4) dx = x^3y^2 + xy^4 + g(y)$$

where $g(y)$ can be any function of y . Finding

$$\frac{\partial f}{\partial y} = 2x^3y + 4xy^3 + g'(y)$$

from this representation implies that

$$g'(y) = 0 \text{ so } g(y) = C.$$

$$\text{So } f(x, y) = x^3y^2 + xy^4 + C = xy^2(x^2 + y^2) + C.$$

30. (E)

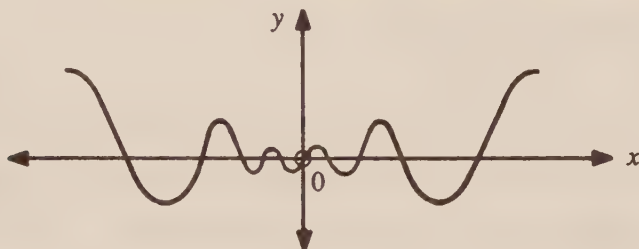
The distance (directed) from the yz plane to any point (x, y, z) is the coordinate $x = r \cos \theta$ in cylindrical coordinates. The representation of dV is $rdzdrd\theta$. The integrand must be $r^2 \cos \theta$.

31. (C)

The equation is separable as $x dx = \frac{dy}{(y^2 + 1)}$ and solved as $\frac{x^2}{2} = \tan^{-1} y + C$ where $y(0) = -1$ implies $C = \frac{\pi}{4}$. Then $x = \sqrt{\pi}$ implies $\tan^{-1} y = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$ and $y = 1$.

32. (B)

The point $(0, 0)$ connects the well known graph of $x \sin\left(\frac{1}{x}\right)$ as shown. Since the domain is $(-\infty, \infty)$, the graph cannot be compact. It is a closed subset since the only missing limit point on the graph of $x \sin\left(\frac{1}{x}\right)$ is $(0, 0)$.



33. (A)

As subtasks we may first choose any one of 13 denominations and then find 4 different subsets of three of the same denomination. Finally the number of ways to choose the remaining two cards from the 48 not in the selected denomination is $48 \cdot 47/2$. Division by two is required as subsets are desired and so orders of selection should not be counted. The counting principle for products implies the solution $13 \cdot 4 \cdot 24 \cdot 47 = 52 \cdot 24 \cdot 47$.

34. (A)

In standard form, $y' - \frac{2}{x}y = x$ and so the integrating factor is

$$e^{\int -\frac{2}{x} dx} = e^{-2 \ln x} = e^{\ln x^{-2}} = x^{-2}$$

35. (C)

Since $(3, -2, 1) = 3(1, 0, 0) - 2(0, 1, 0) + 1(0, 0, 1)$ its image must be $3(1, 2, 3) - 2(2, 3, 1) + (1, 1, -2) = (0, 1, 5)$.

36. (B)

The 9th partial sum of a convergent alternating series with decreasing absolute terms has absolute error less than the 10th term which is $1/100$. Since the approximation ends with a negative term it is less than the sum of the series and so $0 < E < .01$.

37. (B)

In powers of 2 we find $73 = 64 + 8 + 1 = 2^6 + 2^3 + 2^0$, so the binary representaiton is 1 0 0 1 0 0 1.

38. (C)

$f'(x) = 3x^2 - 6x = 3x(x - 2)$ and so $x = 2$ is the only critical point in $[1, 3]$. Now $f(1) = k - 2$, $f(2) = k - 4$, and $f(3) = k$. We must require $k = k - 4$ or $k = 4 - k$ and only the latter can be solved with $k = 2$.

39. (D)

In general an n -gon must have $\binom{n}{2} - n = \frac{n(n-3)}{2}$ diagonals. Note that $\binom{n}{2}$ counts the number of pairs of vertices and all but n of these pairs (sides) represent diagonals. Now to have 10 times as many diagonals requires $\frac{(n-3)}{2} = 10$ and $n = 23$.

40. (A)

All three triangles are similar since they are right triangles sharing an acute angle. The dimensions of the small triangles are in ratio 3:4 as seen in the hypotenuses. Hence the areas must be in ratio

$$\left(\frac{3}{4}\right)^2 = \frac{9}{16}.$$

41. (E)

Not considering the circular layout there are $6! = 720$ permutations. These can be grouped into 6 cyclic permutations per group each representing equivalent circular arrangements. So we get $\frac{6!}{6} = 120$.

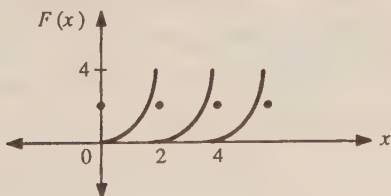
42. (D)

The average value is defined as the area under the curve (in this case a quarter circle of radius 2) divided by the base interval length. We get $\frac{\pi 2^2}{8} = \frac{\pi}{2}$. The integration approach to the area is much harder than this approach.

43. (D)

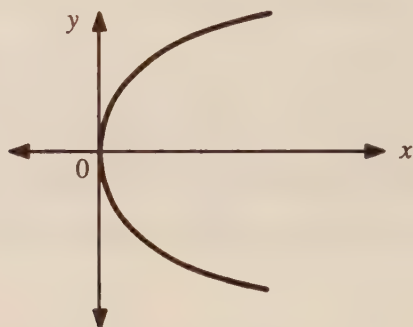
It is well known that $F(x)$ converges to the average of its left and right hand limits at all points. Here,

$$F(4) = \frac{F(4^-) + F(4^+)}{2} = \frac{0 + 4}{2} = 2.$$



44. (A)

Of course many functions satisfy the coordinate criteria but all must have limit zero at $x = 0$. An example where this is the only limit is $f(x) = \sqrt{x}$ if x is rational but $f(x) = -\sqrt{x}$ if x is irrational. All points lie on the parabola $x = y^2$.



45. (D)

Bounded sequences always contain convergent subsequences and the related sequence in (C) is always convergent. Any convergent subsequence contains infinitely many subsequences all converging to the same limit. Since $\{x^n\}$ is divergent we must have different limits for $\min_{k \leq n} x_k$ and $\max_{k \leq n} x_k$.

46. (E)

" $\bar{r}$ implies $\bar{s}$ " is the negation of " r implies s " and is equivalent to its converse which is not generally true. The contrapositive of " $\bar{p}$ implies $\bar{q}$ and $\bar{r}$ " is " q or r implies p " and these are equivalent. So q implies p as q satisfies " q or r " and hence I. is true. Also r implies p and, via the transitivity property, since p implies q we must have " r implies q " and III. is true.

47. (C)

I. follows as true since in general $|ab| = |a||b|$.

II. is also true since $|x| \neq |y|$ implies that $|x + y| > 0$.

III. is not generally true. The triangle inequality states that

$$|x + y| \leq |x| + |y| \text{ in general but } |x - y| \geq \frac{1}{(|x| + |y|)}$$

fails to imply that $|x - y| \geq \frac{1}{|x + y|} \geq \frac{1}{(|x| + |y|)}$.

For example, $x = \frac{4}{5}$ and $y = -\frac{3}{5}$ finds $|x - y| < \frac{1}{|x + y|}$.

Similarly, $|x - y| \geq \frac{1}{4}$ fails to imply $|x - y| \geq \frac{1}{(|x| + |y|)}$

so IV. is false.

48. (E)

Let N represent the event "not one, two, or six" and let S represent the event "six." Then the probability is the sum of the probabilities of the sequences of events, $NN \dots NS$, where the number of N 's varies as $0, 1, 2, \dots, k, \dots$. The sequence probabilities are $\left(\frac{1}{2}\right)^k \left(\frac{1}{6}\right)$ and the sum is geometric with value $\left(\frac{1}{6}\right) \left(1 - \frac{1}{2}\right) = \frac{1}{3}$.

49. (D)

Using $F(x, y, z) = x^2z - 2yz^2 + xy$ we find

$$\frac{\partial x}{\partial z} = -\frac{F_z}{F_x} = -\frac{[x^2 - 4yz]}{[2xz + y]}.$$

At $(1, 1, 1)$ the value is $-\frac{(-3)}{3} = 1$.

50. (C)

The general idea is that inverses of products are products of inverses in the reverse order.

51. (E)

A left ideal is a subring with the additional property that for each $r \in R$ and $h \in H$ we have $rh \in H$ so $RH \subseteq H$. Also, $H = R$. So, $HR = RR = R = H$.

52. (C)

This is a fundamental result of real analysis. The example with $C_n = \left[0, \frac{1}{n}\right]$ yields singleton $S = \{0\}$ which is a counterexample to (D). Boundedness is necessary or $C_n = [n, \infty)$ implies $S = 0$.

53. (E)

Only 10 values of $X(I)$ are examined since $J = 11$ initiates a return. $M = 0$ is replaced only when $X(J) > 0$ so negative entries in the array are not further processed. Whenever a greater positive value of $X(J)$ occurs, M takes this value. Hence the last statement is strongest as minimums or maximums in general may be negative and a strictly positive array is not required as in (C).

54. (D)

Simplest is a conditional probability approach on the second ball drawn when two colors are permitted (20 balls) out of the remaining 29 balls following the first draw.

55. (D)

By inspection, the sum of the coefficients is zero so 1 is a root. Synthetic division implies the quadratic quotient is $x^2 - 2x + 3$ with roots

$$\frac{(2 \pm \sqrt{-8})}{2} = 1 \pm \sqrt{2}i.$$

56. (E)

The main theoretical point is that the coefficient of $(x-2)^n$ is $\frac{f^{(n)}(2)}{n!}$ and here, with $n=5$, we get

$$\frac{1}{32 \cdot 6} = \frac{f^{(5)}(2)}{5!} \text{ so } f^{(5)}(2) = \frac{120}{32 \cdot 6} = \frac{5}{8}.$$

57. (A)

Since our function is “even” we know that $f(-7) = f(7)$.

58. (D)

$$\left| 2(AB)^{-1} \right| = \frac{2^4}{|AB|} = \frac{2^4}{(|A||B|)} = \frac{2^4}{6} = \frac{8}{3}.$$

59. (B)

Using modular arithmetic, we find that

$7x + 3y \equiv 0 \pmod{13}$ and $8x + ky \equiv 0 \pmod{13}$,
so by subtraction

$$x + (k-3)y \equiv 0 \pmod{13}.$$

Multiplying by 6, we have

$$6x + 6(k-3)y \equiv 0 \pmod{13}.$$

Adding $7x + 3y \equiv 0 \pmod{13}$ gives

$$\begin{aligned}13x + 6(k-3)y + 3y &\equiv 0 \pmod{13} \\ \Rightarrow 13x + [6(k-3) + 3]y &\equiv 0 \pmod{13} . \\ \Rightarrow (6k-15)y &\equiv 0 \pmod{13} \\ \Rightarrow (6k-2)y &\equiv 0 \pmod{13}\end{aligned}$$

Since y can vary, we require

$$(6k-2) \equiv 0 \pmod{13}$$

and by inspection of the choices, we find that $k = 9$ works, since

$$6 \cdot 9 - 2 = 54 - 2 = 52 = 13 \cdot 4 \equiv 0 \pmod{13} .$$

60. (A)

The approximating form is $y_{n+1} = y_n + y'(x_n)\Delta x$.

In this case

$$\Delta x = 0.1, x_0 = y_0 = 1 \text{ and } y' = \frac{(x+y)}{x} .$$

Then

$$\begin{aligned}y_1 &= 1 + (2)(0.1) \\ &= 1.2 .\end{aligned}$$

So

$$\begin{aligned}y_2 &= 1.2 + \left[\frac{(1.1 + 1.2)}{1.1} \right] (0.1) \\ &= \frac{6}{5} + \frac{23}{100} \\ &= \frac{155}{110} \\ &= \frac{31}{22} .\end{aligned}$$

61. (B)

Since 1440 has prime factorization $2^5 \cdot 3^2 \cdot 5^1$, the number of divisors is $(6)(3)(2) = 36$. This follows by a simple counting scheme in

that divisors of 1440 must have from 0 to 5 factors that are 2, 0 to 2 factors that are 3, and 0 or 1 factors which are 5. The divisor, 1, corresponds to the case when no prime factors are chosen.

62. (D)

The smallest intersection that is nonempty will contain one point, x , and so we only need count the number of subsets of S that contain x . Each such set will be the union of $\{x\}$ with any subset containing any remaining elements. There are $2^9 = 512$ such subsets. So, the answer is (D).

63. (D)

Linear combinations provide all solutions in the form $c_1f(x) + c_2g(x)$ where c_1, c_2 are constants.

64. (C)

The characteristic polynomial of A is $|\lambda I - A| = \lambda^2 - 5\lambda - 2$. The Cayley-Hamilton Theorem implies that $A^2 - 5A - 2I = 0$.

65. (D)

The translation $(x, y) \rightarrow (x, y + 1)$ works for both I and II. The rotation through one radian maps $(\cos n, \sin n)$ into $(\cos(n + 1), \sin(n + 1))$ leaving the point $(1, 0)$ out of the image. No point can map onto the image since n radians cannot equal $2k\pi$ radians as π is irrational.

66. (B)

An eigenvalue, λ , must satisfy the equation $|A - \lambda I| = 0$. We see that $A - (-2I)$ has all rows equal (all 3's) and so its determinant is immediately zero.

The Graduate Record Examination in
MATHEMATICS

Practice Test 5

The Graduate Record Examination in **MATHEMATICS** Answer Sheet Test 5

- | | | |
|-------------------------|-------------------------|-------------------------|
| 1. (A) (B) (C) (D) (E) | 23. (A) (B) (C) (D) (E) | 45. (A) (B) (C) (D) (E) |
| 2. (A) (B) (C) (D) (E) | 24. (A) (B) (C) (D) (E) | 46. (A) (B) (C) (D) (E) |
| 3. (A) (B) (C) (D) (E) | 25. (A) (B) (C) (D) (E) | 47. (A) (B) (C) (D) (E) |
| 4. (A) (B) (C) (D) (E) | 26. (A) (B) (C) (D) (E) | 48. (A) (B) (C) (D) (E) |
| 5. (A) (B) (C) (D) (E) | 27. (A) (B) (C) (D) (E) | 49. (A) (B) (C) (D) (E) |
| 6. (A) (B) (C) (D) (E) | 28. (A) (B) (C) (D) (E) | 50. (A) (B) (C) (D) (E) |
| 7. (A) (B) (C) (D) (E) | 29. (A) (B) (C) (D) (E) | 51. (A) (B) (C) (D) (E) |
| 8. (A) (B) (C) (D) (E) | 30. (A) (B) (C) (D) (E) | 52. (A) (B) (C) (D) (E) |
| 9. (A) (B) (C) (D) (E) | 31. (A) (B) (C) (D) (E) | 53. (A) (B) (C) (D) (E) |
| 10. (A) (B) (C) (D) (E) | 32. (A) (B) (C) (D) (E) | 54. (A) (B) (C) (D) (E) |
| 11. (A) (B) (C) (D) (E) | 33. (A) (B) (C) (D) (E) | 55. (A) (B) (C) (D) (E) |
| 12. (A) (B) (C) (D) (E) | 34. (A) (B) (C) (D) (E) | 56. (A) (B) (C) (D) (E) |
| 13. (A) (B) (C) (D) (E) | 35. (A) (B) (C) (D) (E) | 57. (A) (B) (C) (D) (E) |
| 14. (A) (B) (C) (D) (E) | 36. (A) (B) (C) (D) (E) | 58. (A) (B) (C) (D) (E) |
| 15. (A) (B) (C) (D) (E) | 37. (A) (B) (C) (D) (E) | 59. (A) (B) (C) (D) (E) |
| 16. (A) (B) (C) (D) (E) | 38. (A) (B) (C) (D) (E) | 60. (A) (B) (C) (D) (E) |
| 17. (A) (B) (C) (D) (E) | 39. (A) (B) (C) (D) (E) | 61. (A) (B) (C) (D) (E) |
| 18. (A) (B) (C) (D) (E) | 40. (A) (B) (C) (D) (E) | 62. (A) (B) (C) (D) (E) |
| 19. (A) (B) (C) (D) (E) | 41. (A) (B) (C) (D) (E) | 63. (A) (B) (C) (D) (E) |
| 20. (A) (B) (C) (D) (E) | 42. (A) (B) (C) (D) (E) | 64. (A) (B) (C) (D) (E) |
| 21. (A) (B) (C) (D) (E) | 43. (A) (B) (C) (D) (E) | 65. (A) (B) (C) (D) (E) |
| 22. (A) (B) (C) (D) (E) | 44. (A) (B) (C) (D) (E) | 66. (A) (B) (C) (D) (E) |

GRE MATHEMATICS TEST V

TIME: 2 hours and 50 minutes
66 Questions

DIRECTIONS: Choose the best answer for each question and mark the letter of your selection on the corresponding answer sheet.

1. $\lim_{n \rightarrow \infty} \left(\prod_{m=2}^n \left(1 - \frac{1}{m} \right) \right)$ is equal to

(A) 1

(D) e^{-1}

(B) e

(E) 0

(C) π

2. If $n = 2^{20}$, then what is the sum of all integer divisors d of n , $1 \leq d \leq n$?

(A) $2^{22} + 1$

(D) $2^{21} + 1$

(B) $2^{22} - 1$

(E) $2^{20} + 1$

(C) $2^{21} - 1$

3. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and A^{-1} is the inverse of A , what is the determinant of A^{-1} ?

- (A) -2 (D) $-\frac{1}{2}$
(B) -5 (E) 2
(C) $\frac{1}{5}$

4. Which of the following complex numbers is equal to $(1+i)^{\frac{4}{3}}$?

- (A) $2^{\frac{2}{3}}\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$ (D) $2^{\frac{2}{3}}\left(\sqrt{\frac{3}{2}} - i\frac{1}{2}\right)$
(B) $2^{\frac{2}{3}}\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$ (E) $2^{\frac{2}{3}}\left(\sqrt{\frac{2}{3}} + i\frac{1}{\sqrt{3}}\right)$
(C) $2^{\frac{2}{3}}\left(\sqrt{\frac{3}{2}} + i\frac{1}{2}\right)$

5. What is the value of $5 \log_{15} (15x) - \log_{15} x^5$?

- (A) 5 (D) $5 \log_{15} \frac{15}{x}$
(B) $105x^2$ (E) $5 \log_{15} 4x$
(C) 105

6. Find the value of $\lim_{x \rightarrow 0} \frac{1}{x} \int_0^x (1 + \sin t)^t dt$.

- (A) 1 (D) $\frac{2}{3}$
(B) $\frac{3}{2}$ (E) e^{-1}
(C) e

7. What is the value of $\lim_{h \rightarrow 0} \frac{\sqrt{3+h} - \sqrt{3}}{h}$?

- (A) 0 (D) $\frac{\sqrt{3}}{6}$
(B) $\frac{\sqrt{3}}{3}$ (E) ∞
(C) Undefined

8. How many solutions are there of the equation

$$\cos^2 x = \cos x \text{ if } 0 \leq x \leq 2\pi?$$

- (A) No solutions (D) 3
(B) 1 (E) 4
(C) 2

9. Given a set S of strictly negative real numbers, what is the greatest lower bound of the set $\{x \text{ real} \mid x \text{ is an upper bound of } S\}$?

(A) 0 (D) $-\infty$
 (B) Infimum of s (E) Does not exist
 (C) supremum of s

10. The partial derivative $\frac{\partial}{\partial y} \left[\int_0^1 e^{y \sin x} dx \right]$ is equal to

(A) $\int_0^1 e^{y \sin x} dx$ (D) $\int_0^1 e^{y \sin x} (\sin y) dy$
 (B) $\int_0^1 \cos x e^{y \sin x} dx$ (E) $\int_0^1 y \sin x e^{y \sin x} dx$
 (C) $\int_0^1 \sin x e^{y \sin x} dx$

11. If $f(x, y) = \frac{2}{x^2} + 3xy$ for $x \neq 0$, and the gradient of f at (r, s) has length r , then which of the following equations is satisfied by r and s ?

(A) $16 + 24 r^3 s + 9 r^6 s^2 + 8 r^8 = 0$
 (B) $24 + 9 r^3 s + r^6 s^2 + 8 r^8 = 0$
 (C) $16 - r^3 s + 9 r^6 s^2 + 8 r^8 = 0$

(D) $16 - 24 r^3 s + 9 r^6 s^2 + r^8 = 0$

(E) $16 - 24 r^3 s + 9 r^6 s^2 + 8 r^8 = 0$

12. Given $f(x) = \frac{x}{x-1}$, find an expression of $f(3x)$ in term of $f(x)$.

(A) $\frac{3f(x)}{3f(x)-1}$

(D) $\frac{3f(x)}{2f(x)+1}$

(B) $\frac{3f(x)}{3f(x)-3}$

(E) $\frac{3f(x)}{2f(x)-1}$

(C) $3f(x)-1$

13. Let $S_n = -1 + 2\left(\frac{2}{3}\right) - 3\left(\frac{2}{3}\right)^2 + \dots + (-1)^n n \left(\frac{2}{3}\right)^{n-1}$.

What is $\lim_{n \rightarrow \infty} S_n$?

(A) $-\frac{9}{25}$

(D) $-\frac{25}{9}$

(B) $\frac{9}{25}$

(E) $\frac{5}{3}$

(C) $\frac{25}{9}$

14. If the domain of function $y = f(x)$ is $[0, 1]$, then what is the domain of function $f(x + \frac{1}{4}) + f(x - \frac{1}{4})$?

(A) $[0, 1]$

(D) $[\frac{1}{4}, \frac{3}{4}]$

(B) $(0, 1)$

(E) $(\frac{1}{4}, \frac{3}{4})$

(C) $[0, \frac{1}{2}]$

15. $[\sqrt{2}(1-i)]^{48} =$

(A) 2^{24}

(D) -2^{48}

(B) -2^{24}

(E) $2^{24}(1-i)$

(C) 2^{48}

16. The solution of differential equation $y dx + \sqrt{x^2 + 1} dy = 0$ is:

(A) $y(x + \sqrt{x^2 + 1}) = c$

(B) $y(1 + \sqrt{x^2 + 1}) = c$

(C) $xy + \sqrt{x^2 + 1} = c$

(D) $yx + \frac{y}{\sqrt{x^2 + 1}} = c$

(E) $yx + \frac{x}{\sqrt{x^2 + 1}} = c$

17. The length of the curve $x(t) = e^t \cos t$, $y(t) = -e^t \sin t$ for $0 \leq t \leq 1$ is

(A) $2(e - 1)$

(D) $2e$

(B) $\sqrt{2}(e - 1)$

(E) $\sqrt{2}$

(C) e

18. The normal line to the graph of $3x^2 + 4x^2y + xy^2 = 8$ at $(1, 1)$ intersects the x -axis at $x =$

(A) $\frac{3}{2}$

(D) $-\frac{2}{3}$

(B) $-\frac{3}{2}$

(E) $-\frac{2}{5}$

(C) $\frac{5}{2}$

19. Let $F(uv) = uv$ where $u = u(t)$ and $v = v(t)$. If $u(1) = 1$, $v(1) = 2$, $v'(1) = 1$ and $u'(1) = 2$, then $\frac{dF}{dt}$ at $t = 1$ is equal to

(A) 0

(D) 4

(B) 1

(E) 5

(C) 3

20. The number of points of discontinuity of the function

$$f(x) = \begin{cases} x + 2 & \text{if } x \leq 0 \\ 1 & \text{if } 0 < x \leq 2 \\ x - 6 & \text{if } 2 < x \leq 5 \\ (6 - x)^2 & \text{if } 5 < x \end{cases} \quad \text{is equal to}$$

(A) 5 (D) 2

(B) 4 (E) 1

(C) 3

21. The solution set for the inequality $\frac{1}{x-2} < \frac{1}{x+3}$ is

(A) $(-3, -2)$ (D) $(-2, 2)$

(B) $(-3, 2)$ (E) $(0, 2)$

(C) $(2, 3)$

22. If $g\left(\frac{3+2x}{4}\right) = 1-x$, for $-\infty < x < \infty$, then $g\left(\frac{7z-8}{4}\right)$ is equal to

(A) $-\frac{13+7z}{4}$ (D) $\frac{7+13z}{2}$

(B) $\frac{13}{2} - \frac{7z}{2}$ (E) $7z + 13$

(C) $\frac{7-13z}{2}$

23. If $f'(e^x) = 1 + x$, then $f(x) =$

- (A) $1 + e^x + c$ (D) $x + \ln x + c$
(B) $1 + \ln x + c$ (E) $x \ln x + c$
(C) $\ln x + c$

24. The iterated integral $\int_0^1 \int_{\frac{y}{2}}^1 e^{x^2} dx dy$ can be expressed as

- (A) $\int_0^1 \int_0^{2x} e^{x^2} dy dx$ (C) $\int_0^1 \int_0^{2y} e^{x^2} dx dy$
(B) $\int_{\frac{y}{2}}^1 \int_0^1 e^{x^2} dy dx$ (D) $\int_0^1 \int_0^{2y} e^{x^2} dy dx$
(E) $\int_0^1 \int_y^1 e^{x^2} dy dx + \int_0^1 \int_0^y e^{x^2} dx dy$

25. The series $\sum_{n=1}^{\infty} \frac{3^n}{n} (x-2)^n$ converges for x in the interval

- (A) $\left(\frac{5}{3}, \frac{7}{3}\right)$ (D) $\left[\frac{5}{3}, \frac{7}{3}\right)$
(B) $\left(\frac{5}{3}, \frac{7}{3}\right]$ (E) $\left(0, \frac{7}{3}\right)$
(C) $\left[\frac{5}{3}, \frac{7}{3}\right]$

26. In R^3 , an equation of the tangent plane to the surface $xz - yz^3 - yz^2 = 378$ at $(-3, 2, -6)$ is

(A) $2x + 60y + 65z = 516$

(B) $-2x + 60y + 65z + 516 = 0$

(C) $2x - 60y + 65z = 0$

(D) $x - 60y + z = 516$

(E) $2x - 60y + 65z + 516 = 0$

27. If $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, then $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$ is equal to

(A) $\frac{\pi^2}{12}$

(D) $\frac{\pi^2}{7}$

(B) $\frac{\pi^2}{36}$

(E) $\frac{\pi^2}{8}$

(C) $\frac{2\pi^2}{9}$

28. The area of the triangle in R^3 with vertices at $A = (2, 1, 5)$, $B = (4, 0, 2)$ and $C = (-1, 0, -1)$ is

(A) $\frac{475}{2}$

(D) $\frac{\sqrt{475}}{2}$

(B) 475

(E) $\sqrt{\frac{475}{2}}$

(C) $\sqrt{475}$

29. The order of the permutation $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix}$ is

(A) 1 (D) 4

(B) 2 (E) 5

(C) 3

30. Let T be a linear transformation from R^3 to R^3 . If u and v are two orthogonal vectors in R^3 , which of the following pairs of vectors must be orthogonal to one another?

(A) ru and sv for all real r and s

(B) $u + v$ and $u - v$ (D) Tv and v

(C) Tu and u (E) Tu and Tv

31. If $x \in A \cap B$, which of the following is (are) true?

I. $x \in A$

II. $x \in B$

III. $x \in A' \cup B'$, where A' is the complement of A and B' is the complement of B

(A) I only

(B) II only

(C) I, II and III

(D) I and II only

(E) III only

32. Let R be the set of real numbers. Define $a * b = a + b + ab$ for a, b in R . The solution of $5 * x * 3 = 7$ is

(A) $\frac{7}{15}$

(D) $\frac{3}{2}$

(B) $\frac{15}{7}$

(E) $-\frac{2}{3}$

(C) $\frac{2}{3}$

33. If $(1 + 2x + 3x^2 + \dots + (n + 1)x^n \dots)^2 = \sum_{k=0}^{\infty} b_k x^k$ for $|x| < 1$ then b_k is

(A) $k(k + 1)(k + 2)(k + 3)$

(B) $\frac{k(k + 1)(k + 2)}{6}$

(C) $\frac{(k + 1)(k + 2)(k + 3)}{6}$

(D) $\frac{(k + 1)(k + 2)}{3}$

(E) $(k + 2)(k + 3)$

34. Suppose $g'(x)$ exists for all real x and $g(a) = g(b) = g(c) = 0$ where $a < b < c$. The minimum possible number of zeros for $g'(x)$ is

(A) 1 (D) 4

(B) 2 (E) 5

(C) 3

35. $\int_0^{\frac{\pi}{2}} |\sin x - \cos x| dx =$

(A) $2\sqrt{2} - 2$ (D) $2\sqrt{2}$

(B) 0 (E) $2 - 2\sqrt{2}$

(C) 1

36. Which of the following are groups?

- I. All integers under subtraction
- II. All non-zero real numbers under division
- III. All even integers under addition
- IV. All integers which are multiples of 13 under addition

(A) I and II only (D) IV only

(B) II and III only (E) III and IV only

(C) III only

37. $(1 + \sqrt{-1})^{8n} - (1 - \sqrt{-1})^{8n}$ is equal to

(A) 2^{4n}

(D) 0

(B) $(-1)^{n+1} 2^{4n}$

(E) $-e^{-2}$

(C) 2^{4n+1}

38. Given two vectors $\vec{U} = 2\vec{i} - 3\vec{j} + 5\vec{k}$, $\vec{V} = -\vec{i} + 4\vec{j} + 2\vec{k}$ then their vector product $\vec{U} \times \vec{V} =$

(A) $\vec{i} + \vec{j} + 7\vec{k}$

(D) $3\vec{i} + 4\vec{j} + 5\vec{k}$

(B) $-2\vec{i} - 12\vec{j} + 10\vec{k}$

(E) $3\vec{i} - 4\vec{j} - 5\vec{k}$

(C) $-26\vec{i} - 9\vec{j} + 5\vec{k}$

39. For what value(s) of p does the system of equations

$$px + y = 1$$

$$x + py = 2$$

$$y + pz = 3$$

have no solution?

(A) 0, 1

(D) -2

(B) 1, -1

(E) 0, 1, -1

(C) -1

40. If the determinant of the matrix:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ x & 2 & 0 & 0 & 0 \\ x^2 & x^3 & 3 & 0 & 0 \\ x^3 & x^4 & x^5 & 4 & 0 \\ x^4 & x^5 & x^6 & x^7 & x \end{bmatrix}$$

is zero, how many values of x are possible?

- (A) 0 (D) 3
(B) 1 (E) ∞
(C) 2
41. The n th derivative of $f(x) = \frac{1}{1-x^2}$ is
- (A) $\frac{1}{(1-x^2)^{n+1}}$ (C) $\frac{(-1)^n n!}{(1-x^2)^{n+1}}$
(B) $\frac{x^n}{(1-x^2)^{n+1}}$ (D) $\frac{n!}{(1-x^2)^{n+1}}$
(E) $\frac{n!}{2} \left[\frac{1}{(1-x)^{n+1}} + \frac{(-1)^n}{(1+x)^{n+1}} \right]$

42. How many different partial derivatives of order k are possible for a function $f(x_1, \dots, x_n)$ of n variables?

- (A) 2^{n+k} (B) $\binom{n+k-1}{n}$

$$(C) \quad \binom{n+k-1}{k}$$

$$(D) \quad \binom{n+k}{k}$$

$$(E) \quad \binom{n+k}{n-1}$$

43. Let $T: C^\infty(R) \rightarrow C^\infty(R)$ be a linear map such that $T(e^{2x}) = \sin x$, $T(e^{3x}) = \cos 4x$ and $T(1) = e^{5x}$, where $C^\infty(R)$ is the vector space infinitely differentiable functions on the real numbers R . Then $T(4e^{2x} + 7e^{3x} - 5)$ is equal to

$$(A) \quad 4 \sin x + 7 \cos 4x - 5e^{5x} \quad (D) \quad 7 \cos 4x - 5e^{5x}$$

$$(B) \quad 4 \sin x + 7 \cos 4x - 5 \quad (E) \quad \sin x + 7 \cos x - 5e^{5x}$$

$$(C) \quad 4 \sin x + 7 \cos 4x$$

44. How many ways can 8 teachers be divided among 4 schools if each school must receive 2 teachers?

$$(A) \quad 520$$

$$(D) \quad 225$$

$$(B) \quad 250$$

$$(E) \quad 2^4$$

$$(C) \quad 2520$$

45. Let C be the circle $|z| = 3$, described in counterclockwise orientation, and write

$$g(w) = \int_C \frac{2z^2 - 2 - z}{z - w} dz$$

Then $g(2)$ is equal to

- (A) 1
(B) $2\pi i$
(C) 0
(D) $4\pi i$
(E) $8\pi i$

46. Given that

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots$$

$$\text{then } -2 + 1 - \frac{2}{3} + \frac{2}{4} - \frac{2}{5} + \frac{2}{6} - \dots$$

is equal to

- (A) $-2 \ln 2$
(B) $-\ln 2$
(C) $2 \ln 2$
(D) $\ln 3$
(E) $-3 \ln 3$

47. Let

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ -3 & -2 & 1 & 0 \\ -4 & -3 & -2 & 2 \end{bmatrix}$$

be a 4×4 matrix viewed as a linear transformation from R^4 to R^4 . What is the dimension of the eigenspace corresponding to the eigenvalue $\lambda = 1$?

- (A) 4 (D) 1
(B) 3 (E) 0
(C) 2

48. Let T be a linear transformation from a vector space V of dimension 11 onto a vector space W of dimension 7. What is the dimension of the null space of T ?

- (A) 0 (D) 4
(B) 2 (E) 5
(C) 3

49. In an $x - y$ plane, if $\vec{a} = (x, y)$, $\vec{b} = (x', y')$, then their scalar product $\vec{a} \cdot \vec{b} =$

- (A) $xx' + yy'$ (B) $xx' - yy'$

(C) $xy - x'y'$

(D) $xy' + x'y$

(E) $xy' - x'y$

50. What is the mean of the random variable $\bar{X}$ whose distribution function is defined by

$$F(x) = \begin{cases} \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

(A) $\frac{2}{\pi}$

(D) 0

(B) $\frac{1}{\sqrt{\pi}}$

(E) Does not exist

(C) $\sqrt{\frac{2}{\pi}}$

51. If $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, a 2×2 matrix, which of the following is true?

(A) $A^2 - 3A = 0$

(D) $A^2 - 3A + I = 0$

(B) $I - 3A = 0$

(E) None of the above

(C) $A^2 + I = 0$

52. The equation of the tangent line to the graph of the equation

$$\begin{cases} x = t^3 - 4 \\ y = 2t^2 + 1 \end{cases} \text{ at } t = 2 \text{ is}$$

(A) $2x - 3y - 19 = 0$ (D) $3x - 2y + 6 = 0$

(B) $2x - 3y + 19 = 0$ (E) $3x + 2y - 6 = 0$

(C) $3x - 2y - 6 = 0$

53. If $f(0) = 1$, $f(2) = 3$ and $f'(2) = 5$, then $\int_0^1 xf''(2x) dx =$

(A) 0 (D) -1

(B) 1 (E) -2

(C) 2

54. Let P denote the product of any four consecutive integers. Then $1 + P$ is

(A) a multiple of 5 (D) a complete square

(B) a multiple of 4 (E) none of the above

(C) a prime number

55. The solution of the differential equation $y'' + 5y' + 6y = 0$ satisfying the initial conditions $y(0) = 0$ and $y'(0) = 1$ is

(A) $e^{2x} + e^{3x}$

(D) $e^{-2x} + e^{-3x}$

(B) $e^{2x} - e^{3x}$

(E) $e^{-2x} - e^{-3x}$

(C) $e^{-2x} + e^{3x}$

56. Which of the following functions is (are) analytic?

I. $\bar{z}$

II. $\bar{z} \sin z$

III. $z + \sin z$

IV. $z + \bar{z}$

V. $z e^z$

(A) I only

(D) IV only

(B) II and I only

(E) None of the above

(C) III and V only

57. For any positive integer n , $n^7 - n$ is divisible by...

(A) 4

(D) 14

(B) 6

(E) 18

(C) 7

58. Consider the permutation $f = (1478)(265)(39)$ in S_9 . $f^{-1} =$

(A) $(1874)(256)(39)$ (D) $(1874)(265)(39)$

(B) $(1874)(265)(39)$ (E) None of the above

(C) $(1847)(256)(39)$

59. For what value of c is the function

$$f(x) = \begin{cases} cxe^{-x^2}, & 0 < x < \infty \\ 0 & , \text{ elsewhere} \end{cases}$$

the probability density function of a random variable $\underline{X}$?

(A) 1 (D) π

(B) 2 (E) 2π

(C) 3

60. The negation of $\forall x \exists y (P(x,y) \wedge \neg Q(x,y))$ is

(A) $\forall x \exists y (P(x,y) \rightarrow Q(x,y))$

(B) $\forall x \exists y (Q(x,y) \rightarrow P(x,y))$

(C) $\exists x \forall y (P(x,y) \rightarrow Q(x,y))$

(D) $\exists x \forall y (Q(x,y) \rightarrow P(x,y))$

(E) None of the above

61. If A is a countable subset of the interval $[0, 1]$, then the Lebesgue measure of A is equal to

- (A) $\frac{1}{2}$ (D) 3^{-1}
 (B) 0 (E) None of the above
 (C) $\frac{2}{3}$

62. What is the interval of convergence of the series

$$\sum_{n=0}^{\infty} (3-x)(6x-7)^n ?$$

- (A) $[1, \frac{4}{3}]$ (D) $(1, \frac{4}{3})$
 (B) $(1, \frac{4}{3}]$ (E) $(0, \frac{4}{3})$
 (C) $[1, \frac{4}{3})$

63. Given

$$A = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} \end{pmatrix}$$

with $\det(A) \neq 0$, the system of equations

$$x_1 + \frac{1}{2}x_2 + \frac{1}{3}x_3 + \frac{1}{4}x_4 = 1$$

$$\frac{1}{2}x_1 + \frac{1}{3}x_2 + \frac{1}{4}x_3 + \frac{1}{5}x_4 = 2$$

$$\frac{1}{3}x_1 + \frac{1}{4}x_2 + \frac{1}{5}x_3 + \frac{1}{6}x_4 = 3$$

$$\frac{1}{4}x_1 + \frac{1}{5}x_2 + \frac{1}{6}x_3 + \frac{1}{7}x_4 = 4$$

has

- (A) No solutions (D) 3 solutions
- (B) A unique solution (E) Infinitely many solutions
- (C) 2 solutions

64. Suppose $f(x) = x^4 + x^6 + \sin(x^2) + e^{x^3}$ and $f(x) = f_1(x) + f_2(x)$, where $f_1(x) = f_1(-x)$ and $f_2(-x) = -f_2(x)$. Then $f_2(x)$ is equal to

- (A) $\frac{1}{2}(e^{x^3} - e^{-x^3})$ (D) $\sin(x^2)$
- (B) $\frac{1}{2}(e^{x^3} + e^{-x^3})$ (E) $x^4 + x^6 + \sin(x^2) + e^{x^3}$
- (C) $\frac{1}{2}(x^4 + x^6)$

65. If w is an n^{th} root of unity other than one, then the sum $w + w^2 + \dots + w^{n-1}$ is equal to

(A) 1

(D) 3

(B) 0

(E) -2

(C) -1

66. If $P(A) = 0.7$, $P(B) = 0.5$ and $P([A \cup B]^c) = 0.1$ then $P(A|B)$ is

(A) $\frac{3}{7}$

(D) $\frac{7}{9}$

(B) $\frac{3}{5}$

(E) 1

(C) $\frac{5}{7}$

GRE MATHEMATICS

TEST V

ANSWER KEY

1.	E	23.	E	45.	E
2.	C	24.	A	46.	A
3.	D	25.	D	47.	D
4.	B	26.	E	48.	D
5.	A	27.	E	49.	A
6.	A	28.	D	50.	B
7.	D	29.	D	51.	D
8.	E	30.	A	52.	B
9.	C	31.	D	53.	C
10.	C	32.	E	54.	D
11.	E	33.	C	55.	E
12.	D	34.	B	56.	C
13.	A	35.	A	57.	C
14.	D	36.	E	58.	A
15.	C	37.	D	59.	B
16.	A	38.	C	60.	C
17.	B	39.	E	61.	B
18.	B	40.	E	62.	D
19.	E	41.	E	63.	B
20.	C	42.	C	64.	A
21.	B	43.	A	65.	C
22.	B	44.	C	66.	B

GRE MATHEMATICS TEST V

DETAILED EXPLANATIONS OF ANSWERS

1. (E)

$$\text{Let } L = \lim_{n \rightarrow \infty} \left(\prod_{m=2}^n \left(1 - \frac{1}{m} \right) \right) = \lim_{n \rightarrow \infty} \left(\prod_{m=2}^n \left(\frac{m-1}{m} \right) \right)$$

Taking the natural logarithm of both sides we have

$$\ln L = \ln \lim_{n \rightarrow \infty} \left(\prod_{m=2}^n \frac{m-1}{m} \right) = \lim_{n \rightarrow \infty} \ln \left(\prod_{m=2}^n \left(\frac{m-1}{m} \right) \right)$$

We can interchange limit with $\ln$ because $\ln x$ is continuous. Using $\ln(ab) = \ln a + \ln b$ we get

$$\begin{aligned} \ln L &= \lim_{n \rightarrow \infty} \sum_{m=2}^n [\ln(m-1) - \ln m] \\ &= \lim_{n \rightarrow \infty} (\ln 1 - \ln 2) + (\ln 2 - \ln 3) + \dots + (\ln(n-1) - \ln n) \\ &= \lim_{n \rightarrow \infty} (\ln 1 - \ln n) \\ &= \lim_{n \rightarrow \infty} (-\ln n) \\ &= -\infty, \end{aligned}$$

since $\ln 1 = 0$:

$$\Rightarrow L = \lim_{b \rightarrow \infty} e^{-b} = \lim_{b \rightarrow \infty} \frac{1}{e^b} = 0.$$

So $L = 0$.

2. (C)

All of the integer divisors of 2^{20} are of the form 2^k , $0 \leq k \leq 20$. Therefore, the sum of these divisors is

$$\sum_{k=0}^{20} 2^k = \frac{2^{21} - 1}{2 - 1} = 2^{21} - 1$$

using the fact that the sum of the geometric sequence

$$\sum_{k=0}^m r^k = \frac{r^{m+1} - 1}{r - 1}.$$

3. (D)

The determinant of A is $\det(A) = 1.4 - 2.3 = 4 - 6 = -2$. Now $\det(A^{-1}) = [\det(A)]^{-1}$ using the fact that the determinant of a product is the product of their determinants.

$$\text{So } \det(A^{-1}) = (-2)^{-1} = -\frac{1}{2}.$$

4. (B)

The polar form of a complex number $a + ib = re^{i\theta}$ where $r = \sqrt{a^2 + b^2}$ and $\theta = \tan^{-1}\left(\frac{b}{a}\right)$.

Therefore $1 + i = 2^{\frac{1}{2}} e^{i\frac{\pi}{4}}$ and, taking the $4/3$ power, one obtains

$$\begin{aligned}(1 + i)^{\frac{4}{3}} &= \left(2^{\frac{1}{2}} e^{i\frac{\pi}{4}}\right)^{\frac{4}{3}} \\ &= \left(2^{\frac{1}{2}}\right)^{\frac{4}{3}} \left(e^{i\frac{\pi}{4}}\right)^{\frac{4}{3}} \\ &= 2^{\frac{2}{3}} e^{i\frac{\pi}{3}}\end{aligned}$$

$$= 2^{\frac{2}{3}} \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$= 2^{\frac{2}{3}} \left(\frac{1}{2} + \frac{i\sqrt{3}}{2} \right).$$

5. (A)

$$\begin{aligned} 5 \log_{15} 15x - \log_{15} x^5 &= 5 [\log_{15} 15 + \log_{15} x] - 5 \log_{15} x \\ &= 5 [1 + \log_{15} x] - 5 \log_{15} x \\ &= 5 + 5 \log_{15} x - 5 \log_{15} x \\ &= 5 \end{aligned}$$

using the properties of log:

$$\log_a xy = \log_a x + \log_a y,$$

$$\log_a x^n = n \log_a x \text{ and } \log_a a = 1$$

6. (A)

By L'Hopital's rule,

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1}{x} \int_0^x (1 + \sin t)^t dt &= \lim_{x \rightarrow 0} \frac{\int_0^x (1 + \sin t)^t dt}{x} \\ &= \lim_{x \rightarrow 0} \frac{\frac{d}{dx} \int_0^x (1 + \sin t)^t dt}{\frac{dx}{dx}} \\ &= \lim_{x \rightarrow 0} \frac{(1 + \sin x)^x}{1} \end{aligned}$$

using the fundamental theorem of calculus:

$$\frac{d}{dx} \int_a^x f(t) dt = f(x) .$$

Let
$$L = \lim_{x \rightarrow 0} (1 + \sin x)^x .$$

Taking $\ln$ of both sides, one gets

$$\begin{aligned} \ln L &= \ln \lim_{x \rightarrow 0} (1 + \sin x)^x \\ &= \ln \lim_{x \rightarrow 0} x \ln(1 + \sin x) \\ &= \lim_{x \rightarrow 0} x \cdot \lim_{x \rightarrow 0} \ln(1 + \sin x) \\ &= 0 \cdot \ln 1 \\ &= 0 \cdot 0 \\ &= 0 \end{aligned}$$

Therefore $L = e^0 = 1$.

7. (D)

Let $f(x) = \sqrt{x}$. The derivative of f at 3 is

$$f'(3) = \frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{6} .$$

But

$$f'(3) = \lim_{h \rightarrow 0} \frac{\sqrt{3+h} - \sqrt{3}}{h} ,$$

by the definition of derivative. Therefore,

$$\lim_{h \rightarrow 0} \frac{\sqrt{3+h} - \sqrt{3}}{h} = f'(3) = \frac{\sqrt{3}}{6} .$$

8. (E)

$\cos^2 x = \cos x$ or $\cos^2 x - \cos x = 0$ or $\cos x (\cos x - 1) = 0$ which implies $\cos x = 0$ or $\cos x - 1 = 0$. $\cos x = 0$ for $x = \frac{\pi}{2}, \frac{3\pi}{2}$ and $\cos x = 1$ for $x = 0$ and 2π .

9. (C)

Let $B = \{ x \in R \mid x \text{ is an upper bound for } S \}$ where S is a set of strictly negative real numbers. B is non-empty since 0 is in B and every x in S is a lower bound of B . So B has an infimum in R ; call it β . If $\gamma > \beta$, then γ is not a lower bound of B , so $\gamma \notin S$. It follows that $s \leq \beta$ for every s in S . Thus $\beta \in B$. If $\alpha < \beta$ then $\alpha \notin B$ since β is a lower bound of B . We have shown that $\beta \in B$ but $\alpha \notin B$ if $\alpha < \beta$. In other words, β is an upper bound of S , but α is not if $\alpha < \beta$. This means that $\beta = \sup S$.

10. (C)

$$\begin{aligned} \frac{\partial}{\partial y} \left(\int_0^1 e^{y \sin x} dx \right) &= \int_0^1 \frac{\partial}{\partial y} (e^{y \sin x}) dx \\ &= \int_0^1 \sin x e^{y \sin x} dx, \end{aligned}$$

using the chain rule.

11. (E)

The gradient of f , $\nabla f(x, y)$, is equal to $\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$.

Differentiating $f(x, y) = \frac{2}{x^2} + 3xy$ with respect to x and y , respectively, we get

$$\frac{\partial f}{\partial x}(x, y) = -\frac{4}{x^3} + 3y \text{ and } \frac{\partial f}{\partial y}(x, y) = 3x.$$

Now

$$r^2 = \|\nabla f(r, s)\|^2 = \left(\frac{\partial f}{\partial x}(r, s)\right)^2 + \left(\frac{\partial f}{\partial y}(r, s)\right)^2,$$

by the definition of the length of a vector

$$\begin{aligned} &= \left(-\frac{4}{r^3} + 3s\right)^2 + (3r)^2 \\ &= \frac{16}{r^6} - 24\frac{s}{r^3} + 9s^2 + 9r^2 \end{aligned}$$

Multiplying both sides by r^6 we get

$$r^8 = 16 - 24r^3s + 9r^6s^2 + 9r^8.$$

That is,

$$16 - 24r^3s + 9r^6s^2 + 8r^8 = 0.$$

12. (D)

$$f(3x) = \frac{3x}{3x-1}.$$

$$\text{For (A), } \frac{3f(x)}{3f(x)-1} = \frac{\frac{3x}{x-1}}{\frac{3x}{x-1}-1} = \frac{3x}{2x-1}$$

$$\text{For (B), } \frac{3f(x)}{3f(x)-3} = \frac{\frac{3x}{x-1}}{\frac{3x}{x-1}-3} = \frac{3x}{3} = x$$

$$\text{For (C), } 3f(x)-1 = \frac{3x}{x-1}-1 = \frac{3x-x+1}{x-1} = \frac{2x-1}{x-1}$$

$$\text{For (D), } \frac{3f(x)}{2f(x)+1} = \frac{\frac{3x}{x-1}}{\frac{2x}{x-1}+1} = \frac{3x}{2x+x-1} = \frac{3x}{3x-1}$$

$$\text{For (E), } \frac{3f(x)}{2f(x)-1} = \frac{\frac{3x}{x-1}}{\frac{2x}{x-1}-1} = \frac{3x}{2x-x+1} = \frac{3x}{x+1}$$

So, (D) is the right answer.

13. (A)

The geometric series $\sum_{k=0}^{\infty} (-x)^k$ converges to $\frac{1}{1+x}$ for $|x| < 1$.

Differentiating $\frac{1}{1+x} = \sum_{k=0}^{\infty} (-1)^k x^k$ one gets

$$-\frac{1}{(1+x^2)} = \sum_{k=1}^{\infty} k(-1)^k x^{k-1}.$$

Set $x = \frac{2}{3}$; then

$$\begin{aligned} -\frac{1}{\left(1 + \frac{2}{3}\right)^2} &= \sum_{k=1}^{\infty} k(-1)^k \left(\frac{2}{3}\right)^{k-1} \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n k(-1)^k \left(\frac{2}{3}\right)^{k-1} \end{aligned}$$

or

$$-\frac{9}{25} = \lim_{n \rightarrow \infty} \left[-1 + 2\left(\frac{2}{3}\right) - 3\left(\frac{2}{3}\right)^2 + \dots + (-1)^n n \left(\frac{2}{3}\right)^{n-1} \right] = \lim_{n \rightarrow \infty} S_n.$$

14. (D)

Since the domain of $y = f(x)$ is $0 \leq x \leq 1$, the domain of

$$f\left(x + \frac{1}{4}\right) + f\left(x - \frac{1}{4}\right) \text{ is } \begin{cases} 0 \leq x + \frac{1}{4} \leq 1 \\ 0 \leq x - \frac{1}{4} \leq 1, \end{cases}$$

$$i.e., \begin{cases} -\frac{1}{4} \leq x \leq \frac{3}{4} \\ \frac{1}{4} \leq x \leq \frac{5}{4} \end{cases}$$

$$\left[\frac{1}{4}, \frac{3}{4} \right].$$

So, the domain is

15. (C) $x + iy = z = re^{i\theta}$ where $r = \sqrt{x^2 + y^2}$

Using the polar form

$$\tan^{-1}\left(\frac{y}{x}\right) = \theta, \text{ we get } 1 - i = \sqrt{2} e^{-i\frac{\pi}{4}}. \text{ So}$$

and

$$\begin{aligned} [\sqrt{2}(1 - i)]^{48} &= [\sqrt{2} \cdot \sqrt{2} \cdot e^{-i\frac{\pi}{4}}]^{48} \\ &= [2e^{-i\frac{\pi}{4}}]^{48} \\ &= 2^{48} e^{-i12\pi} \\ &= 2^{48} (\cos(-12\pi) + i \sin(-12\pi)) \\ &= 2^{48}. \end{aligned}$$

16. (A)

$$ydx + \sqrt{x^2 + 1} dy = 0$$

$$\Rightarrow \frac{dx}{\sqrt{x^2 + 1}} + \frac{dy}{y} = 0$$

$$\Rightarrow \ln(x + \sqrt{x^2 + 1}) + \ln y = \ln C$$

$$\Rightarrow y(x + \sqrt{x^2 + 1}) = C.$$

17. (B)

The length of the arc is

$$\int_0^1 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Now differentiating $x(t) = e^t \cos t$ one obtains $\frac{dx}{dt} = e^t \cos t - e^t \sin t$,

using the product rule. Similarly $\frac{dy}{dt} = -e^t \sin t - e^t \cos t$.

So

$$\begin{aligned} & \int_0^1 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \\ &= \int_0^1 \sqrt{e^{2t}(\cos t - \sin t)^2 + e^{2t}(\cos t + \sin t)^2} dt \\ &= \int_0^1 \sqrt{e^{2t}(\cos^2 t - 2 \sin t \cos t + \sin^2 t + \cos^2 t + 2 \sin t \cos t + \sin^2 t)} dt \\ &= \int_0^1 \sqrt{2} e^t dt, \text{ after simplifying} \\ &= \sqrt{2}(e - 1). \end{aligned}$$

18. (B)

Differentiating $3x^2 + 4x^2y + xy^2 = 8$ with respect to x we get

$$6x + 8xy + 4x^2 \frac{dy}{dx} + y^2 + 2xy \frac{dy}{dx} = 0.$$

So evaluating at $(1, 1)$ $\frac{dy}{dx} = -\frac{15}{6} = -\frac{5}{2}$. Thus the equation of the

normal line at $(1, 1)$ is $y - 1 = \frac{2}{5}(x - 1)$ (Slope of the normal line is the negative reciprocal of the slope of tangent line). To find where this line intersects the x -axis we set $y = 0$. So

$$-1 = \frac{2}{5}(x - 1) \Rightarrow -5 = 2x - 2 \Rightarrow 2x = -3 \Rightarrow x = -\frac{3}{2}.$$

19. (E)

$$\frac{dF}{dt}(t) = \frac{du}{dt}(t) v(t) + u(t) \frac{dv}{dt}(t),$$

by the product rule. So

$$\begin{aligned}\frac{dF}{dt}(1) &= \frac{du}{dt}(1) \cdot v(1) + u(1) \frac{dv}{dt}(1) \\ &= 2 \cdot 2 + 1 \cdot 1 \\ &= 5.\end{aligned}$$

20. (C)

The function

$$f(x) = \begin{cases} x + 2 & \text{if } x \leq 0 \\ 1 & \text{if } 0 < x \leq 2 \\ x - 6 & \text{if } 2 < x \leq 5 \\ (6 - x)^2 & \text{if } 5 < x \end{cases}$$

is continuous everywhere except at the three points $x = 0, 2$ and 5 . So the number of discontinuities of f is 3.

21. (B)

$$\frac{1}{x-2} < \frac{1}{x+3} \quad \text{or} \quad \frac{1}{x-2} - \frac{1}{x+3} < 0, \quad \text{by adding } -\frac{1}{x+3}$$

$$\text{to both sides or } \frac{x+3-x+2}{(x-2)(x+3)} < 0 \quad \text{that is, } \frac{5}{(x-2)(x+3)} < 0$$

which implies

$$\text{either } x-2 > 0 \text{ and } x+3 < 0 \quad (1)$$

$$\text{or } x-2 < 0 \text{ and } x+3 > 0 \quad (2)$$

From relation (1) we have $x > 2$ and $x < -3$ which is impossible. From (2) we get $x < 2$ and $x > -3$. Thus the solution set is $(-3, 2)$.

22. (B)

Given that $g\left(\frac{3+2x}{4}\right) = 1-x$, $-\infty < x < \infty$. Put $y = \frac{3+2x}{4}$

which implies $x = \frac{4y-3}{2}$. So

$$\begin{aligned}g(y) &= 1-x \\&= 1 - \frac{(4y-3)}{2} \\&= \frac{2-4y+3}{2} \\&= \frac{5-4y}{2}\end{aligned}$$

$$\begin{aligned}\text{Thus, } g\left(\frac{7z-8}{4}\right) &= \frac{1}{2} \left[5 - 4\left(\frac{7z-8}{4}\right) \right], \quad \text{by letting } y = \frac{7z-8}{4} \\&= \frac{1}{2} [5 - 7z + 8] \\&= \frac{1}{2} [13 - 7z] \\&= \frac{13}{2} - \frac{7z}{2}.\end{aligned}$$

23. (E)

Let $e^x = u$, then $f'(u) = 1 + \ln u$.

$$\text{So } f(u) = \int (1 + \ln u) du$$

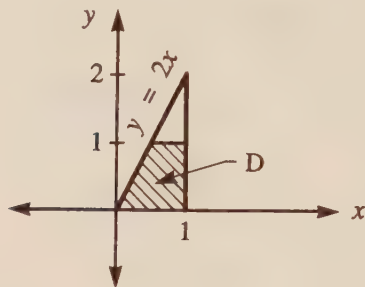
$$= u \ln u + C.$$

Therefore, $f(x) = x \ln x + C$.

24. (A)

Let D be the region of the integration. Then

$$D = \left\{ (x, y): 0 \leq y \leq 1, \frac{y}{2} \leq x \leq 1 \right\}$$



So the iterated integral $\int_0^1 \int_{\frac{y}{2}}^1 e^{x^2} dx dy$ can be written as

$$\int_0^1 \int_0^{2x} e^{x^2} dy dx = \int_0^1 \int_{\frac{y}{2}}^1 e^{x^2} dx dy .$$

25. (D)

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \frac{\left| \frac{3^{n+1}}{n+1} (x-2)^{n+1} \right|}{\left| \frac{3^n}{n} (x-2)^n \right|} \\ &= \lim_{n \rightarrow \infty} |x-2| 3 \left(\frac{n}{n+1} \right) \\ &= 3 |x-2| \end{aligned}$$

By the ratio test, the series converges for

$$\begin{aligned} 3|x-2| &< 1 \Leftrightarrow |x-2| < \frac{1}{3} \\ &\Leftrightarrow -\frac{1}{3} < x-2 < \frac{1}{3} \end{aligned}$$

$$\Leftrightarrow 2 - \frac{1}{3} < x < 2 + \frac{1}{3}$$

$$\Leftrightarrow \frac{5}{3} < x < \frac{7}{3}.$$

When $x = \frac{5}{3}$, $\sum_{n=1}^{\infty} \frac{3^n}{n} \left(\frac{5}{3} - 2\right)^n = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

which converges by the alternating series test. When

$$x = \frac{7}{3}, \sum_{n=1}^{\infty} \frac{3^n}{n} \left(\frac{7}{3} - 2\right)^n = \sum_{n=1}^{\infty} \frac{1}{n}$$

which is a divergent harmonic series. So the series

$$\sum_{n=1}^{\infty} \frac{3^n}{n} (x - 2)^n$$

converges for x in the interval $\left[\frac{5}{3}, \frac{7}{3}\right)$.

26. (E)

The equation of the tangent plane is

$$\nabla f(-3, 2, -6) \cdot (x + 3, y - 2, z + 6) = 0$$

where ∇f is the gradient of f and $f(x, y, z) = xz - yz^3 - yz^2 - 378$. Now differentiating f with respect to x , y and z respectively, we get

$$\frac{\partial f}{\partial x} = z$$

$$\frac{\partial f}{\partial y} = -z^3 - z^2$$

$$\frac{\partial f}{\partial z} = -3yz^2 - 2yz + x$$

Evaluating these derivatives at $(-3, 2, -6)$ one sees that $\nabla f(-3, 2, -6) = (-6, 180, -195)$. Therefore, the equation of the tangent plane is $(-6, 180, -195) \cdot (x + 3, y - 2, z + 6) = 0$ that is, $-6x - 18 + 180y - 360 - 195z - 1170 = 0$ or $-6x + 180y - 195z - 1548 = 0$. Dividing throughout by -3 , we have $2x - 60y + 65z + 516 = 0$.

27. (E)

Notice that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} + \sum_{n=1}^{\infty} \frac{1}{(2n)^2}$$

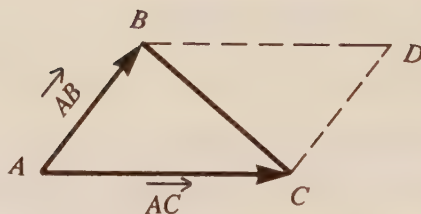
("odd terms" + "even terms"). So

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} &= \sum_{n=1}^{\infty} \frac{1}{n^2} - \frac{1}{4} \sum_{n=1}^{\infty} \frac{1}{n^2} \\ &= \frac{3}{4} \sum_{n=1}^{\infty} \frac{1}{n^2} \end{aligned}$$

Therefore,

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{3}{4} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{8}$$

28. (D)



The area of triangle $\triangle ABC$ is $1/2$ of the area of the parallelogram $ABDC$, which is equal to $\frac{1}{2} \|\vec{AB} \times \vec{AC}\|$, i.e., $1/2$ (the magnitude of the cross product of the vectors $\vec{AB}$ and $\vec{AC}$). Now

$$\begin{aligned} \vec{AB} &= B - A = (4, 0, 2) - (2, 1, 5) \\ &= (2, -1, -3) \end{aligned}$$

and

$$\begin{aligned} \vec{AC} &= C - A = (-1, 0, -1) - (2, 1, 5) \\ &= (-3, -1, -6) \end{aligned}$$

So

$$\begin{aligned}\vec{AB} \times \vec{AC} &= \det \begin{pmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & -3 \\ -3 & -1 & -6 \end{pmatrix} \\ &= 3\vec{i} - (-21)\vec{j} + (-5)\vec{k}\end{aligned}$$

Therefore,

$$\begin{aligned}\|\vec{AB} \times \vec{AC}\| &= \sqrt{3^2 + 21^2 + 5^2} \\ &= \sqrt{9 + 441 + 25} \\ &= \sqrt{475}.\end{aligned}$$

Hence the area of the triangle is $\frac{\sqrt{475}}{2}$.

29. (D)

The order of a permutation σ is the least positive integer n such that $\sigma^n = \text{identity}$. Now

$$\sigma^2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix}$$

Similarly

$$\sigma^3 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 1 & 5 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 4 & 1 & 3 \end{pmatrix}$$

and

$$\sigma^4 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 4 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix}.$$

Therefore the order of σ is 4.

30. (A)

If $\vec{u}$ and $\vec{v}$ are orthogonal vectors, then their dot product is zero, that is $\vec{ru} \cdot \vec{sv} = 0$ and therefore $\vec{ru} \cdot \vec{sv} = (rs) (\vec{u} \cdot \vec{v}) = 0$ for all real r and s .

31. (D)

$x \in A \cap B$ then $x \in A$ and $x \in B$.

32. (E)

First observe that R is an abelian group under the $*$ operation with identity element 0 and inverse of every $a \neq -1$ in R given by $-\frac{a}{1+a}$. Therefore

$$\begin{aligned} 7 &= 5 * x * 3 \\ &= x * 5 * 3 \quad \text{by using the abelian property} \\ &= x * (5 + 3 + 15) \quad \text{by using the definition of } * . \\ &= x * (23) . \end{aligned}$$

Thus

$$\begin{aligned} 7 * \left(-\frac{23}{24}\right) &= x * 23 * \left(-\frac{23}{24}\right) \\ &= x * 0 \quad \text{as } 23 * \left(-\frac{23}{24}\right) = 23 - \frac{23}{24} - \frac{23 \cdot 23}{24} = 0 \\ &= x . \end{aligned}$$

Hence
$$\begin{aligned} x &= 7 * \left(-\frac{23}{24}\right) = 7 - \frac{23}{24} - 7 \cdot \frac{23}{24} \\ &= \frac{168 - 23 - 161}{24} \end{aligned}$$

by simplifying
$$= -\frac{16}{24} = -\frac{2}{3} .$$

33. (C)

$$\frac{1}{1-x} = 1 + x + x^2 + \dots + x^k \dots \quad \text{for } |x| < 1.$$

Differentiating both sides we get

$$\frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \dots + kx^{k-1} + (k+1)x^k + \dots \quad \text{for } |x| < 1.$$

Differentiating two more times we obtain

$$\frac{2}{(1-x)^3} = 2 + 3 \cdot 2x + \dots + (k+1)kx^{k-1} + (k+2)(k+1)x^k + \dots$$

and

$$\frac{6}{(1-x)^4} = 3 \cdot 2 + 4 \cdot 3 \cdot 2x + \dots + (k+3)(k+2)(k+1)x^k + \dots$$

or

$$\begin{aligned} \frac{1}{(1-x)^4} &= \frac{3 \cdot 2}{6} + \frac{4 \cdot 3 \cdot 2}{6}x + \dots \\ &\quad + \frac{(k+3)(k+2)(k+1)}{6}x^k + \dots \end{aligned}$$

by dividing by 6. Now

$$\begin{aligned} (1 + 2x + 3x^2 + \dots + (n+1)x^n + \dots)^2 \\ &= \left[\frac{1}{(1-x)^2} \right]^2 \\ &= \frac{1}{(1-x)^4} \\ &= \sum_{k=0}^{\infty} \frac{(k+3)(k+2)(k+1)}{6} x^k \end{aligned}$$

Therefore

$$b_k = \frac{(k+3)(k+2)(k+1)}{6}.$$

34. (B)

g is continuous since $g'(x)$ exists for all real x . By the mean value theorem, there exist x_1 and x_2 , $a < x_1 < b < x_2 < c$ such that

$$g'(x_1) = \frac{g(b) - g(a)}{b - a}$$

and

$$g'(x_2) = \frac{g(c) - g(b)}{c - b}.$$

But $g(a) = g(b) = g(c) = 0$, therefore $g'(x_1) = 0 = g'(x_2)$ and hence the minimum possible zeros for $g'(x)$ is 2.

35. (A)

Notice that

$$\text{if } 0 \leq x \leq \frac{\pi}{4}, \text{ then } \sin x - \cos x \leq 0,$$

$$\text{if } \frac{\pi}{4} \leq x \leq \frac{\pi}{2}, \text{ then } \sin x - \cos x \geq 0,$$

$$\begin{aligned} \int_0^{\frac{\pi}{2}} |\sin x - \cos x| dx &= \int_0^{\frac{\pi}{4}} (\cos x - \sin x) dx + \\ &\quad \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (\sin x - \cos x) dx \\ &= 2\sqrt{2} - 2. \end{aligned}$$

36. (E)

- I. The set of integers under subtraction is not a group because $a - (b - c) \neq (a - b) - c$, that is associativity fails.
- II. The set of non-zero real numbers under division is not a group.
- III. The set of even integers under addition is a group.
- IV. The set of all integer multiples of 13 is group under addition.

37. (D)

$$(1 + \sqrt{-1})^{8n} - (1 - \sqrt{-1})^{8n} = (1 + i)^{8n} - (1 - i)^{8n}$$

i is an imaginary number

$$\begin{aligned} &= (\sqrt{2}e^{i(\frac{\pi}{4})})^{8n} - (\sqrt{2}e^{i(\frac{-\pi}{4})})^{8n} \\ &= 2^{4n}e^{i2\pi n} - 2^{4n}e^{i(-2\pi n)} \\ &= 2^{4n}(1 - 1) \\ &= 2^{4n}(0) \\ &= 0 \end{aligned}$$

38. (C)

$$\begin{aligned} \vec{u} \times \vec{v} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -3 & 5 \\ -1 & 4 & 2 \end{vmatrix} = -\vec{6i} + \vec{8k} - \vec{5j} - \vec{3k} - \vec{20i} - \vec{4j} \\ &= -\vec{26i} - \vec{9j} + \vec{5k} \end{aligned}$$

39. (E)

The system of equations

$$px + y = 1$$

$$x + py = 2$$

$$y + pz = 3$$

may have no solution if for some values of p , the determinant of the coefficient matrix is zero. Thus

$$\begin{aligned} \det \begin{pmatrix} p & 1 & 0 \\ 1 & p & 0 \\ 0 & 1 & p \end{pmatrix} &= p(p^2 - 0) - 1(p - 0) + 0(1 - 0) \\ &= p(p^2 - 1) \\ &= p(p + 1)(p - 1) \end{aligned}$$

which is zero when $p = 0, 1$ or -1 .

40. (A)

The determinant of the given matrix is always equal to x for any value of x , because the determinant of a lower triangular matrix is the product of its diagonal elements, which in this case is

$$1 \cdot 2 \cdot 3 \cdot 4 \cdot x = 24x.$$

Since the determinant is given equal to zero, then $24x = 0$, or $x = 0$.

41. (E)

By partial fractions decomposition

$$f(x) = \frac{1}{1-x^2} = \frac{1}{2} \left[\frac{1}{1-x} + \frac{1}{1+x} \right].$$

The n th derivative of $\frac{1}{1-x}$ is $\frac{n!}{(1-x)^{n+1}}$ and the n th derivative

of $\frac{1}{1+x}$ is $\frac{(-1)^n n!}{(1+x)^{n+1}}$. Therefore,

$$f^{(n)}(x) = \frac{n!}{2} \left[\frac{1}{(1-x)^{n+1}} + \frac{(-1)^n}{(1+x)^{n+1}} \right]$$

42. (C)

The number of different partial derivatives of order k is equal to the number of distinct nonnegative integer-valued vectors $(\infty_1, \infty_2, \dots, \infty_n)$ satisfying $\infty_1 + \infty_2 + \dots + \infty_n = k$. There are

$$\binom{k+n-1}{k}$$

such vectors. Thus the number of different partial derivatives of order k is

$$\binom{k+n-1}{k}.$$

43. (A)

$$\begin{aligned} T(4e^{2x} + 7e^{3x} - 5) &= 4T(e^{2x}) + 7T(e^{3x}) - 5T(1) \text{ by linearity} \\ &= 4 \sin x + 7 \cos 4x - 5e^{5x}. \end{aligned}$$

44. (C)

First school has $\binom{8}{2}$ choices, second school has $\binom{6}{2}$ choices, third school has $\binom{4}{2}$ choices and last school has to accept the remaining 2. Therefore 8 teachers can be divided into 4 schools in

$$\binom{8}{2} \binom{6}{2} \binom{4}{2} \binom{2}{2} = 2520$$

ways if each school must receive 2 teachers.

45. (E)

The Cauchy integral formula states that

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz$$

where z_0 is an interior point of C . In the case at hand $f(z) = 2z^2 - z - 2$ which is analytic and $z_0 = 2$ which lies inside the circle $|z| = 3$. Observe that $f(2) = 4$. Therefore,

$$g(2) = \int_C \frac{f(z)}{z - 2} dz = 2\pi i f(2) = 8\pi i.$$

46. (A)

Factoring out -2 in $-2 + 1 - \frac{2}{3} + \frac{2}{4} - \frac{2}{5} + \frac{2}{6} \dots$ one gets

$$\begin{aligned}
& -2 + 1 - \frac{2}{3} + \frac{2}{4} - \frac{2}{5} + \frac{2}{6} - \dots \\
& = -2 \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots \right) \\
& = -2 \ln 2.
\end{aligned}$$

47. (D)

The dimension of the eigenspace corresponding to the eigenvalue $\lambda = 1$ is the number of linearly independent solutions of the homogeneous equations $(A - I) \underline{\bar{X}} = 0$. This is equal to

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ -3 & -2 & 0 & 0 \\ -4 & -3 & -2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

That is, using matrix multiplication this reduces to

$$\begin{aligned}
0 &= 0 \\
-2x &= 0 \\
-3x - 2y &= 0 \\
-4x - 3y - 2z + w &= 0.
\end{aligned}$$

Now $-2x = 0$ implies $x = 0$ and $-3x - 2y = 0$ and $x = 0$ implies $y = 0$. Substituting $x = 0$ and $y = 0$, in the fourth equation we have $w = 2z$. Thus there is only one solution

$$\begin{pmatrix} 0 \\ 0 \\ 2 \\ 1 \end{pmatrix}.$$

Therefore the dimension of the eigenspace is 1.

48. (D)

Since T is onto, the dimension of the range space of T is equal to the dimension of W , which is 7. By the rank-nullity theorem we have $\dim V = \dim$ of the null space + $\dim$ of range space. Therefore the dimension of the null space of T is equal to $11 - 7 = 4$.

49. (A)

$$\begin{aligned}\vec{u} \cdot \vec{v} &= (x, y) \cdot (x', y') \\ &= xx' + yy' .\end{aligned}$$

50. (B)

Differentiating

$$F(x) = \begin{cases} \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt , & x > 0 \\ 0 & , \quad x \leq 0 \end{cases}$$

with respect to x we get the density function

$$f(x) = F'(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}$$

for $x > 0$ and 0 for $x < 0$. The mean is

$$\begin{aligned}E(\bar{X}) &= \int_{-\infty}^{\infty} x f(x) dx \\ &= \frac{1}{\sqrt{\pi}} \int_0^{\infty} 2x e^{-x^2} dx \\ \int_0^{\infty} 2x e^{-x^2} dx &= \lim_{b \rightarrow \infty} \int_0^b 2x e^{-x^2} dx \\ &= \lim_{b \rightarrow \infty} 1 - e^{-b^2} = 1\end{aligned}$$

Therefore the mean is $\frac{1}{\sqrt{\pi}}$.

51. (D)

The characteristic polynomial of A is given by

$$\begin{aligned} P_A(x) &= \det(A - xI) = \det \begin{bmatrix} 1-x & 1 \\ 1 & 2-x \end{bmatrix} \\ &= (1-x)(2-x) - 1 \\ &= 2 - 3x + x^2 - 1 \\ &= x^2 - 3x + 1. \end{aligned}$$

By the Cayley-Hamilton theorem $P_A(A) = 0$; that is $A^2 - 3A + I = 0$.

52. (B)

When $t = 2$, $x = 2^3 - 4 = 4$, $y = 2 \cdot 2^2 + 1 = 9$. The slope of the tangent line is $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{4}{3t}.$$

So $\left. \frac{dy}{dx} \right|_{t=2} = \frac{2}{3}$, this equation is $y - 9 = \frac{2}{3}(x - 4)$

or $2x - 3y + 19 = 0$.

53. (C)

$$\begin{aligned} \int_0^1 x f''(2x) dx &= \frac{1}{2} \int_0^1 x d[f'(2x)] \\ &= \frac{1}{2} \left[x f'(2x) \Big|_0^1 - \int_0^1 f'(2x) dx \right] \\ &= \frac{1}{2} \left[5 - \frac{1}{2} f(2x) \Big|_0^1 \right] \\ &= \frac{1}{2} \left[5 - \frac{1}{2} f(2) + \frac{1}{2} f(0) \right] \\ &= 2 \end{aligned}$$

54. (D)

Let four consecutive integers be $n, n + 1, n + 2$, and $n + 3$, then

$$\begin{aligned}n(n + 1)(n + 2)(n + 3) + 1 &= n^4 + 6n^3 + 11n^2 + 6n + 1 \\&= n^4 + 2n^2(3n + 1) + (9n^2 + 6n + 1) \\&= n^4 + 2n^2(3n + 1) + (3n + 1)^2 \\&= (n^2 + 3n + 1)^2.\end{aligned}$$

55. (E)

The auxiliary equation is $r^2 + 5r + 6 = 0$ whose solutions are $r = -2$ and -3 . Thus the solution of the differential equation is $y(x) = c_1 e^{-2x} + c_2 e^{-3x}$. Using the initial conditions we get $c_1 + c_2 = 0$ and $-2c_1 - 3c_2 = 1$, whose solution is $c_1 = 1$ and $c_2 = -1$. Hence the solution of the differential equation satisfying the initial conditions is given by $y(x) = +e^{-2x} - e^{-3x}$.

56. (C)

$z + \sin z$ and ze^z are analytic functions.

57. (C)

We know that if p is a prime number, then $n^p \equiv n \pmod{p}$ for any positive integer n . i.e., $n^p - n \equiv 0 \pmod{p}$. In this problem, 7 is prime, so the answer is (C).

58. (A)

$$f = (1478) (265) (39) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 4 & 6 & 9 & 7 & 2 & 5 & 8 & 1 & 3 \end{pmatrix}$$

$$\begin{aligned} \therefore f^{-1} &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 8 & 5 & 9 & 1 & 6 & 2 & 4 & 7 & 3 \end{pmatrix} \\ &= (1874) (256) (39) . \end{aligned}$$

59. (B)

If f is a probability density function then

$$\int_{-\infty}^{\infty} f(x) dx = 1 .$$

So

$$c \int_0^{\infty} x e^{-x^2} dx = 1 . \text{ Now } \int_0^{\infty} x e^{-x^2} dx = \frac{1}{2} \int_0^{\infty} 2x e^{-x^2} dx .$$

Let $u = x^2$ then $du = 2x dx$ and

$$\int_0^{\infty} x e^{-x^2} dx = \frac{1}{2} \int_0^{+\infty} e^{-u} du = \frac{1}{2} .$$

Therefore $\frac{c}{2} = 1$ which implies $c = 2$.

60. (C)

$$\begin{aligned} &\neg [\forall x \exists y (P(x, y) \wedge \neg Q(x, y))] \\ &= \exists x \forall y (\neg P(x, y) \vee Q(x, y)) \\ &= \exists x \forall y (P(x, y) \rightarrow Q(x, y)) . \end{aligned}$$

61. (B)

The Lebesgue measure of a countable set is always zero.

62. (D)

The ratio test gives

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \left| \frac{(3-x)(6x-7)^{n+1}}{(3-x)(6x-7)^n} \right| \\ &= |6x-7|. \end{aligned}$$

So the series converges for $|6x-7| < 1$, that is $-1 < 6x-7 < 1$. Then dividing by 6 we have $1 < x < \frac{4}{3}$. Now we check the end points.

When $x = 1$ the series

$$\sum_{n=0}^{\infty} (3-1)(6-7)^n = \sum_{n=0}^{\infty} 2(-1)^n$$

diverges by the n th term test. When $x = \frac{4}{3}$, the series

$$\sum_{n=0}^{\infty} \left(3 - \frac{4}{3}\right) \left(6 \cdot \frac{4}{3} - 7\right)^n = \sum_{n=0}^{\infty} \frac{5}{3} (1)^n$$

which diverges again by the n th term test. Thus, the interval of convergence is $(1, \frac{4}{3})$.

63. (B)

Since the coefficient matrix A has non-zero determinant, the system has a unique solution.

64. (A)

$$f(x) = f_1(x) + f_2(x) \quad \text{where} \quad f_1(x) = \frac{1}{2} [f(x) + f(-x)]$$

$$\text{and } f_2(x) = \frac{1}{2} [f(x) - f(-x)] .$$

Thus for the given function $f(x) = x^4 + x^6 + \sin(x^2) + e^{x^3}$. We have

$$\begin{aligned} f_2(x) &= \frac{1}{2} [x^4 + x^6 + \sin(x^2) + e^{x^3} - x^4 - x^6 - \sin((-x)^2) - e^{-x^3}] \\ &= \frac{1}{2} (e^{x^3} - e^{-x^3}) . \end{aligned}$$

65. (C)

The sum of the geometric sequence $1 + w + w^2 + \dots + w^{n-1} = \frac{1 - w^n}{1 - w}$ and hence using the fact that w is n th root of unity, $w^n = 1$, we have $1 + w + w^2 + \dots + w^{n-1} = 0$. Therefore $w + w^2 + \dots + w^{n-1} = -1$.

66. (B)

$$\begin{aligned} P(A \cup B) &= 1 - P([A \cup B]^c) \\ &= 1 - 0.1 \\ &= 0.9 \end{aligned}$$

Now $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ and hence

$$\begin{aligned} P(A \cap B) &= P(A) + P(B) - P(A \cup B) \\ &= 0.7 + 0.5 - 0.9 \\ &= 0.3 . \end{aligned}$$

Therefore
$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = \frac{3}{5} .$$

The Graduate Record Examination in

MATHEMATICS

Practice Test 6

The Graduate Record Examination in **MATHEMATICS** Answer Sheet Test 6

- | | | |
|-------------------------|-------------------------|-------------------------|
| 1. (A) (B) (C) (D) (E) | 23. (A) (B) (C) (D) (E) | 45. (A) (B) (C) (D) (E) |
| 2. (A) (B) (C) (D) (E) | 24. (A) (B) (C) (D) (E) | 46. (A) (B) (C) (D) (E) |
| 3. (A) (B) (C) (D) (E) | 25. (A) (B) (C) (D) (E) | 47. (A) (B) (C) (D) (E) |
| 4. (A) (B) (C) (D) (E) | 26. (A) (B) (C) (D) (E) | 48. (A) (B) (C) (D) (E) |
| 5. (A) (B) (C) (D) (E) | 27. (A) (B) (C) (D) (E) | 49. (A) (B) (C) (D) (E) |
| 6. (A) (B) (C) (D) (E) | 28. (A) (B) (C) (D) (E) | 50. (A) (B) (C) (D) (E) |
| 7. (A) (B) (C) (D) (E) | 29. (A) (B) (C) (D) (E) | 51. (A) (B) (C) (D) (E) |
| 8. (A) (B) (C) (D) (E) | 30. (A) (B) (C) (D) (E) | 52. (A) (B) (C) (D) (E) |
| 9. (A) (B) (C) (D) (E) | 31. (A) (B) (C) (D) (E) | 53. (A) (B) (C) (D) (E) |
| 10. (A) (B) (C) (D) (E) | 32. (A) (B) (C) (D) (E) | 54. (A) (B) (C) (D) (E) |
| 11. (A) (B) (C) (D) (E) | 33. (A) (B) (C) (D) (E) | 55. (A) (B) (C) (D) (E) |
| 12. (A) (B) (C) (D) (E) | 34. (A) (B) (C) (D) (E) | 56. (A) (B) (C) (D) (E) |
| 13. (A) (B) (C) (D) (E) | 35. (A) (B) (C) (D) (E) | 57. (A) (B) (C) (D) (E) |
| 14. (A) (B) (C) (D) (E) | 36. (A) (B) (C) (D) (E) | 58. (A) (B) (C) (D) (E) |
| 15. (A) (B) (C) (D) (E) | 37. (A) (B) (C) (D) (E) | 59. (A) (B) (C) (D) (E) |
| 16. (A) (B) (C) (D) (E) | 38. (A) (B) (C) (D) (E) | 60. (A) (B) (C) (D) (E) |
| 17. (A) (B) (C) (D) (E) | 39. (A) (B) (C) (D) (E) | 61. (A) (B) (C) (D) (E) |
| 18. (A) (B) (C) (D) (E) | 40. (A) (B) (C) (D) (E) | 62. (A) (B) (C) (D) (E) |
| 19. (A) (B) (C) (D) (E) | 41. (A) (B) (C) (D) (E) | 63. (A) (B) (C) (D) (E) |
| 20. (A) (B) (C) (D) (E) | 42. (A) (B) (C) (D) (E) | 64. (A) (B) (C) (D) (E) |
| 21. (A) (B) (C) (D) (E) | 43. (A) (B) (C) (D) (E) | 65. (A) (B) (C) (D) (E) |
| 22. (A) (B) (C) (D) (E) | 44. (A) (B) (C) (D) (E) | 66. (A) (B) (C) (D) (E) |

GRE MATHEMATICS TEST VI

TIME: 2 hours and 50 minutes
66 Questions

DIRECTIONS: Choose the best answer for each question and mark the letter of your selection on the corresponding answer sheet.

1. Let $f(x) = \left(\frac{\sin x}{x}\right)^{(1/x^2)}$. Evaluate $\lim_{x \rightarrow 0^+} f(x)$.

(A) $-1/6$

(D) $\ln(1/6)$

(B) 0

(E) 1

(C) $e^{-1/6}$

2. Suppose $\vec{a}$ and $\vec{b}$ are vectors in R^2 and further, that $\vec{a}$ and $\vec{b}$ are linearly independent.

$$\text{Let } \vec{c} = \frac{\vec{a} \cdot \vec{b}}{(\|\vec{a}\|)^2} \vec{a} - \vec{b}.$$

(A) $\vec{a}$ and $\vec{c}$ are linearly dependent.

(B) $\vec{a}$ and $\vec{c}$ are orthogonal.

(C) $\vec{b}$ and $\vec{c}$ are linearly dependent.

(D) $\vec{b}$ and $\vec{c}$ are orthogonal.

(E) $\{\vec{a}, \vec{b}, \vec{c}\}$ form a linearly independent set.

3. Let $A = \bigcap_{n=0}^{\infty} A_n$, where A_n are the intervals given by
- $$\left(1 - \frac{1}{2^n}, 2 + \frac{1}{2^n}\right).$$

Then the set A is the interval

- (A) $(1, 2)$ (D) $(0, 3)$
(B) $(1/2, 3/2)$ (E) None of the above
(C) $[1, 2]$

4. The sum of the infinite series $\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}$ is

- (A) $\frac{1}{2}$ (D) $\frac{3}{4}$
(B) $\frac{1}{4}$ (E) 1
(C) $\frac{3}{2}$

5. Suppose $f(x) = \sum_{n=1}^{\infty} e^{-n \sin x} / n^2$. For what values of x does the series converge?

(A) $x > 0$

(B) $x \in (n\pi, (n+1)\pi), n = 0, \pm 1, \dots$

(C) $x \in [n\pi, (n+1)\pi], n = 0, \pm 1, \dots$

(D) $x \in (2n\pi, (2n+1)\pi), n = 0, \pm 1, \dots$

(E) $x \in [2n\pi, (2n+1)\pi], n = 0, \pm 1, \dots$

6. Let V_5 be the multiplicative group of invertible elements of Z_5 . Let $v \in V_5$. How many roots does the equation $v^4 = \bar{1}$ have in V_5 ?

(A) 1

(D) 4

(B) 2

(E) 5

(C) 3

7. Suppose $u(x, y)$ is harmonic in a domain D and $v(x, y)$ is an harmonic conjugate of u . Let $f(z) = u(x, y) + iv(x, y)$. Which statements are true?

I. $g(z) = v - iu$ is analytic in D .

II. $f'(z) = u_x + iv_y$.

III. $v(x, y) + x + y$ satisfies Laplace's equation in D .

(A) I only

(D) I and II only

(B) II only

(E) I and III only

(C) III only

8. Let $f(x) = x^x$ with $0 < x < 1$. Determine where the minimum value of f on the interval occurs.

(A) $1/2$

(D) $1/4$

(B) $1/e$

(E) 0

(C) $1/\pi$

9. Which curve in Figure 1 cannot be a mapping of the unit circle $|z| = 1$ under the transformation $w = (az + b)/(cz + d)$ where a , b , c and d are complex constants, and $ad - bc \neq 0$?

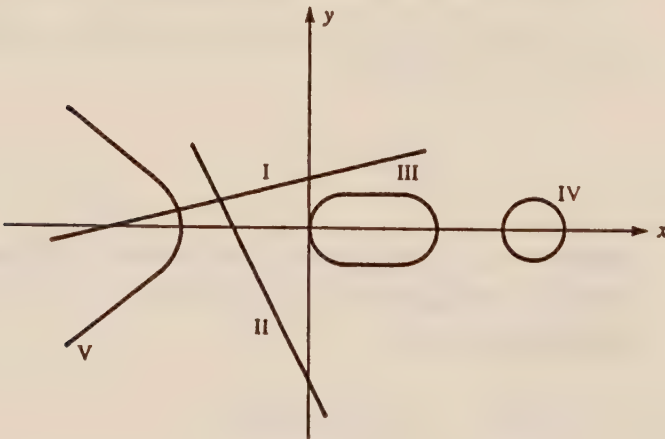


Figure 1

- (A) I only (D) I, II, III, and V
- (B) I and II only (E) IV only
- (C) I, II, and III only
10. Let $p(\lambda)$ be the characteristic polynomial for an $(n \times n)$ matrix A . Suppose that $\lambda = 0$ is a root of $p(\lambda)$. Which statements are always false?
- I. A has n linearly independent eigenvectors.
 II. $\det [p(A)] = 0$.
 III. A is invertible.
- (A) I only (D) I and II only
- (B) II only (E) I and III only
- (C) III
11. Determine the complex roots of the equation $e^{3z} + i = 0$.
- (A) $(i/3)(-\pi/2 + 2n\pi)$, n an integer.
 (B) $(i/3)(\pi/2 + 2n\pi)$, n an integer.
 (C) $(i/3)(\pi/2 + n\pi)$, n an integer.
 (D) $(i/2)(\pi/2 + 2n\pi)$, n an integer.
 (E) $(i/3)(\pi + 2n\pi)$, n an integer.

12. Determine the Laurent series for $f(z) = 1/(z - 2)$ which converges in the annulus $1 < |z - 3| < \infty$.

(A) $\sum_{n=0}^{\infty} (z - 3)^n$ (D) $\sum_{n=0}^{\infty} (-1)^n (z - 3)^{-n}$

(B) $\sum_{n=0}^{\infty} (z - 3)^{-n}$ (E) $\sum_{n=1}^{\infty} (-1)^n (z - 3)^{-n}$

(C) $\sum_{n=0}^{\infty} (-1)^n (z - 3)^{-n-1}$

13. Which of the following could be the equation for the curve in Figure 2?

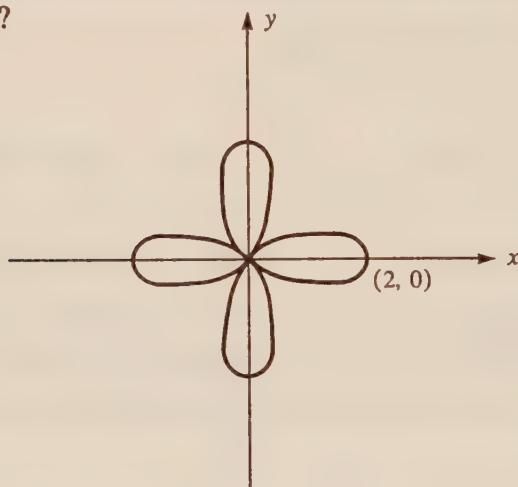


Figure 2:

(A) $r = 1 - 2 \sin 2\theta$ (D) $r = 2 \sin 4\theta$

(B) $r = 1 + 2 \sin 2\theta$ (E) $r = 2 \cos 4\theta$

(C) $r = 4 \cos 2\theta$

14. Suppose $\phi'(x)$ exists and $\phi' \neq 0$ on $[\alpha, \beta]$. Further, suppose there exists a $\gamma \in [\alpha, \beta]$ such that $\phi(\gamma) = 0$. Let γ_0 in $[\alpha, \beta]$ be chosen arbitrarily. Let γ_1 be the point at which the tangent line to $\phi(x)$ at $(\gamma_0, \phi(\gamma_0))$ crosses the x -axis. For each $n \geq 0$, let γ_n be the x -intercept of the line tangent to ϕ at $(\gamma_{n-1}, \phi(\gamma_{n-1}))$. Which formula describes this method for approximateing the root of $\phi(x)$?

(A) $\gamma_{n+1} = \gamma_n - \phi^n(\gamma_n)/\phi(\gamma_{n+1})$

(B) $\gamma_{n+1} = \gamma_n - \phi(\gamma_n)(\gamma_n - \gamma_{n-1})/(\phi(\gamma_n - \gamma_{n-1}))$

(C) $\gamma_{n+1} = \gamma_n + \phi(\gamma_n)/\phi'(\gamma_n)$

(D) $\gamma_{n+1} = \gamma_n - \phi'(\gamma_n)/\phi(\gamma_n)$

(E) $\gamma_{n+1} = \gamma_n - \phi(\gamma_n)/\phi'(\gamma_n)$

15. Let $P(S)$ denote the power set of any set S . Suppose $A = \{a, b, c\}$, $B = \{d, e\}$, and $C = P(A) \times B$. How many elements are in $P(C)$?

(A) 8

(D) 2^{16}

(B) 32

(E) 16

(C) 2^{10}

16. What is the indicial equation for the ordinary differential equation $x^2 y'' + 4xy' + 4y = 0$?

(A) $r^2 + 2r + 4$

(D) $r^2 + 3r + 4$

(B) $r^2 + 4r + 4$

(E) $r^2 + 4xr + 4$

(C) $r^2 + r + 1$

17. Which elements of the modular ring Z_9 are not invertible?

(A) $\bar{0}, \bar{3}, \bar{6}$

(D) $\bar{0}, \bar{2}$

(B) $\bar{0}, \bar{4}, \bar{8}$

(E) $\bar{0}, \bar{7}$

(C) $\bar{0}, \bar{3}$

18. A witness to a robbery told the police that the license number of the car contained three digits, the first of which was a 9, followed by three letters, the last of which was an A. The witness cannot remember the second and third digit, nor the first and second letter, but is positive that all the numbers were different and that the first two letters were the same, but different from the last letter. How many possible license plates match the description given by the witness?

(A) 1872

(D) 729

(B) 2600

(E) 1800

(C) 2106

19. A warm object is placed into a special refrigerator. The rate of change of the temperature of the object is proportional (with proportionality constant k) to the difference between its temperature and the ambient temperature in the refrigerator. Suppose the ambient temperature is 0 degrees, and further, that after one hour, the object's temperature is half its initial value. What is the value of k ?

(A) $\ln 2$ (D) 2
(B) $1/2$ (E) None of the above
(C) e^2

20. Let G be a group with m elements g satisfying $g^2 = e$ where e is the group identity. How many elements of order 2 are there in G ?

(A) $m + 1$ (D) 1
(B) m (E) None of the above
(C) $m - 1$

21. Which functions have unique fixed points on the stated intervals?

I. $f(x) = e^{-x}, x \in [1/3, 1]$
II. $g(x) = \pi + 1/2 \sin x, x \in [0, 2\pi]$
III. $h(x) = x^3 - 1, x \in [1, 2]$

- (A) I only (D) II and III
(B) I and II only (E) None of the above
(C) I, II, and III

22. Let P and Q be logical propositions. Which of the following forms are tautologies?

- I. $(P \vee Q) \wedge \neg (P \wedge Q)$
II. $(P \wedge \neg P) \Rightarrow Q$
III. $(P \vee Q) \Rightarrow \neg (Q \wedge P)$

- (A) I and II only (D) II
(B) I and III only (E) III
(C) II and III only

23. In R^3 , let $\alpha_1 = (1, 0, 1)$, $\alpha_2 = (0, 1, -2)$ and $\alpha_3 = (-1, -1, 0)$. If $f(\alpha)$ is a linear functional in R^3 such that $f(\alpha_1) = 1$, $f(\alpha_2) = -1$ and $f(\alpha_3) = 3$, and if $\alpha = (a, b, c)$, the $f(\alpha)$ can be expressed as

- (A) $-4a - 7b + 3c$ (D) $3a - 6b - 2c$
(B) $4a - 7b - 3c$ (E) $4a - 7b + 3c$
(C) $3a - 6b - 3c$

24. Let $U = \{0, 1, c\}$ be a ring with three elements (1 is the unity). Which statements are true?

I. $1 + 1 + 1 = 0$

II. $1 + 1 = c$

III. $c^2 = 1$

(A) I only

(D) II and III only

(B) II only

(E) I, II, and III

(C) I and II only

25. Which equation describes the plane passing through the points $(2, 0, 1)$, $(0, 6, -2)$, and $(-2, 3, 1)$?

(A) $3x + 4y - 6z = 12$

(D) $x + 2y - 3z = 7$

(B) $2x + 2y - 3z = 9$

(E) $y - z = 4$

(C) $x + 2y + z = 9$

26. Let $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 2 & 5 & 4 \end{pmatrix}$ be an element of the symmetric group S_5 . What is the order of β ?

(A) 2

(D) 5

(B) 3

(E) 6

(C) 4

27. If $A = \begin{pmatrix} 3 & 1 \\ -1 & -1 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, which matrix polynomial vanishes?

(A) $A^2 - 2A - 4I$

(D) $A^2 - 2A - 2I$

(B) $A^2 - 2A + 4I$

(E) $A^2 - A + 2I$

(C) $A^2 + A + 2I$

28. Let the binary operation $\circ$ be defined on all integers by $a \circ b = a + 2b + ab$. Which statements are false?

I. $\circ$ is associative.

II. $\circ$ is commutative.

III. For every a , there is an inverse a^{-1} such that $a \circ a^{-1} = 1$.

(A) I only

(D) I, II, and III

(B) II only

(E) None of the above

(C) I and II only

29. A random variable X has mean μ , variance σ^2 , and an unknown density function. Determine the constant c so that $P(|X - \mu| \geq c) \leq P_0$, where P_0 is a given constant probability.

(A) σ

(B) $\sigma/\sqrt{P_0}$

(C) $P_0\sigma$

(D) σ/P_0

(E) σ^2/P_0^2

30. A coin is biased so that a tail is twice as likely to occur as a head. What is the expected number of tails if the coin is tossed twice?

(A) $4/3$

(D) $1/2$

(B) $8/9$

(E) 2

(C) $4/9$

31. Which of the following matrices is orthogonal?

(A) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

(D) $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$

(B) $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

(E) $\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$

32. Evaluate $\lim_{x \rightarrow 0} \frac{\int_{-x}^0 \sin t^3 dt}{x^4}$.

(A) $-1/2$

(B) $1/2$

(C) $1/4$

(D) $-1/4$

(E) $\pi/2$

33. Which of the following subsets of $C[-1, 1]$ are not vector spaces?

(A) $\{f(x) \in C[-1, 1] : f(-1) = f(1)\}$

(B) $\{f(x) \in C[-1, 1] : f(x) = 0 \text{ if } x \in [-1/2, 1/2]\}$

(C) $\{f(x) \in C[-1, 1] : f(1) = 1\}$

(D) $\{f(x) \in C[-1, 1] : f(1) = 0\}$

(E) $\{f(x) \in C[-1, 1] : \int_{-1}^1 f(x) dx = 0\}$

34. A real, symmetric matrix is called positive definite if $x^T Ax > 0$ for all x in R^n . Which of the following is positive definite?

(A) $\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}$

(D) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

(B) $\begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$

(E) $\begin{pmatrix} 1 & -2 \\ -2 & 0 \end{pmatrix}$

(C) $\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$

35. Let A and B be $n \times n$ symmetric matrices. Which of the following is a necessary and sufficient condition for AB to be symmetric?

- (A) BA is skew-symmetric
- (B) A and B are nonsingular
- (C) $|AB| = |BA|$
- (D) A and B commute
- (E) B is Hermitian

36. Evaluate $\lim_{x \rightarrow \infty} \sqrt{x^2 + 2x} - x$.

- (A) 0
- (B) 1
- (C) $\sqrt{2}$
- (D) 2
- (E) ∞

37. Let $f(x) = \sin^{-1} x + \cos^{-1} x$, where $-\pi/2 \leq \sin^{-1} x \leq \pi/2$ and $0 \leq \cos^{-1} x \leq \pi$. Which statements are true?

- I. $f'(x) \leq 0$ for all x
- II. $f(1/2) = \pi/2$
- III. $f(x)$ is an even function

- (A) I only
- (B) II only
- (C) III only
- (D) II and III only
- (E) I, II, and III

38. Consider the following boundary value problem for $y(x)$:

$$d^2y/dx^2 + \lambda^2 y = 0, y(0) = 0, y'(1) = 0,$$

where $y' = dy/dx$ and l is a parameter. For what values of λ does this boundary value problem have nontrivial solutions?

- (A) $n\pi, n = 1, 2, \dots$
- (B) $\pm n\pi, n = 1, 2, \dots$
- (C) $\frac{2n+1}{2}\pi, n = 0, 1, 2, \dots$
- (D) $\pm \frac{2n+1}{2}\pi, n = 0, 1, 2, \dots$
- (E) $\frac{2n-1}{2}\pi, n = 0, 1, 2, \dots$

39. If $f(x)$ is everywhere differentiable on the closed interval $[a, b]$, then

- (A) $f'(x)$ is Riemann integrable
- (B) $f''(x)$ exists

- (C) $f'(x)$ is continuous
- (D) $f(x)$ may be unbounded
- (E) $f(x)$ is uniformly continuous on the interval
40. Which function is not periodic?
- (A) $\sin(e^{\sin x})$ (D) $\cos[\cos(e^x)]$
- (B) $\tan h[\cos(\sin x)]$ (E) $\ln[1/(2 + \cos x)]$
- (C) $\sqrt{\left|\tan\left(\frac{1}{2}\cos x\right)\right|}$
41. Let $a_n = \left(\frac{n+2}{n+1}\right)^{2n+3}$. Determine $\lim_{n \rightarrow \infty} a_n$.
- (A) 1 (D) e^3
- (B) e (E) 0
- (C) e^2
42. Determine the maximum value of $|A|$ where
- $$A = \begin{pmatrix} 1 & 1 & x \\ 1 & -1 & x^2 \\ -1 & 0 & 1 \end{pmatrix}$$

(A) $-7/4$

(D) $1/2$

(B) $-1/2$

(E) $3/4$

(C) 0

43. Consider the flow of heat in a thin, uniform bar of length L . The temperature distribution $u(x, t)$ in the bar obeys the partial differential equation

$$u_t = k u_{xx},$$

where k is a constant. Suppose the temperature of the bar at $x=0$ is fixed at 0 degrees, while the temperature at $x=L$ is fixed at 10 degrees. At time $t=0$, the temperature distribution is $u(x, 0) = f(x)$. What is the average temperature in the bar in the limit as $t \rightarrow \infty$?

(A) 0

(C) 5

(B) $\frac{1}{10} \int_0^L f(x) dx$

(D) 10

(E) $\sum_{n=1}^{\infty} A_n \cos(n\pi x/L) \exp(-kn^2\pi^2 t/L)$

44. The differential equation

$$dP/dt = P(1-P)(P-2),$$

possesses the three equilibrium solutions $P_1 = 0$, $P_2 = 1$, and $P_3 = 2$. Which of these are stable?

(A) P_1 and P_2 only

(D) P_2 only

(B) P_2 and P_3 only

(E) P_3 only

(C) P_1 only

45. Let $f(z) = 1/(z^2 + 1)$. Evaluate $\oint_{\Gamma} f(z) dz$ where Γ is the curve shown in Figure 3.

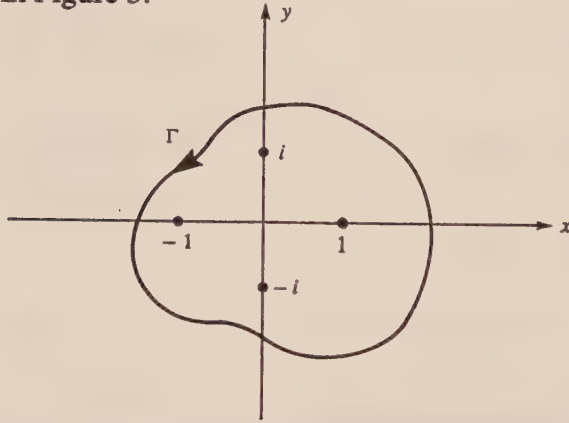


Figure 3:

(A) 0

(D) 2π

(B) π

(E) $2\pi i$

(C) πi

46. Determine an harmonic conjugate $v(x, y)$ for the harmonic function $u(x, y) = y + 3xy^2 - x^3$.

(A) $-y + 3xy^2 - x^3$

(B) $x + 3xy^2 - x^3$

(C) $y^3 - 3x^2y - x$

(D) $y^3 - 3x^2y + x$

(E) $y^3 - 3x^2y + x$

47. Let $A = \begin{pmatrix} 2 & 1 \\ -1 & x \end{pmatrix}$. For what values of x does A possess a repeated eigenvalue?

(A) $\{0, 4\}$

(D) $\{1, 3\}$

(B) $\{0, 3\}$

(E) $\{1, 2\}$

(C) $\{1, 4\}$

48. Consider the sequence $a_{n+1} = \sqrt{b + a_n}$ where $a_1 = 1$. What is the smallest positive value of b such that b is divisible by 7 and that the limit of the sequence is an integer?

(A) 7

(D) 84

(B) 42

(E) 98

(C) 49

49. What is the volume of the body of revolution formed by rotating the curve $y = e^{-x}$, $0 \leq x \leq \infty$ about the x -axis?

(A) $\pi/2$

(B) $1/2$

(C) π

(D) 1

(E) 2π

50. Suppose $\int_0^{\infty} f(x) dx$ exists. Which statements are false?

I. $\lim_{x \rightarrow \infty} f(x) = 0$

II. $\int_0^{\infty} |f(x)| dx$ exists

III. $\int_0^{\infty} [f(x)]^2 dx$ exists

(A) I only

(D) II and III only

(B) II only

(E) I, II, and III

(C) I and II only

51. Suppose $f(z)$ is a nonconstant entire function. Which statement is always true?

(A) $\lim_{z \rightarrow \infty} f(z) = 0$

(B) $\lim_{z \rightarrow 0} f(z) = 0$

(C) $f'(z)$ may not be entire

(D) $\oint f(z) dz = 2\pi i$ for every simple closed curve in the complex plane

(E) None of the above

52. If $y = x$ is a solution of the differential equation

$$y'' + xy' - y = 0,$$

another, linearly-independent solution is given by

(A) x^2

(D) $x \int^x e^{-t^2} dt$

(B) $x \int^x \frac{e^{-t^2/2}}{t^2} dt$

(E) $x \int^x \frac{e^{-t^2/2}}{t^2} dt$

(C) $x \int^x e^{-t^2} / t^2 dt$

53. Which of the following could be the equation for the curve shown in Figure 4?

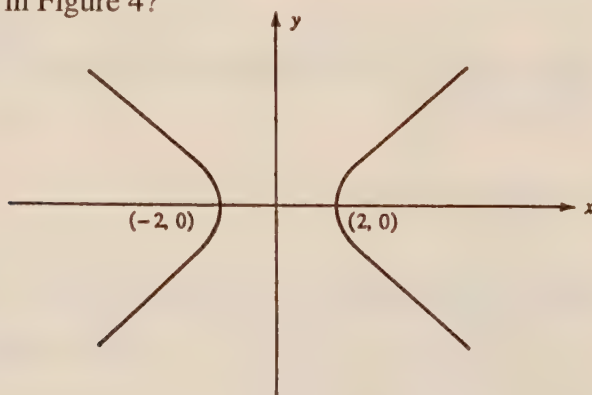


Figure 4

(A) $x^2/4 + y^2 = 1$

(D) $y^2 - x^2/2 = 1$

(B) $x^2/2 - y^2 = 1$

(E) $x^2/4 - y^2 = 1$

(C) $x^2/2 + y^2 = 1$

54. If $z = i^i$, then z can assume the real value

(A) -1

(D) $e^{2\pi}$

(B) 1

(E) $\pi/2$

(C) $e^{3\pi/2}$

55. Let $f(x)$ be a real-valued function defined on $[a, b]$. Which of the following conditions is not sufficient to ensure that

$\int_a^b f(x) dx$ exists?

(A) $f(x)$ is monotonic

(B) $f(x)$ is continuous

(C) $f(x)$ is bounded and piecewise smooth

(D) $f(x)$ has only a finite number of discontinuities

(E) None of the above

56. Let $f(x)$ be a function defined for all real x such that the coordinates of each point of its graph satisfy $|y| = x^2$. The total number of points at which $f(x)$ must be differentiable is

(A) none

(D) 4

(B) 1

(E) infinite

(C) 2

57. Let $f(x) = x^{1/\ln x}$, $x > 0$, and $x \neq 1$. If $f(x)$ is continuous at $x = 1$, then $f(1)$ is

(A) 0

(D) e

(B) 1

(E) 4

(C) 2

58. The series $\sum_{n=2}^{\infty} \frac{(x+2)^n}{n^3 3^{n+1}}$ has the interval of convergence

(A) $-3 < x < 3$

(D) $-5 \leq x \leq 1$

(B) $-3 \leq x \leq 3$

(E) $-5 \leq x \leq 3$

(C) $-5 \leq x < 1$

59. For the algorithm described below, what is the complete output

sequence if the input value is 13?

```
procedure test;  
variable x : real;  
begin{test}  
  input(x);  
  repeat  
    if x mod 2 = 0 then x := x/2  
      else x := 3 * x + 1;  
    output(x);  
  until x = 1;  
end.{test}
```

(A) 13, 40, 20, 10, 5, 16, 8, 4, 2, 1

(B) 13, 40, 20, 10, 5, 1

(C) 40, 20, 10, 5, 16, 8, 4, 2, 1

(D) 13, 39, 40, 20, 10, 5, 1

(E) 40, 20, 10, 5, 20, 10, 8, 4, 2, 1

60. Which of the following sets in R^2 are compact?

(A) $\{x, y \mid x \geq 0, y \geq 0\}$

(B) $\{x, y \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}$

(C) $\{x, y \mid |x - y| < 2\}$

(D) $\{x, y \mid x^2 + y^2 < 2\} \cap \{x, y \mid x^2 + y^2 > 1\}$

(E) $\{x, y \mid |x + y| < 1\}$

61. If $6x + 5y$ is a multiple of 13, what must b be so that $8x + by$ is also a multiple of 13?

(A) 0

(D) 9

(B) 5

(E) 11

(C) 7

62. Which statements about fields are always true?

I. Every field contains at least one finite subfield.

II. The transcendental real numbers form a field.

III. Every field is an integral domain.

(A) I only

(D) I and II

(B) II only

(E) I and III

(C) III only

63. Inside the registers of a microprocessor, two numbers are ANDed by forming the logical AND of their respective binary digits. What is the result of ANDing 27 and 52?

(A) 16

(D) 79

(B) 22

(E) 127

(C) 63

64. Which of the following could be a density function of a continuous random variable X ?

(A) $f(x) = \frac{1}{1+x^2}$

(B) $f(x) = \begin{cases} 0, & x < 0 \\ 2e^{-2x}, & x \geq 0 \end{cases}$

(C) $f(x) = \begin{cases} 0, & x < 1 \\ 1, & 1 \leq x \leq 2 \\ 0, & x > 2 \end{cases}$

(D) $f(x) = \begin{cases} 0, & x < 2 \\ 1/x^2, & x \geq 2 \end{cases}$

(E) $f(x) = \begin{cases} 0, & x \text{ rational} \\ 1, & x \text{ irrational} \end{cases}$

65. Which of the following is the contrapositive of the statement $P \wedge Q \Rightarrow \neg R \wedge S$?

(A) $\neg R \wedge S \Rightarrow P \wedge Q$

(B) $P \wedge Q \Rightarrow R \vee \neg S$

(C) $R \vee \neg S \Rightarrow \neg P \vee \neg Q$

(D) $\neg P \vee \neg Q \Rightarrow R \vee \neg S$

(E) $P \vee Q \Rightarrow \neg R \vee S$

66. Suppose $A = \begin{pmatrix} 5 & -2 \\ 6 & -2 \end{pmatrix}$. Which matrix below can be used to diagonalize A ?

(A) $\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$

(D) $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$

(B) $\begin{pmatrix} 6 & -2 \\ 5 & -2 \end{pmatrix}$

(E) $\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$

(C) $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$

GRE MATHEMATICS

TEST VI

ANSWER KEY

1.	C	23.	B	45.	A
2.	B	24.	E	46.	C
3.	C	25.	A	47.	A
4.	A	26.	E	48.	B
5.	E	27.	D	49.	A
6.	D	28.	D	50.	E
7.	E	29.	B	51.	E
8.	B	30.	A	52.	B
9.	D	31.	A	53.	E
10.	C	32.	C	54.	C
11.	A	33.	C	55.	D
12.	C	34.	A	56.	B
13.	E	35.	D	57.	D
14.	E	36.	B	58.	D
15.	D	37.	E	59.	C
16.	D	38.	D	60.	B
17.	A	39.	E	61.	E
18.	E	40.	D	62.	C
19.	A	41.	C	63.	A
20.	C	42.	A	64.	B
21.	C	43.	C	65.	C
22.	D	44.	D	66.	A

GRE MATHEMATICS TEST VI

DETAILED EXPLANATIONS OF ANSWERS

1. (C)

Note that the limit takes on the indeterminate form 1^∞ . Thus, we apply L'Hopital's rule several times to the logarithm of $f(x)$:

$$\begin{aligned}\ln f(x) &= \frac{\ln \frac{\sin x}{x}}{x^2} \sim \frac{x \cos x - \sin x}{2x^2 \sin x} \sim -\frac{\sin x}{4 \sin x + 2x \cos x} \\ &\sim -\frac{\cos x}{6 \cos x - 2x \sin x} \sim -\frac{1}{6}.\end{aligned}$$

Thus, $\ln f(x) \rightarrow -1/6$ or $f(x) \rightarrow -e^{-1/6}$.

2. (B)

A direct calculation shows that $\vec{a} \cdot \vec{c} = 0$, which implies that $\vec{a}$ and $\vec{c}$ are orthogonal. The other answers are all false.

3. (C)

Note that the intervals are nested, i.e., $A_{n+1} \subset A_n$. The limiting interval is therefore $(1, 2)$. However, the endpoints of this interval are in every A_n . Consequently, the intersection of the nested intervals is $[1, 2]$.

4. (A)

By partial fraction decomposition,

$$\frac{1}{4n^2 - 1} = \frac{1}{2} \left(\frac{1}{2n - 1} - \frac{1}{2n + 1} \right).$$

The partial sums of the series form a telescoping sum, so that

$$\begin{aligned} \sum_{n=1}^m \frac{1}{4n^2 - 1} &= \frac{1}{2} \left[(-1 + 1) + \left(1 - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) \right. \\ &\quad \left. + \dots + \left(\frac{1}{2m - 1} - \frac{1}{2m + 1}\right) \right] \\ &= \frac{1}{2} \left(1 - \frac{1}{2m + 1}\right), \end{aligned}$$

and the partial sums $\rightarrow \frac{1}{2}$ as $m \rightarrow \infty$.

5. (E)

Applying the ratio test, it can be shown that

$$\left| \frac{a_{n+1}}{a_n} \right| \rightarrow e^{-\sin x}.$$

Convergence therefore follows when $\sin x > 0$. This holds whenever $2n\pi < x < (2n+1)\pi$ for any integer n . At the endpoints of these intervals, $\sin x = 0$, and each term of the series is $1/n^2$. In that case, the series also converges. Thus, the closed intervals given by (E) represent the allowable values for x .

6. (D)

Note that $V_5 = \{1, 2, 3, 4\}$. Since $1^4 = 1$, $2^4 = 16 = \overline{1}$, $4^2 = 16 = \overline{1}$, and $3^2 = 9 = \overline{4}$, it follows that every element of V_5 is a solution to the given equation.

7. (E)

Note first that $-if(z) = g(z)$. Since $f(z)$ is analytic, so is $g(z)$; thus I is true. However, $f'(z) = u_x + iv_x$, and so II is false. Finally, since $v(x, y)$ is harmonic, it satisfies Laplace's equation. The second partials of $(x + y)$ vanish, and so III is true.

8. (B)

Taking logarithmic derivatives, we find that $f'(x) = x^x(\ln x + 1)$. This vanishes when $x = 1/e$. Since $f(x) \rightarrow 1$ as $x \rightarrow 0$ or 1 , the minimum of $f(x)$ occurs at $1/e$.

9. (D)

The indicated mapping is a linear fractional transformation which always maps circles (and straight lines) into circles and straight lines. The ellipse (III) and hyperbola (V) cannot be images of a circle under a linear fractional transformation.

10. (C)

Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. This matrix has eigenvalues $\{0, 1\}$ and eigenvectors $(1, 0)^T$ and $(0, 1)^T$. These eigenvectors are clearly linearly independent. Thus, I is *sometimes* true. By the Cayley-Hamilton theorem, every square matrix A satisfies its characteristic polynomial. Thus $p(A)$ is the zero matrix and II is *always* true. Finally, if $\lambda = 0$ is an eigenvalue of A , the $\det(A - \lambda I) = \det(A) = 0$, which implies that A is singular and therefore not invertible. III is *never* true.

11. (A)

Rewrite the equation as

$$e^{3z} = -i = e^{i(-\pi/2 + 2n\pi)}, n \text{ an integer,}$$

which implies that

$$z = (i/3)(\pi/2 + 2n\pi).$$

12. (C)

Observe that

$$\frac{1}{z-2} = \frac{1}{1+z-3} = \frac{1}{z-3} \frac{1}{1 - \frac{-1}{z-3}},$$

and expanding this last term in a geometric series yields the Laurent series given by (C).

13. (E)

The four petals in the figure indicate that the equation must contain an argument proportional to 4θ . Of the two possibilities, only (E) intersects the point $(2, 0)$, which is on the x -axis.

14. (E)

The algorithm described in Newton's method whose correct formula is given by (E).

15. (D)

$P(A)$ contains $2^3 = 8$ elements and B contains 2 elements, so that

C contains 16 elements. $P(C)$ therefore contains 2^{16} elements.

16. (D)

The differential equation is of the Cauchy-Euler type. The indicial equation is found by making the substitution $y = x^r$ which yields

$$r(r-1)x^2x^{r-2} + 4rx x^{r-1} + 4x^r = 0.$$

Then we cancel out the common factor x^r from each term.

17. (A)

Invertible elements are relatively prime to the modulus. Since $3^2 = 9 \equiv 0 \pmod{9}$ and $6^2 = 36 \equiv 0 \pmod{9}$, the correct answer is (A).

18. (E)

There are 9 possibilities for the second digit and 8 possibilities for the third digit. In addition, there are 25 possibilities for the double set of letters. Thus, there are $9 \cdot 8 \cdot 25 = 1800$ possible license plates which match the stated description.

19. (A)

Let T be the temperature and t be time. From the problem statement, the rate of change of temperature, which is the derivative dT/dt , is proportional to the difference between the object's temperature T and the ambient temperature which is zero. Thus, $dT/dt = -kT$, where $k > 0$ (the object must cool; thus the negative sign). The solution to this differential equation is $T = T_0 e^{-kt}$ where T_0 is the initial temperature. When $t = 1$, $T = T_0/2$, or $T_0/2 = T_0 e^{-k}$. Solving for k , we obtain $k = \ln 2$.

20. (C)

An element g is of order 2 if $g^2 = e$ but $g \neq e$. Since e is one of the m elements satisfying the indicated relation, there must be exactly $m - 1$ elements of order 2 in the group.

21. (C)

The function $f(x)$ has a unique fixed point when it maps $[a, b]$ into $[a, b]$ and $|f'(x)| \leq k \leq 1$. The functions in I and II satisfy these conditions. Though III is not a contraction mapping, it does have a fixed point.

22. (D)

A tautology is a logical form which is always true, regardless of the truth value of its components. In II, $(P \wedge \neg P)$ is always false. Theorems with a false hypothesis are always true, independent of the truth value of the conclusion. On the other hand, in I, if P and Q are both false, the statement is false, while if P and Q are both true, the statement is true. Thus I cannot be a tautology. Similarly, in III, if P and Q are both false, the statement is true, while if P and Q are both true, the statement is false. Thus, II is the only tautology.

23. (B)

A linear functional in R^3 must have the form $f(\alpha) = ax + by + cz$ where $\alpha = (a, b, c)$ and x, y , and z are real constants. Using the information given in the problem statement, a system of three equations for x, y , and z are obtained, with solution $x = 4, y = -7$, and $z = -3$. Thus, the correct form of the functional is given by (B).

24. (E)

The three elements of U form an abelian group under addition. The only possible value for $1 + 1$ is c , so II holds. By the same token, $c + 1 = 0$ and, using II, I is also true. Finally, by constructing the multiplicative table for U , $c^2 = c \cdot c = 1$ must also hold. Thus, all three statements are true.

25. (A)

The cross product of any two (non-parallel) vectors in the plane must be normal to the plane. Two such vectors are those joining the given points, namely $\vec{v} = (-2, 6, -3)$ and $\vec{w} = (-4, 3, 0)$. Thus $\vec{v} \times \vec{w} = (9, 12, 18)$. If $P_0 = (x_0, y_0, z_0)$ is a point in the plane, then the equation of the plane is given by

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0,$$

where (A, B, C) is normal to the plane. Using our computed results for the normal, and any of the three given points for P_0 we obtain (A) as the equation for the plane.

26. (E)

Since $1 \mapsto 3 \mapsto 2 \mapsto 1$ and $5 \mapsto 4 \mapsto 5$, it is straightforward to show that $\beta^7 = \beta$. Thus β^6 is the identity, and so the order of β is six.

27. (D)

The characteristic polynomial of A is found from the equation $|A - \lambda I| = 0$:

$$\begin{vmatrix} 3 - \lambda & 1 \\ -1 & -1 - \lambda \end{vmatrix} = 0.$$

This determinant evaluates to the polynomial $\lambda^2 - 2\lambda - 2 = 0$. This

polynomial also annihilates A , i.e., $A^2 - 2A - 2I = 0$.

28. (D)

First, note that $(a \circ b) - (b \circ a) = b - a$, so the operation is not commutative. Further, $[a \circ (b \circ c)] - [(a \circ b) \circ c] = 2c - ac$. Thus the operator is not associative. Finally, it is easy to show that inverses in general do not exist. Let $a = 2$. Then $a^{-1} = -1/4$ satisfies the equation, but is not an integer. All three statements are false.

29. (B)

First, note that

$$P(|x - \mu| \geq c) = 1 - P(-c + \mu \leq x \leq c + \mu).$$

By Chebyshev's theorem, $P(\mu - k\sigma < x < \mu + k\sigma) \geq 1 - 1/k^2$. Letting $k = \sigma/c$, we find that

$$1 - P(-c + \mu \leq x \leq c + \mu) \leq \sigma^2/c^2.$$

Setting the right half of this inequality equal to P_0 yields $c = \sigma/\sqrt{P_0}$.

30. (A)

The probability of a head on a single toss is $1/3$ and of a tail is $2/3$. For two tosses, the outcomes are $\{HH, TH, HT, TT\}$. If X is the number of tails, then $P(X=0) = P(HH) = 1/9$, $P(X=1) = P(HT) + P(TH) = 4/9$, and $P(X=2) = P(TT) = 4/9$. The probability distribution is then

$$f(x) = \begin{cases} 1/9, & \text{if } x = 0; \\ 4/9, & \text{if } x = 1; \\ 4/9, & \text{if } x = 2, \end{cases}$$

and

$$E(X) = \sum_{i=1}^3 x_i f(x_i) = 4/3.$$

31. (A)

A matrix is orthogonal if $AA^T = I$. We can verify that

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

so that (A) is the correct choice.

32. (C)

Using L'Hopital's rule, we find that

$$\frac{\int_{-x}^0 \sin t^3 dt}{x^4} \rightarrow -\frac{\int_0^{-x} \sin t^3 dt}{x^4} \rightarrow \frac{\sin x^3}{4x^3} \rightarrow 1/4,$$

since $\sin x/x \rightarrow 1$.

33. (C)

Every vector space possesses a zero element. In the case of $C[-1, 1]$, the zero element is simply the function $f(x) = 0$. Since every subset of $C[-1, 1]$ that is a vector space must also contain this function, it is clear that the set in (C), which excludes the zero element, cannot be a vector space.

34. (A)

Let $x^T = (x_1, x_2)$. For the matrix in (A),

$$x^T A x = 4x_1^2 + 4x_1x_2 + x_2^2 = (2x_1 + x_2)^2 \geq 0,$$

with equality only if x_1 and x_2 are both zero.

35. (D)

Since A and B are symmetric, we have that $(AB)^T = B^T A^T = BA$. Thus AB is symmetric if and only if $AB = BA$, i.e., that A and B commute.

36. (B)

Note that

$$\sqrt{x^2 + 2x} - x \cdot \frac{\sqrt{x^2 + 2x} + x}{\sqrt{x^2 + 2x} + x} = \frac{2}{\sqrt{1 + 2/x} + 1},$$

and the last quantity evidently tends to 1 as $x \rightarrow \infty$.

37. (E)

Using the sum-of-angles identity, note that

$$\begin{aligned} \sin[f(x)] &= x^2 + \cos(\sin^{-1}x)\sin(\cos^{-1}x) \\ &= x^2 + (\sqrt{1-x^2})^2 = 1. \end{aligned}$$

Thus, $f(x) = \pi/2$ for every value of x . I, II, and III are all true for constant functions.

38. (D)

The general solution to the differential equation is $y(x) = c_1 \sin |\lambda x| + c_2 \cos |\lambda x|$ where c_1 and c_2 are arbitrary constants. Now, $y(0) = 0$ implies that $c_2 = 0$. In addition, $y'(1) = 0$ implies that $\cos |\lambda| = 0$. Thus, $|\lambda| = \pi/2 + n\pi$ for every integer n , or

$$\lambda = \pm \frac{2n+1}{2} \pi, \quad n = 0, 1, 2, \dots$$

39. (E)

If $f(x)$ is everywhere differentiable, it is continuous, and since the interval is closed and bounded, $f(x)$ is uniformly continuous. Note that the existence of $f'(x)$ implies nothing about the continuity of the derivative nor the existence of the second derivative. In fact, $f'(x)$ may exist, but not be Riemann-integrable.

40. (D)

A function is periodic if $f(x+T) = f(x)$ for every x and some $T > 0$. For all the indicated functions, the innermost (first evaluated) function in the compositions is periodic, except for (D), which is not periodic. For example $\lim_{x \rightarrow -\infty} \cos[\cos(e^x)] = \cos(1)$, a constant, which cannot happen for a nonconstant periodic function.

41. (C)

First, note that

$$\left(\frac{n+2}{n+1}\right)^{2n+3} = \left(1 + \frac{1}{n+1}\right)^{2(n+1)+1}$$

Since $[1 + 1/(n+1)]^{n+1} \rightarrow e$, the right term above converges to e^2 .

42. (A)

Evaluating the determinant yields the polynomial $|A| = -x^2 - x - 2$. By setting the first derivative to zero, we find that an extreme value is achieved at $x = -1/2$. This must be a maximum since the second derivative is a negative constant. Finally, note that for this value of x , $|A| = -5/4$.

43. (C)

As $t \rightarrow \infty$ the temperature approaches a steady state because there are no sources of heat and the boundary conditions are time-independent. Under this condition, the governing equation reduces to $u_{xx} = 0$, which is in fact an ordinary differential equation. The solution to the heat equation in this limit is easily seen to be $u = 10x/L$, obtained by integrating the ODE and applying the boundary conditions. The average value of u is then found from the mean value theorem for integrals to be 5.

44. (D)

First, let $P = \hat{P}_i$, where $i = 1, 2, 3$. Now, substituting this change on dependent variable into the differential equation, and linearizing gives the result that $\hat{P}_1$ and $\hat{P}_3$ become unbounded, but the $\hat{P}_2$ decays to zero. Thus, P_2 is the only stable equilibrium point.

45. (A)

By the residue theorem, $\oint_{\Gamma} f(z) dz = 2\pi i \sum \text{Res}[f(z)]$. Evaluating the residues at each singularity of $f(z)$, we find that $\text{Res}[f(z)]_{z=i} = -i/2$ and $\text{Res}[f(z)]_{z=-i} = i/2$. Since the sum of the residues vanishes, the integral evaluates to zero.

46. (C)

By the Cauchy Riemann equations, $u_x = v_y$ and $u_y = -v_x$. Thus, $v_y = 3y^2 - 3x^2$ so that $v(x, y) = y^3 - 3x^2y + f(x)$. From this, $v_x = -6xy + f'(x)$ which must equal $-u_y$. This last condition implies that $f'(x) = -x$. Consequently, $v(x, y) = y^3 - 3x^2y - x$.

47. (A)

Evaluating the expression $|A - \lambda I| = 0$ yields the following polynomial for λ :

$$\lambda^2 - (2 + x)\lambda + 2x + 1 = 0,$$

which has the solution

$$\lambda = \frac{2 + x \pm \sqrt{(2 + x)^2 - 4(2x + 1)}}{2}.$$

There will be a double eigenvalue when the discriminant is zero, that is when

$$(2 + x)^2 - 4(2x + 1) = 0.$$

The roots to this last equation are at $x = 0$ and $x = 4$.

48. (B)

Since the sequence converges, let $a = \lim_{n \rightarrow \infty} a_n$. Then $a = \sqrt{b + a}$. Because the sequence is monotone increasing, a must be positive. Solving for a , we find that

$$a = \frac{1 + \sqrt{1 + 4b}}{2}.$$

For a to be an integer, $1 + 4b$ must be an odd perfect square. Examining

the odd perfect squares from 9 upward, the first value of b to occur which is a multiple of 7 is 42.

49. (A)

Using the disk method, the infinitesimal volume dV of a disk centered at x is $dV = \pi[r(x)]^2 dx$, where $r(x) = e^{-x}$. Thus,

$$V = \int_0^{\infty} dV = \int_0^{\infty} \pi e^{-2x} dx = \pi/2 .$$

50. (E)

Counterexamples exist for all three statements. For example, the

Fresnel integral $\int_0^{\infty} \sin(x^2) dx$, converges, but $\sin(x^2)$ does not converge at all as $x \rightarrow \infty$. Thus, I may be false. By the same token, the integral just mentioned is not absolutely integrable, so II is also false. In fact, the Fresnel integral is not a square integrable either, and III is false as well.

51. (E)

Since $f(z)$ is entire, it is analytic everywhere in the complex plane. Thus $\bar{f}(z)$ is also entire. In addition, $f(z)$ has a singularity at infinity, so that no limit exists as $z \rightarrow \infty$. Also, every contour integral vanishes because $f(z)$ is everywhere analytic. $f(z) = e^z$ does not vanish at the origin.

52. (B)

Using the technique of reduction of order, a second linearly-independent solution has the form

$$y = x \left(\int^x \left[\frac{1}{s^2} \exp \left(- \int^s t dt \right) \right] ds \right).$$

This evaluates to $y = x \int^x \frac{e^{-t^2/2}}{t^2} dt$.

53. (E)

The curves are recognized as a pair of hyperbola. The general equation for this conic section is $\pm x^2/a^2 \mp y^2/b^2 = 1$. Since the hyperbolas have vertices at $(\pm 2, 0)$, it must hold that $a^2 = 4$. The only equation with the correct form is (E).

54. (C)

Note that

$$i^i = e^{(i\pi/2 + 2n\pi)i} = e^{-\pi/2 + 2n\pi},$$

where n is any integer. For $n = 1$, $i^i = e^{3\pi/2}$.

55. (D)

Monotonicity and continuity (or a closed and bounded interval) are sufficient conditions for integrability. In addition, all bounded and piecewise smooth functions are also integrable. However, consider the function

$$f(x) = \begin{cases} 0, & \text{if } x = 0 \\ 1/x, & \text{if } 0 < x \leq 1. \end{cases}$$

This function is clearly not integrable, but has only one discontinuity.

56. (B)

At worst, y can be discontinuous (hence not differentiable) at every point other than zero. For example, $y = x^2$ if x is rational and $y = -x^2$ if x is irrational. However, y is differentiable at the origin. To see this, examine the difference quotient at $x = 0$:

$$-x \leq (y - 0)/(x - 0) \leq x.$$

As $x \rightarrow 0$, the difference quotient goes to 0 as well, showing that y' exists at the origin.

57. (D)

Take the logarithm of $f(x)$:

$$\ln f(x) = \frac{1}{\ln x} \cdot \ln x = 1,$$

so that $f(x) = e$ for $x \neq 1$. Clearly $f(1) = e$ also.

58. (D)

Applying the ratio test, we obtain

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{|x + 2|}{3}.$$

Since the limit must be less than unity for convergence, we have that $|x + 2| < 3$ or $-5 < x < 2$. At the right and left end of the interval, the series converges by the comparison test.

59. (C)

Note that this algorithm implements the formula

$$a_{n+1} = \begin{cases} a_n/2, & \text{if } a_n \text{ is even;} \\ 3a_n + 1, & \text{if } a_n \text{ is odd.} \end{cases}$$

Since the first value output is made *after* the input is changed, the first output values must be 40. The remaining output values are obtained by applying the above formula until the output value 1 is produced at which point the algorithm terminates.

60. (B)

In R^2 , sets are compact if and only if they are both closed and bounded. (A) is closed but not bounded. (B) is closed and bounded. Therefore, it is the answer. (C), (D), and (E) are open sets.

61. (E)

In Z_{13} , we have two simultaneous equations

$$\bar{6}x + \bar{5}y = \bar{0} \text{ and } \bar{8}x + by = \bar{0}.$$

Eliminating x , we find that the remaining equation for y is satisfied only if

$$\bar{3}b = \bar{7}.$$

Since $33 \equiv 7 \pmod{13}$, $b = 11$.

62. (C)

That every field is also an integral domain follows from the definition of a field. However, not every field has a finite subfield. For example, the rational numbers possess no finite subfield. The transcendental real numbers are not a field because 0 (the additive identity) is not in the set.

63. (A)

Note that $27 = 2^4 + 2^3 + 2^1 + 2^0 = (011011)_2$ and $52 = 2^5 + 2^4 + 2^2 = (110100)_2$. The result digit is 1 if and only if both respective digits are 1. The result is therefore $(010000)_2 = 2^4 = 16$.

64. (B)

Every density function must integrate to unity. (E) is not integrable, while (A), (C), and (D) integrate to something other than 1. (B), in fact, is an example of an exponential distribution.

65. (C)

The contraposition of an implication is that the negation of the conclusion implies the negation of the hypothesis. Thus,

$$\neg(\neg R \wedge S) = R \vee \neg S \Rightarrow \neg(P \wedge Q) = \neg P \vee \neg Q.$$

66. (A)

First, note that the eigenvalues of A are $\lambda_1 = 2$ and $\lambda_2 = 1$. Corresponding eigenvectors are $x_1^T = (2, 3)$ and $x_2^T = (1, 2)$. These eigenvalues form the columns of a matrix S which can be used to diagonalize A . The similarity matrix is therefore given by (A).

The Graduate Record Examination in
MATHEMATICS

Answer Sheets

The Graduate Record Examination in **MATHEMATICS** Answer Sheet Test

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| 1. (A) (B) (C) (D) (E) | 23. (A) (B) (C) (D) (E) | 45. (A) (B) (C) (D) (E) |
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The Graduate Record Examination in **MATHEMATICS** Answer Sheet Test

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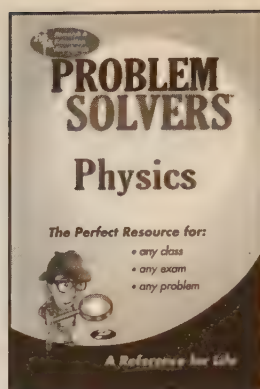
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